

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\104D2A-104-D2B-STACKEDR1.pvdb

Date of Analysis : Aug 12, 2026 2:45pm

PV Elite 28 U2, April 2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 7610.000 x 5600.000 mm ID	
Vessel Design Length, Tangent to Tangent	305.00 cm
Distance of Bottom Tangent above Grade	670.00 cm
Distance of Base above Grade	30.00 cm
Specified Datum Line Distance	640.00 cm
Internal Design Temperature	275 °C
Internal Design Pressure	50.897 kgf/cm ²
External Design Temperature	0 °C
Maximum Allowable Working Pressure	51.1 kgf/cm ²
Shop Test Pressure	67.214 kgf/cm ²
Field Test Pressure	66.166 kgf/cm ²
Required Minimum Design Metal Temperature	-12.0 °C
Warmest Computed Minimum Design Metal Temperature	48.0 °C

Warning: Computed overall MDMT was higher than the required value !

Wind Design Code	ASCE/SEI 7-16
Earthquake Design Code	ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-387 11	2	...	K11789	+IT
Head	SA-387 11	2	...	K11789	
Skirt	SA-516 70	...	...	K02700	
Skirt	SA-387 11	2	...	K11789	
Nozzle	SA-182 F11	2	...	K11572	
Nozzle	SA-335 P11	...	...	K11597	+IT
Nozzle Flg	SA-182 F11	2	...	K11572	
Nozzle Flg	SA-105	...	...	K03504	
Basering	SA-36	...	...	K02600	
Base Bolting	SA-307 A	...	6 <= t <= 100		

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (kgf/cm² & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Skirt-3	...	...	...	1.6000	N/A	No
Skirt-2	...	...	...	1.6000	N/A	No
Skirt-1	...	...	...	1.6000	N/A	No
Bottom Head	50.897	0.00	51.100	3.0000	N/A	No
Shell 1	50.897	0.00	51.100	3.0000	N/A	No
Top Head	50.897	0.00	51.100	3.0000	N/A	No
INT-SKIRT-1	...	...	...	1.6000	N/A	No
INT-SKIRT-2	...	...	...	1.6000	N/A	No
INT-SKIRT-3	...	...	...	3.0000	N/A	No
INT-SKIRT-4	...	...	...	1.6000	N/A	No
INT-SKIRT-5	...	...	...	3.0000	N/A	No
B-DISHED-END-102A	50.897	0.00	52.100	3.0000	N/A	No
SHELL-104-D2A	50.897	0.00	51.200	3.0000	N/A	No
T-DISHED-END-102A	50.897	0.00	52.100	3.0000	N/A	No



Element Types and Properties:

Element Type	"To" Elev cm	Element Length cm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Skirt	-385.0	255.0	34.0	34.0	...	...	0.70	0.70
Skirt	-100.0	285.0	34.0	34.0	...	...	0.70	0.70
Skirt	0.0	100.0	34.0	34.0	...	...	0.70	0.50
Sphere	0.0	0.0	74.0	66.0	65.8	...	1.00	1.00
Cylinder	205.0	205.0	130.0	130.0	129.6	...	1.00	1.00
Sphere	205.0	0.0	74.0	66.0	65.8	...	1.00	1.00
Skirt	337.5	132.5	34.0	34.0	...	...	0.70	0.70
Skirt	637.5	300.0	34.0	34.0	...	...	0.70	0.70
Skirt	937.5	300.0	34.0	34.0	...	...	0.70	0.70
Skirt	1237.5	300.0	34.0	34.0	...	...	0.70	0.70
Skirt	1370.0	132.5	34.0	34.0	...	...	0.70	0.70
Sphere	1370.0	0.0	58.0	52.0	50.8	...	1.00	1.00
Cylinder	1470.0	100.0	100.0	100.0	99.5	...	1.00	1.00
Sphere	1470.0	0.0	58.0	52.0	50.8	...	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N2 36"	WRC-107/537	0.993	n/a	Passed
N1 36"	WRC-107/537	0.993	n/a	Passed
N1-104-D2A	WRC-107/537	0.973	n/a	Passed

Wind/Earthquake Shear, Bending:

From	To	Distance to Support cm	Cumulative Wind Shear kgf	Earthquake Shear kgf	Wind Bending kgf-m.	Earthquake Bending kgf-m.
10	20	127.5	9904.39	156871	117262	2289322
20	30	397.5	8991.28	156474	93170.4	1889806
30	40	590	7854.67	155353	69164.9	1445453
40	50	639.999	7496.59	154829	61489.5	1290366
50	60	742.5	7022.27	128737	61489.3	1290362
60	70	844.999	6240.84	103864	47894.8	1051949
70	80	911.25	6240.82	91981.7	47894.6	1051946
80	90	1127.5	5754.31	90910.2	39947.8	930780
90	100	1427.5	4420.47	87894.6	24685.6	662573
100	110	1727.5	3190.48	83729.2	13269.1	405137
110	120	1943.75	2074.11	79425.4	5372.26	160404
120	130	2010	1702.19	77680.5	2870.51	56323.9
130	140	2060	1702.18	42390.3	2870.45	56322.1
140	150	2110	1145.39	19924.1	1446.7	25165.5

Abs Max of the all of the Stress Ratio's : 0.9728

Basing Data : Continuous Top Ring W/Gussets

Thickness of Basing	44.0000	mm
Inside Diameter of Basing	7312.0000	mm
Outside Diameter of Basing	7799.9995	mm
Nominal Diameter of Bolts	48.0000	mm
Diameter of Bolt Circle	7660.0000	mm
Number of Bolts	32	

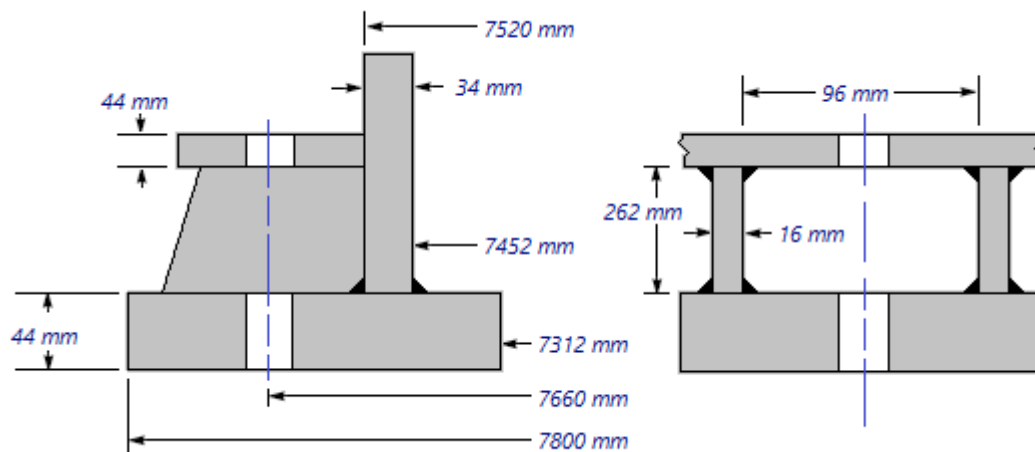
Thickness of Gusset Plates	16.0000	mm
Average Width of Gusset Plates	140.0000	mm
Height of Gussets	262.0000	mm
Distance between Gussets	96.0000	mm
Thickness of Top Plate or Ring	44.0000	mm
Radial Width of the Top Plate	140.0000	mm



Number of Gussets

64

Basering Sketch for Cylindrical Skirts:



Basering Cross Section View

Loads for Foundation/Support Design:

Total Wind Shear on Support	28458.	kgf
Total Earthquake Shear on Support	175425.	kgf
Wind Moment on Support	519453.	kgf-m.
Earthquake Moment on Support	2691513.	kgf-m.
Maximum Initial Bolt Load	13070.	kgf

Wind and Earthquake moments include the effects of user defined forces and moments if any exist in the job and were specified to act (compute loads and stresses) during these cases. Also included are moment effects due to eccentric weights if any are present in the input.

Weights:

Fabricated - Bare w/o removable internals	474361.3	kg
Shop Test - Fabricated + water (full)	893501.5	kg
Shipping - Fab. + rem. intls.+ shipping app.	507913.6	kg
Erected - Fab. + rem. intls.+ insul. (etc)	511988.5	kg
Empty - Fab. + intls. + details + wghts.	511988.5	kg
Operating - Empty + operating liquid (no ca)	1000454.0	kg
Field Test - Empty weight + test medium (full)	927053.8	kg

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Nozzle Calculation Summary (kgf/cm² & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N2 36"	51.1	...	...	OK	19.30	OK	Passed
UC1- 24"	51.1	...	...	OK	13.87	OK	Passed
H1 8"	51.1	...	...	OK	10.16	OK	Passed
K4 3"	51.1	...	...	OK	7.80	OK	Passed
N1 36"	51.1	...	...	OK	19.30	OK	Passed
K1 2"	51.1	...	...	OK	6.42	OK	Passed
M1 24"	51.1	...	...	OK	13.87	OK	Passed
K5 4"	51.1	...	...	OK	8.26	OK	Passed
K2 2"	51.1	...	...	OK	6.42	OK	Passed
N2-104-D2A	52.1	...	...	OK	19.30	OK	Passed
UC1-104-D2A	52.1	...	...	OK	13.87	OK	Passed
H1-104-D2A	52.1	...	...	...		OK	Passed
K4-104-D2A	52.1	...	...	OK	7.80	OK	Passed
K2	51.2	...	...	OK	6.42	OK	Passed
N1-104-D2A	52.1	...	...	OK	19.30	OK	Passed
M1-104-D2A	51.2	...	...	...		OK	Passed
M1-104-D2A	52.1	...	...	...		OK	Passed
K1-104-D2A	52.1	...	...	OK	6.42	OK	Passed
K3-104-D2A	52.1	...	...	OK	6.42	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 51.1 Nozzle : K2 2"
 Minimum MAWP Shells/Flanges : 51.1 Element : Shell 1
 Minimum MAPnc Shells/Flanges : 52.3 Element : Shell 1

Computed Vessel M.A.W.P. : 51.1 kgf/cm²

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	Y Coordinate mm	Layout Angle deg	Dia. Limit mm
40	N2 36"	0.000	0.000	1744.800
40	UC1- 24"	0.000	180.000	1167.200
40	H1 8"	0.000	0.000	397.343
40	K4 3"	0.000	150.000	214.900
60	N1 36"	0.000	90.000	1744.800
60	K1 2"	0.000	0.000	186.325
60	M1 24"	0.000	180.000	1135.200
60	K5 4"	0.000	60.000	240.300
60	K2 2"	0.000	0.000	186.325
120	N2-104-D2A	0.000	0.000	1744.800
120	UC1-104-D2A	0.000	270.000	1167.200
120	H1-104-D2A	0.000	180.000	405.700
120	K4-104-D2A	0.000	0.000	186.900
130	K2	14200.029	180.000	254.325
140	N1-104-D2A	0.000	0.000	1760.800
140	M1-104-D2A	0.000	0.000	1231.200
140	K1-104-D2A	0.000	180.000	158.325
140	K3-104-D2A	0.000	270.000	158.325

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.
 lc - Arc length along the inside vessel surface in the circ. direction



If any interferences/violations are found, they will be noted below.

Diameter Limit clash between : H1-104-D2A and N2-104-D2A.

The Centerline Dist. is 1022.12 while the sum of the limit Radii is 1075.25 mm.

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
K1 2"	2.000 in	160	WNF	2.375	8.738	...	...	6747.05	600
K2 2"	2.000 in	160	WNF	2.375	8.738	...	...	683.35	600
K2	2.000 in	160	WNF	2.375	8.738	...	...	600.94	600
K1-104-D2A	2.000 in	160	WNF	2.375	8.738	...	...	4755.94	600
K3-104-D2A	2.000 in	160	WNF	2.375	8.738	...	...	981.85	600
K4 3"	3.000 in	160	WNF	3.500	11.125	...	...	774.17	600
K4-104-D2A	3.000 in	160	WNF	3.500	11.125	...	...	1104.17	600
K5 4"	4.000 in	160	WNF	4.500	13.487	...	...	448.18	600
H1 8"	8.000 in	100	WNF	8.625	15.088	...	...	775.52	600
H1-104-D2A	8.000 in	80	WNF	8.625	12.700	...	...	2502.00	600
UC1- 24"	24.000 in	Actual	WNF	24.000	16.000	...	...	2866.00	600
M1 24"	24.000 in	Actual	WNF	24.000	24.000	...	...	595.90	600
UC1-104-D2A	24.000 in	Actual	WNF	24.000	16.000	...	...	2502.00	600
M1-104-D2A	24.000 in	Actual	WNF	25.575	20.000	...	...	1117.30	600
N2 36"	36.000 in	Actual	WNF	36.000	24.000	...	...	244.30	600
N1 36"	36.000 in	Actual	WNF	36.000	24.000	...	...	666.00	600
N2-104-D2A	36.000 in	Actual	WNF	36.000	24.000	...	...	239.26	600
N1-104-D2A	36.000 in	Actual	WNF	36.000	20.000	...	...	952.00	600

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
K1 2"	SA-182 F11	60.000	14.000	...	...	14.000
K2 2"	SA-182 F11	60.000	12.000	...	...	...
K2	SA-182 F11	52.000	10.000	...	...	...
K1-104-D2A	SA-182 F11	52.000	10.000	...	...	10.000
K3-104-D2A	SA-182 F11	52.000	10.000	...	...	...
K4 3"	SA-182 F11	60.000	10.000	...	...	...
K4-104-D2A	SA-335 P11	52.000	10.000	...	...	...
K5 4"	SA-182 F11	60.000	10.000	...	...	...
H1 8"	SA-182 F11	60.000	10.000	...	...	...
H1-104-D2A	SA-335 P11	52.000	10.000	...	...	14.000
UC1- 24"	SA-182 F11	60.000	10.000	...	...	14.000
M1 24"	SA-182 F11	60.000	10.000	...	...	...
UC1-104-D2A	SA-182 F11	52.000	10.000	...	...	14.000
M1-104-D2A	SA-182 F11	52.000	10.000	...	...	...
N2 36"	SA-182 F11	60.000	10.000	...	...	...
N1 36"	SA-182 F11	60.000	10.000	...	...	14.000
N2-104-D2A	SA-182 F11	52.000	10.000	...	...	...
N1-104-D2A	SA-182 F11	52.000	10.000	...	...	14.000

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum cm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
K1 2"	...	0.0	300.00	6375.00	Top Head



K2 2"	...	0.0	226.85	0.00	Top Head
K2	1420.003	180.0	500.00	0.00	SHELL-104-D2A
K1-104-D2A	...	180.0	900.00	3800.00	T-DISHED-END-102
K3-104-D2A	...	270.0	900.00	0.00	T-DISHED-END-102
K4 3"	...	150.0	707.90	0.00	Bottom Head
K4-104-D2A	...	0.0	800.00	0.00	B-DISHED-END-102
K5 4"	...	60.0	300.00	0.00	Top Head
H1 8"	...	0.0	707.90	0.00	Bottom Head
H1-104-D2A	...	180.0	1850.00	600.00	B-DISHED-END-102
UC1- 24"	...	180.0	2500.00	300.00	Bottom Head
M1 24"	...	180.0	300.00	0.00	Top Head
UC1-104-D2A	...	270.0	1850.00	600.00	B-DISHED-END-102
M1-104-D2A	...	0.0	700.00	0.00	T-DISHED-END-102
N2 36"	...	0.0	150.00	0.00	Bottom Head
N1 36"	...	90.0	300.00	300.00	Top Head
N2-104-D2A	...	0.0	150.00	0.00	B-DISHED-END-102
N1-104-D2A	...	0.0	600.00	300.00	T-DISHED-END-102



Bill of Materials:

QTY	DESCRIPTION	MATERIAL
2	SPHERICAL HEAD: 74.0mm THK X 7414.0mm ID X 0.0cm	SA-387 11
2	SPHERICAL HEAD: 58.0mm THK X 5648.0mm ID X 0.0cm	SA-387 11
6	PLATFORM: 1000mm X 1500mm WIDE	
1	LADDER: 3800.0mm	...
1	LADDER: 6200.0mm	...
2	INSULATION: 3707mm X 100mm THK	
1	INSULATION: 2050mm X 100mm THK	
3	LADDER: 4000.0mm	...
1	INSULATION: 2824mm X 100mm THK	
1	LADDER: 2700.0mm	...
1	INSULATION: 1000mm X 100mm THK	
1	PLATFORM: 1000mm X 5000mm WIDE	
1	LADDER: 3200.0mm	...
1	INSULATION: 2800mm X 100mm THK	
3	CLASS 600 GR 1.9, 24.0" BLIND FLANGE(S)	SA-182 F11
2	CLASS 600 GR 1.9, 8.0" BLIND FLANGE(S)	SA-182 F11
2	CLASS 600 GR 1.9, 3.0" BLIND FLANGE(S)	SA-182 F11
4	CLASS 600 GR 1.9, 2.0" BLIND FLANGE(S)	SA-182 F11
1	CLASS 600 GR 1.9, 4.0" BLIND FLANGE(S)	SA-182 F11
1	CLASS 600 GR 1.9, 24.0" BLIND FLANGE(S)	SA-105
1	CLASS 600 GR 1.9, 2.0" BLIND FLANGE(S)	SA-105
1	BASERING: 7800.0mm OD X 7312.0 ID X 44.0mm	SA-36
32	BASERING BOLTS: 48.0mm DIA	SA-307 A
32	BASERING BOLT NUTS: 48.0mm DIA	...
1	BASERING TOP RING: 7800.0mm OD X 7520.0 ID X 44.0mm	...
64	BASERING GUSSET PLATES: 140.0mm X 140.0mm X 262.0mm X 16.0mm	...
5	CYLINDRICAL SKIRT SEGMENT 34.0mm THK X 3966.8mm X 255.0cm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 34.0mm THK X 3576.6mm X 255.0cm	SA-516 70
5	CYLINDRICAL SKIRT SEGMENT 34.0mm THK X 3966.8mm X 285.0cm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 34.0mm THK X 3576.6mm X 285.0cm	SA-516 70
5	CYLINDRICAL SKIRT SEGMENT 34.0mm THK X 4014.4mm X 100.0cm	SA-387 11
1	CYLINDRICAL SKIRT SEGMENT 34.0mm THK X 3338.9mm X 100.0cm	SA-387 11
5	CYLINDRICAL SEGMENT 130.0mm THK X 4000.4mm X 205.0cm	SA-387 11



1	CYLINDRICAL SEGMENT 130.0mm THK X 3088.1mm X 205.0cm	SA-387 11
7	CYLINDRICAL SKIRT SEGMENT 34.0mm THK X 3064.8mm X 132.5cm	SA-387 11
1	CYLINDRICAL SKIRT SEGMENT 34.0mm THK X 21453.6mm X 132.5cm	SA-387 11
6	CYLINDRICAL SKIRT SEGMENT 34.0mm THK X 3040.8mm X 300.0cm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 34.0mm THK X 3105.5mm X 300.0cm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 34.0mm THK X 21350.4mm X 300.0cm	SA-516 70
6	CON SKIRT SEGMENT 34.0mm THK X 2330.9mm X 300.0cm X 3040.8mm	SA-516 70
1	CON SKIRT SEGMENT 34.0mm THK X 2380.4mm X 300.0cm X 3105.5mm	SA-516 70
1	CON SKIRT SEGMENT 34.0mm THK X 16365.6mm X 300.0cm X 21350.4mm	SA-516 70
3	CYLINDRICAL SKIRT SEGMENT 34.0mm THK X 3025.2mm X 300.0cm	SA-516 70
2	CYLINDRICAL SKIRT SEGMENT 34.0mm THK X 3074.7mm X 300.0cm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 34.0mm THK X 2628.4mm X 300.0cm	SA-516 70
5	CYLINDRICAL SKIRT SEGMENT 34.0mm THK X 3048.0mm X 132.5cm	SA-387 11
1	CYLINDRICAL SKIRT SEGMENT 34.0mm THK X 2613.6mm X 132.5cm	SA-387 11
5	CYLINDRICAL SEGMENT 100.0mm THK X 3003.4mm X 100.0cm	SA-387 11
1	CYLINDRICAL SEGMENT 100.0mm THK X 2575.4mm X 100.0cm	SA-387 11
1	NAMEPLATE	...



Minimum Design Metal Temperature Results Summary (°C):

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Bottom Head	[9]	A	42	42		0.999	66.000	1.00	Yes
Shell 1	[8]	!	-12	-12		1.000	130.000	1.00	Yes
Top Head	[9]	A	42	42		0.999	66.000	1.00	Yes
B-DISHED-END-1	[9]	A	37	36		0.979	52.000	1.00	Yes
SHELL-104-D2A	[8]	A	48	48		0.998	100.000	1.00	Yes
T-DISHED-END-1	[9]	A	37	36		0.979	52.000	1.00	Yes
N2 36"	[1]	A	42	42		0.996	66.000	1.00	Yes
Nozzle Flg	[5]	B	-18	-48					
UC1- 24"	[1]	A	42	42		0.996	66.000	1.00	Yes
Nozzle Flg	[5]	B	-18	-48					
H1 8"	[1]	A	42	42		0.996	66.000	1.00	Yes
Nozzle Flg	[5]	B	-18	-48					
K4 3"	[1]	A	-29	-104		0.235	9.735	1.00	Yes
Nozzle Flg	[5]	B	-18	-48					
N1 36"	[1]	A	42	42		0.996	66.000	1.00	Yes
Nozzle Flg	[5]	B	-18	-48					
K1 2"	[1]	A	-29	-104		0.203	7.645	1.00	Yes
Nozzle Flg	[5]	B	-18	-48					
M1 24"	[1]	A	42	42		0.996	66.000	1.00	Yes
Nozzle Flg	[5]	B	-18	-48					
K5 4"	[1]	A	-24	-104		0.203	11.801	1.00	Yes
Nozzle Flg	[5]	B	-18	-48					
K2 2"	[1]	A	-29	-104		0.203	7.645	1.00	Yes
Nozzle Flg	[5]	B	-18	-48					
N2-104-D2A	[1]	A	37	36		0.976	52.000	1.00	Yes
Nozzle Flg	[5]	A	-18	-48					
UC1-104-D2A	[1]	A	37	36		0.976	52.000	1.00	Yes
Nozzle Flg	[5]	A	-18	-48					
H1-104-D2A	[1]	A	37	36		0.976	52.000	1.00	Yes
Nozzle Flg	[5]	B	-18	-48					
K4-104-D2A	[1]	A	37	36		0.976	52.000	1.00	Yes
Nozzle Flg	[5]	B	-18	-48					
K2	[1]	A	-8	-104		0.203	7.645	1.00	Yes
Nozzle Flg	[5]	A	-18	-48					
N1-104-D2A	[1]	A	37	36		0.976	52.000	1.00	Yes
Nozzle Flg	[5]	A	-18	-48					
M1-104-D2A	[1]	A	37	36		0.976	52.000	1.00	Yes
Nozzle Flg	[5]	B	-18	-48					
K1-104-D2A	[1]	A	-29	-104		0.203	7.645	1.00	Yes
Nozzle Flg	[5]	B	-18	-48					
K3-104-D2A	[1]	A	-29	-104		0.203	7.645	1.00	Yes
Nozzle Flg	[5]	B	-18	-48					
Nozl. Bolting[22]			-48			1.000	64.000	1.00	No

Warmest MDMT: 48 48

Required Minimum Design Metal Temperature -12.0 °C
 Warmest Computed Minimum Design Metal Temperature 48.0 °C

Warning: Computed overall MDMT was higher than the required value !

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3



- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.

UCS-66(g) was not considered.

UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.

UCS-66(i) applies to impact tested materials not by specification and

UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).

The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

Note	30	50	There is a high jump in the Joint Eff.
Note	60	80	There is a high jump in the Joint Eff.
Note	110	130	There is a high jump in the Joint Eff.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Units used in this Analysis (Kgfc^{m2}):

Name	System Unit	Constant	User Unit
Length	Feet	30.4800	cm
Force	Pounds	0.4536	kgf
Mass	Pounds	0.4536	kg
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	kgf-m.
Stress	lbs./sq.in.	0.0703	kgf/cm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0703	kgf/cm ²
Elast. Modulus	lbs./sq.in.	0.0703	kgf/cm ²
Pipe Density	lbs./cu.in.	0.0277	kg/cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Fluid Density	lbs./cu.ft.	16.0183	kg/m ³
Wind Speed	miles/hr	1.6093	km/hr
Tray Weight	lbs./sq.ft.	0.0005	kgf/cm ²
Inertia	in. ⁴	41.6231	cm ⁴
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8824	kgf/m ²
Elevation	Feet	0.3048	m
Volume	in. ³	16.3871	cm ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Design Internal Pressure (for Hydrotest)	50.897	kgf/cm ²
Design Internal Temperature	275.0	°C
Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Minimum Design Metal Temperature	-12.0	°C
Type of Construction	Welded	
Special Service	None	
Degree of Radiography	RT-1	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(c)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No

Field Pressure Test:

Type of Pressure Test	UG-99(b) Note [35]
Pressure Test Position	Vertical
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+1.1WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+1.1WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+1.1WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+1.1WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW



Load Case 16		IP+VE+EW	
Load Case 17		NP+VO+OW	
Load Case 18		FS+BS+IP+OW	
Load Case 19		FS+BS+EP+OW	
Load Case 20		BL+IP+OW	
Wind Design Code		ASCE/SEI 7-16	
Wind Load Reduction Scale Factor		1.100	
Basic Wind Speed	[V]	97.2	km/hr
Surface Roughness Category		C: Open Terrain	
Type of Surface		Moderately Smooth	
Base Elevation		30	cm
Percent Wind for Hydrotest		50.0	
Using User defined Wind Press. Vs Elev.		N	
Height of Hill or Escarpment	H or Hh	0	cm
Distance Upwind of Crest	Lh	0	cm
Distance from Crest to the Vessel	x	0	cm
Type of Terrain (Hill, Escarpment)		Flat	
Damping Factor (Beta) for Wind (Ope)		0.0100	
Damping Factor (Beta) for Wind (Empty)		0.0000	
Damping Factor (Beta) for Wind (Filled)		0.0000	
Site Elevation above Sea Level		0	cm
Seismic Design Code		ASCE/SEI 7-16	
Seismic Load Reduction Scale Factor		1.000	
Importance Factor		2.000	
Table Value Fa		1.600	
Table Value Fv		2.400	
Max. Mapped Res. Acceleration	[Ss]	0.147	
Max. Eff. Ground Acceleration	[S]	0.065	
Force Modification Factor R		2.000	
Site Class		D	
Component Elevation Ratio	z/h	0.000	
Amplification Factor	Ap	2.500	
Force Factor		0.000	
Consider Vertical Acceleration		Yes	
Minimum Acceleration Multiplier		0.000	
User Value of Sds (used if > 0)		0.000	
User Value of Sd1 (used if > 0)		0.000	
Moment Reduction Factor Tau		1.000	
Design Pressure + Static Head		Y	
Consider MAP New and Cold in Noz. Design		N	
Consider External Loads for Nozzle Des.		Y	
Use ASME VIII-1 Appendix 1-9		N	
Perform Blast Load Analysis		No	
Material Database Year	Current w/Addenda or Code Year		

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/14
Element From Node	10



Element To Node	20	
Element Type	Skirt Support	
Description	Skirt-3	
Distance "FROM" to "TO"	255	cm
Skirt Outside Diameter	7520	mm
Diameter of Skirt at Base	7520	mm
Skirt Thickness	34	mm
Internal Corrosion Allowance	0.8	mm
Nominal Thickness	34	mm
External Corrosion Allowance	0.8	mm
Design Temperature Internal Pressure	50	°C
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Allowable Stress, Ambient	1407.1	kgf/cm²
Allowable Stress, Operating	1407.1	kgf/cm²
Allowable Stress, Hydrotest	2386	kgf/cm²
Material Density	0.00775	kg/cm³
P Number Thickness	32.004	mm
Yield Stress, Operating	2606.3	kgf/cm²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	0.7	
Efficiency, Head-to-Skirt or Circ. Seam	0.7	
Weld is pre-Heated	No	
Element From Node	10	
Detail Type	Weight	
Detail ID	Tailing+Bracing	
Dist. from "FROM" Node / Offset dist	30	cm
Miscellaneous Weight	2550	kgf
Offset from Element Centerline	3200	mm

Element#	2/14	
Element From Node	20	
Element To Node	30	
Element Type	Skirt Support	
Description	Skirt-2	
Distance "FROM" to "TO"	285	cm
Skirt Outside Diameter	7520	mm
Diameter of Skirt at Base	7520	mm
Skirt Thickness	34	mm
Internal Corrosion Allowance	0.8	mm
Nominal Thickness	34	mm
External Corrosion Allowance	0.8	mm
Design Temperature Internal Pressure	50	°C
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	0.7	
Efficiency, Head-to-Skirt or Circ. Seam	0.7	
Weld is pre-Heated	No	
Element From Node	20	
Detail Type	Platform	
Detail ID	PLTF-1	
Dist. from "FROM" Node / Offset dist	100	cm
Platform Start Angle (degrees)	180.0	
Platform End Angle (degrees)	270.0	
Platform Wind Area	15000	cm²
Platform Weight	2038.5	kgf
Platform Railing Weight	0	kgf/cm
Platform Grating Weight	0.017	kgf/cm²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	185	mm



Platform Force Coefficient	1.0	
Ladder Layout Angle	270.0	
Ladder Start Elevation	0	cm
Ladder End Elevation	380	cm
Unit Weight of Ladder	0.3773	kgf/cm
Platform Length (top head platform)	0	mm

Element#	3/14	
Element From Node	30	
Element To Node	40	
Element Type	Skirt Support	
Description	Skirt-1	
Distance "FROM" to "TO"	100	cm
Skirt Outside Diameter	7520	mm
Diameter of Skirt at Base	7520	mm
Skirt Thickness	34	mm
Internal Corrosion Allowance	0.8	mm
Nominal Thickness	34	mm
External Corrosion Allowance	0.8	mm
Design Temperature Internal Pressure	275	°C
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-387 11	
Allowable Stress, Ambient	1509.1	kgf/cm ²
Allowable Stress, Operating	1509.1	kgf/cm ²
Allowable Stress, Hydrotest	2844.9	kgf/cm ²
Material Density	0.00775	kg/cm ³
P Number Thickness	16.002	mm
Yield Stress, Operating	2539	kgf/cm ²
UCS-66 Chart Curve Designation	A	
External Pressure Chart Name	CS-3	
UNS Number	K11789	
Class / Thickness / Grade	2 ::	
Product Form	Plate	
Efficiency, Longitudinal Seam	0.7	
Efficiency, Head-to-Skirt or Circ. Seam	0.5	
Weld is pre-Heated	No	

Element#	4/14	
Element From Node	40	
Element To Node	50	
Element Type	Spherical	
Description	Bottom Head	
Distance "FROM" to "TO"	0	cm
Inside Diameter	7414	mm
Element Thickness	66	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	74	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	50.897	kgf/cm ²
Design Temperature Internal Pressure	275	°C
Design External Pressure	0	kgf/cm ²
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.23	
Material Name	SA-387 11	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	40	
Detail Type	Platform	
Detail ID	PLTF-2	
Dist. from "FROM" Node / Offset dist	0	cm
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	360.0	
Platform Wind Area	55197	cm ²



Platform Weight	7780.9	kgf
Platform Railing Weight	0	kgf/cm
Platform Grating Weight	0.017	kgf/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	185	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	cm
Ladder End Elevation	620	cm
Unit Weight of Ladder	0.3773	kgf/cm
Platform Length (top head platform)	0	mm

Element From Node	40	
Detail Type	Insulation	
Detail ID	Ins: 40	
Dist. from "FROM" Node / Offset dist	-370.7	cm
Height/Length of Insulation	370.7	cm
Thickness of Insulation	100	mm
Density	0.000352	kg/cm ³

Element From Node	40	
Detail Type	Nozzle	
Detail ID	N2 36"	
Dist. from "FROM" Node / Offset dist	0	mm
Nozzle Diameter	36	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	5478.3	kgf
Grade of Attached Flange	GR 1.9	
Nozzle Matl	SA-182 F11	

Element From Node	40	
Detail Type	Nozzle	
Detail ID	UC1- 24"	
Dist. from "FROM" Node / Offset dist	2500	mm
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	180.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	2220.5	kgf
Grade of Attached Flange	GR 1.9	
Nozzle Matl	SA-182 F11	

Element From Node	40	
Detail Type	Nozzle	
Detail ID	H1 8"	
Dist. from "FROM" Node / Offset dist	2000	mm
Nozzle Diameter	8	in.
Nozzle Schedule	100	
Nozzle Class	600	
Layout Angle	0.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	0	kgf
Grade of Attached Flange	GR 1.9	
Nozzle Matl	SA-182 F11	

Element From Node	40	
Detail Type	Nozzle	
Detail ID	K4 3"	
Dist. from "FROM" Node / Offset dist	2000	mm
Nozzle Diameter	3	in.
Nozzle Schedule	160	
Nozzle Class	600	
Layout Angle	150.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	0	kgf



Grade of Attached Flange	GR 1.9	
Nozzle Matl	SA-182 F11	
Element From Node	40	
Detail Type	Weight	
Detail ID	HH Hinge	
Dist. from "FROM" Node / Offset dist	-200	cm
Miscellaneous Weight	99.998	kgf
Offset from Element Centerline	2000	mm
Element From Node	40	
Detail Type	Weight	
Detail ID	UC1 Hinge	
Dist. from "FROM" Node / Offset dist	-200	cm
Miscellaneous Weight	99.998	kgf
Offset from Element Centerline	1500	mm
Element From Node	40	
Detail Type	Weight	
Detail ID	N2 inner dish	
Dist. from "FROM" Node / Offset dist	-250	cm
Miscellaneous Weight	5009.9	kgf
Offset from Element Centerline	0	mm
Element From Node	40	
Detail Type	Weight	
Detail ID	WIRE SCREEN	
Dist. from "FROM" Node / Offset dist	0	cm
Miscellaneous Weight	99.998	kgf
Offset from Element Centerline	2000	mm
Element From Node	40	
Detail Type	Weight	
Detail ID	SF weld	
Dist. from "FROM" Node / Offset dist	-5	cm
Miscellaneous Weight	4544.9	kgf
Offset from Element Centerline	2200	mm
Element From Node	40	
Detail Type	Weight	
Detail ID	Support balls	
Dist. from "FROM" Node / Offset dist	-141.5	cm
Miscellaneous Weight	205000	kgf
Offset from Element Centerline	0	mm

Element#	5/14	
Element From Node	50	
Element To Node	60	
Element Type	Cylinder	
Description	Shell 1	
Distance "FROM" to "TO"	205	cm
Element Outside Diameter	7610	mm
Element Thickness	130	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	130	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	50.897	kgf/cm ²
Design Temperature Internal Pressure	275	°C
Design External Pressure	0	kgf/cm ²
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.23	
Material Name	SA-387 11 +IT	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	50
Detail Type	Insulation



Detail ID	Ins: 50	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Insulation	205	cm
Thickness of Insulation	100	mm
Density	0.000352	kg/cm ³
Element From Node	50	
Detail Type	Weight	
Detail ID	Catalyst	
Dist. from "FROM" Node / Offset dist	103	cm
Miscellaneous Weight	164717	kgf
Offset from Element Centerline	0	mm
Element From Node	50	
Detail Type	Weight	
Detail ID	Split Flange	
Dist. from "FROM" Node / Offset dist	10	cm
Miscellaneous Weight	349.99	kgf
Offset from Element Centerline	2000	mm
Element From Node	50	
Detail Type	Weight	
Detail ID	Cleat Weight	
Dist. from "FROM" Node / Offset dist	50	cm
Miscellaneous Weight	3554.9	kgf
Offset from Element Centerline	0	mm
Element From Node	50	
Detail Type	Weight	
Detail ID	Lifting Lug-1	
Dist. from "FROM" Node / Offset dist	110	cm
Miscellaneous Weight	2550	kgf
Offset from Element Centerline	3200	mm
Element From Node	50	
Detail Type	Weight	
Detail ID	Lifting Lug-2	
Dist. from "FROM" Node / Offset dist	110	cm
Miscellaneous Weight	2550	kgf
Offset from Element Centerline	3200	mm
Element From Node	50	
Detail Type	Weight	
Detail ID	MISC WEIGHT	
Dist. from "FROM" Node / Offset dist	102.5	cm
Miscellaneous Weight	9999.8	kgf
Offset from Element Centerline	0	mm

Element#	6/14	
Element From Node	60	
Element To Node	70	
Element Type	Spherical	
Description	Top Head	
Distance "FROM" to "TO"	0	cm
Inside Diameter	7414	mm
Element Thickness	66	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	74	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	50.897	kgf/cm ²
Design Temperature Internal Pressure	275	°C
Design External Pressure	0	kgf/cm ²
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.23	
Material Name	SA-387 11	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	



Element From Node	60	
Detail Type	Insulation	
Detail ID	Ins: 60	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Insulation	370.7	cm
Thickness of Insulation	100	mm
Density	0.000352	kg/cm ³

Element From Node	60	
Detail Type	Nozzle	
Detail ID	N1 36"	
Dist. from "FROM" Node / Offset dist	10	mm
Nozzle Diameter	36	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	2194.5	kgf
Grade of Attached Flange	GR 1.9	
Nozzle Matl	SA-182 F11	

Element From Node	60	
Detail Type	Nozzle	
Detail ID	K1 2"	
Dist. from "FROM" Node / Offset dist	1500	mm
Nozzle Diameter	2	in.
Nozzle Schedule	160	
Nozzle Class	600	
Layout Angle	0.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	0	kgf
Grade of Attached Flange	GR 1.9	
Nozzle Matl	SA-182 F11	

Element From Node	60	
Detail Type	Nozzle	
Detail ID	M1 24"	
Dist. from "FROM" Node / Offset dist	2000	mm
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	180.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	1881.3	kgf
Grade of Attached Flange	GR 1.9	
Nozzle Matl	SA-182 F11	

Element From Node	60	
Detail Type	Nozzle	
Detail ID	K5 4"	
Dist. from "FROM" Node / Offset dist	2000	mm
Nozzle Diameter	4	in.
Nozzle Schedule	160	
Nozzle Class	600	
Layout Angle	60.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	0	kgf
Grade of Attached Flange	GR 1.9	
Nozzle Matl	SA-182 F11	

Element From Node	60	
Detail Type	Nozzle	
Detail ID	K2 2"	
Dist. from "FROM" Node / Offset dist	3500	mm
Nozzle Diameter	2	in.
Nozzle Schedule	160	
Nozzle Class	600	
Layout Angle	0.0	
Blind Flange (Y/N)	Y	



Weight of Nozzle (Used if > 0)	0	kgf
Grade of Attached Flange	GR 1.9	
Nozzle Matl	SA-182 F11	
Element From Node	60	
Detail Type	Weight	
Detail ID	Manway Davit	
Dist. from "FROM" Node / Offset dist	282.9	cm
Miscellaneous Weight	99.998	kgf
Offset from Element Centerline	2000	mm
Element From Node	60	
Detail Type	Weight	
Detail ID	N1 Vanes	
Dist. from "FROM" Node / Offset dist	300	cm
Miscellaneous Weight	299.99	kgf
Offset from Element Centerline	10	mm
Element From Node	60	
Detail Type	Weight	
Detail ID	Grating	
Dist. from "FROM" Node / Offset dist	80	cm
Miscellaneous Weight	99.998	kgf
Offset from Element Centerline	0	mm
Element From Node	60	
Detail Type	Weight	
Detail ID	Support Balls	
Dist. from "FROM" Node / Offset dist	40	cm
Miscellaneous Weight	30750	kgf
Offset from Element Centerline	0	mm
Element From Node	60	
Detail Type	Weight	
Detail ID	SF weld	
Dist. from "FROM" Node / Offset dist	5	cm
Miscellaneous Weight	4544.9	kgf
Offset from Element Centerline	3000	mm
Element From Node	60	
Detail Type	Weight	
Detail ID	GAS WIPPER RING	
Dist. from "FROM" Node / Offset dist	100	cm
Miscellaneous Weight	299.99	kgf
Offset from Element Centerline	0	mm
Element From Node	60	
Detail Type	Weight	
Detail ID	WIRE SCREEN-1	
Dist. from "FROM" Node / Offset dist	60	cm
Miscellaneous Weight	99.998	kgf
Offset from Element Centerline	0	mm
Element From Node	60	
Detail Type	Weight	
Detail ID	WIRE SCREEN-2	
Dist. from "FROM" Node / Offset dist	20	cm
Miscellaneous Weight	99.998	kgf
Offset from Element Centerline	0	mm
Element From Node	60	
Detail Type	Weight	
Detail ID	N1-ELBOW	
Dist. from "FROM" Node / Offset dist	370.7	cm
Miscellaneous Weight	2911.9	kgf
Offset from Element Centerline	0	mm
Element From Node	60	
Detail Type	Weight	
Detail ID	K4-elbow	



Dist. from "FROM" Node / Offset dist	370.7	cm
Miscellaneous Weight	250	kgf
Offset from Element Centerline	2000	mm

Element From Node	60	
Detail Type	Weight	
Detail ID	K1-ELBOW	
Dist. from "FROM" Node / Offset dist	370.7	cm
Miscellaneous Weight	150	kgf
Offset from Element Centerline	1500	mm

Element From Node	60	
Detail Type	For./Mom.	
Detail ID	N2 F&M	
Dist. from "FROM" Node / Offset dist	300	cm
Force in X Direction	6559.6	kgf
Force in Y Direction	-9081.9	kgf
Force in Z Direction	6559.6	kgf
Moment about X Axis	18164	kgf-m.
Moment about Y Axis	0	kgf-m.
Moment about Z Axis	18164	kgf-m.
Force/Moment Combination Method	SRSS	

Element#	7/14	
Element From Node	70	
Element To Node	80	
Element Type	Skirt Support	
Description	INT-SKIRT-1	
Distance "FROM" to "TO"	132.5	cm
Skirt Inside Diameter	7455	mm
Diameter of Skirt at Base	7455	mm
Skirt Thickness	34	mm
Internal Corrosion Allowance	1.6	mm
Nominal Thickness	34	mm
External Corrosion Allowance	0	mm
Design Temperature Internal Pressure	275	°C
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.23	
Material Name	SA-387 11	
Efficiency, Longitudinal Seam	0.7	
Efficiency, Head-to-Skirt or Circ. Seam	0.7	
Weld is pre-Heated	No	

Element#	8/14	
Element From Node	80	
Element To Node	90	
Element Type	Skirt Support	
Description	INT-SKIRT-2	
Distance "FROM" to "TO"	300	cm
Skirt Inside Diameter	7414	mm
Diameter of Skirt at Base	7414	mm
Skirt Thickness	34	mm
Internal Corrosion Allowance	1.6	mm
Nominal Thickness	34	mm
External Corrosion Allowance	0	mm
Design Temperature Internal Pressure	275	°C
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.23	
Material Name	SA-516 70	
Allowable Stress, Ambient	1407.1	kgf/cm²
Allowable Stress, Operating	1396.9	kgf/cm²
Allowable Stress, Hydrotest	2386	kgf/cm²
Material Density	0.00775	kg/cm³
P Number Thickness	32.004	mm
Yield Stress, Operating	2141.3	kgf/cm²
UCS-66 Chart Curve Designation	B	



External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	0.7	
Efficiency, Head-to-Skirt or Circ. Seam	0.7	
Weld is pre-Heated	No	
Element From Node	80	
Detail Type	Platform	
Detail ID	PLTF-3	
Dist. from "FROM" Node / Offset dist	62.5	cm
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	180.0	
Platform Wind Area	30000	cm ²
Platform Weight	185.68	kgf
Platform Railing Weight	0.01734	kgf/cm
Platform Grating Weight	0	kgf/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	185	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	180.0	
Ladder Start Elevation	0	cm
Ladder End Elevation	400	cm
Unit Weight of Ladder	0.3773	kgf/cm
Platform Length (top head platform)	0	mm

Element#	9/14	
Element From Node	90	
Element To Node	100	
Element Type	Skirt Support	
Description	INT-SKIRT-3	
Distance "FROM" to "TO"	300	cm
Skirt Inside Diameter	5683	mm
Diameter of Skirt at Base	7414	mm
Skirt Thickness	34	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	34	mm
External Corrosion Allowance	0	mm
Design Temperature Internal Pressure	275	°C
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.23	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	0.7	
Efficiency, Head-to-Skirt or Circ. Seam	0.7	
Weld is pre-Heated	No	

Element From Node	90	
Detail Type	Platform	
Detail ID	PLTF-4	
Dist. from "FROM" Node / Offset dist	162.5	cm
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	180.0	
Platform Wind Area	30000	cm ²
Platform Weight	3365.4	kgf
Platform Railing Weight	0	kgf/cm
Platform Grating Weight	0.017	kgf/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	385	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	cm
Ladder End Elevation	400	cm
Unit Weight of Ladder	0.3773	kgf/cm
Platform Length (top head platform)	0	mm



Element#	10/14	
Element From Node	100	
Element To Node	110	
Element Type	Skirt Support	
Description	INT-SKIRT-4	
Distance "FROM" to "TO"	300	cm
Skirt Inside Diameter	5683	mm
Diameter of Skirt at Base	5683	mm
Skirt Thickness	34	mm
Internal Corrosion Allowance	1.6	mm
Nominal Thickness	34	mm
External Corrosion Allowance	0	mm
Design Temperature Internal Pressure	275	°C
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.23	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	0.7	
Efficiency, Head-to-Skirt or Circ. Seam	0.7	
Weld is pre-Heated	No	

Element From Node	100	
Detail Type	Platform	
Detail ID	PLTF-5	
Dist. from "FROM" Node / Offset dist	262.5	cm
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	180.0	
Platform Wind Area	30000	cm ²
Platform Weight	3205.1	kgf
Platform Railing Weight	0	kgf/cm
Platform Grating Weight	0.017	kgf/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	185	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	180.0	
Ladder Start Elevation	0	cm
Ladder End Elevation	400	cm
Unit Weight of Ladder	0.3773	kgf/cm
Platform Length (top head platform)	0	mm

Element#	11/14	
Element From Node	110	
Element To Node	120	
Element Type	Skirt Support	
Description	INT-SKIRT-5	
Distance "FROM" to "TO"	132.5	cm
Skirt Inside Diameter	5683	mm
Diameter of Skirt at Base	5683	mm
Skirt Thickness	34	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	34	mm
External Corrosion Allowance	0	mm
Design Temperature Internal Pressure	275	°C
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.23	
Material Name	SA-387 11	
Allowable Stress, Ambient	1509.1	kgf/cm ²
Allowable Stress, Operating	1509.1	kgf/cm ²
Allowable Stress, Hydrotest	2844.9	kgf/cm ²
Material Density	0.00775	kg/cm ³
P Number Thickness	16.002	mm
Yield Stress, Operating	2539	kgf/cm ²
UCS-66 Chart Curve Designation	A	
External Pressure Chart Name	CS-3	
UNS Number	K11789	
Class / Thickness / Grade	2 ::	
Product Form	Plate	



Efficiency, Longitudinal Seam	0.7
Efficiency, Head-to-Skirt or Circ. Seam	0.7
Weld is pre-Heated	No

Element#	12/14
Element From Node	120
Element To Node	130
Element Type	Spherical
Description	B-DISHED-END-102A
Distance "FROM" to "TO"	0 cm
Inside Diameter	5648 mm
Element Thickness	52 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	58 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	50.897 kgf/cm ²
Design Temperature Internal Pressure	275 °C
Design External Pressure	0 kgf/cm ²
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.23
Material Name	SA-387 11
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	120
Detail Type	Insulation
Detail ID	INSULATION
Dist. from "FROM" Node / Offset dist	-282.4 cm
Height/Length of Insulation	282.4 cm
Thickness of Insulation	100 mm
Density	0.000352 kg/cm ³

Element From Node	120
Detail Type	Nozzle
Detail ID	N2-104-D2A
Dist. from "FROM" Node / Offset dist	0 mm
Nozzle Diameter	36 in.
Nozzle Schedule	None
Nozzle Class	600
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	5207.5 kgf
Grade of Attached Flange	GR 1.9
Nozzle Matl	SA-182 F11

Element From Node	120
Detail Type	Nozzle
Detail ID	UC1-104-D2A
Dist. from "FROM" Node / Offset dist	1800 mm
Nozzle Diameter	24 in.
Nozzle Schedule	None
Nozzle Class	600
Layout Angle	270.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	2330.4 kgf
Grade of Attached Flange	GR 1.9
Nozzle Matl	SA-182 F11

Element From Node	120
Detail Type	Nozzle
Detail ID	H1-104-D2A
Dist. from "FROM" Node / Offset dist	1000 mm
Nozzle Diameter	8 in.
Nozzle Schedule	80
Nozzle Class	600
Layout Angle	180.0
Blind Flange (Y/N)	Y



Weight of Nozzle (Used if > 0)	0	kgf
Grade of Attached Flange	GR 1.9	
Nozzle Matl	SA-335 P11 +IT	
Element From Node	120	
Detail Type	Nozzle	
Detail ID	K4-104-D2A	
Dist. from "FROM" Node / Offset dist	2500	mm
Nozzle Diameter	3	in.
Nozzle Schedule	160	
Nozzle Class	600	
Layout Angle	0.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	118.5	kgf
Grade of Attached Flange	GR 1.9	
Nozzle Matl	SA-335 P11 +IT	
Element From Node	120	
Detail Type	Weight	
Detail ID	HH Hinge	
Dist. from "FROM" Node / Offset dist	-200	cm
Miscellaneous Weight	99.998	kgf
Offset from Element Centerline	2000	mm
Element From Node	120	
Detail Type	Weight	
Detail ID	UC1 Hinge	
Dist. from "FROM" Node / Offset dist	-200	cm
Miscellaneous Weight	99.998	kgf
Offset from Element Centerline	1500	mm
Element From Node	120	
Detail Type	Weight	
Detail ID	N2 inner dish	
Dist. from "FROM" Node / Offset dist	-250	cm
Miscellaneous Weight	5009.9	kgf
Offset from Element Centerline	0	mm
Element From Node	120	
Detail Type	Weight	
Detail ID	Wire Screen	
Dist. from "FROM" Node / Offset dist	0	cm
Miscellaneous Weight	99.998	kgf
Offset from Element Centerline	2000	mm
Element From Node	120	
Detail Type	Weight	
Detail ID	SF weld	
Dist. from "FROM" Node / Offset dist	0	cm
Miscellaneous Weight	2000	kgf
Offset from Element Centerline	2200	mm
Element From Node	120	
Detail Type	Weight	
Detail ID	Support balls	
Dist. from "FROM" Node / Offset dist	-141.5	cm
Miscellaneous Weight	81999	kgf
Offset from Element Centerline	0	mm

Element#	13/14	
Element From Node	130	
Element To Node	140	
Element Type	Cylinder	
Description	SHELL-104-D2A	
Distance "FROM" to "TO"	100	cm
Inside Diameter	5600	mm
Element Thickness	100	mm
Internal Corrosion Allowance	3	mm



Nominal Thickness	100	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	50.897	kgf/cm ²
Design Temperature Internal Pressure	275	°C
Design External Pressure	0	kgf/cm ²
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.23	
Material Name	SA-387 11	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	130	
Detail Type	Platform	
Detail ID	PLTF-6	
Dist. from "FROM" Node / Offset dist	99.9	cm
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	180.0	
Platform Wind Area	30000	cm ²
Platform Weight	3175.6	kgf
Platform Railing Weight	0	kgf/cm
Platform Grating Weight	0.017	kgf/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	185	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	cm
Ladder End Elevation	270	cm
Unit Weight of Ladder	0.3773	kgf/cm
Platform Length (top head platform)	0	mm

Element From Node	130	
Detail Type	Insulation	
Detail ID	INSULATION	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Insulation	100	cm
Thickness of Insulation	100	mm
Density	0.000352	kg/cm ³

Element From Node	130	
Detail Type	Nozzle	
Detail ID	K2	
Dist. from "FROM" Node / Offset dist	50.002	cm
Nozzle Diameter	2	in.
Nozzle Schedule	160	
Nozzle Class	600	
Layout Angle	180.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	0	kgf
Grade of Attached Flange	GR 1.9	
Nozzle Matl	SA-182 F11	

Element From Node	130	
Detail Type	Weight	
Detail ID	Catalyst-D2A	
Dist. from "FROM" Node / Offset dist	50.003	cm
Miscellaneous Weight	54549	kgf
Offset from Element Centerline	0	mm

Element From Node	130	
Detail Type	Weight	
Detail ID	Split Flange-D2	
Dist. from "FROM" Node / Offset dist	10.002	cm
Miscellaneous Weight	349.99	kgf
Offset from Element Centerline	2000	mm

Element From Node	130	
Detail Type	Weight	
Detail ID	Cleat Weight-D2	



Dist. from "FROM" Node / Offset dist	50.002	cm
Miscellaneous Weight	2275	kgf
Offset from Element Centerline	0	mm

Element From Node	130	
Detail Type	Weight	
Detail ID	Lifting Lug-2-D	
Dist. from "FROM" Node / Offset dist	60.002	cm
Miscellaneous Weight	1270	kgf
Offset from Element Centerline	2600	mm

Element#	14/14	
Element From Node	140	
Element To Node	150	
Element Type	Spherical	
Description	T-DISHED-END-102A	
Distance "FROM" to "TO"	0	cm
Inside Diameter	5648	mm
Element Thickness	52	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	58	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	50.897	kgf/cm ²
Design Temperature Internal Pressure	275	°C
Design External Pressure	0	kgf/cm ²
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.23	
Material Name	SA-387 11	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	140	
Detail Type	Platform	
Detail ID	PLTF-7	
Dist. from "FROM" Node / Offset dist	320	cm
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	0.0	
Platform Wind Area	50000	cm ²
Platform Weight	155.41	kgf
Platform Railing Weight	0.01734	kgf/cm
Platform Grating Weight	0	kgf/cm ²
Platform Width	5000	mm
Platform Height	1000	mm
Platform Clearance or End Offset	2500	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	180.0	
Ladder Start Elevation	0	cm
Ladder End Elevation	320	cm
Unit Weight of Ladder	0.3773	kgf/cm
Platform Length (top head platform)	5000	mm

Element From Node	140	
Detail Type	Insulation	
Detail ID	INSULATION	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Insulation	280	cm
Thickness of Insulation	100	mm
Density	0.000352	kg/cm ³

Element From Node	140	
Detail Type	Nozzle	
Detail ID	N1-104-D2A	
Dist. from "FROM" Node / Offset dist	0	mm
Nozzle Diameter	36	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	0.0	



Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	2322.9	kgf
Grade of Attached Flange	GR 1.9	
Nozzle Matl	SA-182 F11	
Element From Node	140	
Detail Type	Nozzle	
Detail ID	M1-104-D2A	
Dist. from "FROM" Node / Offset dist	1450	mm
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	0.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	1723.1	kgf
Grade of Attached Flange	GR 1.9	
Nozzle Matl	SA-182 F11	
Element From Node	140	
Detail Type	Nozzle	
Detail ID	K1-104-D2A	
Dist. from "FROM" Node / Offset dist	1050	mm
Nozzle Diameter	2	in.
Nozzle Schedule	160	
Nozzle Class	600	
Layout Angle	180.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	0	kgf
Grade of Attached Flange	GR 1.9	
Nozzle Matl	SA-182 F11	
Element From Node	140	
Detail Type	Nozzle	
Detail ID	K3-104-D2A	
Dist. from "FROM" Node / Offset dist	1050	mm
Nozzle Diameter	2	in.
Nozzle Schedule	160	
Nozzle Class	600	
Layout Angle	270.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	0	kgf
Grade of Attached Flange	GR 1.9	
Nozzle Matl	SA-182 F11	
Element From Node	140	
Detail Type	Weight	
Detail ID	MW-DAVIT-M1-D2A	
Dist. from "FROM" Node / Offset dist	280	cm
Miscellaneous Weight	99.998	kgf
Offset from Element Centerline	2000	mm
Element From Node	140	
Detail Type	Weight	
Detail ID	N1 Vanes	
Dist. from "FROM" Node / Offset dist	250	cm
Miscellaneous Weight	299.99	kgf
Offset from Element Centerline	0	mm
Element From Node	140	
Detail Type	Weight	
Detail ID	Grating	
Dist. from "FROM" Node / Offset dist	80	cm
Miscellaneous Weight	99.998	kgf
Offset from Element Centerline	0	mm
Element From Node	140	
Detail Type	Weight	
Detail ID	Support Balls	
Dist. from "FROM" Node / Offset dist	40	cm
Miscellaneous Weight	18450	kgf



Offset from Element Centerline	0	mm
Element From Node	140	
Detail Type	Weight	
Detail ID	SF weld	
Dist. from "FROM" Node / Offset dist	0	cm
Miscellaneous Weight	2000	kgf
Offset from Element Centerline	0	mm
Element From Node	140	
Detail Type	Weight	
Detail ID	PSV WEIGHT	
Dist. from "FROM" Node / Offset dist	280	cm
Miscellaneous Weight	15000	kgf
Offset from Element Centerline	3000	mm
Element From Node	140	
Detail Type	Weight	
Detail ID	GAS WIPER RING	
Dist. from "FROM" Node / Offset dist	100	cm
Miscellaneous Weight	299.99	kgf
Offset from Element Centerline	0	mm
Element From Node	140	
Detail Type	Weight	
Detail ID	Wire Screen1	
Dist. from "FROM" Node / Offset dist	60	cm
Miscellaneous Weight	99.998	kgf
Offset from Element Centerline	0	mm
Element From Node	140	
Detail Type	Weight	
Detail ID	Wire Screen 2	
Dist. from "FROM" Node / Offset dist	20	cm
Miscellaneous Weight	99.998	kgf
Offset from Element Centerline	0	mm
Element From Node	140	
Detail Type	For./Mom.	
Detail ID	N2-104-D2A	
Dist. from "FROM" Node / Offset dist	280	cm
Force in X Direction	6559.6	kgf
Force in Y Direction	-9081.9	kgf
Force in Z Direction	6559.6	kgf
Moment about X Axis	18164	kgf-m.
Moment about Y Axis	0	kgf-m.
Moment about Z Axis	18164	kgf-m.
Force/Moment Combination Method	SRSS	



XY Coordinate Calculations:

From	To	X (Horiz.) cm	Y (Vert.) cm	DX (Horiz.) cm	DY (Vert.) cm
Skirt-3	...	...	255	...	255
Skirt-2	...	...	540	...	285
Skirt-1	...	...	640	...	100
Bottom Head	...	...	640	...	...
Shell 1	...	...	845	...	205
Top Head	...	...	845	...	...
INT-SKIRT-1	...	...	977.5	...	132.5
INT-SKIRT-2	...	...	1277.5	...	300
INT-SKIRT-3	...	...	1577.5	...	300
INT-SKIRT-4	...	...	1877.5	...	300
INT-SKIRT-5	...	...	2010	...	132.5
B-DISHED-END-10	...	...	2010	...	...
SHELL-104-D2A	...	...	2110	...	100
T-DISHED-END-10	...	...	2110	...	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd kgf/cm ²	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) kgf/cm ²	Element#
Skirt-3		...	34	1.6	7520	...	1
Skirt-2		...	34	1.6	7520	...	2
Skirt-1		...	34	1.6	7520	...	3
Bottom Head		50.897	74	3	7414	1509.1	4
Shell 1		50.897	130	3	7610	1509.1	5
Top Head		50.897	74	3	7414	1509.1	6
INT-SKIRT-1		...	34	1.6	7455	...	7
INT-SKIRT-2		...	34	1.6	7414	...	8
INT-SKIRT-3		...	34	3	5683	...	9
INT-SKIRT-4		...	34	1.6	5683	...	10
INT-SKIRT-5		...	34	3	5683	...	11
B-DISHED-END-102		50.897	58	3	5648	1509.1	12
SHELL-104-D2A		50.897	100	3	5600	1509.1	13
T-DISHED-END-102		50.897	58	3	5648	1509.1	14

Element Required Thickness and MAWP :

From	To	Design Pressure kgf/cm ²	M.A.W.P. Corroded kgf/cm ²	M.A.P. New & Cold kgf/cm ²	Minimum Thickness mm	Required Thickness mm	Element#
Skirt-3		...	No Calc	No Calc	34	No Calc	1
Skirt-2		...	No Calc	No Calc	34	No Calc	2
Skirt-1		...	No Calc	No Calc	34	No Calc	3
Bottom Head		50.897	51.1	53.5	66	65.7748	4
Shell 1		50.897	51.1	52.3	130	129.622	5
Top Head		50.897	51.1	53.5	66	65.7748	6
INT-SKIRT-1		...	No Calc	No Calc	34	No Calc	7
INT-SKIRT-2		...	No Calc	No Calc	34	No Calc	8
INT-SKIRT-3		...	No Calc	No Calc	34	No Calc	9
INT-SKIRT-4		...	No Calc	No Calc	34	No Calc	10
INT-SKIRT-5		...	No Calc	No Calc	34	No Calc	11
B-DISHED-END-102		50.897	52.1	55.4	52	50.834	12
SHELL-104-D2A		50.897	51.2	52.8	100	99.4885	13
T-DISHED-END-102		50.897	52.1	55.4	52	50.834	14

Minimum 51.1 52.3

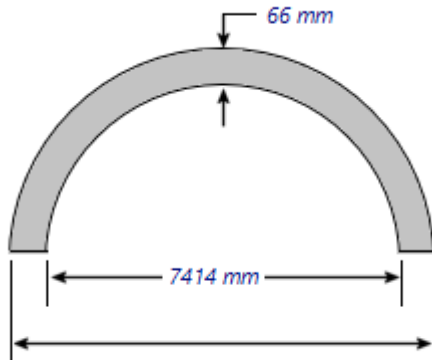
MAWP: 51.1 kgf/cm², limited by: Shell 1.

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Bottom Head

Spherical Head Design Information:					
Design Pressure	P	50.897 kgf/cm²	Int. Design Temperature	275 °C	
Material	SA-387 11		Ext. Design Temperature	0 °C	
Length	0.000 cm		External Pressure Chart	CS-3	
			UNS Number	K11789	
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	1509.1 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1509.1 kgf/cm²
Inside Diameter	Di	7414.000 mm	Actual Stress	1503.7 kgf/cm²	
Spec. Min. Thk t _e , or s _a	66.0000 mm		Spec. Nominal Thk.	74.000 mm	
Surface Area	898241.9 cm²		SG of Contents	0.000	
Element Volume	106690664.0 cm³		Weight of Contents	0.0 kgf	
Empty Weight	50515.0 kgf		Operating Weight	50515.0 kgf	
			Comp. All. Ope. Stress	950.4 kgf/cm²	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 92.675 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 46.338 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ per UG-27 (d)}$$

$$= (50.9 \cdot 3710) / (2 \cdot 1509 \cdot 1 - 0.2 \cdot 50.9)$$

$$= 62.7748 + 3.0000 = 65.7748 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (2 \cdot S \cdot E \cdot t) / (R + 0.2 \cdot t) \text{ per UG-27 (d)}$$

$$= (2 \cdot 1509 \cdot 1 \cdot 63) / (3710 + 0.2 \cdot 63)$$

$$= 51.1 \text{ kgf/cm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (R + 0.2 \cdot t) \text{ per UG-27 (d)}$$

$$= (2 \cdot 1509 \cdot 1 \cdot 66) / (3707 + 0.2 \cdot 66)$$

$$= 53.5 \text{ kgf/cm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (50.9 \cdot (3710 + 0.2 \cdot 63)) / (2 \cdot 1 \cdot 63)$$



= 1503.724 kgf/cm²

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 1.482 %

MDMT Calculations in the Spherical Portion:

Govrn. thk, tg = 66, tr = 63, c = 3 mm, E* = 1

Thickness Ratio = $tr \cdot E^*/(tg - c) = 1$, Temp. Reduction = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 42 °C

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	51.051 kgf/cm ²
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	66.000 mm	Basic MDMT	42 °C
Stress/Thk. Ratio	0.999	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	42 °C
		Joint Efficiency E*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 92.675 mm

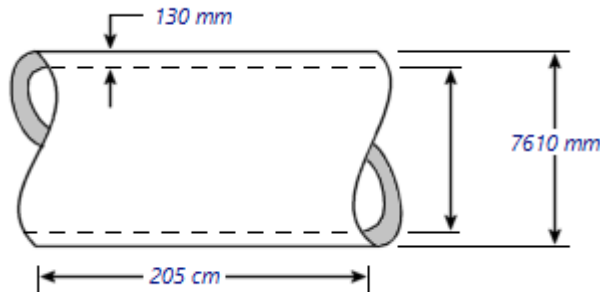
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 46.338 mm

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell 1

Cylindrical Shell Design Information:					
Design Pressure	P	50.897 kgf/cm²	Int. Design Temperature	275 °C	
Material		SA-387 11	Ext. Design Temperature	0 °C	
Length		205.000 cm	External Pressure Chart	CS-3	
			UNS Number	K11789	
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	1509.1 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1509.1 kgf/cm²
Outside Diameter	Do	7610.000 mm	Actual Stress	1504.5 kgf/cm²	
Spec. Min. Thk t _e , or s _a		130.0000 mm	Spec. Nominal Thk.	130.000 mm	
Surface Area		490104.2 cm²	SG of Contents	0.000	
Element Volume		86979808.0 cm³	Weight of Contents	0.0 kgf	
Empty Weight		48536.8 kgf	Operating Weight	48536.8 kgf	
			Comp. All. Ope. Stress	1072.2 kgf/cm²	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R_o) / (S \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= (50.9 \cdot 3805) / (1509 \cdot 1 + 0.4 \cdot 50.9)$$

$$= 126.6222 + 3.0000 = 129.6222 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R_o - 0.4 \cdot t) \text{ per Appendix 1-1 (a)(1)}$$

$$= (1509 \cdot 1 \cdot 127) / (3805 - 0.4 \cdot 127)$$

$$= 51.1 \text{ kgf/cm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R_o - 0.4 \cdot t) \text{ per Appendix 1-1 (a)(1)}$$

$$= (1509 \cdot 1 \cdot 130) / (3805 - 0.4 \cdot 130)$$

$$= 52.3 \text{ kgf/cm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R_o - 0.4 \cdot t)) / (E \cdot t)$$

$$= (50.9 \cdot (3805 - 0.4 \cdot 127)) / (1 \cdot 127)$$

$$= 1504.547 \text{ kgf/cm}^2$$

% Elongation per Table UG-79-1 $(50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o))$ 1.738 %

Minimum Design Metal Temperature Results:



Note: This Element/Detail was specified as being Impact Tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

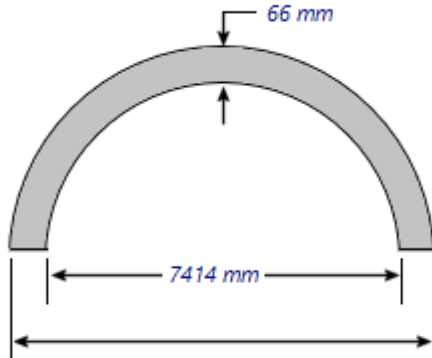
Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Top Head

Spherical Head Design Information:					
Design Pressure	P	50.897 kgf/cm²	Int. Design Temperature	275 °C	
Material		SA-387 11	Ext. Design Temperature	0 °C	
Length		0.000 cm	External Pressure Chart	CS-3	
			UNS Number	K11789	
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	1509.1 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1509.1 kgf/cm²
Inside Diameter	Di	7414.000 mm	Actual Stress	1503.7 kgf/cm²	
Spec. Min. Thk t,e, or sa		66.0000 mm	Spec. Nominal Thk.	74.000 mm	
Surface Area		898241.9 cm²	SG of Contents	0.000	
Element Volume		106690664.0 cm³	Weight of Contents	0.0 kgf	
Empty Weight		50515.0 kgf	Operating Weight	50515.0 kgf	
			Comp. All. Ope. Stress	950.4 kgf/cm²	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 92.675 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 46.338 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ per UG-27 (d)}$$

$$= (50.9 \cdot 3710) / (2 \cdot 1509 \cdot 1 - 0.2 \cdot 50.9)$$

$$= 62.7748 + 3.0000 = 65.7748 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (2 \cdot S \cdot E \cdot t) / (R + 0.2 \cdot t) \text{ per UG-27 (d)}$$

$$= (2 \cdot 1509 \cdot 1 \cdot 63) / (3710 + 0.2 \cdot 63)$$

$$= 51.1 \text{ kgf/cm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (R + 0.2 \cdot t) \text{ per UG-27 (d)}$$

$$= (2 \cdot 1509 \cdot 1 \cdot 66) / (3707 + 0.2 \cdot 66)$$

$$= 53.5 \text{ kgf/cm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (50.9 \cdot (3710 + 0.2 \cdot 63)) / (2 \cdot 1 \cdot 63)$$



= 1503.724 kgf/cm²

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 1.482 %

MDMT Calculations in the Spherical Portion:

Govrn. thk, tg = 66, tr = 63, c = 3 mm, E* = 1

Thickness Ratio = $tr \cdot E^*/(tg - c) = 1$, Temp. Reduction = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 42 °C

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	51.051 kgf/cm ²
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	66.000 mm	Basic MDMT	42 °C
Stress/Thk. Ratio	0.999	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	42 °C
		Joint Efficiency E*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

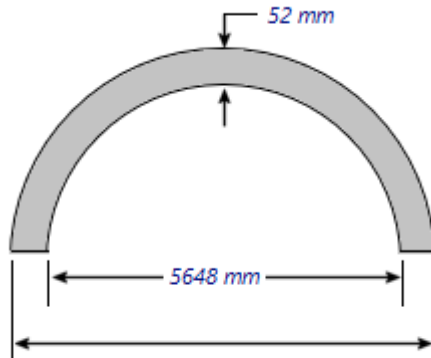
Head inner surface maximum deviation outside the specified shape, 1.25% of D: 92.675 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 46.338 mm

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: B-DISHED-END-102A

Spherical Head Design Information:					
Design Pressure	P	50.897 kgf/cm ²	Int. Design Temperature	275 °C	
Material		SA-387 11	Ext. Design Temperature	0 °C	
Length		0.000 cm	External Pressure Chart	CS-3	
			UNS Number	K11789	
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	1509.1 kgf/cm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1509.1 kgf/cm ²
Inside Diameter	Di	5648.000 mm	Actual Stress	1473.3 kgf/cm ²	
Spec. Min. Thk t _e , or s _a		52.0000 mm	Spec. Nominal Thk.	58.000 mm	
Surface Area		521876.6 cm ²	SG of Contents	0.000	
Element Volume		47168576.0 cm ³	Weight of Contents	0.0 kgf	
Empty Weight		22990.5 kgf	Operating Weight	22990.5 kgf	
			Comp. All. Ope. Stress	954.0 kgf/cm ²	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 70.600 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 35.300 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ per UG-27 (d)}$$

$$= (50.9 \cdot 2827) / (2 \cdot 1509 \cdot 1 - 0.2 \cdot 50.9)$$

$$= 47.8340 + 3.0000 = 50.8340 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (2 \cdot S \cdot E \cdot t) / (R + 0.2 \cdot t) \text{ per UG-27 (d)}$$

$$= (2 \cdot 1509 \cdot 1 \cdot 49) / (2827 + 0.2 \cdot 49)$$

$$= 52.1 \text{ kgf/cm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (R + 0.2 \cdot t) \text{ per UG-27 (d)}$$

$$= (2 \cdot 1509 \cdot 1 \cdot 52) / (2824 + 0.2 \cdot 52)$$

$$= 55.4 \text{ kgf/cm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (50.9 \cdot (2827 + 0.2 \cdot 49)) / (2 \cdot 1 \cdot 49)$$



= 1473.312 kgf/cm²

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 1.525 %

MDMT Calculations in the Spherical Portion:

Govrn. thk, tg = 52, tr = 48, c = 3 mm, E* = 1

Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.98$, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 37 °C

Min Metal Temp. at Required thickness (UCS 66.1) 36 °C

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	51.051 kgf/cm ²
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	52.000 mm	Basic MDMT	37 °C
Stress/Thk. Ratio	0.979	Temperature Difference	1 °C
UCS-66(c) red.	No	Computed Min. Temp.	36 °C
		Joint Efficiency E*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

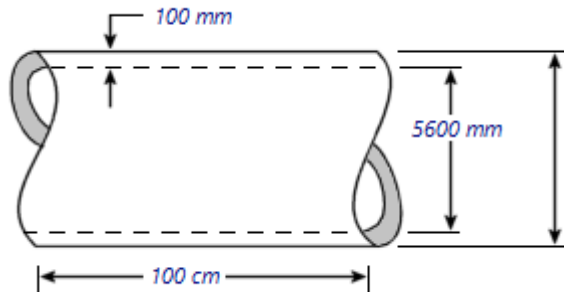
Head inner surface maximum deviation outside the specified shape, 1.25% of D: 70.600 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 35.300 mm

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: SHELL-104-D2A

Cylindrical Shell Design Information:					
Design Pressure	P	50.897 kgf/cm²	Int. Design Temperature	275 °C	
Material		SA-387 11	Ext. Design Temperature	0 °C	
Length		100.000 cm	External Pressure Chart	CS-3	
			UNS Number	K11789	
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	1509.1 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1509.1 kgf/cm²
Inside Diameter	Di	5600.000 mm	Actual Stress	1501.3 kgf/cm²	
Spec. Min. Thk t _e , or s _a		100.0000 mm	Spec. Nominal Thk.	100.000 mm	
Surface Area		182212.4 cm²	SG of Contents	0.000	
Element Volume		24630086.0 cm³	Weight of Contents	0.0 kgf	
Empty Weight		13878.7 kgf	Operating Weight	13878.7 kgf	
			Comp. All. Ope. Stress	1072.6 kgf/cm²	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (50.9 \cdot 2803) / (1509 \cdot 1 - 0.6 \cdot 50.9)$$

$$= 96.4885 + 3.0000 = 99.4885 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1509 \cdot 1 \cdot 97) / (2803 + 0.6 \cdot 97)$$

$$= 51.2 \text{ kgf/cm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1509 \cdot 1 \cdot 100) / (2800 + 0.6 \cdot 100)$$

$$= 52.8 \text{ kgf/cm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (50.9 \cdot (2803 + 0.6 \cdot 97)) / (1 \cdot 97)$$

$$= 1501.304 \text{ kgf/cm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 1.754 %

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 100, t_r = 96.8, c = 3 mm, E* = 1
 Thickness Ratio = t_r * E* / (t_g - c) = 1, Temp. Reduction = 0 °C



Min Metal Temp. w/o impact per UCS-66, Curve A 48 °C
Min Metal Temp. at Required thickness (UCS 66.1) 48 °C

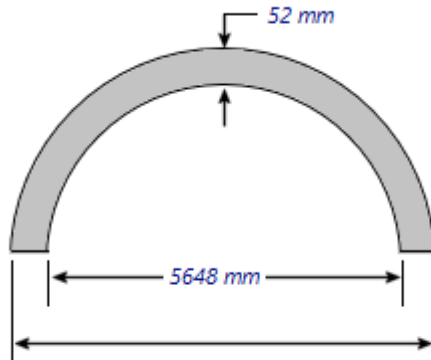
Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	51.051 kgf/cm ²
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	100.000 mm	Basic MDMT	48 °C
Stress/Thk. Ratio	0.998	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	48 °C
		Joint Efficiency E*	100 %

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: T-DISHED-END-102A

Spherical Head Design Information:					
Design Pressure	P	50.897 kgf/cm ²	Int. Design Temperature		275 °C
Material		SA-387 11	Ext. Design Temperature		0 °C
Length		0.000 cm	External Pressure Chart		CS-3
			UNS Number		K11789
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	1509.1 kgf/cm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1509.1 kgf/cm ²
Inside Diameter	Di	5648.000 mm	Actual Stress		1473.3 kgf/cm ²
Spec. Min. Thk t,e, or sa		52.0000 mm	Spec. Nominal Thk.		58.000 mm
Surface Area		521876.6 cm ²	SG of Contents		0.000
Element Volume		47168576.0 cm ³	Weight of Contents		0.0 kgf
Empty Weight		22990.5 kgf	Operating Weight		22990.5 kgf
			Comp. All. Ope. Stress		954.0 kgf/cm ²
Radiography Information:					
User Defined			User Defined		
Circ. Joint Efficiency		100 %	Long. Joint Eff.	E or v	100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 70.600 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 35.300 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ per UG-27 (d)}$$

$$= (50.9 \cdot 2827) / (2 \cdot 1509 \cdot 1 - 0.2 \cdot 50.9)$$

$$= 47.8340 + 3.0000 = 50.8340 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (2 \cdot S \cdot E \cdot t) / (R + 0.2 \cdot t) \text{ per UG-27 (d)}$$

$$= (2 \cdot 1509 \cdot 1 \cdot 49) / (2827 + 0.2 \cdot 49)$$

$$= 52.1 \text{ kgf/cm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (R + 0.2 \cdot t) \text{ per UG-27 (d)}$$

$$= (2 \cdot 1509 \cdot 1 \cdot 52) / (2824 + 0.2 \cdot 52)$$

$$= 55.4 \text{ kgf/cm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (50.9 \cdot (2827 + 0.2 \cdot 49)) / (2 \cdot 1 \cdot 49)$$



= 1473.312 kgf/cm²

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 1.525 %

MDMT Calculations in the Spherical Portion:

Govrn. thk, tg = 52, tr = 48, c = 3 mm, E* = 1

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.98$, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 37 °C

Min Metal Temp. at Required thickness (UCS 66.1) 36 °C

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	51.051 kgf/cm ²
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	52.000 mm	Basic MDMT	37 °C
Stress/Thk. Ratio	0.979	Temperature Difference	1 °C
UCS-66(c) red.	No	Computed Min. Temp.	36 °C
		Joint Efficiency E*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 70.600 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 35.300 mm

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	66.366 kgf/cm ²
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	66.166 kgf/cm ²
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	67.214 kgf/cm ²
UG100	= 1.10 * M.A.W.P. * Sa/S	56.156 kgf/cm ²
PED	= max(1.43*DP, 1.25*DP*ratio)	72.783 kgf/cm ²
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	72.783 kgf/cm ²
App 27-4	= M.A.W.P.	51.051 kgf/cm ²

Horizontal Test performed per: UG-99c

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (kgf/cm²):

From To	Stress	Allowable	Ratio	Pressure
Bottom Head	1915.2	2844.9	0.673	67.96
Shell 1	1962.4	2844.9	0.690	67.98
Top Head	1915.2	2844.9	0.673	67.96
B-DISHED-END-102A	1847.2	2844.9	0.649	67.78
SHELL-104-D2A	1938.3	2844.9	0.681	67.77
T-DISHED-END-102A	1847.2	2844.9	0.649	67.78

Stress ratios for Nozzle and Pad Materials (kgf/cm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N2 36"	Nozzle	1407.00	1407.00	1.000
UC1- 24"	Nozzle	1407.00	1407.00	1.000
H1 8"	Nozzle	1407.00	1407.00	1.000
K4 3"	Nozzle	1407.00	1407.00	1.000
N1 36"	Nozzle	1407.00	1407.00	1.000
K1 2"	Nozzle	1407.00	1407.00	1.000
M1 24"	Nozzle	1407.00	1407.00	1.000
K5 4"	Nozzle	1407.00	1407.00	1.000
K2 2"	Nozzle	1407.00	1407.00	1.000
N2-104-D2A	Nozzle	1407.00	1407.00	1.000
UC1-104-D2A	Nozzle	1407.00	1407.00	1.000
H1-104-D2A	Nozzle	1203.00	1127.00	1.067
K4-104-D2A	Nozzle	1203.00	1127.00	1.067
K2	Nozzle	1407.00	1407.00	1.000
N1-104-D2A	Nozzle	1407.00	1407.00	1.000
M1-104-D2A	Nozzle	1407.00	1407.00	1.000
K1-104-D2A	Nozzle	1407.00	1407.00	1.000
K3-104-D2A	Nozzle	1407.00	1407.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (kgf/cm²):

Description	Ambient	Operating	Ratio
Bottom Head	1509.10	1509.10	1.000
Shell 1	1509.10	1509.10	1.000
Top Head	1509.10	1509.10	1.000
B-DISHED-END-102A	1509.10	1509.10	1.000
SHELL-104-D2A	1509.10	1509.10	1.000
T-DISHED-END-102A	1509.10	1509.10	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (kgf/cm²):

Description	Ambient	Operating	Ratio
-------------	---------	-----------	-------



N2 36"	1267.37	2524.00	0.502
UC1- 24"	1267.37	2524.00	0.502
H1 8"	536.66	2524.00	0.213
K4 3"	283.12	2524.00	0.112
N1 36"	1267.37	2524.00	0.502
K1 2"	240.91	2524.00	0.095
M1 24"	835.85	2524.00	0.331
K5 4"	301.90	2524.00	0.120
K2 2"	240.91	2524.00	0.095
N2-104-D2A	1264.08	2524.00	0.501
UC1-104-D2A	1264.08	2524.00	0.501
H1-104-D2A	641.00	1881.00	0.341
K4-104-D2A	282.38	1881.00	0.150
K2	240.27	2524.00	0.095
N1-104-D2A	1522.32	2524.00	0.603
M1-104-D2A	1073.62	2524.00	0.425
K1-104-D2A	240.29	2524.00	0.095
K3-104-D2A	240.29	2524.00	0.095

Elements Suitable for Test Pressure.

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	66.366 kgf/cm ²
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	66.166 kgf/cm ²
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	67.380 kgf/cm ²
UG100	= 1.10 * M.A.W.P. * Sa/S	56.156 kgf/cm ²
PED	= max(1.43*DP, 1.25*DP*ratio)	72.783 kgf/cm ²
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	72.783 kgf/cm ²
App 27-4	= M.A.W.P.	51.051 kgf/cm ²

UG-99(b) Note 35, Test Pressure Calculation:
 = Test Factor • Design Pressure • Stress Ratio
 = 1.3 • 50.9 • 1
 = 66.166 kgf/cm²

Vertical Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (kgf/cm²):

From To	Stress	Allowable	Ratio	Pressure
Bottom Head	1982.8	2844.9	0.697	67.11
Shell 1	1972.9	2844.9	0.694	66.74
Top Head	1965.8	2844.9	0.691	66.54
B-DISHED-END-102A	1934.5	2844.9	0.680	66.83
SHELL-104-D2A	1963.0	2844.9	0.690	66.55
T-DISHED-END-102A	1923.5	2844.9	0.676	66.45

Stress ratios for Nozzle and Pad Materials (kgf/cm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N2 36"	Nozzle	1407.00	1407.00	1.000
UC1- 24"	Nozzle	1407.00	1407.00	1.000
H1 8"	Nozzle	1407.00	1407.00	1.000
K4 3"	Nozzle	1407.00	1407.00	1.000
N1 36"	Nozzle	1407.00	1407.00	1.000
K1 2"	Nozzle	1407.00	1407.00	1.000
M1 24"	Nozzle	1407.00	1407.00	1.000
K5 4"	Nozzle	1407.00	1407.00	1.000
K2 2"	Nozzle	1407.00	1407.00	1.000
N2-104-D2A	Nozzle	1407.00	1407.00	1.000
UC1-104-D2A	Nozzle	1407.00	1407.00	1.000
H1-104-D2A	Nozzle	1203.00	1127.00	1.067
K4-104-D2A	Nozzle	1203.00	1127.00	1.067
K2	Nozzle	1407.00	1407.00	1.000
N1-104-D2A	Nozzle	1407.00	1407.00	1.000
M1-104-D2A	Nozzle	1407.00	1407.00	1.000
K1-104-D2A	Nozzle	1407.00	1407.00	1.000
K3-104-D2A	Nozzle	1407.00	1407.00	1.000

Minimum

1.000

Stress ratios for Pressurized Vessel Elements (kgf/cm²):

Description	Ambient	Operating	Ratio
Bottom Head	1509.10	1509.10	1.000
Shell 1	1509.10	1509.10	1.000
Top Head	1509.10	1509.10	1.000
B-DISHED-END-102A	1509.10	1509.10	1.000
SHELL-104-D2A	1509.10	1509.10	1.000
T-DISHED-END-102A	1509.10	1509.10	1.000



Minimum

1.000

Hoop Stress in Nozzle Wall during Pressure Test (kgf/cm²):

Description	Ambient	Operating	Ratio
N2 36"	1251.64	2524.00	0.496
UC1- 24"	1251.64	2524.00	0.496
H1 8"	530.00	2524.00	0.210
K4 3"	279.60	2524.00	0.111
N1 36"	1240.91	2524.00	0.492
K1 2"	235.88	2524.00	0.093
M1 24"	818.40	2524.00	0.324
K5 4"	295.60	2524.00	0.117
K2 2"	235.88	2524.00	0.093
N2-104-D2A	1246.39	2524.00	0.494
UC1-104-D2A	1246.39	2524.00	0.494
H1-104-D2A	632.03	1881.00	0.336
K4-104-D2A	278.43	1881.00	0.148
K2	235.93	2524.00	0.093
N1-104-D2A	1492.43	2524.00	0.591
M1-104-D2A	1052.54	2524.00	0.417
K1-104-D2A	235.57	2524.00	0.093
K3-104-D2A	235.57	2524.00	0.093

Elements Suitable for Test Pressure.

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length cm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B kgf/cm ²
10	20	1805	...	...	No Calc	No Calc
20	30	1805	...	...	No Calc	No Calc
30	40	1805	...	...	No Calc	No Calc
40	50	No Calc	7546	63	0.0020872	1301.96
50	60	452.133	7610	127	0.0051698	1367.25
60	70	No Calc	7546	63	0.0020872	1301.96
70	80	1805	...	...	No Calc	No Calc
80	90	1805	...	...	No Calc	No Calc
90	100	1805	...	...	No Calc	No Calc
100	110	1805	...	...	No Calc	No Calc
110	120	1805	...	...	No Calc	No Calc
120	130	No Calc	5752	49	0.0021297	1303.41
130	140	288.267	5800	97	0.0063681	1382.26
140	150	No Calc	5752	49	0.0021297	1303.41

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure kgf/cm ²	External M.A.W.P. kgf/cm ²
10	20	...	No Calc	...	No Calc
20	30	...	No Calc	...	No Calc
30	40	...	No Calc	...	No Calc
40	50	66	No Calc	...	21.7
50	60	130	No Calc	...	30.4
60	70	66	No Calc	...	21.7
70	80	...	No Calc	...	No Calc
80	90	...	No Calc	...	No Calc
90	100	...	No Calc	...	No Calc
100	110	...	No Calc	...	No Calc
110	120	...	No Calc	...	No Calc
120	130	52	No Calc	...	22.2
130	140	100	No Calc	...	30.8
140	150	52	No Calc	...	22.2

Minimum

21.7

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners cm	Allowable Length Bet. Stiffeners cm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	1805	No Calc	No Calc	No Calc
20	30	1805	No Calc	No Calc	No Calc
30	40	1805	No Calc	No Calc	No Calc
40	50	No Calc	No Calc	No Calc	No Calc
50	60	452.133	No Calc	No Calc	No Calc
60	70	No Calc	No Calc	No Calc	No Calc
70	80	1805	No Calc	No Calc	No Calc
80	90	1805	No Calc	No Calc	No Calc
90	100	1805	No Calc	No Calc	No Calc
100	110	1805	No Calc	No Calc	No Calc
110	120	1805	No Calc	No Calc	No Calc
120	130	No Calc	No Calc	No Calc	No Calc
130	140	288.267	No Calc	No Calc	No Calc
140	150	No Calc	No Calc	No Calc	No Calc



Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kg	Element ID Volume cm ³	Corroded Metal Wgt. kg	Corroded ID Volume cm ³	Extra due Misc % kg
10	20	19218.2	...	18474.5	...	...
20	30	17662.2	...	16831.1	...	...
30	40	6197.28	...	5905.61	...	...
40	50	50515	106690664	48505.9	106949888	...
50	60	48536.8	86979808	47435.7	87121872	...
60	70	50515	106690664	48505.9	106949888	...
70	80	8214.64	...	7829.75	...	...
80	90	18497.4	...	17630.7	...	...
90	100	17018.2	...	16238.4	...	...
100	110	14198.4	...	13534	...	...
110	120	6270.97	...	5720.66	...	...
120	130	22990.5	47168576	21824.2	47319056	...
130	140	13878.7	24630086	13469.4	24682894	...
140	150	22990.5	47168576	21824.2	47319056	...
Total		316703	0.419E+09	303729	0.420E+09	0

Weight of Details:

From	Type	Weight of Detail kg	X Offset, Dtl. Cent. cm	Y Offset, Dtl. Cent. cm	Z Offset Dtl. Cent. cm	Description
10	Wght	2549.96	-320	30	...	Tailing+Bracing
20	Plat	2038.54	-299.289	100	299.289	PLTF-1
40	Plat	7780.9	0.41462E-05	...	...	PLTF-2
40	Insl	3557.38	...	-185.35	...	Ins: 40
40	Nozl	5478.34	...	-374.901	...	N2 36"
40	Nozl	2220.52	-250	-380.412	...	UC1- 24"
40	Nozl	254.285	200	-344.214	...	H1 8"
40	Nozl	139.911	-173.205	-574.513	-100	K4 3"
40	Wght	99.9982	200	-200	...	HH Hinge
40	Wght	99.9982	-150	-200	...	UC1 Hinge
40	Wght	5009.91	...	-250	...	N2 inner dish
40	Wght	99.9982	200	...	...	WIRE SCREEN
40	Wght	4544.92	-220	-5	...	SF weld
40	Wght	205000	...	-141.5	...	Support balls
50	Insl	1749.85	...	102.5	...	Ins: 50
50	Wght	164717	...	103	...	Catalyst
50	Wght	349.994	200	10	...	Split Flange
50	Wght	3554.94	...	50	...	Cleat Weight
50	Wght	2549.96	-320	110	...	Lifting Lug-1
50	Wght	2549.96	320	110	...	Lifting Lug-2
50	Wght	9999.82	...	102.5	...	MISC WEIGHT
60	Insl	3557.38	...	185.35	...	Ins: 60
60	Nozl	2194.54	...	367.399	-1	N1 36"
60	Nozl	366.454	150	31.9466	...	K1 2"
60	Nozl	1881.32	-200	323.819	...	M1 24"
60	Nozl	71.1046	100	323.819	-173.205	K5 4"
60	Nozl	35.6367	350	130.184	...	K2 2"
60	Wght	99.9982	-200	282.9	...	Manway Davit
60	Wght	299.995	...	300	-1	N1 Vanes
60	Wght	99.9982	...	80	...	Grating
60	Wght	30750	...	40	...	Support Balls
60	Wght	4544.92	-300	5	...	SF weld
60	Wght	299.995	...	100	...	GAS WIPPER RING
60	Wght	99.9982	...	60	...	WIRE SCREEN-1
60	Wght	99.9982	...	20	...	WIRE SCREEN-2
60	Wght	2911.95	...	370.7	...	N1-ELBOW
60	Wght	249.996	100	370.7	-173.205	K4-elbow
60	Wght	149.997	150	370.7	...	K1-ELBOW
60	Forc	...	...	300	...	N2 F&M
80	Plat	185.678	...	62.5	-298.09	PLTF-3



90	Plat	3365.36	...	162.5	-256.154	PLTF-4
100	Plat	3205.06	...	262.5	-243.58	PLTF-5
120	Insl	2063.93	...	-141.2	...	INSULATION
120	Nozl	5207.49	...	-287.301	...	N2-104-D2A
120	Nozl	2330.4	...	-277.501	180	UC1-104-D2A
120	Nozl	628.513	-100	-324.002	...	H1-104-D2A
120	Nozl	118.505	250	-388.354	...	K4-104-D2A
120	Wght	99.9982	200	-200	...	HH Hinge
120	Wght	99.9982	-150	-200	...	UC1 Hinge
120	Wght	5009.91	...	-250	...	N2 inner dish
120	Wght	99.9982	200	...	...	Wire Screen
120	Wght	1999.96	220	...	...	SF weld
120	Wght	81998.6	...	-141.5	...	Support balls
130	Plat	3175.64	...	99.9	-240.973	PLTF-6
130	Insl	653.198	...	50	...	INSULATION
130	Nozl	32.5917	-305	50.0025	...	K2
130	Wght	54549	...	50.0025	...	Catalyst-D2A
130	Wght	349.994	200	10.0025	...	Split Flange-D2
130	Wght	2274.96	...	50.0025	...	Cleat Weight-D2
130	Wght	1269.98	260	60.0025	...	Lifting Lug-2-D
140	Plat	155.412	...	320	...	PLTF-7
140	Insl	2063.93	...	140	...	INSULATION
140	Nozl	2322.94	...	294.8	...	N1-104-D2A
140	Nozl	1723.12	145	274.732	...	M1-104-D2A
140	Nozl	230.444	-105	114.554	...	K1-104-D2A
140	Nozl	26.8879	...	304.554	105	K3-104-D2A
140	Wght	99.9982	-200	280	...	MW-DAVIT-M1-D2A
140	Wght	299.995	...	250	...	N1 Vanes
140	Wght	99.9982	...	80	...	Grating
140	Wght	18449.7	...	40	...	Support Balls
140	Wght	1999.96	...	...	...	SF weld
140	Wght	14999.7	300	280	...	PSV WEIGHT
140	Wght	299.995	...	100	...	GAS WIPER RING
140	Wght	99.9982	...	60	...	Wire Screen1
140	Wght	99.9982	...	20	...	Wire Screen 2
140	Forc	...	...	280	...	N2-104-D2A

Total Weight of Each Detail Type:

Platforms	19906.6
Insulation	13645.7
Nozzles	25263.0
Weights	624935.0

Sum of the Detail Weights 683750.2 kg

Weight Summation Results: (kg)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	316703.6	316703.6	316703.6	316703.6	316703.6	316703.6
Nozzles	25263.0	25263.0	25263.0	25263.0	25263.0	25263.0
Wld Weights	132394.8	132394.8	132394.8	132394.8	132394.8	132394.8
Platforms	...	...	19906.6	19906.6	19906.6	19906.6
Insulation	...	...	13645.7	13645.7	13645.7	13645.7
Empty Weights	...	...	...	4074.9	4074.9	...
Ope Weights	...	...	...	...	...	492540.4
Test Liquid	...	419140.1	...	...	...	...
Totals	474361.3	893501.5	507913.6	511988.5	511988.5	1000454.0

Field Installation Options:

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	474361.3 kg
Shop Test Wt.	- Fabricated weight + water (full)	893501.5 kg
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	507913.6 kg
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	511988.5 kg



Op. Wt. no Liq - Fab. weight + internals. + details + weights	511988.5 kg
Operating Wt. - Empty weight + operating liq. uncorroded	1000454.0 kg
Field Test Wt. - Empty weight + test medium (full)	927053.8 kg

Weights used in Vortex Shedding Calculations:

Mass of the Upper 1/3 of the Vertical Vessel (operating)	302093.6 kg
Mass of the Upper 1/3 of the Vertical Vessel (filled w/water)	324690.2 kg
Mass of the Upper 1/3 of the Vertical Vessel (empty)	218496.0 kg

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	602432	596984
20	30	673306	667218
30	40	236248	234112
40	50	898242	863426
50	60	490104	473360
60	70	898242	863426
70	80	313153	310323
80	90	705162	698753
90	100	542019	535610
100	110	542019	535610
110	120	239392	236561
120	130	521877	501083
130	140	182212	175929
140	150	521877	501083

Total	7366283.5	7193477.0 cm ²
Total	7929.0	7743.0 ft ²

Element and Detail Weights:

From	To	Total Ele. Empty Wgt. kg	Total. Ele. Oper. Wgt. kg	Total. Ele. Hydro. Wgt. kg	Total Dtl. Offset Mom. kgf-m.	Oper. Wgt. No Liquid kg
10	20	21768.2	21768.2	21768.2	...	21768.2
20	30	19700.8	19700.8	19700.8	6101.13	19700.8
30	40	6197.28	6197.28	6197.28	...	6197.28
40	50	79801.1	284801	186344	139.911	284801
50	60	69291.3	234008	155882	...	234008
60	70	67478.2	98228.2	173521	581.108	98228.2
70	80	8214.64	8214.64	8214.64	...	8214.64
80	90	18683.1	18683.1	18683.1	553.488	18683.1
90	100	20383.6	20383.6	20383.6	8620.5	20383.6
100	110	17403.5	17403.5	17403.5	7806.89	17403.5
110	120	6270.97	6270.97	6270.97	...	6270.97
120	130	122648	122648	169695	4194.72	122648
130	140	21635	76184	43629.1	7652.46	76184
140	150	32513.2	65962.5	79360.6	28.2324	65962.5

Empty Support Force + the Sum of the Y forces	530152.44	kgf
Operating Support Force + the Sum of the Y forces	1018617.81	kgf
Hydro Support Force + the Sum of the Y forces	945217.56	kgf

Cumulative Vessel Weight

From	To	Cumulative Ope Wgt. No Liquid kg	Cumulative Oper. Wgt. kg	Cumulative Hydro. Wgt. kg
10	20	1000454	1000454	927054
20	30	978686	978686	905286
30	40	958985	958985	885585
40	50	952788	952788	879388



50	60	667987	667987	693044
60	70	433978	433978	537162
70	80	335750	335750	363641
80	90	327535	327535	355426
90	100	308852	308852	336743
100	110	288469	288469	316359
110	120	271065	271065	298956
120	130	264794	264794	292685
130	140	142147	142147	122990
140	150	65962.5	65962.5	79360.6

The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.

Cumulative Vessel Moment

From	To	Cumulative Empty Mom. kgf-m.	Cumulative Oper. Mom. kgf-m.	Cumulative Hydro. Mom. kgf-m.
10	20	35678.4	35678.4	35678.4
20	30	35678.4	35678.4	35678.4
30	40	29577.3	29577.3	29577.3
40	50	29577.3	29577.3	29577.3
50	60	29437.4	29437.4	29437.4
60	70	29437.4	29437.4	29437.4
70	80	28856.3	28856.3	28856.3
80	90	28856.3	28856.3	28856.3
90	100	28302.8	28302.8	28302.8
100	110	19682.3	19682.3	19682.3
110	120	11875.4	11875.4	11875.4
120	130	11875.4	11875.4	11875.4
130	140	7680.7	7680.7	7680.7
140	150	28.2324	28.2324	28.2324



Nozzle Flange MAWP Results: (kgf/cm² & °C)

Nozzle Description	Flange Ope.	Rating Ambient	Des. Temp	Class	Grade/Group	Equiv. Press	- - - UG-44	Max Pressure 50%	DNV
N2 36"	93.1	105.4	275	600	GR 1.9	14.4	90.9	93.1	93.1
UC1- 24"	93.10	105.44	275	600	GR 1.9	...	...	...	...
H1 8"	93.10	105.44	275	600	GR 1.9	...	...	...	...
K4 3"	93.10	105.44	275	600	GR 1.9	...	...	...	...
N1 36"	93.1	105.4	275	600	GR 1.9	14.4	90.9	93.1	93.1
K1 2"	93.10	105.44	275	600	GR 1.9	...	...	...	...
M1 24"	93.10	105.44	275	600	GR 1.9	...	...	...	...
K5 4"	93.10	105.44	275	600	GR 1.9	...	...	...	...
K2 2"	93.10	105.44	275	600	GR 1.9	...	...	...	...
N2-104-D2A	93.10	105.44	275	600	GR 1.9	...	...	...	...
UC1-104-D2A	93.10	105.44	275	600	GR 1.9	...	...	...	...
H1-104-D2A	93.10	105.44	275	600	GR 1.9	...	...	...	...
K4-104-D2A	93.10	105.44	275	600	GR 1.9	...	...	...	...
K2	93.10	105.44	275	600	GR 1.9	...	...	...	...
N1-104-D2A	93.1	105.4	275	600	GR 1.9	14.4	90.8	78.7	93.1
M1-104-D2A	93.10	105.44	275	600	GR 1.9	...	...	...	...
K1-104-D2A	93.10	105.44	275	600	GR 1.9	...	...	...	...
K3-104-D2A	93.10	105.44	275	600	GR 1.9	...	...	...	...
Min Rating	90.850	105.441, [for Core Elements]					...	...	...

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME kgf-m.	Axial Force FE kgf	MAWP PD kgf/cm ²	Maximum Pressure Pmax kgf/cm ²	Pass Fail
N2 36"	965.200	0.13	23354	8257	51.1	90.9	Pass
N1 36"	965.200	0.13	23354	8257	51.1	90.9	Pass
N1-104-D2A	965.200	0.13	23355	8257	51.1	90.8	Pass

Evaluating Flange on Nozzle [N2 36"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(23354) + 4 \cdot \text{abs}(8257)965.200 \leq \pi \cdot 965.200^3 \cdot ((93.103 - 51.051) + .130 \cdot 93.103)$$

$$= 0.40554E+06 \leq 0.15298E+07 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N2 36" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(93.103, 0.130 \cdot 93.103 + 93.103 - 405544.3 / (\pi \cdot 965.200^3))$$

$$= \min(93.103, 90.850)$$

$$= 90.850 \text{ kgf/cm}^2$$

Evaluating Flange on Nozzle [N1 36"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(23354) + 4 \cdot \text{abs}(8257)965.200 \leq \pi \cdot 965.200^3 \cdot ((93.103 - 51.051) + .130 \cdot 93.103)$$

$$= 0.40554E+06 \leq 0.15298E+07 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N1 36" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(93.103, 0.130 \cdot 93.103 + 93.103 - 405544.3 / (\pi \cdot 965.200^3))$$

$$= \min(93.103, 90.850)$$

$$= 90.850 \text{ kgf/cm}^2$$



Evaluating Flange on Nozzle [N1-104-D2A]:

$$\begin{aligned} &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\ &= 16(23355) + 4 \cdot \text{abs}(8257)965.200 \leq \pi \cdot 965.200^3 [(93.103 - 51.051) + .130 \cdot 93.103] \\ &= 0.40555E+06 \leq 0.15298E+07 \quad \text{Check Passes} \end{aligned}$$

Maximum Pressure for Flange on Nozzle N1-104-D2A [Pmax]:

$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(93.103, 0.130 \cdot 93.103 + 93.103 - 405553.9 / (\pi \cdot 965.200^3)) \\ &= \min(93.103, 90.850) \\ &= 90.850 \text{ kgf/cm}^2 \end{aligned}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Large Diameter Flange Pressure Ratings are per ASME B16.47 2025 Edition

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The Natural Frequencies for the vessel have been computed iteratively by solving a system of matrices. These matrices describe the mass and the stiffness of the vessel. This is the generalized eigenvalue/eigenvector problem and is referenced in some mathematical texts.

The Natural Frequency for the Vessel (Empty.) is 4.81579 Hz.

The Natural Frequency for the Vessel (Ope...) is 4.81579 Hz.

The Natural Frequency for the Vessel (Filled) is 4.72519 Hz.

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Forces/Moments Applied to Vessel (Combined w/Wind Loads)

From	To	X and Z Dir Force Res. kgf	X,Z Moment and For Res kgf-m.
10	20	...	...
20	30	...	...
30	40	...	...
40	50	...	...
50	60	...	...
60	70	9276.71	53518.3
70	80	...	...
80	90	...	...
90	100	...	...
100	110	...	...
110	120	...	...
120	130	...	...
130	140	...	...
140	150	9276.71	51663

Forces/Moments Applied to Vessel (Combined w/Seismic Loads)

From	To	X and Z Dir Force Res. kgf	X,Z Moment and For Res kgf-m.
10	20	...	...
20	30	...	...
30	40	...	...
40	50	...	...
50	60	...	...
60	70	9276.71	53518.3
70	80	...	...
80	90	...	...
90	100	...	...
100	110	...	...
110	120	...	...
120	130	...	...
130	140	...	...
140	150	9276.71	51663

User Input Forces and Moments:

From Node	Distance From	- - - - Fx	Forces - - - - Fy	- - - - Fz	- - - - Mx	Moments - - - - My	- - - - Mz
60	300.00	6560.	-9082.	6560.	18164.		18164.
140	280.00	6560.	-9082.	6560.	18164.		18164.



Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	97.2 km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	30 cm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 cm
Distance Upwind of Crest Lh	0 cm
Distance from Crest to the Vessel x	0 cm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 cm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot g_Q \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot g_V \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.19 \cdot 0.91) / (1 + 1.7 \cdot 3.4 \cdot 0.19)))$
 $= \min(0.85, 0.88)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	4.816 Hz
Natural Frequency of Vessel (Empty)	4.816 Hz
Natural Frequency of Vessel (Test)	4.725 Hz

User Entered Importance Factor is	1.000
Force Coefficient	[Cf] 0.543
Structure Height to Diameter ratio	3.596
Height to top of Structure	2407.600 cm

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Z_g]$:
Exposure Category: C from Table 26.11-1
 $\alpha = 9.5$: $Z_g = 27432$ cm

Effective Height $[z]$:
 $=$ Centroid Height + Vessel Base Elevation
 $= 128 + 30 = 158$ cm
 $= 5.17$ ft. Imperial Units

Velocity Pressure coefficient evaluated at height z $[K_z]$:
Because z (5.17 ft.) < 15 ft.
 $= 2.01(15 / Z_g)^{2/\alpha}$
 $= 2.01(15/900)^{2/9.5}$
 $= 0.85$

Type of Hill: No Hill

Wind Directionality Factor $[K_d]$:
 $= 1$, per Table 26.6-1

As there is No Hill Present: $[K_{zt}]$:



K1 = 0, K2 = 0, K3 = 0

Topographical Factor [Kzt]:

$$= (1 + K1 \cdot K2 \cdot K3)^2$$

$$= (1 + 0 \cdot 0 \cdot 0)^2$$

$$= 1$$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$$

$$= \max(16, 0.00256 \cdot 0.85 \cdot 1 \cdot 1 \cdot 1 \cdot 60.4^2)$$

$$= 16 \text{ psf } [78.1] \text{ kgf/m}^2$$

Force on the first element [F]:

$$= qz \cdot G \cdot Cf \cdot \text{WindArea}$$

$$= 16 \cdot 0.85 \cdot 0.54 \cdot 248$$

$$= 1830 \text{ lbs. } [830] \text{ kgf}$$

Element	Hgt (z) cm	K1	K2	K3	Kz	Kzt	qz kgf/m ²
Skirt-3	157.5	0.000	0.000	0.000	0.849	1.000	78.119
Skirt-2	427.5	0.000	0.000	0.000	0.849	1.000	78.119
Skirt-1	620.0	0.000	0.000	0.000	0.905	1.000	78.119
Bottom Head	670.0	0.000	0.000	0.000	0.920	1.000	78.119
Shell 1	772.5	0.000	0.000	0.000	0.948	1.000	78.119
Top Head	875.0	0.000	0.000	0.000	0.973	1.000	78.119
INT-SKIRT-1	941.3	0.000	0.000	0.000	0.988	1.000	78.119
INT-SKIRT-2	1157.5	0.000	0.000	0.000	1.032	1.000	78.119
INT-SKIRT-3	1450.9	0.000	0.000	0.000	1.083	1.000	78.119
INT-SKIRT-4	1757.5	0.000	0.000	0.000	1.127	1.000	78.119
INT-SKIRT-5	1973.8	0.000	0.000	0.000	1.155	1.000	78.119
B-DISHED-END-10	2040.0	0.000	0.000	0.000	1.163	1.000	78.119
SHELL-104-D2A	2090.0	0.000	0.000	0.000	1.169	1.000	78.119
T-DISHED-END-10	2266.3	0.000	0.000	0.000	1.189	1.000	78.119

Wind Vibration Calculations

This evaluation is based on work by Kanti Mahajan and Ed Zorilla

Nomenclature

- Cf - Correction factor for natural frequency
- D - Average outer insulated diameter of vessel, cm
- Df - Damping Factor < 0.75 Unstable, > 0.95 Stable
- Dr - Average outer insulated of top half of vessel, cm
- f - Natural frequency of vibration (Hertz)
- f1 - Natural frequency of bare vessel based on a unit value of (D/L²)(104)
- L - Total height of structure cm
- Lc - Total length of conical section(s) of vessel, cm
- tb - Uncorroded plate thickness at bottom of vessel, mm
- V30 - Design Wind Speed provided by user, km/hr
- Vc - Critical wind velocity, km/hr
- Vw - Maximum wind speed at top of structure, km/hr
- W - Total corroded weight of structure, kgf
- Ws - Cor. vessel weight excl. weight of parts which do not effect stiff. kgf
- Z - Maximum amplitude of vibration at top of vessel, mm
- Dl - Logarithmic decrement (taken as 0.03 for Welded Structures)
- Vp - Vib. Chance, <= 0.000320369 (High); 0.000320369 < 0.000400461 (Probable)
- P30 - wind pressure 30 feet above the base

Check other Conditions and Basic Assumptions:

:: Total Cone Length / Total Length < 0.5
 :: 0/2110 = 0

:: [D / L²] • 10⁴ < 8 (English Units, ft.)
 :: [22.9/69.2²] • 10⁴ = 47.8 [Geometry Violation]

Compute the vibration possibility. If Vp > 0.000400461 no chance. [Vp]:

$$= W / (L \cdot Dr^2)$$



$$= 987480 / (2110 \cdot 613^2)$$

$$= 0.12441E-02$$

Since Vp is > 0.000400461 no further vibration analysis is required !

Platform Load Calculations

ID	Wind Area cm ²	Elevation cm	Pressure kgf/m ²	Force kgf	Cf
PLTF-1	15000.00	385.00	78.12	117.18	1.00
PLTF-2	55196.80	670.00	78.12	431.19	1.00
PLTF-3	30000.00	1070.00	78.12	234.36	1.00
PLTF-4	30000.00	1470.00	78.12	234.36	1.00
PLTF-5	30000.00	1870.00	78.12	234.36	1.00
PLTF-6	30000.00	2139.90	78.12	234.36	1.00
PLTF-7	50000.00	2460.00	78.12	390.59	1.00

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation cm	Pressure kgf/m ²	Force kgf
Tailing+Braci	0.00	60.00	78.12	0.00
HH Hinge	0.00	470.00	78.12	0.00
UC1 Hinge	0.00	470.00	78.12	0.00
N2 inner dish	0.00	420.00	78.12	0.00
WIRE SCREEN	0.00	670.00	78.12	0.00
SF weld	0.00	665.00	78.12	0.00
Support balls	0.00	528.50	78.12	0.00
Catalyst	0.00	773.00	78.12	0.00
Split Flange	0.00	680.00	78.12	0.00
Cleat Weight	0.00	720.00	78.12	0.00
Lifting Lug-1	0.00	780.00	78.12	0.00
Lifting Lug-2	0.00	780.00	78.12	0.00
MISC WEIGHT	0.00	772.50	78.12	0.00
Manway Davit	0.00	1157.90	78.12	0.00
N1 Vanes	0.00	1175.00	78.12	0.00
Grating	0.00	955.00	78.12	0.00
Support Balls	0.00	915.00	78.12	0.00
SF weld	0.00	880.00	78.12	0.00
GAS WIPPER RI	0.00	975.00	78.12	0.00
WIRE SCREEN-1	0.00	935.00	78.12	0.00
WIRE SCREEN-2	0.00	895.00	78.12	0.00
N1-ELBOW	0.00	1245.70	78.12	0.00
K4-elbow	0.00	1245.70	78.12	0.00
K1-ELBOW	0.00	1245.70	78.12	0.00
HH Hinge	0.00	1840.00	78.12	0.00
UC1 Hinge	0.00	1840.00	78.12	0.00
N2 inner dish	0.00	1790.00	78.12	0.00
Wire Screen	0.00	2040.00	78.12	0.00
SF weld	0.00	2040.00	78.12	0.00
Support balls	0.00	1898.50	78.12	0.00
Catalyst-D2A	0.00	2090.00	78.12	0.00
Split Flange-	0.00	2050.00	78.12	0.00
Cleat Weight-	0.00	2090.00	78.12	0.00
Lifting Lug-2	0.00	2100.00	78.12	0.00
MW-DAVIT-M1-D	0.00	2420.00	78.12	0.00
N1 Vanes	0.00	2390.00	78.12	0.00
Grating	0.00	2220.00	78.12	0.00
Support Balls	0.00	2180.00	78.12	0.00
SF weld	0.00	2140.00	78.12	0.00
PSV WEIGHT	0.00	2420.00	78.12	0.00
GAS WIPER RIN	0.00	2240.00	78.12	0.00
Wire Screen1	0.00	2200.00	78.12	0.00
Wire Screen 2	0.00	2160.00	78.12	0.00

The Natural Frequency for the Vessel (Ope...) is 4.81579 Hz.



Wind Load Calculation:

From	To	Wind Height cm	Wind Diameter cm	Wind Area cm ²	Wind Pressure kgf/m ²	Element Wind Load kgf
10	20	157.5	902.4	230112	78.1189	913.107
20	30	427.5	902.4	257184	78.1189	1136.61
30	40	620	902.4	90240	78.1189	358.081
40	50	669.999	952.758	2.90401	78.1189	474.32
50	60	772.5	960.63	196929	78.1189	781.434
60	70	874.999	952.758	2.90401	78.1189	0.011523
70	80	941.25	925.329	122606	78.1189	486.513
80	90	1157.5	920.286	276086	78.1189	1333.84
90	100	1450.89	813.83	244149	78.1189	1229.99
100	110	1757.5	707.373	212212	78.1189	1116.37
110	120	1973.75	707.373	93726.9	78.1189	371.918
120	130	2040	732.096	2.23143	78.1189	0.0088545
130	140	2090	738	73800	78.1189	556.792
140	150	2266.3	732.096	171118	78.1189	1145.39

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	2.000
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	2.000
Site Class	D
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	2.500
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Seismic Analysis Results:

$$\begin{aligned}
 SMS &= Fa * Ss = 1.6 * 0.15 = 0.24 \\
 SM1 &= Fv * S1 = 2.4 * 0.065 = 0.16 \\
 SDS &= 2/3 * Sms = 2/3 * 0.24 = 0.16 \\
 SD1 &= 2/3 * Sm1 = 2/3 * 0.16 = 0.1
 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 = $C_t \cdot h_{nx}$ where $C_t = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 = $0.020(78.50.75)$
 = 0.527 seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 = (1/Natural Frequency) = (1/4.82)
 = 0.208

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 = Minimum Value of ($1.7 \cdot 0.53$, 0.21) per 12.8.2
 = 0.208

Seismic Response Coefficient per equation 12.8-2 [Cs]:
 = $Sds / (R / I)$
 = $0.16 / (2/2)$
 = 0.157

Maximum value of C_s per equation 12.8-3 [Cs]:
 = $Sd1 / (T(R / I_e))$
 = $0.1 / (0.21 (2/2))$
 = 0.501

Minimum value of C_s per equation 15.4-1 [Cs]:
 = $\max((0.044 \cdot SDS \cdot I_e), 0.030)$
 = $\max((0.044 \cdot 0.16 \cdot 2), 0.030)$
 = 0.030

Total Base Shear [V]:
 = $C_s \cdot W$ (Equation 12.8-1):
 = $0.16 \cdot 1000454$
 = 156871.172 kgf

Vertical load per 12.4-4, [Ev]:
 = $0.2 \cdot Sds \cdot W$
 = $0.2 \cdot 0.16 \cdot 1000454$



= 31374.2 kgf

Distribute the Base shear force to each element according to the equations

$F_x = C_{vx} * V$ (eqn. 12.8-11) and the vertical distribution factor

$C_{vx} = W_x * h_x^k / (\text{Sum of } W_i * h_i^k)$ and k is an exponent which is related to the period of Vibration.

In this case, the value of k was 1

The Natural Frequency for the Vessel (Ope...) is 4.81579 Hz.

Earthquake Load Calculation:

From	To	Earthquake Height cm	Earthquake Weight kgf	Element Ope Load kgf	Element Emp Load kgf	Element Vertical Load kgf
10	20	127.5	21768.2	397.311	351.772	13210.4
20	30	397.5	19700.8	1121.04	992.545	12527.7
30	40	590	6197.28	523.422	463.428	11909.9
40	50	639.999	284801	26092.7	6473.18	11715.6
50	60	742.5	234008	24872.9	6520.86	2784.21
60	70	844.999	98228.2	11882	7226.85	-4554.29
70	80	911.25	8214.64	1071.58	948.758	1447.2
80	90	1127.5	18683.1	3015.53	2669.89	1189.58
90	100	1427.5	20383.6	4165.38	3687.96	603.685
100	110	1727.5	17403.5	4303.81	3810.51	-35.5432
110	120	1943.75	6270.97	1744.91	1544.92	-581.316
120	130	2010	122648	35290.2	31245.3	-777.974
130	140	2060	76184	22466.2	5648.77	-4624.21
140	150	2110	65962.5	19924.1	8695.03	-7013.34



Bending Moments due to user defined forces and moments.

User Force/Moment Shear and Bending

From	To	Distance to Support cm	Cumulative Shr Wind Cas kgf	Cumulative Shr Eqk Cas kgf	Wind Bending kgf-m.	Earthquake Bending kgf-m.
10	20	127.5	18553.4	18553.4	379308	379308
20	30	397.5	18553.4	18553.4	331997	331997
30	40	590	18553.4	18553.4	279119	279119
40	50	639.999	18553.4	18553.4	260566	260566
50	60	742.5	18553.4	18553.4	260566	260566
60	70	844.999	18553.4	18553.4	222532	222532
70	80	911.25	9276.71	9276.71	169013	169013
80	90	1127.5	9276.71	9276.71	156722	156722
90	100	1427.5	9276.71	9276.71	128891	128891
100	110	1727.5	9276.71	9276.71	101061	101061
110	120	1943.75	9276.71	9276.71	73231.1	73231.1
120	130	2010	9276.71	9276.71	60939.7	60939.7
130	140	2060	9276.71	9276.71	60939.4	60939.4
140	150	2110	9276.71	9276.71	51663	51663

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
 Wind: 1.100; Earthquake: 1.000.

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The following table is for the Operating Case.

Wind/Earthquake Shear, Bending:

From	To	Distance to Support cm	Cumulative Wind Shear kgf	Earthquake Shear kgf	Wind Bending kgf-m.	Earthquake Bending kgf-m.
10	20	127.5	9904.39	156871	117262	2289322
20	30	397.5	8991.28	156474	93170.4	1889806
30	40	590	7854.67	155353	69164.9	1445453
40	50	639.999	7496.59	154829	61489.5	1290366
50	60	742.5	7022.27	128737	61489.3	1290362
60	70	844.999	6240.84	103864	47894.8	1051949
70	80	911.25	6240.82	91981.7	47894.6	1051946
80	90	1127.5	5754.31	90910.2	39947.8	930780
90	100	1427.5	4420.47	87894.6	24685.6	662573
100	110	1727.5	3190.48	83729.2	13269.1	405137
110	120	1943.75	2074.11	79425.4	5372.26	160404
120	130	2010	1702.19	77680.5	2870.51	56323.9
130	140	2060	1702.18	42390.3	2870.45	56322.1
140	150	2110	1145.39	19924.1	1446.7	25165.5

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
 Wind: 1.100; Earthquake: 1.000.

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Wind Deflection Calculations:

The following table is for Wind Deflection; Applied Forces Case.

Wind Load Deflection:

From	To	Cumulative Shear kgf	Centroid Deflection mm	Elem. End Deflection mm	Elem. Ang. Rotation
10	20	18553.4	0.0026829	0.010504	0.80596E-05
20	30	18553.4	0.024905	0.04482	0.15799E-04
30	40	18553.4	0.053024	0.061823	0.18180E-04
40	50	18553.4	0.061824	0.061824	0.18181E-04
50	60	18553.4	0.080767	0.1003	0.19324E-04
60	70	18553.4	0.1003	0.1003	0.19324E-04
70	80	9276.71	0.11342	0.12718	0.21226E-04
80	90	9276.71	0.16056	0.19684	0.25091E-04
90	100	9276.71	0.23634	0.2793	0.29693E-04
100	110	9276.71	0.326	0.37661	0.34907E-04
110	120	9276.71	0.40006	0.42409	0.36705E-04
120	130	9276.71	0.42409	0.42409	0.36705E-04
130	140	9276.71	0.4425	0.461	0.37090E-04
140	150	9276.71	0.461	0.461	0.37090E-04

Allowable deflection at the Tower Top (For)(6.000"/100ft. Criteria)

Allowable deflection : 105.500 Actual deflection : 0.461 mm

The following table is for Wind Deflection; Operating Case.

Wind Load Deflection:

From	To	Cumulative Shear kgf	Centroid Deflection mm	Elem. End Deflection mm	Elem. Ang. Rotation
10	20	9904.39	0.00081982	0.0031615	0.23883E-05
20	30	8991.28	0.0073703	0.013043	0.44443E-05
30	40	7854.67	0.015346	0.017805	0.50671E-05
40	50	7496.59	0.017805	0.017805	0.50671E-05
50	60	7022.27	0.023076	0.02849	0.53462E-05
60	70	6240.84	0.02849	0.028491	0.53463E-05
70	80	6240.82	0.03213	0.035952	0.58998E-05
80	90	5754.31	0.045209	0.055166	0.68349E-05
90	100	4420.47	0.065784	0.077005	0.76424E-05
100	110	3190.48	0.088753	0.10094	0.82268E-05
110	120	2074.11	0.10641	0.11192	0.83450E-05
120	130	1702.19	0.11192	0.11192	0.83450E-05
130	140	1702.18	0.1161	0.12028	0.83606E-05
140	150	1145.39	0.12028	0.12028	0.83606E-05

Critical Wind Velocity for Tower Vibration:

From	To	1st Crit. Wind Speed km/hr	2nd Crit. Wind Speed km/hr
10	20	780.132	4875.82
20	30	780.132	4875.82
30	40	780.132	4875.82
40	50	823.667	5147.92
50	60	830.472	5190.45
60	70	823.667	5147.92
70	80	799.954	4999.71
80	90	795.594	4972.47
90	100	703.562	4397.26
100	110	611.529	3822.06



110	120	611.529	3822.06
120	130	632.903	3955.64
130	140	638.007	3987.54
140	150	632.903	3955.64

Allowable deflection at the Tower Top (Ope)(6.000"/100ft. Criteria)
Allowable deflection : 105.500 Actual deflection : 0.120 mm

Total Deflection in the Operating Condition + Applied Forces:
Allowable deflection : 105.500 Actual deflection : 0.581 mm

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Longitudinal Stress Constants:

From	To	Metal Area New cm ²	Metal Area Corroded cm ²	Section Modulus New mm ³	Section Modulus Corroded mm ³
10	20	7996.08	7619.81	1489729408	1419928704
20	30	7996.08	7619.81	1489729408	1419928704
30	40	7996.08	7619.81	1489729408	1419928704
40	50	15509.4	14810.4	2.875E+09	2.748E+09
50	60	30548.8	29855.8	5.617E+09	5.494E+09
60	70	15509.4	14810.4	2.875E+09	2.748E+09
70	80	7999.29	7624.52	1490926464	1421677568
80	90	7955.47	7582.76	1474611840	1406124288
90	100	6357.14	5799.41	903260864	824907200
100	110	6106.56	5820.8	867652992	827511360
110	120	6106.56	5570.68	867652992	792337536
120	130	9311.68	8779.08	1315026048	1241105920
130	140	17907.1	17379	2.509E+09	2.437E+09
140	150	9311.68	8779.08	1315026048	1241105920



Longitudinal Allowable Stresses:

From	To	Tensile kgf/cm ²	Hydrotest Tensile kgf/cm ²	Compressive kgf/cm ²	Hydrotest Compressive kgf/cm ²
10	20	1181.99	2004.25	-1071.37	-1071.37
20	30	1181.99	2004.25	-1071.37	-1071.37
30	40	905.46	1706.92	-831.893	-1317.98
40	50	1810.92	3413.83	-950.407	-1397.82
50	60	1810.92	3413.83	-1072.2	-1449.65
60	70	1810.92	3413.83	-950.407	-1397.82
70	80	1267.64	2389.68	-831.783	-1317.17
80	90	1173.44	2004.25	-832.771	-1073.14
90	100	1173.44	2004.25	-872.245	-1149.5
100	110	1173.44	2004.25	-880.18	-1164.06
110	120	1267.64	2389.68	-872.245	-1365.09
120	130	1810.92	3413.83	-953.985	-1399.33
130	140	1810.92	3413.83	-1072.57	-1449.81
140	150	1810.92	3413.83	-953.985	-1399.33



Longitudinal Stress Report



Longitudinal Operating and Empty Stresses are computed in the corroded condition. Stresses due to loads in the hydrostatic test cases have also been computed in the corroded condition.

Longitudinal Pressure Stresses due to:

From	To	Longitudinal Stress Internal Pressure kgf/cm ²	Longitudinal Stress External Pressure kgf/cm ²	Longitudinal Stress Hydrotest Pressure kgf/cm ²
10	20	...	...	...
20	30	...	...	...
30	40	...	...	...
40	50	1488.45	...	1934.99
50	60	726.825	...	944.873
60	70	1488.45	...	1934.99
70	80	...	...	...
80	90	...	...	...
90	100	...	...	...
100	110	...	...	...
110	120	...	...	...
120	130	1458.04	...	1895.46
130	140	725.204	...	942.765
140	150	1458.04	...	1895.46

Longitudinal Stresses due to Weight Loads for these Conditions:

From	To	Wght. Str. Empty kgf/cm ²	Wght. Str. Operating kgf/cm ²	Wght. Str. Hydrotest kgf/cm ²	Wght. Str. Emp. Mom. kgf/cm ²	Wght. Str. Opr. Mom. kgf/cm ²
10	20	-131.297	-131.297	-121.664	2.51269	2.51269
20	30	-128.44	-128.44	-118.807	2.51269	2.51269
30	40	-125.854	-125.854	-116.222	2.08301	2.08301
40	50	-64.3325	-64.3325	-64.3325	1.07643	1.07643
50	60	-22.3738	-22.3738	-22.3738	0.53584	0.53584
60	70	-29.3023	-29.3023	-29.3023	1.07134	1.07134
70	80	-44.0356	-44.0356	-47.6936	2.02974	2.02974
80	90	-43.1948	-43.1948	-46.873	2.05219	2.05219
90	100	-49.7412	-49.7412	-54.5504	2.386	2.386
100	110	-49.5583	-49.5583	-54.3499	2.37849	2.37849
110	120	-48.6594	-48.6594	-53.6661	1.49878	1.49878
120	130	-30.162	-30.162	-30.162	0.95684	0.95684
130	140	-8.17921	-8.17921	-8.17921	0.31516	0.31516
140	150	-7.51361	-7.51361	-7.51361	0.0022748	0.0022748

Longitudinal Stresses due to Weight Loads and Bending for these Conditions:

From	To	Wght. Str. Hyd. Mom. kgf/cm ²	Bend. Str. Oper. Wind kgf/cm ²	Bend. Str. Oper. Equ. kgf/cm ²	Bend. Str. Hyd. Wind kgf/cm ²	Bend. Str. Hyd. Equ. kgf/cm ²
10	20	2.51269	8.25833	161.228	4.12917	...
20	30	2.51269	6.56163	133.092	3.28081	...
30	40	2.08301	4.87101	101.798	2.43551	...
40	50	1.07643	2.23784	46.9614	1.11892	...
50	60	0.53584	1.11928	23.4882	0.55964	...
60	70	1.07134	1.74308	38.2845	0.87154	...
70	80	2.02974	3.36888	73.9933	1.68444	...
80	90	2.05219	2.84099	66.1948	1.42049	...
90	100	2.386	1.60856	49.113	0.80428	...
100	110	2.37849	1.6035	48.9585	0.80175	...
110	120	1.49878	0.67803	20.2445	0.33901	...



120	130	0.95684	0.23129	4.5382	0.11564	...
130	140	0.31516	0.11778	2.31105	0.058891	...
140	150	0.0022748	0.11657	2.02767	0.058283	...

Longitudinal Stresses due to these Conditions:

From	To	Vortex Shedding Operating Case kgf/cm ²	Vortex Shedding Empty Case kgf/cm ²	Vortex Shedding Test Case kgf/cm ²	Earthquake Empty Case kgf/cm ²
10	20	...	...	...	90.4366
20	30	...	...	...	76.051
30	40	...	...	...	60.1079
40	50	...	...	...	28.1975
50	60	...	...	...	14.1033
60	70	...	...	...	23.0691
70	80	...	...	...	44.5862
80	90	...	...	...	39.635
90	100	...	...	...	28.009
100	110	...	...	...	27.9209
110	120	...	...	...	10.5927
120	130	...	...	...	1.8131
130	140	...	...	...	0.9233
140	150	...	...	...	0.88489

Longitudinal Stresses due to Applied Axial Forces:

From	To	Longitudinal Stress Y Forces Wind kgf/cm ²	Longitudinal Stress Y Forces Seismic kgf/cm ²
10	20	-2.38377	1.73369
20	30	-2.38377	1.6441
30	40	-2.38377	1.56302
40	50	-1.22643	0.79104
50	60	-0.60839	0.093255
60	70	-1.22643	-0.30751
70	80	-1.19115	0.18981
80	90	-1.19771	0.15688
90	100	-1.56601	0.10409
100	110	-1.56026	-0.0061062
110	120	-1.63031	-0.10435
120	130	-1.0345	-0.088617
130	140	-0.52258	-0.26608
140	150	-1.0345	-0.79887

Longitudinal Stresses due to User Forces and Moments:

From	To	Wind For/Mom Corroded kgf/cm ²	Earthquake For/Mom Corroded kgf/cm ²	Wind For/Mom No Corrosion kgf/cm ²	Earthquake For/Mom No Corrosion kgf/cm ²
10	20	26.7132	26.7132	25.4615	25.4615
20	30	23.3812	23.3812	22.2857	22.2857
30	40	19.6573	19.6573	18.7362	18.7362
40	50	9.48303	9.48303	9.06283	9.06283
50	60	4.74304	4.74304	4.6391	4.6391
60	70	8.0988	8.0988	7.73994	7.73994
70	80	11.8883	11.8883	11.3361	11.3361
80	90	11.1456	11.1456	10.628	10.628
90	100	12.2512	12.2512	11.1885	11.1885
100	110	12.2127	12.2127	11.6477	11.6477
110	120	9.24241	9.24241	8.44014	8.44014
120	130	4.91011	4.91011	4.63411	4.63411
130	140	2.50051	2.50051	2.42928	2.42928
140	150	4.16266	4.16266	3.92867	3.92867



Bending Stresses due to Blast Load:

From	To	Bending Stress kgf/cm ²
10	20	...
20	30	...
30	40	...
40	50	...
50	60	...
60	70	...
70	80	...
80	90	...
90	100	...
100	110	...
110	120	...
120	130	...
130	140	...
140	150	...



Stress Combination Load Cases for Vertical Vessels:

Load Case Definition Key

IP = Longitudinal Stress due to Internal Pressure
 EP = Longitudinal Stress due to External Pressure
 HP = Longitudinal Stress due to Hydrotest Pressure
 NP = No Pressure
 EW = Longitudinal Stress due to Weight (No Liquid)
 OW = Longitudinal Stress due to Weight (Operating)
 HW = Longitudinal Stress due to Weight (Hydrotest)
 WI = Bending Stress due to Wind Moment (Operating)
 EQ = Bending Stress due to Earthquake Moment (Operating)
 EE = Bending Stress due to Earthquake Moment (Empty)
 HI = Bending Stress due to Wind Moment (Hydrotest)
 HE = Bending Stress due to Earthquake Moment (Hydrotest)
 WE = Bending Stress due to Wind Moment (Empty) (no CA)
 WF = Bending Stress due to Wind Moment (Filled) (no CA)
 CW = Longitudinal Stress due to Weight (Empty) (no CA)
 VO = Bending Stress due to Vortex Shedding Loads (Ope)
 VE = Bending Stress due to Vortex Shedding Loads (Emp)
 VF = Bending Stress due to Vortex Shedding Loads (Test No CA.)
 FW = Axial Stress due to Vertical Forces for the Wind Case
 FS = Axial Stress due to Vertical Forces for the Seismic Case
 BW = Bending Stress due to Lat. Forces for the Wind Case, Corroded
 BS = Bending Stress due to Lat. Forces for the Seismic Case, Corroded
 BN = Bending Stress due to Lat. Forces for the Wind Case, UnCorroded
 BU = Bending Stress due to Lat. Forces for the Seismic Case, UnCorroded
 BL = Bending Stress due to Blast Loads, Corroded

General Notes:

Case types HI and HE are in the Corroded condition.

Case types WE, WF, and CW are in the Un-Corroded condition.

A blank stress and stress ratio indicates that the corresponding stress comprising those components that did not contribute to that type of stress.

An asterisk (*) in the final column denotes overstress.

Analysis of Load Case 1 : NP+EW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		1181.99	-171.16	1071.37		0.1598
20		1181.99	-163.28	1071.37		0.1524
30		905.46	-154.85	831.89		0.1861
40		1810.92	-78.36	950.41		0.0824
50		1810.92	-29.38	1072.20		0.0274
60		1810.92	-41.44	950.41		0.0436
70		1267.64	-62.51	831.78		0.0752
80		1173.44	-60.43	832.77		0.0726
90		1173.44	-67.55	872.25		0.0774
100		1173.44	-67.31	880.18		0.0765
110		1267.64	-61.71	872.25		0.0707
120		1810.92	-37.29	953.98		0.0391
130		1810.92	-11.64	1072.57		0.0108
140		1810.92	-12.83	953.98		0.0134

Analysis of Load Case 2 : NP+EW+EE+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		1181.99	-249.23	1071.37		0.2326
20		1181.99	-228.74	1071.37		0.2135
30		905.46	-206.14	831.89		0.2478
40		1810.92	-102.30	950.41		0.1076
50		1810.92	-41.66	1072.20		0.0389



60	2.63	1810.92	-61.85	950.41	0.0015	0.0651
70	14.66	1267.64	-102.35	831.78	0.0116	0.1230
80	9.79	1173.44	-95.87	832.77	0.0083	0.1151
90		1173.44	-92.28	872.25		0.1058
100		1173.44	-92.08	880.18		0.1046
110		1267.64	-70.10	872.25		0.0804
120		1810.92	-37.93	953.98		0.0398
130		1810.92	-12.18	1072.57		0.0114
140		1810.92	-13.36	953.98		0.0140

Analysis of Load Case 3 : NP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		1181.99	-171.16	1071.37		0.1598
20		1181.99	-163.28	1071.37		0.1524
30		905.46	-154.85	831.89		0.1861
40		1810.92	-78.36	950.41		0.0824
50		1810.92	-29.38	1072.20		0.0274
60		1810.92	-41.44	950.41		0.0436
70		1267.64	-62.51	831.78		0.0752
80		1173.44	-60.43	832.77		0.0726
90		1173.44	-67.55	872.25		0.0774
100		1173.44	-67.31	880.18		0.0765
110		1267.64	-61.71	872.25		0.0707
120		1810.92	-37.29	953.98		0.0391
130		1810.92	-11.64	1072.57		0.0108
140		1810.92	-12.83	953.98		0.0134

Analysis of Load Case 4 : NP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	60.89	1181.99	-320.02	1071.37	0.0515	0.2987
20	32.19	1181.99	-285.78	1071.37	0.0272	0.2667
30		905.46	-247.83	831.89		0.2979
40		1810.92	-121.06	950.41		0.1274
50	6.49	1810.92	-51.05	1072.20	0.0036	0.0476
60	17.84	1810.92	-77.06	950.41	0.0099	0.0811
70	44.07	1267.64	-131.76	831.78	0.0348	0.1584
80	36.35	1173.44	-122.43	832.77	0.0310	0.1470
90	14.11	1173.44	-113.39	872.25	0.0120	0.1300
100	13.99	1173.44	-113.11	880.18	0.0119	0.1285
110		1267.64	-79.75	872.25		0.0914
120		1810.92	-40.66	953.98		0.0426
130		1810.92	-13.57	1072.57		0.0127
140		1810.92	-14.51	953.98		0.0152

Analysis of Load Case 5 : NP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		2004.25	-128.31	1071.37		0.1198
20		2004.25	-124.60	1071.37		0.1163
30		1706.92	-120.74	1317.98		0.0916
40		3413.83	-66.53	1397.82		0.0476
50		3413.83	-23.47	1449.65		0.0162
60		3413.83	-31.25	1397.82		0.0224
70		2389.68	-51.41	1317.17		0.0390
80		2004.25	-50.35	1073.14		0.0469
90		2004.25	-57.74	1149.50		0.0502
100		2004.25	-57.53	1164.06		0.0494
110		2389.68	-55.50	1365.09		0.0407
120		3413.83	-31.23	1399.33		0.0223
130		3413.83	-8.55	1449.81		0.0059
140		3413.83	-7.57	1399.33		0.0054

Analysis of Load Case 6 : NP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
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10	2004.25	-124.18	1071.37	0.1159
20	2004.25	-121.32	1071.37	0.1132
30	1706.92	-118.30	1317.98	0.0898
40	3413.83	-65.41	1397.82	0.0468
50	3413.83	-22.91	1449.65	0.0158
60	3413.83	-30.37	1397.82	0.0217
70	2389.68	-49.72	1317.17	0.0378
80	2004.25	-48.93	1073.14	0.0456
90	2004.25	-56.94	1149.50	0.0495
100	2004.25	-56.73	1164.06	0.0487
110	2389.68	-55.16	1365.09	0.0404
120	3413.83	-31.12	1399.33	0.0222
130	3413.83	-8.49	1449.81	0.0059
140	3413.83	-7.52	1399.33	0.0054

Analysis of Load Case 7 : IP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		1181.99	-171.16	1071.37		0.1598
20		1181.99	-163.28	1071.37		0.1524
30		905.46	-154.85	831.89		0.1861
40	1435.69	1810.92		950.41	0.7928	
50	710.24	1810.92		1072.20	0.3922	
60	1468.84	1810.92		950.41	0.8111	
70		1267.64	-62.51	831.78		0.0752
80		1173.44	-60.43	832.77		0.0726
90		1173.44	-67.55	872.25		0.0774
100		1173.44	-67.31	880.18		0.0765
110		1267.64	-61.71	872.25		0.0707
120	1432.94	1810.92		953.98	0.7913	
130	719.44	1810.92		1072.57	0.3973	
140	1453.78	1810.92		953.98	0.8028	

Analysis of Load Case 8 : IP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	60.89	1181.99	-320.02	1071.37	0.0515	0.2987
20	32.19	1181.99	-285.78	1071.37	0.0272	0.2667
30		905.46	-247.83	831.89		0.2979
40	1482.43	1810.92		950.41	0.8186	
50	733.31	1810.92		1072.20	0.4049	
60	1506.30	1810.92		950.41	0.8318	
70	44.07	1267.64	-131.76	831.78	0.0348	0.1584
80	36.35	1173.44	-122.43	832.77	0.0310	0.1470
90	14.11	1173.44	-113.39	872.25	0.0120	0.1300
100	13.99	1173.44	-113.11	880.18	0.0119	0.1285
110		1267.64	-79.75	872.25		0.0914
120	1438.20	1810.92		953.98	0.7942	
130	721.88	1810.92		1072.57	0.3986	
140	1455.92	1810.92		953.98	0.8040	

Analysis of Load Case 9 : EP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		1181.99	-171.16	1071.37		0.1598
20		1181.99	-163.28	1071.37		0.1524
30		905.46	-154.85	1317.98		0.1175
40		1810.92	-78.36	1397.82		0.0561
50		1810.92	-29.38	1449.65		0.0203
60		1810.92	-41.44	1397.82		0.0296
70		1267.64	-62.51	1317.17		0.0475
80		1173.44	-60.43	1073.14		0.0563
90		1173.44	-67.55	1149.50		0.0588
100		1173.44	-67.31	1164.06		0.0578
110		1267.64	-61.71	1365.09		0.0452
120		1810.92	-37.29	1399.33		0.0267



130	1810.92	-11.64	1449.81	0.0080
140	1810.92	-12.83	1399.33	0.0092

Analysis of Load Case 10 : EP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	60.89	1181.99	-320.02	1071.37	0.0515	0.2987
20	32.19	1181.99	-285.78	1071.37	0.0272	0.2667
30		905.46	-247.83	1317.98		0.1880
40		1810.92	-121.06	1397.82		0.0866
50	6.49	1810.92	-51.05	1449.65	0.0036	0.0352
60	17.84	1810.92	-77.06	1397.82	0.0099	0.0551
70	44.07	1267.64	-131.76	1317.17	0.0348	0.1000
80	36.35	1173.44	-122.43	1073.14	0.0310	0.1141
90	14.11	1173.44	-113.39	1149.50	0.0120	0.0986
100	13.99	1173.44	-113.11	1164.06	0.0119	0.0972
110		1267.64	-79.75	1365.09		0.0584
120		1810.92	-40.66	1399.33		0.0291
130		1810.92	-13.57	1449.81		0.0094
140		1810.92	-14.51	1399.33		0.0104

Analysis of Load Case 11 : HP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		2004.25	-128.31	1071.37		0.1198
20		2004.25	-124.60	1071.37		0.1163
30		1706.92	-120.74	1317.98		0.0916
40	1872.85	3413.83		1397.82	0.5486	
50	923.59	3413.83		1449.65	0.2705	
60	1907.63	3413.83		1397.82	0.5588	
70		2389.68	-51.41	1317.17		0.0390
80		2004.25	-50.35	1073.14		0.0469
90		2004.25	-57.74	1149.50		0.0502
100		2004.25	-57.53	1164.06		0.0494
110		2389.68	-55.50	1365.09		0.0407
120	1866.37	3413.83		1399.33	0.5467	
130	934.96	3413.83		1449.81	0.2739	
140	1888.00	3413.83		1399.33	0.5530	

Analysis of Load Case 12 : HP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		2004.25	-124.18	1071.37		0.1159
20		2004.25	-121.32	1071.37		0.1132
30		1706.92	-118.30	1317.98		0.0898
40	1871.73	3413.83		1397.82	0.5483	
50	923.03	3413.83		1449.65	0.2704	
60	1906.76	3413.83		1397.82	0.5585	
70		2389.68	-49.72	1317.17		0.0378
80		2004.25	-48.93	1073.14		0.0456
90		2004.25	-56.94	1149.50		0.0495
100		2004.25	-56.73	1164.06		0.0487
110		2389.68	-55.16	1365.09		0.0404
120	1866.25	3413.83		1399.33	0.5467	
130	934.90	3413.83		1449.81	0.2739	
140	1887.94	3413.83		1399.33	0.5530	

Analysis of Load Case 13 : IP+WE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		1181.99	-133.81	1071.37		0.1249
20		1181.99	-130.95	1071.37		0.1222
30		905.46	-127.94	831.89		0.1538
40	1425.20	1810.92		950.41	0.7870	
50	704.99	1810.92		1072.20	0.3893	
60	1460.22	1810.92		950.41	0.8063	



70		1267.64	-46.07	831.78		0.0554
80		1173.44	-45.25	832.77		0.0543
90		1173.44	-52.13	872.25		0.0598
100		1173.44	-51.94	880.18		0.0590
110		1267.64	-50.16	872.25		0.0575
120	1428.84	1810.92		953.98	0.7890	
130	717.34	1810.92		1072.57	0.3961	
140	1450.53	1810.92		953.98	0.8010	

Analysis of Load Case 14 : IP+WF+CW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		1181.99	-125.12	1071.37		0.1168
20		1181.99	-122.40	1071.37		0.1142
30		905.46	-119.93	831.89		0.1442
40	1427.02	1810.92		950.41	0.7880	
50	704.96	1810.92		1072.20	0.3893	
60	1460.47	1810.92		950.41	0.8065	
70		1267.64	-41.97	831.78		0.0505
80		1173.44	-41.17	832.77		0.0494
90		1173.44	-45.38	872.25		0.0520
100		1173.44	-47.24	880.18		0.0537
110		1267.64	-44.39	872.25		0.0509
120	1429.61	1810.92		953.98	0.7894	
130	717.27	1810.92		1072.57	0.3961	
140	1450.96	1810.92		953.98	0.8012	

Analysis of Load Case 15 : IP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		1181.99	-133.81	1071.37		0.1249
20		1181.99	-130.95	1071.37		0.1222
30		905.46	-127.94	831.89		0.1538
40	1425.20	1810.92		950.41	0.7870	
50	704.99	1810.92		1072.20	0.3893	
60	1460.22	1810.92		950.41	0.8063	
70		1267.64	-46.07	831.78		0.0554
80		1173.44	-45.25	832.77		0.0543
90		1173.44	-52.13	872.25		0.0598
100		1173.44	-51.94	880.18		0.0590
110		1267.64	-50.16	872.25		0.0575
120	1428.84	1810.92		953.98	0.7890	
130	717.34	1810.92		1072.57	0.3961	
140	1450.53	1810.92		953.98	0.8010	

Analysis of Load Case 16 : IP+VE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		1181.99	-133.81	1071.37		0.1249
20		1181.99	-130.95	1071.37		0.1222
30		905.46	-127.94	831.89		0.1538
40	1425.20	1810.92		950.41	0.7870	
50	704.99	1810.92		1072.20	0.3893	
60	1460.22	1810.92		950.41	0.8063	
70		1267.64	-46.07	831.78		0.0554
80		1173.44	-45.25	832.77		0.0543
90		1173.44	-52.13	872.25		0.0598
100		1173.44	-51.94	880.18		0.0590
110		1267.64	-50.16	872.25		0.0575
120	1428.84	1810.92		953.98	0.7890	
130	717.34	1810.92		1072.57	0.3961	
140	1450.53	1810.92		953.98	0.8010	

Analysis of Load Case 17 : NP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
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10	1181.99	-133.81	1071.37	0.1249
20	1181.99	-130.95	1071.37	0.1222
30	905.46	-127.94	831.89	0.1538
40	1810.92	-65.41	950.41	0.0688
50	1810.92	-22.91	1072.20	0.0214
60	1810.92	-30.37	950.41	0.0320
70	1267.64	-46.07	831.78	0.0554
80	1173.44	-45.25	832.77	0.0543
90	1173.44	-52.13	872.25	0.0598
100	1173.44	-51.94	880.18	0.0590
110	1267.64	-50.16	872.25	0.0575
120	1810.92	-31.12	953.98	0.0326
130	1810.92	-8.49	1072.57	0.0079
140	1810.92	-7.52	953.98	0.0079

Analysis of Load Case 18 : FS+BS+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		984.99	-158.79	892.81		0.1779
20		984.99	-152.69	892.81		0.1710
30		754.55	-146.03	693.24		0.2106
40	1435.47	1509.10		792.01	0.9512	
50	709.82	1509.10		893.50	0.4704	
60	1468.01	1509.10		792.01	0.9728	
70		1056.37	-57.76	693.15		0.0833
80		977.86	-56.24	693.98		0.0810
90		977.86	-64.27	726.87		0.0884
100		977.86	-64.16	733.48		0.0875
110		1056.37	-59.50	726.87		0.0819
120	1433.66	1509.10		794.99	0.9500	
130	719.57	1509.10		893.81	0.4768	
140	1453.90	1509.10		794.99	0.9634	

Analysis of Load Case 19 : FS+BS+EP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		984.99	-158.79	892.81		0.1779
20		984.99	-152.69	892.81		0.1710
30		754.55	-146.03	1098.31		0.1330
40		1509.10	-74.10	1164.85		0.0636
50		1509.10	-27.56	1208.04		0.0228
60		1509.10	-38.78	1164.85		0.0333
70		1056.37	-57.76	1097.64		0.0526
80		977.86	-56.24	894.29		0.0629
90		977.86	-64.27	957.92		0.0671
100		977.86	-64.16	970.05		0.0661
110		1056.37	-59.50	1137.57		0.0523
120		1509.10	-36.12	1166.11		0.0310
130		1509.10	-11.26	1208.18		0.0093
140		1509.10	-12.48	1166.11		0.0107

Analysis of Load Case 20 : BL+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		1181.99	-133.81	1071.37		0.1249
20		1181.99	-130.95	1071.37		0.1222
30		905.46	-127.94	831.89		0.1538
40	1425.20	1810.92		950.41	0.7870	
50	704.99	1810.92		1072.20	0.3893	
60	1460.22	1810.92		950.41	0.8063	
70		1267.64	-46.07	831.78		0.0554
80		1173.44	-45.25	832.77		0.0543
90		1173.44	-52.13	872.25		0.0598
100		1173.44	-51.94	880.18		0.0590
110		1267.64	-50.16	872.25		0.0575
120	1428.84	1810.92		953.98	0.7890	
130	717.34	1810.92		1072.57	0.3961	



140 1450.53 1810.92 953.98 0.8010

Absolute Maximum of the all of the Stress Ratio's 0.9728

Governing Element: Top Head

Governing Load Case 18 : FS+BS+IP+OW

Element Description	Long Pressure Stress	Weight + Axial Load Stress	Eccentric Moment Stress	Wind or Earthquake Stress	Combined Stress	Which Side
Skirt-3	...	-129.6	2.5	26.7	...	+ Tens
Skirt-3	...	-129.6	-2.5	-26.7	-158.8	- Comp
Skirt-2	...	-126.8	2.5	23.4	...	+ Tens
Skirt-2	...	-126.8	-2.5	-23.4	-152.7	- Comp
Skirt-1	...	-124.3	2.1	19.7	...	+ Tens
Skirt-1	...	-124.3	-2.1	-19.7	-146.0	- Comp
Bottom Head	1488.5	-63.5	1.1	9.5	1435.5	+ Tens
Bottom Head	1488.5	-63.5	-1.1	-9.5	...	- Comp
Shell 1	726.8	-22.3	0.5	4.7	709.8	+ Tens
Shell 1	726.8	-22.3	-0.5	-4.7	...	- Comp
Top Head	1488.5	-29.6	1.1	8.1	1468.0	+ Tens
Top Head	1488.5	-29.6	-1.1	-8.1	...	- Comp
INT-SKIRT-1	...	-43.8	2.0	11.9	...	+ Tens
INT-SKIRT-1	...	-43.8	-2.0	-11.9	-57.8	- Comp
INT-SKIRT-2	...	-43.0	2.1	11.1	...	+ Tens
INT-SKIRT-2	...	-43.0	-2.1	-11.1	-56.2	- Comp
INT-SKIRT-3	...	-49.6	2.4	12.3	...	+ Tens
INT-SKIRT-3	...	-49.6	-2.4	-12.3	-64.3	- Comp
INT-SKIRT-4	...	-49.6	2.4	12.2	...	+ Tens
INT-SKIRT-4	...	-49.6	-2.4	-12.2	-64.2	- Comp
INT-SKIRT-5	...	-48.8	1.5	9.2	...	+ Tens
INT-SKIRT-5	...	-48.8	-1.5	-9.2	-59.5	- Comp
B-DISHED-END-102A	1458.0	-30.3	1.0	4.9	1433.7	+ Tens
B-DISHED-END-102A	1458.0	-30.3	-1.0	-4.9	...	- Comp
SHELL-104-D2A	725.2	-8.4	0.3	2.5	719.6	+ Tens
SHELL-104-D2A	725.2	-8.4	-0.3	-2.5	...	- Comp
T-DISHED-END-102A	1458.0	-8.3	...	4.2	1453.9	+ Tens
T-DISHED-END-102A	1458.0	-8.3	...	-4.2	...	- Comp



Skirt Support Analysis Per EN 13445-3 Section 16.12:

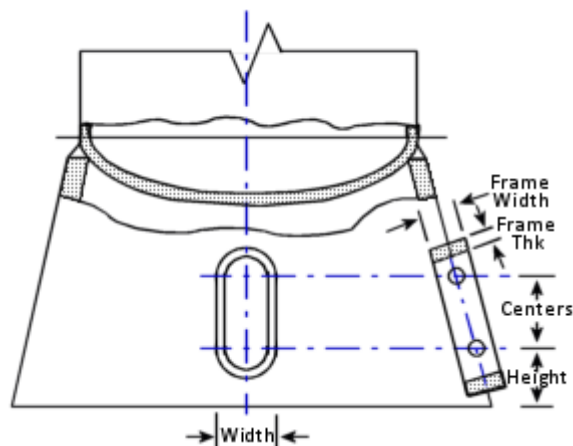
The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.

The access opening(s) is/are in section 4-4 only. The other calculations are for areas where no access openings exist. Please be aware of this when you review the results.

The skirt comprises more than one piece. If this is the case, the Openings will only be considered in this Section of the Skirt! Otherwise, make the whole skirt all one piece.

Item: Skirt-3

Total vessel (empty) weight	Fg	893176.625 kgf
Weight of vessel contents (liquid)	Ff	0.000 kgf
Vessel Gross Weight	Fg + Ff	893176.625 kgf
Skirt Mean Diameter	Dz	7486.000 mm
Skirt Analysis Thickness (Corroded)	ez	32.400 mm
Skirt Height	H	2550.000 mm
Additional Axial Force at section 1-1	F1	-18163.852 kgf
Moment at section 1-1	M1	2221803.500 kgf-m.
Applied Moment - Bottom of Skirt	Mb	2668630.250 kgf-m.
Weight at Bottom of Skirt	W	1000454.000 kgf



Access Opening: 2 Cuts through the top of the skirt.

Note:

If an opening cuts the skirt top or bottom, this is not a problem. This becomes a 'Sentry Box' or 'Mouse Hole' type Opening.

Skirt Access Opening Input Data:

Access #	Diameter mm	Layout Angle deg	Cntr. Dist. From Bottom mm	Oround Cntr-Cntr mm	Frame Width mm	Frame Thickness mm
1	1200.0000	0.0	1400.00	0.00	26.000	26.000
2	1200.0000	180.0	2000.00	0.00	26.000	26.000

Interference check:

Access	Access	Pitch mm	Ligament mm
1	2	11774.0830	10574.0840
1	2	11774.0830	10574.0840

Interference Check: No Overlap Detected between Openings



Minimum Pitch between openings is: 11774.0830 mm
 Minimum Ligament between Openings is: 10574.0840 mm

Recommended minimum Ligament [IMin]:

$$= 2 \cdot \sqrt{(D \cdot t)}$$

$$= 2 \cdot \sqrt{(7486 \cdot 32.4)} \text{ mm}$$

$$= 984.980 \text{ mm}$$

Carefully evaluate this acceptance criteria.

Section properties at least metal area [A]:

Number of Openings At Critical Height = 2

Access No	Layout Angle 1 Deg.	Layout Angle 2 Deg.
1	-7.9938	7.9938
2	172.0116	187.9884

The points for integrating Ixx, Iyy are as follows:

FROM THETA1 Deg	TO THETA2 Deg
7.994	172.012
-172.012	-7.994

Height of Weakest Section	H	1700.966	mm
Rotation of Principle Axes	a	-0.000	deg.
Centroid Position in X direction	g	-0.121	mm
Centroid Position in Y direction	h	-0.000	mm
Principle Moment of Inertia	Ixx	533227264.000	cm ⁴
Principle Moment of Inertia	Iyy	443310784.000	cm ⁴
Distance to Extreme Fiber	ybar	3759.200	mm
Distance to Extreme Fiber	xbar	3759.322	mm
Minimum Section Modulus	W4	1179231872.000	mm ³
Metal Area at the Cutting Plane	A4	6970.301	cm ²

Half Opening Width Subtended Angle at Weakest Section [δ]:

$$= \sin^{-1}(\text{Largest Opening Width} / D_z)$$

$$= \sin^{-1}(1200/7486)$$

$$= 0.16 \text{ rad (9.22 deg)}$$

Note: The largest opening width is considered.

Skirt Mean Radius at Section 4-4 [r]:

$$= D_z / 2$$

$$= 7486/2$$

$$= 3743.0 \text{ mm}$$

Because the Code uses the words 'may be' in the evaluation below, the results should be considered optional.

Check areas for openings per EN 13445 Equation 16.12-76 [A,check]:

At,i	Adelta,i	Status	lt,i <= 8*eat,i
13.5	382.0	...	...
13.5	382.0	...	...

This calculation of ratio below is per an earlier version of EN 13445.

Opening ratio at weakest section: EN 13445 equation 16.12-76 [ratio]:

$$= \delta \cdot \sqrt{(r / e_z)}$$

$$= 0.16 \cdot \sqrt{(3743/32.4)}$$

$$= 1.730$$

Ok as ratio is < 2 - See EN 13445 16.12.76(a).



Analysis of the Section at Weakest Section Modulus [Section 4-4]:

Skirt Corroded Thickness [ez]:

$$\begin{aligned} &= ex(\text{new}) - ci - cext \\ &= 34 - 0.8 - 0.8 \\ &= 32.4 \text{ mm} \end{aligned}$$

Note: At least one of the openings cuts the Top Of the Skirt (Ok)

Find the moment acting at the Critical Height [M4]:

$$\begin{aligned} &= (M_{\text{bottom}} - M_{\text{top}}) \cdot (H - \text{CutHeight}) / H + M_{\text{top}} \\ &= (2668630 - 2221803) \cdot (2550 - 1701) / 2550 + 2221803 \\ &= 2370576 \text{ kgf-m.} \end{aligned}$$

Compute the section modulus [Zxx and Zyy]:

$$\begin{aligned} Z_{xx} &= I_{xx} / y_{\text{bar}} = 533227264 / 3759 = 1418460544 \text{ mm}^3 \\ Z_{yy} &= I_{yy} / x_{\text{bar}} = 443310784 / 3759 = 1179231872 \text{ mm}^3 \\ Z &= \min(Z_{xx}, Z_{yy}) = \min(1418460544, 1179231872) = 1179231872 \text{ mm}^3 \end{aligned}$$

The balance of the calculations are per EN 13445-3, and they may be treated as academic.

Radius of Gyration of Skirt at 4-4 [rg]:

$$\begin{aligned} &= \sqrt{\min(I_{xx}, I_{yy}) / \text{Area}(A_4)} \\ &= \sqrt{\min(533227264, 443310784) / 6970} \\ &= 2522 \text{ mm} \end{aligned}$$

Note: rg is not included in the analysis.

Skirt Slenderness Ratio [rg/L]:

$$\begin{aligned} &= rg / L \\ &= 2522 / 2550 \\ &= 0.989 \end{aligned}$$

Location of the Skirt Centroid at critical height [e]:

$$\begin{aligned} &= \sqrt{ex^2 + ey^2} \\ &= \sqrt{(-0.12)^2 + (-0.62778E-06)^2} \\ &= 0.000 \text{ mm} \end{aligned}$$

$$\text{Where } ex = -0.12 : ey = -0.62778E-06$$

Moment Increment at Centroid [ΔM_4]:

$$\begin{aligned} &= \text{abs}(F_1 - F_f - F_g) \cdot e = F_4 \cdot e \\ &= \text{abs}(-18164 - 893177 - 0) \cdot 0.12 \\ &= 110.580 \text{ kgf-m.} \end{aligned}$$

Check of the Skirt in Regions with Openings, per 16.12.4.3:

Weakening Factor, 16.12-70(b) [psi1]:

$$\begin{aligned} &= \min(1; A_4 / (\pi \cdot D_3 \cdot ea_3)) \\ &= \min(1; 6970 / (\pi \cdot 7486 \cdot 32.4)) \\ &= 0.915 \end{aligned}$$

Weakening Factor, 16.12-70(c) [psi2]:

$$\begin{aligned} &= \min(1; 4 \cdot W_4 / (\pi \cdot D_3^3 / (\pi \cdot D_3 \cdot ea_3))) \\ &= \min(1; 4.0 \cdot 1179231872 / (\pi \cdot 7486^2 \cdot 32.4)) \\ &= 0.827 \end{aligned}$$

Maximum Bending Moment [Mmax]:

$$\begin{aligned} &= \pi / 4 \cdot d_3^2 \cdot ea_3 \cdot \sigma_{\text{call}} \\ &= \pi / 4 \cdot 7486^2 \cdot 32.4 \cdot 1071 \\ &= 15278292.0 \text{ kgf-m.} \end{aligned}$$

Maximum Compressive Force [Fc,max]:

$$\begin{aligned} &= \pi \cdot d_3 \cdot ea_3 \cdot \sigma_{\text{call}} \\ &= \pi \cdot 7486 \cdot 32.4 \cdot 1071 \\ &= 8163651.0 \text{ kgf} \end{aligned}$$

Unity Check of Weakened Section, 16.12-70 [UC1]:

$$\begin{aligned} &= F_4 / (\text{psi1} \cdot F_{c,\text{max}}) + (\text{abs}(M_4) + F_4 \cdot \text{abs}(y_q)) / (\text{psi2} \cdot M_{\text{max}}) \\ &= 911340 / (0.91 \cdot 8163651) + (\text{abs}(2370576) + 911340 \cdot \text{abs}(0.12)) / \end{aligned}$$



($0.83 \cdot 15278292$)
= 0.310 must be $\leq$ to 1, Passed

The stresses are satisfactory in the region with the opening(s).

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Skirt Support Analysis Per EN 13445-3 Section 16.12:

The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.

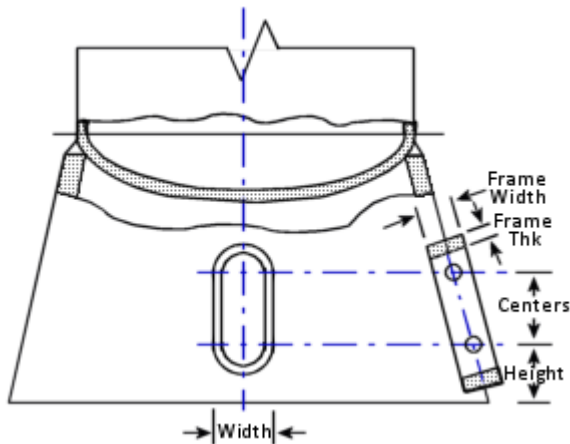
The access opening(s) is/are in section 4-4 only. The other calculations are for areas where no access openings exist. Please be aware of this when you review the results.

Item: Skirt-1

Total vessel (empty) weight	Fg	893176.625 kgf
Head Weight below section 2-2	DeltaFg	284801.094 kgf
Weight of vessel contents (liquid)	Ff	0.000 kgf
Vessel Gross Weight	Fg + Ff	893176.625 kgf

Skirt Mean Diameter	Dz	7486.000 mm
Skirt Analysis Thickness (Corroded)	ez	32.400 mm
Skirt Height	H	1000.000 mm

Additional Axial Force at section 1-1	F1	-18163.852 kgf
Moment at section 1-1	M1	1550932.875 kgf-m.
Applied Moment - Bottom of Skirt	Mb	1724572.375 kgf-m.
Weight at Bottom of Skirt	W	958985.000 kgf



Skirt Access Opening Input Data:

Access #	Diameter mm	Layout Angle deg	Cntr. Dist. From Bottom mm	Obround Cntr-Cntr mm	Frame Width mm	Frame Thickness mm
1	50.0000	0.0	500.00	0.00	26.000	26.000
2	50.0000	90.0	500.00	0.00	26.000	26.000
3	50.0000	180.0	500.00	0.00	26.000	26.000
4	50.0000	270.0	500.00	0.00	26.000	26.000

Interference check:

Access	Access	Pitch mm	Ligament mm
1	2	5879.3926	5829.3926
2	3	5879.3926	5829.3926
3	4	5879.3926	5829.3926
1	4	5879.3926	5829.3926

Interference Check: No Overlap Detected between Openings

Minimum Pitch between openings is: 5879.3926 mm
 Minimum Ligament between Openings is: 5829.3926 mm



Recommended minimum Ligament [IMin]:

$$= 2 \cdot \sqrt{(D \cdot t)}$$

$$= 2 \cdot \sqrt{(7486 \cdot 32.4)} \text{ mm}$$

$$= 984.980 \text{ mm}$$

Carefully evaluate this acceptance criteria.

Section properties at least metal area [A]:

Number of Openings At Critical Height = 4

Access No	Layout Angle 1 Deg.	Layout Angle 2 Deg.
1	-0.3827	0.3827
2	89.6173	90.3827
3	179.6173	180.3827
4	269.6173	270.3827

The points for integrating Ixx, Iyy are as follows:

FROM THETA1 Deg	TO THETA2 Deg
0.383	89.617
90.383	179.617
180.383	269.617
-89.617	-0.383

Height of Weakest Section	H	499.988	mm
Rotation of Principle Axes	a	-0.476	deg.
Centroid Position in X direction	g	0.000	mm
Centroid Position in Y direction	h	-0.000	mm
Principle Moment of Inertia	Ixx	533018688.000	cm ⁴
Principle Moment of Inertia	Iyy	533018720.000	cm ⁴
Distance to Extreme Fiber	ybar	3759.200	mm
Distance to Extreme Fiber	xbar	3759.201	mm
Minimum Section Modulus	W4	1417905664.000	mm ³
Metal Area at the Cutting Plane	A4	7609.100	cm ²

Half Opening Width Subtended Angle at Weakest Section [δ]:

$$= \sin^{-1}(\text{Largest Opening Width} / D_z)$$

$$= \sin^{-1}(50/7486)$$

$$= 0.0067 \text{ rad (0.38 deg)}$$

Skirt Mean Radius at Section 4-4 [r]:

$$= D_z / 2$$

$$= 7486/2$$

$$= 3743.0 \text{ mm}$$

Because the Code uses the words 'may be' in the evaluation below, the results should be considered optional.

Check areas for openings per EN 13445 Equation 16.12-76 [A,check]:

At,i	Adelta,i	Status	lt,i <= 8*eat,i
13.5	7.8	...	...
13.5	7.8	...	...
13.5	7.8	...	...
13.5	7.8	...	...

This calculation of ratio below is per an earlier version of EN 13445.

Opening ratio at weakest section: EN 13445 equation 16.12-76 [ratio]:

$$= \delta \cdot \sqrt{(r / e_z)}$$

$$= 0.0067 \cdot \sqrt{(3743/32.4)}$$

$$= 0.072$$

Ok as ratio is < 2 - See EN 13445 16.12.76(a).



Analysis of the Section at Weakest Section Modulus [Section 4-4]:

Skirt Corroded Thickness [ez]:

$$\begin{aligned} &= ex(\text{new}) - ci - cext \\ &= 34 - 0.8 - 0.8 \\ &= 32.4 \text{ mm} \end{aligned}$$

Find the moment acting at the Critical Height [M4]:

$$\begin{aligned} &= (M_{\text{bottom}} - M_{\text{top}}) \cdot (H - \text{CutHeight}) / H + M_{\text{top}} \\ &= (1724572 - 1550932) \cdot (1000 - 500) / 1000 + 1550932 \\ &= 1637754 \text{ kgf-m.} \end{aligned}$$

Compute the section modulus [Zxx and Zyy]:

$$\begin{aligned} Z_{xx} &= I_{xx} / y_{\text{bar}} = 533018688 / 3759 = 1417905664 \text{ mm}^3 \\ Z_{yy} &= I_{yy} / x_{\text{bar}} = 533018720 / 3759 = 1417905664 \text{ mm}^3 \\ Z &= \min(Z_{xx}, Z_{yy}) = \min(1417905664, 1417905664) = 1417905664 \text{ mm}^3 \end{aligned}$$

The balance of the calculations are per EN 13445-3, and they may be treated as academic.

Radius of Gyration of Skirt at 4-4 [rg]:

$$\begin{aligned} &= \sqrt{(\min(I_{xx}, I_{yy}) / \text{Area}(A_4))} \\ &= \sqrt{(\min(533018688, 533018720) / 7609)} \\ &= 2647 \text{ mm} \end{aligned}$$

Note: rg is not included in the analysis.

Skirt Slenderness Ratio [rg/L]:

$$\begin{aligned} &= rg / L \\ &= 2647 / 1000 \\ &= 2.647 \end{aligned}$$

Location of the Skirt Centroid at critical height [e]:

$$\begin{aligned} &= \sqrt{(ex^2 + ey^2)} \\ &= \sqrt{(0.00025^2 + -0.00015^2)} \\ &= 0.121 \text{ mm} \end{aligned}$$

$$\text{Where } ex = 0.00025 : ey = -0.00015$$

Moment Increment at Centroid [ΔM_4]:

$$\begin{aligned} &= \text{abs}(F_1 - F_f - F_g) \cdot e = F_4 \cdot e \\ &= \text{abs}(-18164 - 893177 - 0) \cdot 0 \\ &= 0.000 \text{ kgf-m.} \end{aligned}$$

Check of the Skirt in Regions with Openings, per 16.12.4.3:

Weakening Factor, 16.12-70(b) [psi1]:

$$\begin{aligned} &= \min(1; A_4 / (\pi \cdot D_3 \cdot ea_3)) \\ &= \min(1; 7609 / (\pi \cdot 7486 \cdot 32.4)) \\ &= 0.999 \end{aligned}$$

Weakening Factor, 16.12-70(c) [psi2]:

$$\begin{aligned} &= \min(1; 4 \cdot W_4 / (\pi \cdot D^2 / 3 / (\pi \cdot D_3 \cdot ea_3))) \\ &= \min(1; 4.0 \cdot 1417905664 / (\pi \cdot 7486^2 \cdot 32.4)) \\ &= 0.994 \end{aligned}$$

Maximum Bending Moment [Mmax]:

$$\begin{aligned} &= \pi / 4 \cdot d_3^2 \cdot ea_3 \cdot \sigma_{\text{call}} \\ &= \pi / 4 \cdot 7486^2 \cdot 32.4 \cdot 1071 \\ &= 15278292.0 \text{ kgf-m.} \end{aligned}$$

Maximum Compressive Force [Fc,max]:

$$\begin{aligned} &= \pi \cdot d_3 \cdot ea_3 \cdot \sigma_{\text{call}} \\ &= \pi \cdot 7486 \cdot 32.4 \cdot 1071 \\ &= 8163651.0 \text{ kgf} \end{aligned}$$

Unity Check of Weakened Section, 16.12-70 [UC1]:

$$\begin{aligned} &= F_4 / (\psi_1 \cdot F_{c,\text{max}}) + (\text{abs}(M_4) + F_4 \cdot \text{abs}(y_g)) / (\psi_2 \cdot M_{\text{max}}) \\ &= 911340 / (1 \cdot 8163651) + (\text{abs}(1637754) + 911340 \cdot \text{abs}(0)) / \\ &\quad (0.99 \cdot 15278292) \end{aligned}$$



= 0.220 must be $\leq$ to 1, Passed

The stresses are satisfactory in the region with the opening(s).

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Skirt Support Analysis Per EN 13445-3 Section 16.12:

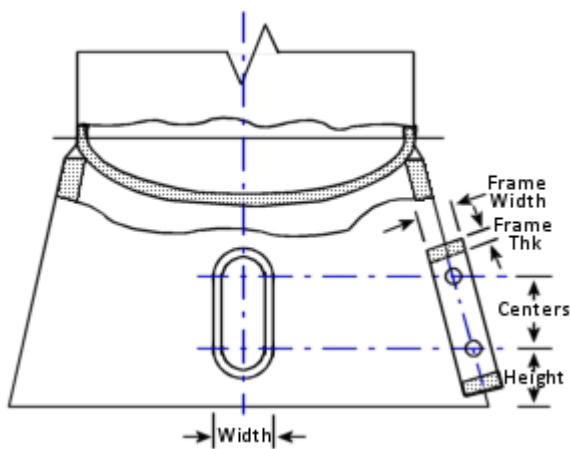
The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.

Item: INT-SKIRT-5

Total vessel (empty) weight	Fg	893176.625 kgf
Head Weight below section 2-2	DeltaFg	122647.805 kgf
Weight of vessel contents (liquid)	Ff	0.000 kgf
Vessel Gross Weight	Fg + Ff	893176.625 kgf
Skirt Mean Diameter	Dz	5720.000 mm
Skirt Analysis Thickness (Corroded)	ez	31.000 mm
Skirt Height	H	1325.000 mm
Additional Axial Force at section 1-1	F1	-18163.852 kgf
Moment at section 1-1	M1	117263.586 kgf-m.
Applied Moment - Bottom of Skirt	Mb	233635.469 kgf-m.
Weight at Bottom of Skirt	W	271065.344 kgf

Structural Type B Skirt per EN 13445 Figure 16.12-2:

Type of Head:		Spherical
Head Mean Diameter at Skirt Top	Db	5703.000 mm
Head Corroded Thickness	eb	49.000 mm



There are no Access Openings in the Skirt - Plain Skirt Considered

Moment of Inertia	Ixx =	227829152.000 cm ⁴
Moment of Inertia	Iyy =	227829152.000 cm ⁴
Section Modulus	W =	796606144.000 mm ³
Section Metal Area	A =	5570.673 cm ²

Analysis of the Section at weakest section Modulus [Z min]:

Radius of Gyration of Skirt at 4-4 [rg]:

$$= \sqrt{\frac{\min(I_{xx}, I_{yy})}{\text{Area}}}$$

$$= \sqrt{\frac{\min(227829152, 227829152)}{5571}}$$

$$= 2022 \text{ mm}$$

Skirt Slenderness Ratio [rg/L]:

$$= \text{rg} / L$$

$$= 2022/1325$$

$$= 1.526$$

Correct the Moment [M4]:

top of skirt is the critical end
 Controlling Moment [M4] = 117264 kgf-m.



Axial Stress at Residual (Weakest) Surface Area [Sigma]:

Downwind Side:

$$= -M / Z - W / A$$

$$= -117264/3.19E+09 - 271065/5571$$

$$= -52.3 \text{ kgf/cm}^2$$

Upwind Side:

$$= +(M) / Z - W / A$$

$$= +117264/3.19E+09 - 271065/5571$$

$$= -45 \text{ kgf/cm}^2$$

16.12.3.4 Checking at connection areas:

Force at the joint, 16.12-1 [Fzp]:

$$= F1 - Fg - Ff + 4 \cdot M1 / Dz$$

$$= -18164 - 893177 - 0 + 4 \cdot 117264/5720$$

$$= -829338 \text{ kgf}$$

Force at the joint, 16.12-2 [Fzq]:

$$= F1 - Fg - Ff - 4 \cdot M1 / Dz$$

$$= -18164 - 893177 - 0 - 4 \cdot 117264/5720$$

$$= -993343 \text{ kgf}$$

16.12.3.4.1 Membrane stresses:

Membrane stress at 1-1, 16.12-3 & 4:

Stress [σ_{1pm}]:

$$= (Fzp + \Delta Fg) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb)$$

$$= (-829338 + 122648 + 0) / (\pi \cdot 5703 \cdot 49) + 50.9 \cdot 5703 / (4 \cdot 49)$$

$$= 1400.45 \text{ kgf/cm}^2$$

Stress [σ_{1qm}]:

$$= (Fzq + \Delta Fg + Ff) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb)$$

$$= (-993343 + 122648 + 0) / (\pi \cdot 5703 \cdot 49) + 50.9 \cdot 5703 / (4 \cdot 49)$$

$$= 1381.77 \text{ kgf/cm}^2$$

Membrane stress at 2-2, 16.12-9:

Stress [σ_{2pm} & σ_{2qm}]:

$$= (Ff + \Delta Fg) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb)$$

$$= (0 + 122648) / (\pi \cdot 5703 \cdot 49) + 50.9 \cdot 5703 / (4 \cdot 49)$$

$$= 1494.92 \text{ kgf/cm}^2$$

Membrane stress at 3-3, 16.12 12 & 13:

Stress [σ_{3pm}]:

$$= Fzp / (\pi \cdot Dz \cdot ez)$$

$$= -829338 / (\pi \cdot 5720 \cdot 31)$$

$$= -148.88 \text{ kgf/cm}^2$$

Stress [σ_{3qm}]:

$$= Fzq / (\pi \cdot Dz \cdot ez)$$

$$= -993343 / (\pi \cdot 5720 \cdot 31)$$

$$= -178.32 \text{ kgf/cm}^2$$

16.12.3.4.2 Bending stresses:

Cosine of angle, 16.12-25 [cos(γ)]:

$$= 1 - 0.5(Db + eb - Dz + ez) / (r + eb)$$

$$= 1 - 0.5(5703 + 49 - 5720 + 31) / (2827 + 49)$$

$$= 0.99$$

$$\gamma = \arccos(\cos(\gamma)) = \arccos(0.99) = 8.49 \text{ deg}$$

Dimension [a]:

$$= 0.5 \cdot \sqrt{(eb^2 + ez^2 + 2 \cdot eb \cdot ez \cdot \cos(\gamma))}$$

$$= 0.5 \cdot \sqrt{(49^2 + 31^2 + 2 \cdot 49 \cdot 31 \cdot 0.99)}$$

$$= 39.9 \text{ mm}$$



Bending moment at verification points p and q [Mp & Mq]:

$$\begin{aligned} M_p &= a \cdot F_{zp} = 39.89 \cdot -829338.1 = -33087.21 \text{ kgf-m.} \\ M_q &= a \cdot F_{zq} = 39.89 \cdot -993342.9 = -39630.34 \text{ kgf-m.} \end{aligned}$$

Correction factor [C]:

$$\begin{aligned} &= 0.63 - 0.057 (e_b/e_z)^2 \\ &= 0.63 - 0.057 (49/31)^2 \\ &= 0.488 \end{aligned}$$

Bending stresses in sections 1-1 to 3-3 at the outer surface (o), 16.12-26 to 29:

Stress [$\sigma_{1pb(o)}$ & $\sigma_{2pb(o)}$]:

$$\begin{aligned} &= C \cdot 6 \cdot M_p / (\pi \cdot D_b \cdot e_b^2) \\ &= 0.49 \cdot 6 \cdot -33087 / (\pi \cdot 5703 \cdot 49^2) \\ &= -225.02 \text{ kgf/cm}^2 \end{aligned}$$

Stress [$\sigma_{1qb(o)}$ & $\sigma_{2qb(o)}$]:

$$\begin{aligned} &= C \cdot 6 \cdot M_q / (\pi \cdot D_b \cdot e_b^2) \\ &= 0.49 \cdot 6 \cdot -39630 / (\pi \cdot 5703 \cdot 49^2) \\ &= -269.52 \text{ kgf/cm}^2 \end{aligned}$$

Stress [$\sigma_{3pb(o)}$]:

$$\begin{aligned} &= C \cdot 6 \cdot M_p / (\pi \cdot D_z \cdot e_z^2) \\ &= 0.49 \cdot 6 \cdot -33087 / (\pi \cdot 5720 \cdot 31^2) \\ &= -560.53 \text{ kgf/cm}^2 \end{aligned}$$

Stress [$\sigma_{3qb(o)}$]:

$$\begin{aligned} &= C \cdot 6 \cdot M_q / (\pi \cdot D_z \cdot e_z^2) \\ &= 0.49 \cdot 6 \cdot -39630 / (\pi \cdot 5720 \cdot 31^2) \\ &= -671.37 \text{ kgf/cm}^2 \end{aligned}$$

Geometry variables, 16.12-32 to 36:

$$\begin{aligned} e_b/D_b &= 1.93/225 = 0.0086 \\ y &= 125 \cdot e_b/D_b = 125 \cdot 49/5703 = 1.07 \\ \alpha &= 0 \end{aligned}$$

Bending stress in sections 1-1 and 2-2 Equation 16.12-31 [$\sigma_{1b(p)}$ & $\sigma_{2b(p)}$]:

$$= 0.0 \text{ kgf/cm}^2 \text{ (16.12-31)}$$

16.12.3.4.3 Total stresses and strength conditions:

Structural shape B and C:

Stress [σ_{1pitot}]:

$$\begin{aligned} &= \sigma_{1pm} - \sigma_{1pb(o)} + \sigma_{1b(p)} \\ &= 1400 - -225 + 0 \\ &= 1625.47 \text{ kgf/cm}^2 \end{aligned}$$

Stress [σ_{1potot}]:

$$\begin{aligned} &= \sigma_{1pm} + \sigma_{1pb(o)} - \sigma_{1b(p)} \\ &= 1400 + -225 - 0 \\ &= 1175.43 \text{ kgf/cm}^2 \end{aligned}$$

Stress [σ_{1qitot}]:

$$\begin{aligned} &= \sigma_{1qm} - \sigma_{1qb(o)} + \sigma_{1b(p)} \\ &= 1382 - -270 + 0 \\ &= 1651.29 \text{ kgf/cm}^2 \end{aligned}$$

Stress [σ_{1qotot}]:

$$\begin{aligned} &= \sigma_{1qm} + \sigma_{1qb(o)} - \sigma_{1b(p)} \\ &= 1382 + -270 - 0 \\ &= 1112.25 \text{ kgf/cm}^2 \end{aligned}$$

Stress [σ_{2pitot}]:

$$\begin{aligned} &= \sigma_{2pm} + \sigma_{2pb(o)} + \sigma_{2b(p)} \\ &= 1495 + -225 + 0 \\ &= 1269.90 \text{ kgf/cm}^2 \end{aligned}$$

Stress [σ_{2potot}]:



$$\begin{aligned}
 &= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)} \\
 &= 1495 - -225 - 0 \\
 &= 1719.94 \text{ kgf/cm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Stress } [\sigma_{2qitot}] &: \\
 &= \sigma_{2qm} + \sigma_{2qb(o)} + \sigma_{2b(p)} \\
 &= 1495 + -270 + 0 \\
 &= 1225.40 \text{ kgf/cm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Stress } [\sigma_{2qotot}] &: \\
 &= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)} \\
 &= 1495 - -270 - 0 \\
 &= 1764.44 \text{ kgf/cm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Stress } [\sigma_{3pitot}] &: \\
 &= \sigma_{3pm} - \sigma_{3pb} \\
 &= -149 - -561 \\
 &= 411.65 \text{ kgf/cm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Stress } [\sigma_{3potot}] &: \\
 &= \sigma_{3pm} + \sigma_{3pb} \\
 &= -149 + -561 \\
 &= -709.40 \text{ kgf/cm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Stress } [\sigma_{3qitot}] &: \\
 &= \sigma_{3qm} - \sigma_{3qb} \\
 &= -178 - -671 \\
 &= 493.06 \text{ kgf/cm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Stress } [\sigma_{3qotot}] &: \\
 &= \sigma_{3qm} + \sigma_{3qb} \\
 &= -178 + -671 \\
 &= -849.69 \text{ kgf/cm}^2
 \end{aligned}$$

Allowable stresses:

$$\begin{aligned}
 \text{For } \sigma_{1pitot} \text{ and } \sigma_{1potot}, \text{ the stress must not exceed:} \\
 &= fs(3 - (1 / 1.5)) \cdot (\sigma_{1pm} / f)^2 \\
 &= 1509 (3 - (1/1.5)) \cdot (1400/1509)^2 \\
 &= 3660.89 \text{ kgf/cm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{For } \sigma_{1qitot} \text{ and } \sigma_{1qotot}, \text{ the stress must not exceed:} \\
 &= fs(3 - (1 / 1.5)) \cdot (\sigma_{1qm} / f)^2 \\
 &= 1509 (3 - (1/1.5)) \cdot (1382/1509)^2 \\
 &= 3683.85 \text{ kgf/cm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{For } \sigma_{2pitot} \text{ and } \sigma_{2potot}, \text{ the stress must not exceed:} \\
 &= fs(3 - (1 / 1.5)) \cdot (\sigma_{2pm} / f)^2 \\
 &= 1509 (3 - (1/1.5)) \cdot (1495/1509)^2 \\
 &= 3540.06 \text{ kgf/cm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{For } \sigma_{2qitot} \text{ and } \sigma_{2qotot}, \text{ the stress must not exceed:} \\
 &= fs(3 - (1 / 1.5)) \cdot (\sigma_{2qm} / f)^2 \\
 &= 1509 (3 - (1/1.5)) \cdot (1495/1509)^2 \\
 &= 3540.06 \text{ kgf/cm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{For } \sigma_{3pitot} \text{ \& } \sigma_{3potot}, \text{ the stress must not exceed:} \\
 &= fs(3 - (1 / 1.5)) \cdot (\sigma_{3pm} / f)^2 \\
 &= 1509 (3 - (1/1.5)) \cdot (-149/1509)^2 \\
 &= 4517.51 \text{ kgf/cm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{For } \sigma_{3qitot} \text{ \& } \sigma_{3qotot}, \text{ the stress must not exceed:} \\
 &= fs(3 - (1 / 1.5)) \cdot (\sigma_{3qm} / f)^2 \\
 &= 1509 (3 - (1/1.5)) \cdot (-178/1509)^2 \\
 &= 4513.25 \text{ kgf/cm}^2
 \end{aligned}$$

Please note: None of the following stresses in the table below are in the region where there is an access opening.

Summary of Stresses: (kgf/cm²)



Stress	Actual	Allowed	Result	Equation
Stot1pi	1625.47	3660.89	Pass	46
Stot1po	1175.43	3660.89	Pass	47
Stot1qi	1651.29	3683.85	Pass	48
Stot1qo	1112.25	3683.85	Pass	49
Stot2pi	1269.90	3540.06	Pass	50
Stot2po	1719.94	3540.06	Pass	51
Stot2qi	1225.40	3540.06	Pass	52
Stot2qo	1764.44	3540.06	Pass	53
Stot3pi	411.65	4517.51	Pass	54
Stot3po	-709.40	4517.51	Pass	55
Stot3qi	493.06	4513.25	Pass	56
Stot3qo	-849.69	4513.25	Pass	57

The stresses are satisfactory in the region without openings.

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Shop/Field Installation Options :

Platform(s) installed in the Shop.
Insulation is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Platforms	1191.460 cm
Center of Gravity of the Insulation	1203.931 cm
Center of Gravity of the Nozzles	1267.154 cm
Center of Gravity of the Added Weights (Operating)	1038.013 cm
Center of Gravity of the Added Weights (Empty)	1531.571 cm
Center of Gravity of Bare Shell New and Cold	1063.236 cm
Center of Gravity of Bare Shell Corroded	1060.254 cm
Vessel CG in the Operating Condition	1056.102 cm
Vessel CG in the Fabricated (Shop/Empty) Condition	1206.867 cm
Vessel CG in the Test Condition	1166.093 cm

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Skirt Data :

Skirt Outside Diameter at Base	SOD, D6	7520.0000	mm
Skirt Thickness	STHK	34.0000	mm
Skirt Internal Corrosion Allowance	SCA	0.8000	mm
Skirt External Corrosion Allowance		0.8000	mm
Skirt Material		SA-516 70	

Basing Input: Type of Geometry: Continuous Top Ring W/Gussets

Thickness of Basing	TBA, ea4	44.0000	mm
Design Temperature of the Basing		38.00	°C
Basing Material		SA-36	
Basing Ope. All. Stress	BASOPE, f4	1162.42	kgf/cm²
Basing Yield Stress		2528.76	kgf/cm²
Inside Diameter of Basing	DI, D4	7312.0000	mm
Outside Diameter of Basing	DOU, D5	7799.9995	mm

Nominal Diameter of Bolts	BND, d(b0)	48.0000	mm
Bolt Corrosion Allowance	BCA	1.5000	mm
Bolt Material		SA-307 A	
Bolt Material Class/Thickness Range		6 <= t <= 100	
Bolt Operating Allowable Stress	SA, fB	1407.16	kgf/cm²
Number of Bolts	RN, nb	32	
Diameter of Bolt Circle	DC, DBC	7660.0000	mm
Bolt Allowable Shear Stress		844.296	kgf/cm²

Ultimate Comp. Strength of Concrete	FPC	418.1	kgf/cm²
Allowable Comp. Strength of Concrete	FC	229.4	kgf/cm²
Modular ratio Steel/Concrete		9.833	

Thickness of Gusset Plates	TGA, e7	16.0000	mm
Width of Gussets at Top Plate	TWDT	140.0000	mm
Width of Gussets at Base Plate	BWDT	140.0000	mm
Gusset Plate Elastic Modulus	E	2057160.0	kgf/cm²
Gusset Plate Yield Stress	SY	2528.8	kgf/cm²
Height of Chair	h1	350.0000	mm
Distance between Gussets	RG, b6	96.0000	mm
Dist. from Bolt Center to Gusset (Rg/2)	CG	48.0000	mm
Number of Gussets per bolt	NG	2	

Thickness of Top Plate or Ring	TTA, e5	44.0000	mm
Radial Width of the Top Plate	TOPWTH, b5	140.0000	mm
Anchor Bolt Hole Dia. in Top Plate	dia,H	64.0000	mm

External Corrosion Allowance	CA	1.6000	mm
------------------------------	----	--------	----

Dead Weight of Vessel	DW	530152.4	kgf
Operating Weight of Vessel	ROW	1018617.8	kgf
Earthquake Moment on Basing	EQMOM	2312204.8	kgf-m.
Wind Moment on Basing	WIMOM	140145.0	kgf-m.

User Moment at the Base [Earthquake Case]		379307.9	kgf-m. (UEQMOM)
User Moment at the Base [WindLoad Case]		379307.9	kgf-m. (UWIMOM)
Percent Bolt Preload	ppl	100.0	

Use AISC A5.2 Increase in Fc and Bolt Stress		No	
Use Allowable Weld Stress per AISC J2.5		No	

Factor for Increase of Allowables	Fact	1.0000	
-----------------------------------	------	--------	--

Results for Brownell and Young Basing Analysis : Analyze Option

Note: This analysis is based on Neutral Axis shift method for Steel on Concrete (or a material with significantly different Young's modulus).

Calculation of Bolt Loads for Multiple Load Conditions:

Bolt Load, Earthquake Operating Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$



$$= ((4 \cdot 2691512/7660) - 987244) / 32$$

$$= 13070.1 \text{ kgf}$$

Bolt Load, Windload Empty Case:

$$= ((4 \cdot M/DC) - W) / nb \text{ per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 519453/7660) - 530152) / 32$$

$$= -8090.6 \text{ kgf}$$

Bolt Load, Windload Vortex Shedding Case:

$$= ((4 \cdot M/DC) - W) / nb \text{ per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 0/7660) - 1018617) / 32$$

$$= -31831.8 \text{ kgf}$$

Governing Bolt Load Condition, Earthquake + Operating Condition:

Area Available in 1 Bolt (corroded)	Abss :	11.5517	cm ²
Area Available in all the Bolts	Abss * RN :	369.6529	cm ²



*There seems to be no tensile load on bolts per the Brownell and Young Method.
 It is possible that this method is encountering a problem. It is strongly
 recommended that you try the simplified analysis method for the basing.*

Check the Bearing Stress in the Concrete [fc(max)]

$$= fc[(2kd + t3) / (2kd)]$$

$$= 31.9[(2 \cdot 1 \cdot 7660 + 244) / (2 \cdot 1 \cdot 7660)]$$

$$= 32.4 \text{ kgf/cm}^2, \text{ Should be } \leq 229 \text{ kgf/cm}^2$$

Values for table 10.3, l = 140, b = 376, l/b = 0.37233111

Maximum Moment per unit width [Mmax]:

$$= \max(Mx, My) = \max(587, 2553) = 2553 \text{ kgf}$$

Reqd Thickness of Basing, Brownell & Young Method [T]:

$$= \sqrt{(6 \cdot Mmax / fallow)} + Ca$$

$$= \sqrt{(6 \cdot 2553 / 1669)} + 1.6$$

$$= 31.9 \text{ mm}$$

Nomenclature:

a = (Dc - Ds) / 2	Skirt Distance to Bolt Circle
P = Sa * Abss	Maximum Load on one Bolt
l	Radial Top Plate Width
lavg	Average Gusset Plate Width
g1 = Gamma 1	Constant Term f(b/l)
g2 = Gamma 2	Constant Term f(b/l)
g = Flat distance / 2	Nut 1/2 Dimension (from Tema)
Fb	Allowable Bending Stress

Values for table 10.6, l = 140, b = 96, b/l = 0.6857143

As b/l (0.69) is less than 1, inverting b/l = 1.46 0.

Moment Term, based on geometry [Mo]:

$$= P / (4 \cdot \pi) [1.3(\ln((2 \sin(\pi \cdot a/l) / (\pi \cdot g))) + 1) - [(0.7 - g2)P / (4\pi)]]$$

$$= 16255 / (4 \cdot \pi) [1.3(\ln((2 \cdot 140 \cdot \sin(\pi \cdot 70/140)) / (\pi \cdot 40))) + 1] - [(0.7 - 0.076) \cdot 16255 / (4 \cdot \pi)]$$

$$= 1832.9821 \text{ kgf}$$

Required Thickness of Continuous Top Ring [Tc]:

$$= \sqrt{(6 \cdot Abs(Mo) / Fb)} + Ca$$

$$= \sqrt{(6 \cdot Abs(1833) / 1686)} + 1.6$$

$$= 27.1415 \text{ mm}$$

Required Gusset Plate Thickness [tg]:

$$= P / (\text{Stress Term} \cdot lavg) + Ca$$

$$= 16255 / (1266 \cdot 140) + 1.6$$

$$= 11.1 \text{ (not less than } 9.52 + 1.6) \text{ mm}$$

Bolt spacing [m]:



$$= \pi \cdot \text{Bolt Circle Diameter} / \text{number of Bolts}$$
$$= \pi \cdot 7660/32 = 752 \text{ mm}$$

Req. Skirt Thk. to withstand Local Bending, (Brownell and Young) [t]:

$$= 1.76(P \cdot a / (m \cdot (h + tba) \cdot 1.5 \cdot Sktope))^{2/3} \cdot r^{1/3} + Ca$$
$$= 1.76(16255 \cdot 70 / (752 \cdot 306 \cdot 2111))^{2/3} \cdot 3760^{1/3} + Ca$$
$$= 10.4 + 1.6 = 12 \text{ mm}$$

Shear Stress in a Single Bolt [tb]:

$$= \text{Shear Force} / (\text{Bolt Area} \cdot \text{Number of Bolts})$$
$$= 175425 / (11.6 \cdot 32)$$
$$= 474.6 \text{ kgf/cm}^2. \text{ Must be less than } 844.3 \text{ kgf/cm}^2.$$

Summary of Basing Thickness Calculations

Required Basing Thickness (tension)	31.8957	mm
Actual Basing Thickness as entered by user	44.0000	mm
Required Thickness of Chair Cap	27.1415	mm
Actual Top Ring Thickness as entered by user	44.0000	mm
Required Gusset thickness, + CA	11.1250	mm
Actual Gusset Thickness as entered by user	16.0000	mm
Required Thickness of Skirt for Local Stress	12.0003	mm
Given Thickness of Skirt	34.0000	mm
Required Gusset Height to meet local stress	11.6513	mm

Weld Size Calculations per Steel Plate Engineering Data - Vol. 2

Compute the Weld load at the Skirt/Base Junction [W]:

$$= \text{SkirtStress} \cdot (\text{SkirtThickness} - CA)$$
$$= 320 \cdot (34 - 1.6)$$
$$= 104 \text{ kgf/mm}$$

Results for Computed Minimum Basing Weld Size [BWeld]:

$$= W / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$
$$= 104 / [(0.4 \cdot 2529) \cdot 2 \cdot 0.707]$$
$$= 7.25 \text{ mm}$$

Results for Computed Minimum Gusset and Top Plate to Skirt Weld Size

Vertical Plate Load [Wv]:

$$= \text{Bolt Load} / (\text{Cmwth} + 2 \cdot (\text{Hg} + \text{Tta}))$$
$$= 16255 / (170 + 2 \cdot (262 + 44))$$
$$= 20.8 \text{ kgf/mm}$$

Horizontal Plate Load [Wh]:

$$= \text{Bolt Load} \cdot e / (\text{Cmwth} \cdot (\text{Hg} + \text{Tta}) + 0.6667 \cdot (\text{Hg} + \text{Tta})^2)$$
$$= 16255 \cdot 70 / (170 \cdot (306) + 0.6667 \cdot (306)^2)$$
$$= 9.94 \text{ kgf/mm}$$

Resultant Weld Load [Wr]:

$$= \sqrt{ Wv^2 + Wh^2 }$$
$$= \sqrt{ 20.8^2 + 9.94^2 }$$
$$= 23 \text{ kgf/mm}$$

Results for Computed Min Gusset and Top Plate to Skirt Weld Size [GsWeld]:

$$= Wr / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$
$$= 23 / [(0.4 \cdot 2529) \cdot 2 \cdot 0.707]$$
$$= 1.61 \text{ mm}$$

Results for Computed Minimum Gusset to Top Plate Weld Size

Weld Load [Wv]:

$$= \text{Bolt Load} / (2 \cdot \text{TopWth})$$
$$= 16255 / (2 \cdot 140)$$
$$= 58.1 \text{ kgf/mm}$$

Weld Load [Wh]:



$$\begin{aligned} &= \text{Bolt Load} \cdot e / (2 \cdot \text{Hgt} \cdot \text{TopWth}) \\ &= 16255 \cdot 70 / (2 \cdot 306 \cdot 140) \\ &= 13.3 \text{ kgf/mm} \end{aligned}$$

Resultant Weld Load [Wr]:

$$\begin{aligned} &= \sqrt{ Wv^2 + Wh^2 } \\ &= \sqrt{ 58.1^2 + 13.3^2 } \\ &= 59.6 \text{ kgf/mm} \end{aligned}$$

Results for Computed Min Gusset to Top Plate Weld Size [GtpWeld]:

$$\begin{aligned} &= W_r / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707] \\ &= 59.6 / [(0.4 \cdot 2529) \cdot 2 \cdot 0.707] \\ &= 4.16 \text{ mm} \end{aligned}$$

Note: The calculated weld sizes need not exceed the component thickness framing into the weld. At the same time, the weld must meet a minimum size specification which is 3/16 in. (4.76 mm) or 1/4 in. (6.35 mm), depending on the component thickness.

Summary of Required Weld Sizes:

Required Basing to Skirt Double Fillet Weld Size	7.2494	mm
Required Gusset to Skirt Double Fillet Weld Size	6.3500	mm
Required Top Plate to Skirt Weld Size	6.3500	mm
Required Gusset to Top Plate Double Fillet Weld Size	4.1638	mm

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: N2 36"

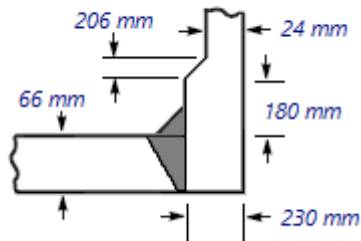
From: 40

Pressure for Reinforcement Calculations	P	50.897	kgf/cm ²
Temperature for Internal Pressure	Temp	275	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	1509.10	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1509.10	kgf/cm ²
Inside Diameter of Hemispherical Head	D	7414.00	mm
Head Finished (Minimum) Thickness	t	66.0000	mm
Head Internal Corrosion Allowance	cas	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1407.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.00	kgf/cm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		36.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	24.0000	mm
Flange Material		SA-182 F11	
Hub Height of Integral Nozzle	h	180.0000	mm
Height of Beveled Transition	L'	206.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		230.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	150.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		60.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.9	
Flange Series		B	
Piping Exit Angle [North Clockwise reference]		90.0	deg
Bend Radius Multiplier		1.50	
Horizontal Run Length		4000.0	mm
Centerline Distance to Tangent		5000.0	mm

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N2 36"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	36.000 in.
Actual Thickness Used in Calculation	0.945 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
 $= P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
 $= 50.9 \cdot 3710 / (2 \cdot 1509 \cdot 1 - 0.2 \cdot 50.9)$
 $= 62.7748 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 50.9 \cdot 457 / (1407 \cdot 1 + 0.4 \cdot 50.9)$
 $= 16.3029 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 206 / 0.5773$
 $= 356.8027 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

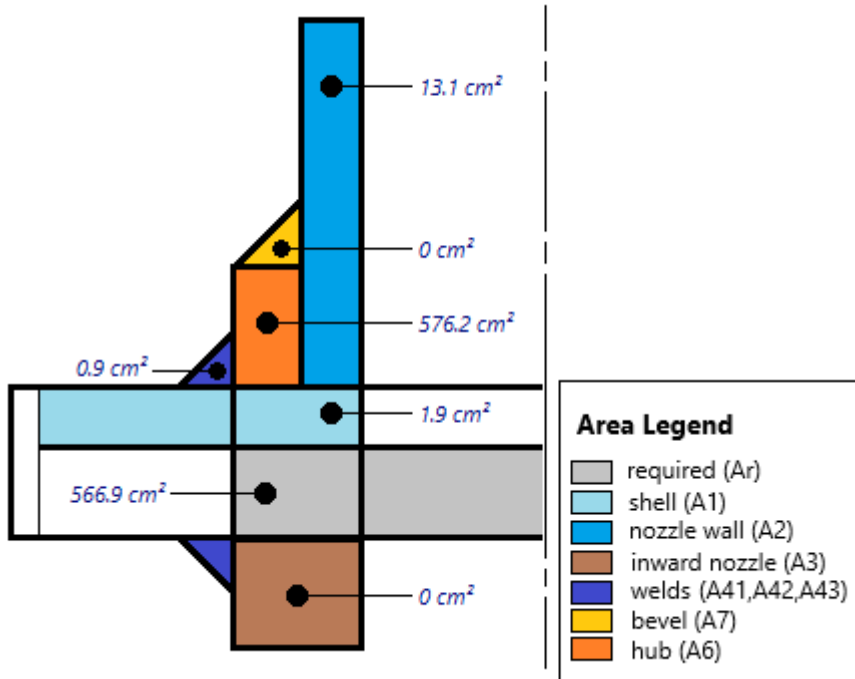
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	1744.8000 mm
Parallel to Vessel Wall, opening length	d	872.4000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	157.5000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1407 / 1509)$
 $= 0.932$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1407 / 1509)$
 $= 0.932$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.93, 1)$
 $= 0.932$



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	566.929	NA	NA
Area in Shell	A1	1.896	NA	NA
Area in Nozzle Wall	A2	13.138	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.932	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	576.188	NA	NA
TOTAL AREA AVAILABLE	Atot	592.154	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (872 \cdot 62.8 \cdot 1 + 2 \cdot 227 \cdot 62.8 \cdot 1 \cdot (1 - 0.93))$
 $= 566.929 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 872(1 \cdot 63 - 1 \cdot 62.8) - 2 \cdot 227$
 $(1 \cdot 63 - 1 \cdot 62.8) \cdot (1 - 0.93)$
 $= 1.896 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot ho)(tn - trn)fr2$
 $= (2 \cdot 150)(21 - 16.3)0.93$
 $= 13.138 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10^2 \cdot 0.93 + (0)^2 \cdot 0.93$
 $= 0.932 \text{ cm}^2$



Area Available in the Hub Section [A6]:

$$\begin{aligned} &= (2 \cdot \min(T_{lp}, h_o, H_{ubht})) \cdot (H_{ubt} - t_n) \cdot f_{r2} \\ &= (2 \cdot \min(150, 150, 180)) \cdot (230 - 24) \cdot 0.93 \\ &= 576.188 \text{ cm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 19.3029 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 65.7748 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 65.7748 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned} &= \min[tb3, \max(tb1, tb2)] \\ &= \min[11.3, \max(65.8, 4.5)] \\ &= 11.3312 \text{ mm} \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= \max(ta, tb) \\ &= \max(19.3, 11.3) \\ &= 19.3029 \text{ mm} \end{aligned}$$

Available Nozzle Neck Thickness = 24.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 683.6,	Allowable	: 1407.0 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 2833.9 kgf/cm ²	Passed
Occasional	: 516.2,	Allowable	: 1871.3 kgf/cm ²	Passed
Shear	: 104.9,	Allowable	: 984.9 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, tg = 24, tr = 16.3, c = 3 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.78, Temp. Reduction = 12 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-2 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-15 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:B, Curve:A)

Govrn. thk, tg = 66, tr = 62.8, c = 3 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 1, Temp. Reduction = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	42 °C
Min Metal Temp. at Required thickness (UCS 66.1)	42 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 42 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 50.90/105.44 = 0.483

Weld Size Calculations, Description: N2 36"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

Required Thickness	Actual Thickness
--------------------	------------------



Nozzle Weld 6.0000 = Min per Code 7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))Sv)
= max(0, (567 - 1.9 + 2 • 227 • 0.93 •
(1 • 63 - 62.8))1509)
= 854129.69 kgf

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A4-(Wi-Can/.707)²•fr2)•Sv
= (589 + 0.93 - 0 • 0.93) • 1509
= 890758.19 kgf

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (589 + 0 + 0.93 + (267)) • 1509
= 1293187.88 kgf

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (589 + 0 + 0.93 + 0 + (267)) • 1509
= 1293187.88 kgf

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.14/2.0) • 914 • 10 • 0.49 • 1407
= 99025. kgf

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.14 • 447) • (230 - 3) • 0.7 • 1407
= 3137496. kgf

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • Wgnvi • 0.74 • Sng
= (3.14/2.0) • 914 • (60 - 0) • 0.74 • 1509
= 962402. kgf

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (99025 + 3137496) = 3236521 kgf

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (99025 + 0 + 962402 + 0) = 1061427 kgf

Failure Path [Path 3-3]:
= (Sonw + Tngw + Sinw)
= (99025 + 962402 + 0) = 1061427 kgf

Summary of Failure Path Calculations:

Path 1-1 = 3236521 kgf, must exceed W = 854129 kgf or W1 = 890758 kgf
Path 2-2 = 1061427 kgf, must exceed W = 854129 kgf or W2 = 1293187 kgf
Path 3-3 = 1061427 kgf, must exceed W = 854129 kgf or W3 = 1293187 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 51.1 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.



The Drop for this Nozzle is : 28.3021 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 244.3020 mm

Input Echo, WRC107/537 Item 1, Description: N2 36" :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	3.0000	mm
Vessel Diameter	Dv	7414.000	mm
Vessel Thickness	Tv	66.000	mm
Design Temperature	T1	275.0	C
Vessel Material		SA-387 11	
Vessel UNS Number		K11789	
Vessel Cold S.I. Allowable	Smc	1509.10	kgf/cm ²
Vessel Hot S.I. Allowable	Smh	1509.10	kgf/cm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	OD	
Corrosion Allowance for Nozzle	Can	3.0000	mm
Nozzle Diameter	Dn	1326.400	mm
Nozzle Thickness	Tn	230.000	mm
Nozzle Material		SA-182 F11	
Nozzle UNS Number		K11572	
Nozzle Cold S.I. Allowable	SNmc	1407.00	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh	1407.00	kgf/cm ²
Design Internal Pressure	Dp	50.897	kgf/cm ²
Include Pressure Thrust		No	

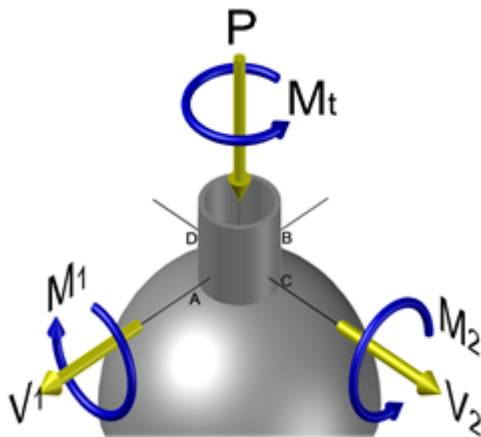
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-8256.9	kgf
Longitudinal Shear (SUS)	(V1) V1	5963.3	kgf
Circumferential Shear (SUS)	(Vc) V2	5963.3	kgf
Circumferential Moment (SUS)	(Mc) M1	16513.9	kgf-m.
Longitudinal Moment (SUS)	(Ml) M2	16513.9	kgf-m.
Torsional Moment (SUS)	Mt	18990.7	kgf-m.

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 1326.4 + 2 \cdot 1.65 \cdot \sqrt{(3741.5 (66.0 - 3.0))}$$

$$= 2928.564 \text{ mm}$$

WRC 107 Stress Calculation for SUSTAINED loads:

Radial Load		P	-8256.9	kgf
Circumferential Shear	(VC)	V2	5963.3	kgf
Longitudinal Shear	(VL)	V1	5963.3	kgf
Circumferential Moment	(MC)	M1	16513.9	kgf-m.
Longitudinal Moment	(ML)	M2	16513.9	kgf-m.
Torsional Moment		MT	18990.7	kgf-m.

Dimensionless Param: U = 1.37 TAU = 5.00 (2.42) RHO = 0.28

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 1	0.02885	(A,B,C,D)
$M(x) / P$	SP 1	0.02905	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.02901	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 1	0.04287	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.02901	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 1	0.04287	(A,B,C,D)
$N(y) * T / P$	SP 1	0.04213	(A,B,C,D)
$M(y) / P$	SP 1	0.00797	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.03904	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 1	0.01270	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.03904	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 1	0.01270	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (kgf/cm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb.	P	6.0	6.0	6.0	6.0	6.0	6.0	6.0	6.0
Rad. Bend.	P	36.3	-36.3	36.3	-36.3	36.3	-36.3	36.3	-36.3
Rad. Memb.	MC	0.0	0.0	0.0	0.0	-24.9	-24.9	24.9	24.9
Rad. Bend.	MC	0.0	0.0	0.0	0.0	-220.5	220.5	220.5	-220.5
Rad. Memb.	ML	-24.9	-24.9	24.9	24.9	0.0	0.0	0.0	0.0
Rad. Bend.	ML	-220.5	220.5	220.5	-220.5	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-203.1	165.3	287.6	-225.9	-203.1	165.3	287.6	-225.9

Tang. Memb. P	8.8	8.8	8.8	8.8	8.8	8.8	8.8	8.8
Tang. Bend. P	9.9	-9.9	9.9	-9.9	9.9	-9.9	9.9	-9.9
Tang. Memb. MC	0.0	0.0	0.0	0.0	-33.5	-33.5	33.5	33.5
Tang. Bend. MC	0.0	0.0	0.0	0.0	-65.3	65.3	65.3	-65.3
Tang. Memb. ML	-33.5	-33.5	33.5	33.5	0.0	0.0	0.0	0.0
Tang. Bend. ML	-65.3	65.3	65.3	-65.3	0.0	0.0	0.0	0.0
Tot. Tang. Str.	-80.0	30.6	117.4	-33.0	-80.0	30.6	117.4	-33.0
Shear VC	4.5	4.5	-4.5	-4.5	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-4.5	-4.5	4.5	4.5
Shear MT	10.9	10.9	10.9	10.9	10.9	10.9	10.9	10.9
Tot. Shear	15.5	15.5	6.4	6.4	6.4	6.4	15.5	15.5
Str. Int.	205.0	167.1	287.8	226.1	203.4	165.6	289.0	227.1

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (kgf/cm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		1498.9	1498.9	1498.9	1498.9	1498.9	1498.9	1498.9	1498.9
Rad. Pl (SUS)		-18.9	-18.9	30.9	30.9	-18.9	-18.9	30.9	30.9
Rad. Q (SUS)		-184.2	184.2	256.7	-256.7	-184.2	184.2	256.7	-256.7
Long. Pm (SUS)		1498.9	1498.9	1498.9	1498.9	1498.9	1498.9	1498.9	1498.9
Long. Pl (SUS)		-24.7	-24.7	42.2	42.2	-24.7	-24.7	42.2	42.2
Long. Q (SUS)		-55.3	55.3	75.2	-75.2	-55.3	55.3	75.2	-75.2
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		4.5	4.5	-4.5	-4.5	-4.5	-4.5	4.5	4.5
Shear Q (SUS)		10.9	10.9	10.9	10.9	10.9	10.9	10.9	10.9
Pm (SUS)		1498.9	1498.9	1498.9	1498.9	1498.9	1498.9	1498.9	1498.9
Pm+Pl (SUS)		1482.5	1482.5	1542.7	1542.7	1482.5	1482.5	1542.7	1542.7
Pm+Pl+Q (Total)		1420.8	1666.0	1786.7	1466.1	1419.2	1664.6	1787.9	1467.2

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1498.92	1509.10	Passed
Pm+Pl (SUS)	1542.73	2263.65	Passed
Pm+Pl+Q (TOTAL)	1787.89	4527.30	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: UC1- 24"

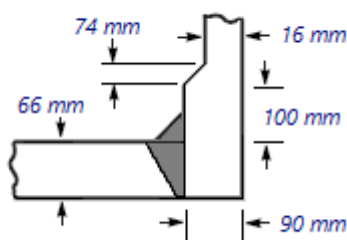
From: 40

Pressure for Reinforcement Calculations	P	50.897	kgf/cm ²
Temperature for Internal Pressure	Temp	275	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	1509.10	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1509.10	kgf/cm ²
Inside Diameter of Hemispherical Head	D	7414.00	mm
Head Finished (Minimum) Thickness	t	66.0000	mm
Head Internal Corrosion Allowance	cas	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	2500.0000	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1407.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.00	kgf/cm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.0000	mm
Flange Material		SA-182 F11	
Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	74.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		90.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	2500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	60.0000	mm
Inside Projection	h	300.0000	mm
Weld leg size, Inside Element to Shell	Wi	14.0000	mm
Flange Class		600	
Flange Grade		GR 1.9	
Flange Series		A	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: UC1- 24"



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 24.000 in.
Actual Thickness Used in Calculation 0.630 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
= $P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
= $50.9 \cdot 3710 / (2 \cdot 1509 \cdot 1 - 0.2 \cdot 50.9)$
= 62.7748 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $50.9 \cdot 305 / (1407 \cdot 1 + 0.4 \cdot 50.9)$
= 10.8686 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 74/0.5773
= 128.1718 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

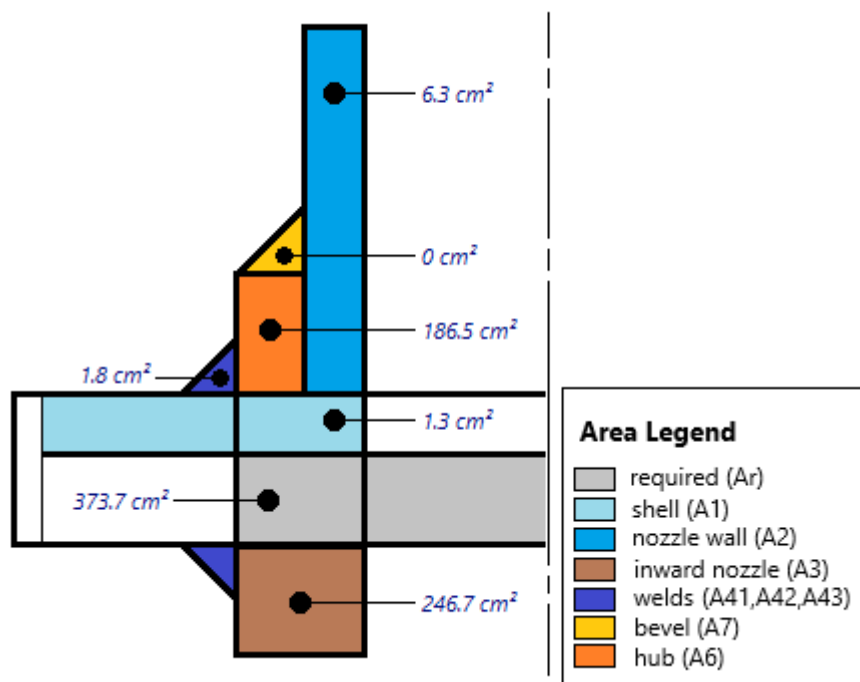
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	1167.2001	mm
Parallel to Vessel Wall, opening length	d	583.6000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	157.5000	mm
Normal to Vessel Wall, Inward		25.0000	mm

Weld Strength Reduction Factor [fr1]:
= min(1, S_n / S_v)
= min(1, 1407/1509)
= 0.932

Weld Strength Reduction Factor [fr2]:
= min(1, S_n / S_v)
= min(1, 1407/1509)
= 0.932

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(0.93, 1)
= 0.932



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	373.744	NA	NA
Area in Shell	A1	1.288	NA	NA
Area in Nozzle Wall	A2	6.260	NA	NA
Area in Inward Nozzle	A3	246.698	NA	NA
Area in Welds	A41+A42+A43	1.820	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	186.504	NA	NA
TOTAL AREA AVAILABLE	Atot	442.569	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (584 \cdot 62.8 \cdot 1 + 2 \cdot 87 \cdot 62.8 \cdot 1 \cdot (1 - 0.93))$$

$$= 373.744 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 584 (1 \cdot 63 - 1 \cdot 62.8) - 2 \cdot 87 (1 \cdot 63 - 1 \cdot 62.8) \cdot (1 - 0.93)$$

$$= 1.288 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp) (tn - trn) fr2$$

$$= (2 \cdot 158) (13 - 10.9) 0.93$$

$$= 6.260 \text{ cm}^2$$

Area Available in Inward Nozzle [A3]:

$$= 2 \cdot ti \cdot \min(h, TI, 2.5 \cdot ti) \cdot fr2$$

$$= 2 \cdot 84 \cdot (158) \cdot 0.93$$

$$= 246.698 \text{ cm}^2$$



Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can / 0.707)^2 \cdot fr_2$$

$$= 10^2 \cdot 0.93 + (9.76)^2 \cdot 0.93$$

$$= 1.820 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{hp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(158, 2500, 100)) \cdot (90 - 16) \cdot 0.93$$

$$= 186.504 \text{ cm}^2 \text{ Includes Bevel Area of } 48.517 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 13.8686 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 65.7748 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 65.7748 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.3, \max(65.8, 4.5)]$$

$$= 11.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(13.9, 11.3)$$

$$= 13.8686 \text{ mm}$$

Available Nozzle Neck Thickness = 16.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, tg = 16, tr = 10.9, c = 3 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.84, Temp. Reduction = 9 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-23 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:B, Curve:A)

Govrn. thk, tg = 66, tr = 62.8, c = 3 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 1, Temp. Reduction = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	42 °C
Min Metal Temp. at Required thickness (UCS 66.1)	42 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 42 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 50.90/105.44 = 0.483

Weld Size Calculations, Description: UC1- 24"

Intermediate Calc. for nozzle/shell Welds	Tmin	19.0000 mm
Intermediate Calc. for Inward Weld	TminIns	19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm
Inward Weld	6.0000 = Min per Code	6.8980 = 0.7 * Wi-Can mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)



Weld Load [W]:

$$\begin{aligned} &= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t-tr)) \cdot Sv) \\ &= \max(0, (374 - 1.29 + 2 \cdot 87 \cdot 0.93 \cdot \\ &\quad (1 \cdot 63 - 62.8)) \cdot 1509) \\ &= 562623.62 \text{ kgf} \end{aligned}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned} &= (A2+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv \\ &= (193 + 1.82 - 95.2 \cdot 0.93) \cdot 1509 \\ &= 292305.81 \text{ kgf} \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned} &= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv \\ &= (193 + 247 + 1.82 + (102)) \cdot 1509 \\ &= 820172.12 \text{ kgf} \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned} &= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S \\ &= (193 + 247 + 1.82 + 0 + (102)) \cdot 1509 \\ &= 820172.12 \text{ kgf} \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw \\ &= (3.14/2.0) \cdot 610 \cdot 10 \cdot 0.49 \cdot 1407 \\ &= 66017. \text{ kgf} \end{aligned}$$

Shear, Inward Nozzle Weld [Sinw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw \\ &= (3.14/2.0) \cdot 610 \cdot 9.76 \cdot 0.49 \cdot 1407 \\ &= 64415. \text{ kgf} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\ &= (3.14 \cdot 298) \cdot (90 - 3) \cdot 0.7 \cdot 1407 \\ &= 802997. \text{ kgf} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng \\ &= (3.14/2.0) \cdot 610 \cdot (60 - 0) \cdot 0.74 \cdot 1509 \\ &= 641602. \text{ kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (66017 + 802997) = 869014 \text{ kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (66017 + 0 + 641602 + 64415) = 772033 \text{ kgf} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (66017 + 641602 + 64415) = 772033 \text{ kgf} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 869013	kgf, must exceed W = 562623 kgf	or W1 = 292305 kgf
Path 2-2 = 772033	kgf, must exceed W = 562623 kgf	or W2 = 820172 kgf
Path 3-3 = 772033	kgf, must exceed W = 562623 kgf	or W3 = 820172 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 51.1 kgf/cm²



Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 12.5519 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 2865.9998 mm

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Input, Nozzle Desc: H1 8"

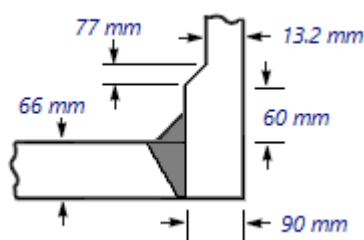
From: 40

Pressure for Reinforcement Calculations	P	50.897	kgf/cm ²
Temperature for Internal Pressure	Temp	275	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	1509.10	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1509.10	kgf/cm ²
Inside Diameter of Hemispherical Head	D	7414.00	mm
Head Finished (Minimum) Thickness	t	66.0000	mm
Head Internal Corrosion Allowance	cas	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	2000.0000	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1407.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.00	kgf/cm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		8.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		100	
Flange Material		SA-182 F11	
Hub Height of Integral Nozzle	h	60.0000	mm
Height of Beveled Transition	L'	77.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		90.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	707.9000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		60.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.9	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: H1 8"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 8.625 in.
Actual Thickness Used in Calculation 0.520 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
= $P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
= $50.9 \cdot 3710 / (2 \cdot 1509 \cdot 1 - 0.2 \cdot 50.9)$
= 62.7748 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $50.9 \cdot 110 / (1407 \cdot 1 + 0.4 \cdot 50.9)$
= 3.9059 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 76.8/0.5773
= 133.0187 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

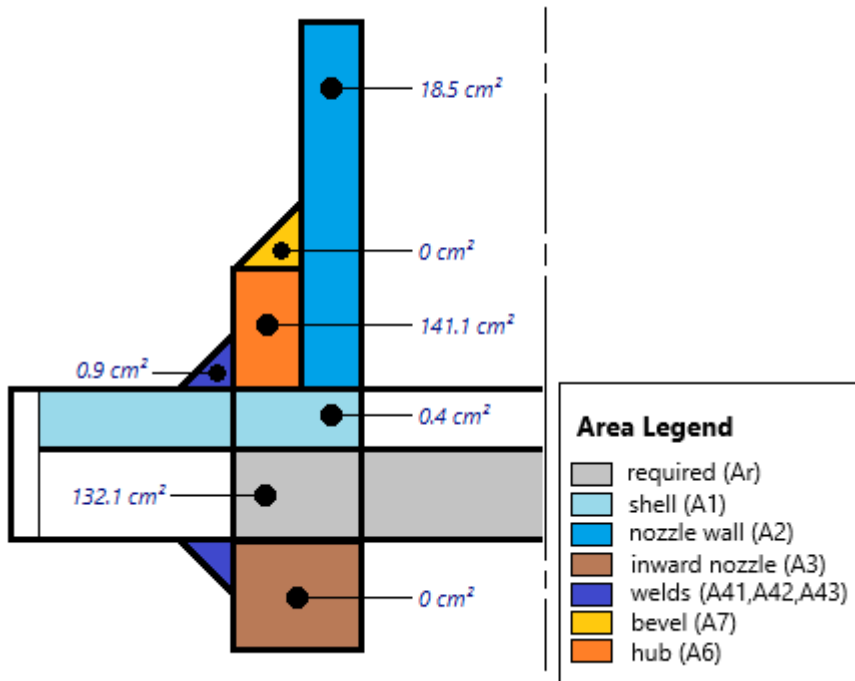
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	397.3434	mm
Parallel to Vessel Wall, opening length	d	198.6717	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	157.5000	mm

Weld Strength Reduction Factor [fr1]:
= min(1, S_n/S_v)
= min(1, 1407/1509)
= 0.932

Weld Strength Reduction Factor [fr2]:
= min(1, S_n/S_v)
= min(1, 1407/1509)
= 0.932

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(0.93, 1)
= 0.932



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	132.106	NA	NA
Area in Shell	A1	0.421	NA	NA
Area in Nozzle Wall	A2	18.490	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.932	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	141.057	NA	NA
TOTAL AREA AVAILABLE	Atot	160.900	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (199 \cdot 62.8 \cdot 1 + 2 \cdot 87 \cdot 62.8 \cdot 1 \cdot (1 - 0.93))$
 $= 132.106 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 199(1 \cdot 63 - 1 \cdot 62.8) - 2 \cdot 87$
 $(1 \cdot 63 - 1 \cdot 62.8) \cdot (1 - 0.93)$
 $= 0.421 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 158)(10.2 - 3.91)0.93$
 $= 18.490 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10^2 \cdot 0.93 + (0)^2 \cdot 0.93$
 $= 0.932 \text{ cm}^2$



Area Available in the Hub Section [A6]:

$$\begin{aligned} &= (2 \cdot \min(T_{lp}, h_o, H_{ubht})) \cdot (H_{ubt} - t_n) \cdot f_{r2} \\ &= (2 \cdot \min(158, 708, 60)) \cdot (90 - 13.2) \cdot 0.93 \\ &= 141.057 \text{ cm}^2 \text{ Includes Bevel Area of } 55.134 \text{ cm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 6.9059 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 65.7748 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 65.7748 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 10.1600 mm

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned} &= \min[tb3, \max(tb1, tb2)] \\ &= \min[10.2, \max(65.8, 4.5)] \\ &= 10.1600 \text{ mm} \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= \max(ta, tb) \\ &= \max(6.91, 10.2) \\ &= 10.1600 \text{ mm} \end{aligned}$$

Available Nozzle Neck Thickness = 13.2016 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, tg = 13.2, tr = 3.91, c = 3 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.38, Temp. Reduction = 55 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-20 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:B, Curve:A)

Govrn. thk, tg = 66, tr = 62.8, c = 3 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 1, Temp. Reduction = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	42 °C
Min Metal Temp. at Required thickness (UCS 66.1)	42 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 42 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 50.90/105.44 = 0.483

Weld Size Calculations, Description: H1 8"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned} &= \max(0, (A - A1 + 2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - tr)) S_v) \\ &= \max(0, (132 - 0.42 + 2 \cdot 87 \cdot 0.93 \cdot (1 \cdot 63 - 62.8)) 1509) \\ &= 199276.64 \text{ kgf} \end{aligned}$$

For hub type nozzles, A2 includes the area of the hub.



Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned} &= (A2 + A4 - (W_i - \text{Can} / .707)^2 \cdot \text{fr}2) \cdot S_v \\ &= (160 + 0.93 - 0 \cdot 0.93) \cdot 1509 \\ &= 242178.61 \text{ kgf} \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned} &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot \text{fr}1)) \cdot S_v \\ &= (160 + 0 + 0.93 + (102)) \cdot 1509 \\ &= 396413.72 \text{ kgf} \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned} &= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot \text{fr}1)) \cdot S \\ &= (160 + 0 + 0.93 + 0 + (102)) \cdot 1509 \\ &= 396413.72 \text{ kgf} \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.14/2.0) \cdot 219 \cdot 10 \cdot 0.49 \cdot 1407 \\ &= 23725. \text{ kgf} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (D_{lr} + D_{lo}) / 4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n \\ &= (3.14 \cdot 104) \cdot (90 - 3) \cdot 0.7 \cdot 1407 \\ &= 281134. \text{ kgf} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_{gnv} \cdot 0.74 \cdot S_{ng} \\ &= (3.14/2.0) \cdot 219 \cdot (60 - 0) \cdot 0.74 \cdot 1509 \\ &= 230576. \text{ kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (23725 + 281134) = 304859 \text{ kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ &= (23725 + 0 + 230576 + 0) = 254300 \text{ kgf} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (\text{Sonw} + \text{Tngw} + \text{Sinw}) \\ &= (23725 + 230576 + 0) = 254300 \text{ kgf} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 304859 kgf, must exceed W = 199276 kgf	or W1 = 242178 kgf
Path 2-2 = 254300 kgf, must exceed W = 199276 kgf	or W2 = 396413 kgf
Path 3-3 = 254300 kgf, must exceed W = 199276 kgf	or W3 = 396413 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 51.1 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 1.6185 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 775.5184 mm

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Input, Nozzle Desc: K4 3"

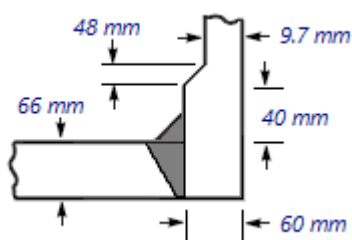
From: 40

Pressure for Reinforcement Calculations	P	50.897	kgf/cm ²
Temperature for Internal Pressure	Temp	275	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	1509.10	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1509.10	kgf/cm ²
Inside Diameter of Hemispherical Head	D	7414.00	mm
Head Finished (Minimum) Thickness	t	66.0000	mm
Head Internal Corrosion Allowance	cas	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	2000.0000	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1407.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.00	kgf/cm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		150.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-182 F11	
Hub Height of Integral Nozzle	h	40.0000	mm
Height of Beveled Transition	L'	48.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		60.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	707.9000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		60.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.9	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K4 3"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 3.500 in.
Actual Thickness Used in Calculation 0.383 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
= $P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
= $50.9 \cdot 3710 / (2 \cdot 1509 \cdot 1 - 0.2 \cdot 50.9)$
= 62.7748 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $50.9 \cdot 44.5 / (1407 \cdot 1 + 0.4 \cdot 50.9)$
= 1.5850 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= $50.3 / 0.5773$
= 87.0623 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

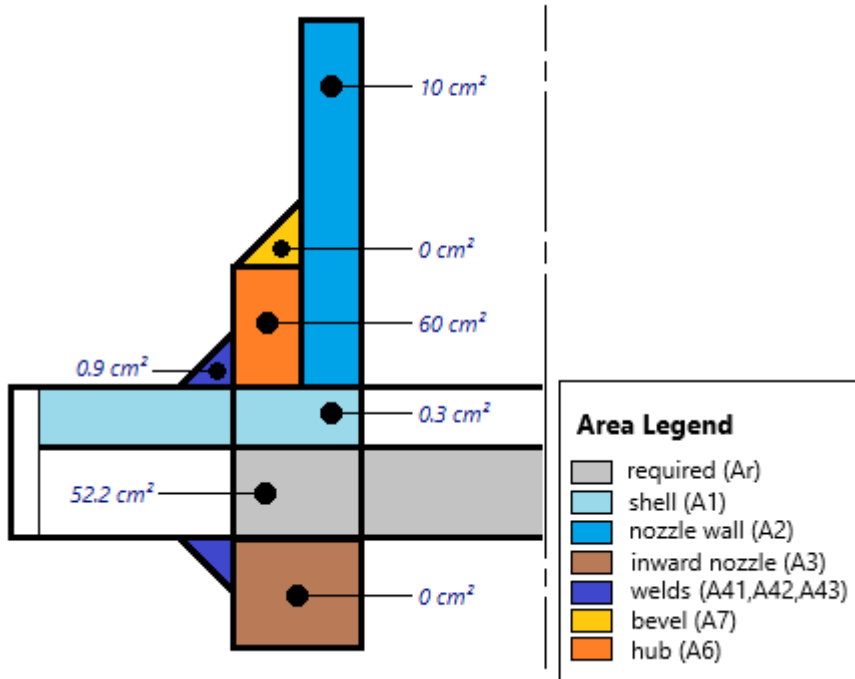
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	214.9000	mm
Parallel to Vessel Wall	Rn+tn+t	107.4500	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	103.8987	mm

Weld Strength Reduction Factor [fr1]:
= min(1, Sn/Sv)
= min(1, 1407/1509)
= 0.932

Weld Strength Reduction Factor [fr2]:
= min(1, Sn/Sv)
= min(1, 1407/1509)
= 0.932

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(0.93, 1)
= 0.932



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	52.193	NA	NA
Area in Shell	A1	0.297	NA	NA
Area in Nozzle Wall	A2	9.977	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.932	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	59.987	NA	NA
TOTAL AREA AVAILABLE	Atot	71.193	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (75.4 \cdot 62.8 \cdot 1 + 2 \cdot 57 \cdot 62.8 \cdot 1 \cdot (1 - 0.93))$
 $= 52.193 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 139(1 \cdot 63 - 1 \cdot 62.8) - 2 \cdot 57(1 \cdot 63 - 1 \cdot 62.8) \cdot (1 - 0.93)$
 $= 0.297 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 104)(6.73 - 1.59)0.93$
 $= 9.977 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10^2 \cdot 0.93 + (0)^2 \cdot 0.93$
 $= 0.932 \text{ cm}^2$



Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot f_{r2}$$

$$= (2 \cdot \min(104, 708, 40)) \cdot (60 - 9.73) \cdot 0.93$$

$$= 59.987 \text{ cm}^2 \text{ Includes Bevel Area of } 22.495 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 4.5850 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 65.7748 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 65.7748 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 7.8000 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[7.8, \max(65.8, 4.5)]$$

$$= 7.8000 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(4.59, 7.8)$$

$$= 7.8000 \text{ mm}$$

Available Nozzle Neck Thickness = 9.7345 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, tg = 9.73, tr = 1.59, c = 3 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.24, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:B, Curve:A)

Govrn. thk, tg = 60, tr = 1.59, c = 3 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.028, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	40 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 50.90/105.44 = 0.483

Weld Size Calculations, Description: K4 3"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - tr)) S_v)$$

$$= \max(0, (52.2 - 0.3 + 2 \cdot 57 \cdot 0.93 \cdot (1 \cdot 63 - 62.8)) 1509)$$

$$= 78678.25 \text{ kgf}$$

For hub type nozzles, A2 includes the area of the hub.



Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
$$= (A2 + A4 - (W_i - \text{Can} / .707)^2 \cdot \text{fr}2) \cdot S_v$$
$$= (70 + 0.93 - 0 \cdot 0.93) \cdot 1509$$
$$= 106988.69 \text{ kgf}$$

Weld Load [W2]:
$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot \text{fr}1)) \cdot S_v$$
$$= (70 + 0 + 0.93 + (67)) \cdot 1509$$
$$= 208039.30 \text{ kgf}$$

Weld Load [W3]:
$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot \text{fr}1)) \cdot S$$
$$= (70 + 0 + 0.93 + 0 + (67)) \cdot 1509$$
$$= 208039.30 \text{ kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$
$$= (3.14/2.0) \cdot 88.9 \cdot 10 \cdot 0.49 \cdot 1407$$
$$= 9627. \text{ kgf}$$

Shear, Nozzle Wall [Snw]:
$$= (\pi \cdot (D_{lr} + D_{lo}) / 4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$$
$$= (3.14 \cdot 41.1) \cdot (60 - 3) \cdot 0.7 \cdot 1407$$
$$= 72456. \text{ kgf}$$

Tension, Shell Groove Weld [Tngw]:
$$= (\pi/2) \cdot D_{lo} \cdot W_{gnv} \cdot 0.74 \cdot S_{ng}$$
$$= (3.14/2.0) \cdot 88.9 \cdot (60 - 0) \cdot 0.74 \cdot 1509$$
$$= 93567. \text{ kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
$$= (\text{SONW} + \text{SNW}) = (9627 + 72456) = 82084 \text{ kgf}$$

Failure Path [Path 2-2]:
$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$
$$= (9627 + 0 + 93567 + 0) = 103194 \text{ kgf}$$

Failure Path [Path 3-3]:
$$= (\text{Sonw} + \text{Tngw} + \text{Sinw})$$
$$= (9627 + 93567 + 0) = 103194 \text{ kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 82083	kgf, must exceed W = 78678 kgf	or W1 = 106988 kgf
Path 2-2 = 103194	kgf, must exceed W = 78678 kgf	or W2 = 208039 kgf
Path 3-3 = 103194	kgf, must exceed W = 78678 kgf	or W3 = 208039 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 51.1 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 0.2663 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 774.1661 mm

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Input, Nozzle Desc: N1 36"

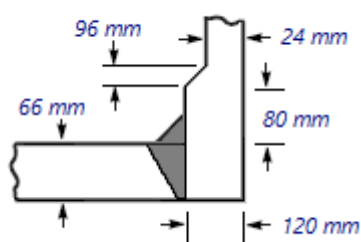
From: 60

Pressure for Reinforcement Calculations	P	50.897	kgf/cm ²
Temperature for Internal Pressure	Temp	275	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	1509.10	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1509.10	kgf/cm ²
Inside Diameter of Hemispherical Head	D	7414.00	mm
Head Finished (Minimum) Thickness	t	66.0000	mm
Head Internal Corrosion Allowance	cas	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	10.0000	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1407.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.00	kgf/cm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		36.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	24.0000	mm
Flange Material		SA-182 F11	
Hub Height of Integral Nozzle	h	80.0000	mm
Height of Beveled Transition	L'	96.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		120.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		60.0000	mm
Inside Projection	h	300.0000	mm
Weld leg size, Inside Element to Shell	Wi	14.0000	mm
Flange Class		600	
Flange Grade		GR 1.9	
Flange Series		B	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle in the Meridional direction.



Reinforcement CALCULATION, Description: N1 36"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 36.000 in.
Actual Thickness Used in Calculation 0.945 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
= $P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
= $50.9 \cdot 3710 / (2 \cdot 1509 \cdot 1 - 0.2 \cdot 50.9)$
= 62.7748 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $50.9 \cdot 457 / (1407 \cdot 1 + 0.4 \cdot 50.9)$
= 16.3029 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= $96 / 0.5773$
= 166.2770 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

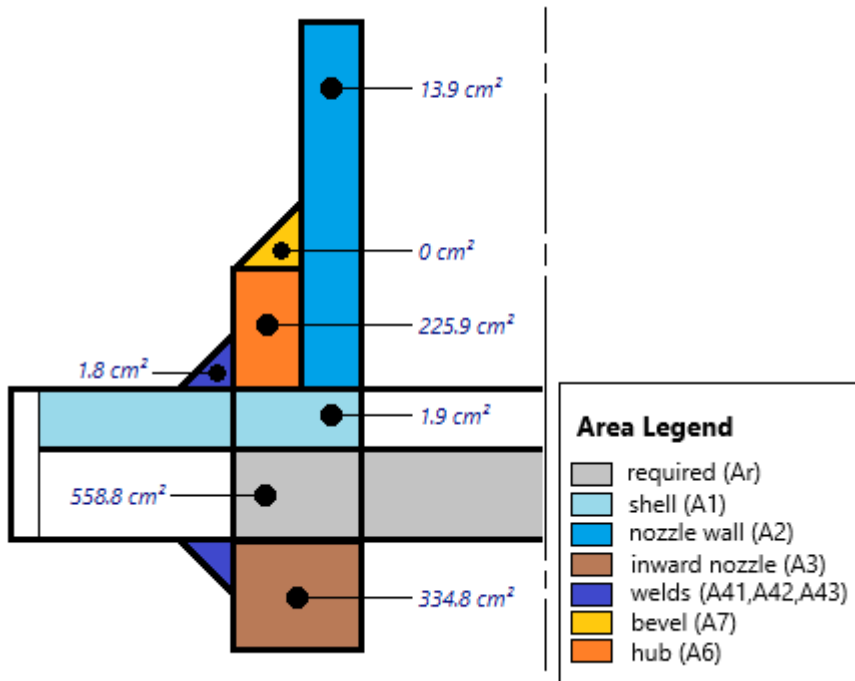
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1748.7843	mm
Parallel to Vessel Wall, opening length	d	874.3922	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	157.5000	mm
Normal to Vessel Wall, Inward		45.0000	mm

Weld Strength Reduction Factor [fr1]:
= min(1, S_n / S_v)
= min(1, $1407 / 1509$)
= 0.932

Weld Strength Reduction Factor [fr2]:
= min(1, S_n / S_v)
= min(1, $1407 / 1509$)
= 0.932

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(0.93, 1)
= 0.932



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	558.836	NA	NA
Area in Shell	A1	1.934	NA	NA
Area in Nozzle Wall	A2	13.884	NA	NA
Area in Inward Nozzle	A3	334.805	NA	NA
Area in Welds	A41+A42+A43	1.820	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	225.942	NA	NA
TOTAL AREA AVAILABLE	Atot	578.384	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 86.13 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (874 \cdot 62.8 \cdot 1 + 2 \cdot 117 \cdot 62.8 \cdot 1 \cdot (1 - 0.93))$$

$$= 558.836 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 874 (1 \cdot 63 - 1 \cdot 62.8) - 2 \cdot 117$$

$$(1 \cdot 63 - 1 \cdot 62.8) \cdot (1 - 0.93)$$

$$= 1.934 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp) (tn - trn) fr2 / \sin(\alpha3)$$

$$= (2 \cdot 158) (21 - 16.3) 0.93 / \sin(96.5)$$

$$= 13.884 \text{ cm}^2$$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Nozzle [A3]:

$$= 2 \cdot ti \cdot \min(h, TI, 2.5 \cdot ti) \cdot fr2$$

$$= 2 \cdot 114 \cdot (158) \cdot 0.93$$



= 334.805 cm²

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i\text{-can}/0.707)^2 \cdot fr_2$$

$$= 10^2 \cdot 0.93 + (9.76)^2 \cdot 0.93$$

$$= 1.820 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(158, 300, 80)) \cdot (120 - 24) \cdot 0.93$$

$$= 225.942 \text{ cm}^2 \text{ Includes Bevel Area of } 82.734 \text{ cm}^2$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, tg = 24, tr = 16.3, c = 3 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.78, Temp. Reduction = 12 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-2 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-15 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:B, Curve:A)

Govrn. thk, tg = 66, tr = 62.8, c = 3 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 1, Temp. Reduction = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	42 °C
Min Metal Temp. at Required thickness (UCS 66.1)	42 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 42 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 50.90/105.44 = 0.483

Weld Size Calculations, Description: N1 36"

Intermediate Calc. for nozzle/shell Welds	Tmin	19.0000 mm
Intermediate Calc. for Inward Weld	TminIns	19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * W _o mm
Inward Weld	6.0000 = Min per Code	6.8980 = 0.7 * W _i -Can mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A_1 + 2 \cdot t_n \cdot fr_1 \cdot (E_1 \cdot t - tr)) \cdot S_v)$$

$$= \max(0, (559 - 1.93 + 2 \cdot 117 \cdot 0.93 \cdot (1 \cdot 63 - 62.8)) \cdot 1509)$$

$$= 841161.81 \text{ kgf}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A_2 + A_4 - (W_i\text{-Can}/0.707)^2 \cdot fr_2) \cdot S_v$$

$$= (240 + 1.82 - 95.2 \cdot 0.93) \cdot 1509$$

$$= 363327.22 \text{ kgf}$$

Weld Load [W2]:

$$= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S_v$$

$$= (240 + 335 + 1.82 + (137)) \cdot 1509$$



= 1077339.38 kgf

Weld Load [W3]:

$$\begin{aligned} &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f r_1)) \cdot S \\ &= (240 + 335 + 1.82 + 0 + (137)) \cdot 1509 \\ &= 1077339.38 \text{ kgf} \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.14/2.0) \cdot 916 \cdot 10 \cdot 0.49 \cdot 1407 \\ &= 99251. \text{ kgf} \end{aligned}$$

Shear, Inward Nozzle Weld [Sinw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.14/2.0) \cdot 916 \cdot 9.76 \cdot 0.49 \cdot 1407 \\ &= 96843. \text{ kgf} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n \\ &= (3.14 \cdot 448) \cdot (120 - 3) \cdot 0.7 \cdot 1407 \\ &= 1620816. \text{ kgf} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_{g n v i} \cdot 0.74 \cdot S_{ng} \\ &= (3.14/2.0) \cdot 916 \cdot (60 - 0) \cdot 0.74 \cdot 1509 \\ &= 964600. \text{ kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (99251 + 1620816) = 1720067 \text{ kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (99251 + 0 + 964600 + 96843) = 1160694 \text{ kgf} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (99251 + 964600 + 96843) = 1160694 \text{ kgf} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 1720067 kgf, must exceed W = 841161 kgf	or W1 = 363327 kgf
Path 2-2 = 1160694 kgf, must exceed W = 841161 kgf	or W2 = 1077339 kgf
Path 3-3 = 1160694 kgf, must exceed W = 841161 kgf	or W3 = 1077339 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 51.1 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: N1 36"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	36.000 in.
Actual Thickness Used in Calculation	0.945 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]

$$\begin{aligned} &= P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P) \text{ per UG-27 (d)} \\ &= 50.9 \cdot 3710 / (2 \cdot 1509 \cdot 1 - 0.2 \cdot 50.9) \\ &= 62.7748 \text{ mm} \end{aligned}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 50.9 \cdot 457 / (1407 \cdot 1 + 0.4 \cdot 50.9)$
 $= 16.3029 \text{ mm}$

Intermediate Hub Nozzle Calculations:

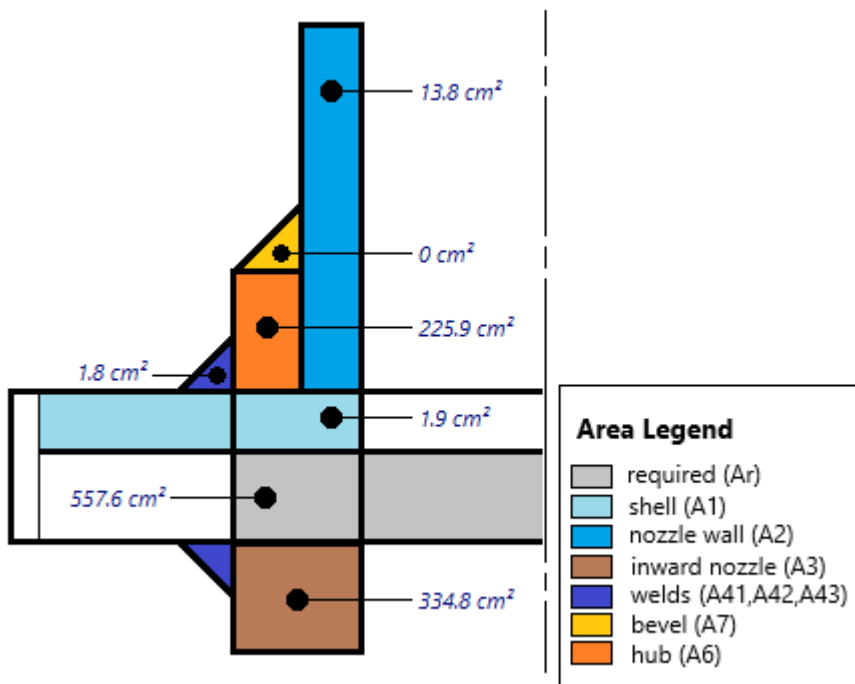
Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 96 / 0.5773$
 $= 166.2770 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1744.8000	mm
Parallel to Vessel Wall, opening length	d	872.4000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	157.5000	mm
Normal to Vessel Wall, Inward		45.0000	mm



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	557.585	NA	NA
Area in Shell	A1	1.929	NA	NA
Area in Nozzle Wall	A2	13.795	NA	NA
Area in Inward Nozzle	A3	334.805	NA	NA
Area in Welds	A41+A42+A43	1.820	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	225.942	NA	NA
TOTAL AREA AVAILABLE	Atot	578.290	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:



$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (872 \cdot 62.8 \cdot 1 + 2 \cdot 117 \cdot 62.8 \cdot 1 \cdot (1 - 0.93))$$

$$= 557.585 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 872(1 \cdot 63 - 1 \cdot 62.8) - 2 \cdot 117$$

$$(1 \cdot 63 - 1 \cdot 62.8) \cdot (1 - 0.93)$$

$$= 1.929 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 158)(21 - 16.3)0.93$$

$$= 13.795 \text{ cm}^2$$

Area Available in Inward Nozzle [A3]:

$$= 2 \cdot ti \cdot \min(h, TI, 2.5 \cdot ti) \cdot fr2$$

$$= 2 \cdot 114 \cdot (158) \cdot 0.93$$

$$= 334.805 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10^2 \cdot 0.93 + (9.76)^2 \cdot 0.93$$

$$= 1.820 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(Tlnp, ho, Hubht)) \cdot (Hubtk - tn) \cdot fr2$$

$$= (2 \cdot \min(158, 300, 80)) \cdot (120 - 24) \cdot 0.93$$

$$= 225.942 \text{ cm}^2 \text{ Includes Bevel Area of } 82.734 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 19.3029 \text{ mm}$
 Wall Thickness per UG16(b), $tr16b = 4.5000 \text{ mm}$
 Wall Thickness, shell/head, internal pressure $trb1 = 65.7748 \text{ mm}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 65.7748 \text{ mm}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 4.5000 \text{ mm}$
 Wall Thickness per table UG-45 $tb3 = 11.3312 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.3, \max(65.8, 4.5)]$$

$$= 11.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(19.3, 11.3)$$

$$= 19.3029 \text{ mm}$$

Available Nozzle Neck Thickness = 24.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 683.6,	Allowable	: 1407.0 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 2833.9 kgf/cm ²	Passed
Occasional	: 516.2,	Allowable	: 1871.3 kgf/cm ²	Passed
Shear	: 104.9,	Allowable	: 984.9 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Weld Size Calculations, Description: N1 36"

Intermediate Calc. for nozzle/shell Welds	Tmin	19.0000	mm
Intermediate Calc. for Inward Weld	TminIns	19.0000	mm

Results Per UW-16.1:

Required Thickness	Actual Thickness
--------------------	------------------



Nozzle Weld 6.0000 = Min per Code 7.0700 = 0.7 * Wo mm
 Inward Weld 6.0000 = Min per Code 6.8980 = 0.7 * Wi-Can mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (558 - 1.93 + 2 \cdot 117 \cdot 0.93 \cdot (1 \cdot 63 - 62.8)) \cdot 1509)$
 $= 839281.31 \text{ kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (240 + 1.82 - 95.2 \cdot 0.93) \cdot 1509$
 $= 363193.09 \text{ kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (240 + 335 + 1.82 + (137)) \cdot 1509$
 $= 1077205.38 \text{ kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (240 + 335 + 1.82 + 0 + (137)) \cdot 1509$
 $= 1077205.38 \text{ kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.14/2.0) \cdot 914 \cdot 10 \cdot 0.49 \cdot 1407$
 $= 99025. \text{ kgf}$

Shear, Inward Nozzle Weld [Sinw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.14/2.0) \cdot 914 \cdot 9.76 \cdot 0.49 \cdot 1407$
 $= 96622. \text{ kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.14 \cdot 447) \cdot (120 - 3) \cdot 0.7 \cdot 1407$
 $= 1617124. \text{ kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$
 $= (3.14/2.0) \cdot 914 \cdot (60 - 0) \cdot 0.74 \cdot 1509$
 $= 962402. \text{ kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SONW + SNW) = (99025 + 1617123) = 1716148 \text{ kgf}$

Failure Path [Path 2-2]:
 $= (Sonw + Tpgw + Tngw + Sinw)$
 $= (99025 + 0 + 962402 + 96622) = 1158049 \text{ kgf}$

Failure Path [Path 3-3]:
 $= (Sonw + Tngw + Sinw)$
 $= (99025 + 962402 + 96622) = 1158049 \text{ kgf}$

Summary of Failure Path Calculations:

Path 1-1 = 1716148 kgf, must exceed W = 839281 kgf or W1 = 363193 kgf
 Path 2-2 = 1158049 kgf, must exceed W = 839281 kgf or W2 = 1077205 kgf



Path 3-3 = 1158049 kgf, must exceed W = 839281 kgf or W3 = 1077205 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 51.1 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 29.5451 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 666.0002 mm

Warning:

The equations for nozzle local stress analysis pertain to radial nozzles only and this nozzle is not specified as radial. Please check the specific analysis code for all assumptions and limitations.

Input Echo, WRC107/537 Item 1, Description: N1 36" :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	3.0000	mm
Vessel Diameter	Dv	7414.000	mm
Vessel Thickness	Tv	66.000	mm
Design Temperature	T1	275.0	C
Vessel Material		SA-387 11	
Vessel UNS Number		K11789	
Vessel Cold S.I. Allowable	Smc	1509.10	kgf/cm ²
Vessel Hot S.I. Allowable	Smh	1509.10	kgf/cm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	OD	
Corrosion Allowance for Nozzle	Can	3.0000	mm
Nozzle Diameter	Dn	1106.400	mm
Nozzle Thickness	Tn	120.000	mm
Nozzle Material		SA-182 F11	
Nozzle UNS Number		K11572	
Nozzle Cold S.I. Allowable	SNmc	1407.00	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh	1407.00	kgf/cm ²
Design Internal Pressure	Dp	50.897	kgf/cm ²
Include Pressure Thrust		No	

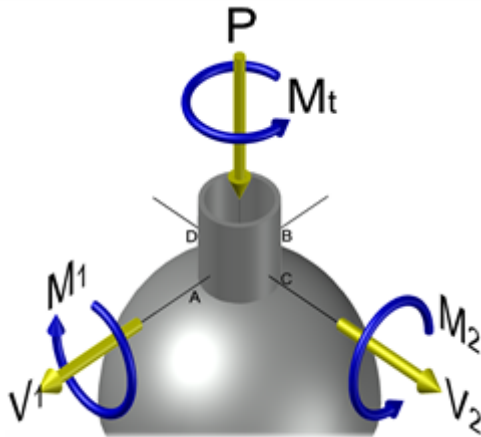
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-8256.9	kgf
Longitudinal Shear (SUS)	(V1) V1	5963.3	kgf
Circumferential Shear (SUS)	(Vc) V2	5963.3	kgf
Circumferential Moment (SUS)	(Mc) M1	16513.9	kgf-m.
Longitudinal Moment (SUS)	(Ml) M2	16513.9	kgf-m.
Torsional Moment (SUS)	Mt	18990.7	kgf-m.

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 1106.4 + 2 \cdot 1.65 \cdot \sqrt{(3741.5 (66.0 - 3.0))}$$

$$= 2708.564 \text{ mm}$$

WRC 107 Stress Calculation for SUSTAINED loads:

Radial Load	P	-8256.9	kgf
Circumferential Shear	(VC) V2	5963.3	kgf
Longitudinal Shear	(VL) V1	5963.3	kgf
Circumferential Moment	(MC) M1	16513.9	kgf-m.
Longitudinal Moment	(ML) M2	16513.9	kgf-m.
Torsional Moment	MT	18990.7	kgf-m.

Dimensionless Param: U = 1.14 TAU = 5.00 (4.23) RHO = 0.54

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 1	0.03557	(A,B,C,D)
$M(x) / P$	SP 1	0.03578	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.04063	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 1	0.05963	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.04063	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 1	0.05963	(A,B,C,D)
$N(y) * T / P$	SP 1	0.05752	(A,B,C,D)
$M(y) / P$	SP 1	0.01047	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.05676	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 1	0.01960	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.05676	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 1	0.01960	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (kgf/cm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb.	P	7.4	7.4	7.4	7.4	7.4	7.4	7.4	7.4
Rad. Bend.	P	44.7	-44.7	44.7	-44.7	44.7	-44.7	44.7	-44.7
Rad. Memb.	MC	0.0	0.0	0.0	0.0	-34.8	-34.8	34.8	34.8
Rad. Bend.	MC	0.0	0.0	0.0	0.0	-306.6	306.6	306.6	-306.6
Rad. Memb.	ML	-34.8	-34.8	34.8	34.8	0.0	0.0	0.0	0.0
Rad. Bend.	ML	-306.6	306.6	306.6	-306.6	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-289.4	234.5	393.5	-309.1	-289.4	234.5	393.5	-309.1

Tang. Memb. P	12.0	12.0	12.0	12.0	12.0	12.0	12.0	12.0
Tang. Bend. P	13.1	-13.1	13.1	-13.1	13.1	-13.1	13.1	-13.1
Tang. Memb. MC	0.0	0.0	0.0	0.0	-48.6	-48.6	48.6	48.6
Tang. Bend. MC	0.0	0.0	0.0	0.0	-100.8	100.8	100.8	-100.8
Tang. Memb. ML	-48.6	-48.6	48.6	48.6	0.0	0.0	0.0	0.0
Tang. Bend. ML	-100.8	100.8	100.8	-100.8	0.0	0.0	0.0	0.0
Tot. Tang. Str.	-124.4	51.1	174.5	-53.2	-124.4	51.1	174.5	-53.2
Shear VC	5.4	5.4	-5.4	-5.4	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-5.4	-5.4	5.4	5.4
Shear MT	15.7	15.7	15.7	15.7	15.7	15.7	15.7	15.7
Tot. Shear	21.1	21.1	10.2	10.2	10.2	10.2	21.1	21.1
Str. Int.	292.1	236.9	394.0	309.5	290.0	235.1	395.5	310.8

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (kgf/cm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		1498.9	1498.9	1498.9	1498.9	1498.9	1498.9	1498.9	1498.9
Rad. Pl (SUS)		-27.4	-27.4	42.2	42.2	-27.4	-27.4	42.2	42.2
Rad. Q (SUS)		-262.0	262.0	351.3	-351.3	-262.0	262.0	351.3	-351.3
Long. Pm (SUS)		1498.9	1498.9	1498.9	1498.9	1498.9	1498.9	1498.9	1498.9
Long. Pl (SUS)		-36.7	-36.7	60.6	60.6	-36.7	-36.7	60.6	60.6
Long. Q (SUS)		-87.7	87.7	113.9	-113.9	-87.7	87.7	113.9	-113.9
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		5.4	5.4	-5.4	-5.4	-5.4	-5.4	5.4	5.4
Shear Q (SUS)		15.7	15.7	15.7	15.7	15.7	15.7	15.7	15.7
Pm (SUS)		1498.9	1498.9	1498.9	1498.9	1498.9	1498.9	1498.9	1498.9
Pm+Pl (SUS)		1474.0	1474.0	1561.0	1561.0	1474.0	1474.0	1561.0	1561.0
Pm+Pl+Q (Total)		1377.2	1735.9	1892.9	1446.1	1375.1	1734.0	1894.5	1447.4

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1498.92	1509.10	Passed
Pm+Pl (SUS)	1561.02	2263.65	Passed
Pm+Pl+Q (TOTAL)	1894.46	4527.30	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: K1 2"

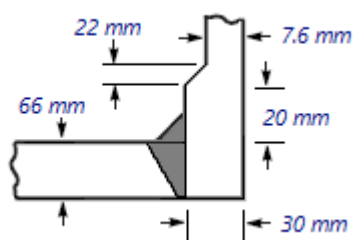
From: 60

Pressure for Reinforcement Calculations	P	50.897	kgf/cm ²
Temperature for Internal Pressure	Temp	275	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	1509.10	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1509.10	kgf/cm ²
Inside Diameter of Hemispherical Head	D	7414.00	mm
Head Finished (Minimum) Thickness	t	66.0000	mm
Head Internal Corrosion Allowance	cas	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	1500.0000	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1407.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.00	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-182 F11	
Hub Height of Integral Nozzle	h	20.0000	mm
Height of Beveled Transition	L'	22.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		30.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	14.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	60.0000	mm
Inside Projection	h	6375.0000	mm
Weld leg size, Inside Element to Shell	Wi	14.0000	mm
Flange Class		600	
Flange Grade		GR 1.9	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle in the Meridional direction.

Reinforcement CALCULATION, Description: K1 2"



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 1.773 in.
Actual Thickness Used in Calculation 0.301 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
= $P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
= $50.9 \cdot 3710 / (2 \cdot 1509 \cdot 1 - 0.2 \cdot 50.9)$
= 62.7748 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $50.9 \cdot 25.5 / (1407 \cdot 1 - 0.6 \cdot 50.9)$
= 0.9435 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 22.4 / 0.5773
= 38.7193 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

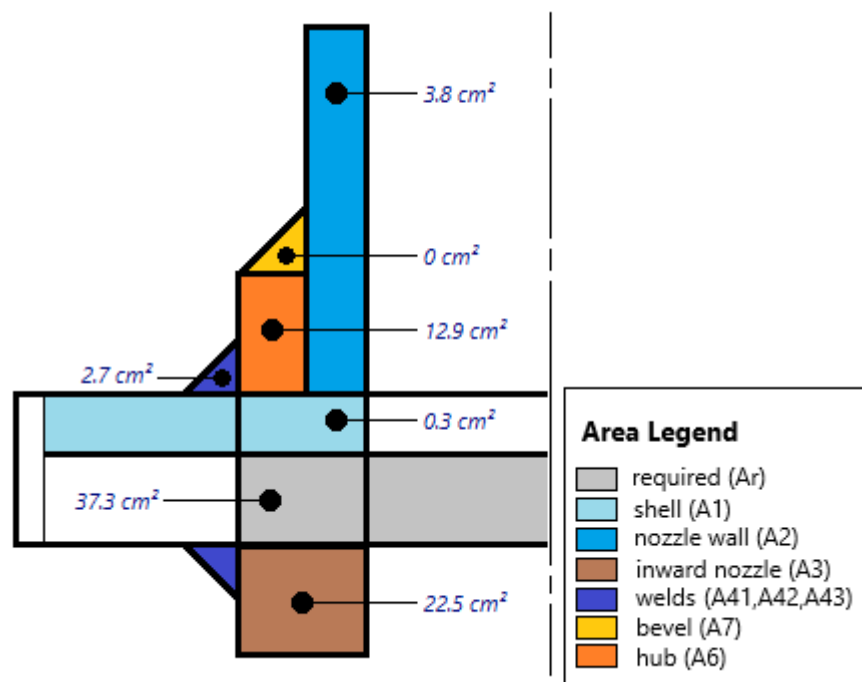
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	190.9995	mm
Parallel to Vessel Wall	Rn+tn+t	95.4997	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	50.3328	mm
Normal to Vessel Wall, Inward		4.1135	mm

Weld Strength Reduction Factor [fr1]:
= min(1, S_n / S_v)
= min(1, 1407/1509)
= 0.932

Weld Strength Reduction Factor [fr2]:
= min(1, S_n / S_v)
= min(1, 1407/1509)
= 0.932

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(0.93, 1)
= 0.932



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	37.264	NA	NA
Area in Shell	A1	0.296	NA	NA
Area in Nozzle Wall	A2	3.775	NA	NA
Area in Inward Nozzle	A3	22.525	NA	NA
Area in Welds	A41+A42+A43	2.715	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	12.922	NA	NA
TOTAL AREA AVAILABLE	Atot	42.233	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 66.36 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (55.7 \cdot 62.8 \cdot 1 + 2 \cdot 27 \cdot 62.8 \cdot 1 \cdot (1 - 0.93))$
 $= 37.264 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 135(1 \cdot 63 - 1 \cdot 62.8) - 2 \cdot 27(1 \cdot 63 - 1 \cdot 62.8) \cdot (1 - 0.93)$
 $= 0.296 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2/\sin(\alpha3)$
 $= (2 \cdot 50.3)(4.65 - 0.94)0.93/\sin(67)$
 $= 3.775 \text{ cm}^2$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Nozzle [A3]:
 $= 2 \cdot ti \cdot \min(h, TI, 2.5 \cdot ti) \cdot fr2$
 $= 2 \cdot 24 \cdot (50.3) \cdot 0.93$



= 22.525 cm²

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr2 + (W_i\text{-can}/0.707)^2 \cdot fr2$$

$$= 14^2 \cdot 0.93 + (9.76)^2 \cdot 0.93$$

$$= 2.715 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr2$$

$$= (2 \cdot \min(50.3, 300, 20)) \cdot (30 - 7.65) \cdot 0.93$$

$$= 12.922 \text{ cm}^2 \text{ Includes Bevel Area of } 4.585 \text{ cm}^2$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, tg = 7.65, tr = 0.94, c = 3 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.2, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:A)

Govrn. thk, tg = 30, tr = 0.94, c = 3 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.035, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	24 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 50.90/105.44 = 0.483

Weld Size Calculations, Description: K1 2"

Intermediate Calc. for nozzle/shell Welds	Tmin	19.0000 mm
Intermediate Calc. for Inward Weld	TminIns	19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	9.8980 = 0.7 * Wo mm
Inward Weld	6.0000 = Min per Code	6.8980 = 0.7 * Wi-Can mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$$

$$= \max(0, (37.3 - 0.3 + 2 \cdot 27 \cdot 0.93 \cdot (1 \cdot 63 - 62.8)) \cdot 1509)$$

$$= 55959.38 \text{ kgf}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A4-(W_i\text{-Can}/.707)^2 \cdot fr2) \cdot Sv$$

$$= (16.7 + 2.72 - 95.2 \cdot 0.93) \cdot 1509$$

$$= 27954.66 \text{ kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$$

$$= (16.7 + 22.5 + 2.72 + (31.7)) \cdot 1509$$



= 111153.00 kgf

Weld Load [W3]:

$$\begin{aligned} &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f r_1)) \cdot S \\ &= (16.7 + 22.5 + 2.72 + 0 + (31.7)) \cdot 1509 \\ &= 111153.00 \text{ kgf} \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.14/2.0) \cdot 115 \cdot 14 \cdot 0.49 \cdot 1407 \\ &= 17383. \text{ kgf} \end{aligned}$$

Shear, Inward Nozzle Weld [Sinw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.14/2.0) \cdot 115 \cdot 9.76 \cdot 0.49 \cdot 1407 \\ &= 12115. \text{ kgf} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n \\ &= (3.14 \cdot 42.6) \cdot (30 - 3) \cdot 0.7 \cdot 1407 \\ &= 35581. \text{ kgf} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng} \\ &= (3.14/2.0) \cdot 115 \cdot (60 - 0) \cdot 0.74 \cdot 1509 \\ &= 120674. \text{ kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (17383 + 35581) = 52964 \text{ kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (17383 + 0 + 120674 + 12115) = 150172 \text{ kgf} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (17383 + 120674 + 12115) = 150172 \text{ kgf} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 52964 kgf, must exceed W = 55959 kgf	or W1 = 27954 kgf
Path 2-2 = 150172 kgf, must exceed W = 55959 kgf	or W2 = 111153 kgf
Path 3-3 = 150172 kgf, must exceed W = 55959 kgf	or W3 = 111153 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 51.1 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: K1 2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.773 in.
Actual Thickness Used in Calculation	0.301 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]

$$\begin{aligned} &= P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P) \text{ per UG-27 (d)} \\ &= 50.9 \cdot 3710 / (2 \cdot 1509 \cdot 1 - 0.2 \cdot 50.9) \\ &= 62.7748 \text{ mm} \end{aligned}$$



Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 50.9 \cdot 25.5 / (1407 \cdot 1 - 0.6 \cdot 50.9)$
 $= 0.9435 \text{ mm}$

Intermediate Hub Nozzle Calculations:

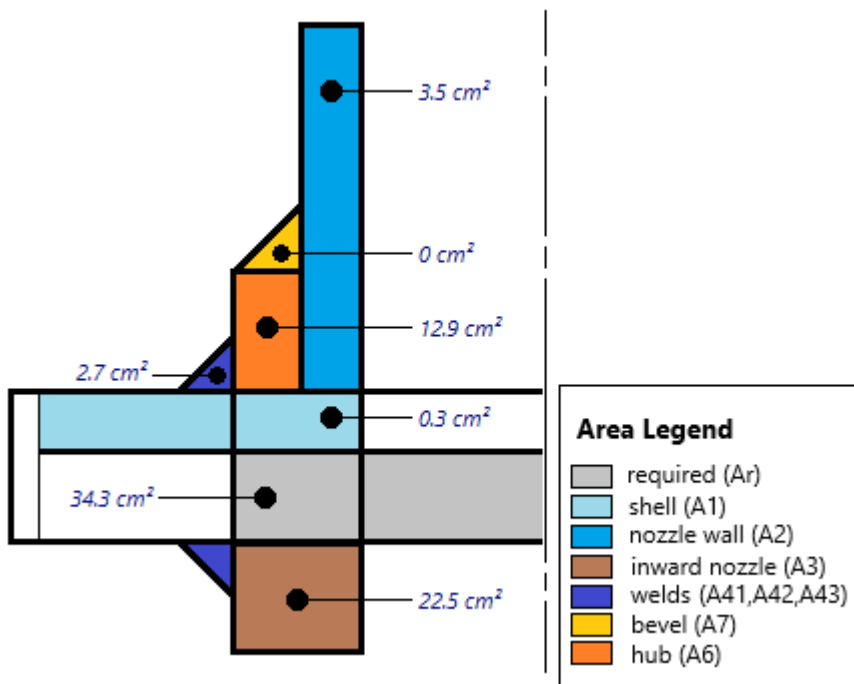
Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 22.4 / 0.5773$
 $= 38.7193 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	186.3250	mm
Parallel to Vessel Wall	Rn+tn+t	93.1625	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	50.3328	mm
Normal to Vessel Wall, Inward		4.1135	mm



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	34.330	NA	NA
Area in Shell	A1	0.296	NA	NA
Area in Nozzle Wall	A2	3.474	NA	NA
Area in Inward Nozzle	A3	22.525	NA	NA
Area in Welds	A41+A42+A43	2.715	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	12.922	NA	NA
TOTAL AREA AVAILABLE	Atot	41.933	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:



$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (51 \cdot 62.8 \cdot 1 + 2 \cdot 27 \cdot 62.8 \cdot 1 \cdot (1 - 0.93)) \\
 &= 34.330 \text{ cm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned}
 &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 135(1 \cdot 63 - 1 \cdot 62.8) - 2 \cdot 27 \\
 &\quad (1 \cdot 63 - 1 \cdot 62.8) \cdot (1 - 0.93) \\
 &= 0.296 \text{ cm}^2
 \end{aligned}$$

Area Available in Nozzle Projecting Outward [A2]:

$$\begin{aligned}
 &= (2 \cdot tlnp)(tn - trn)fr2 \\
 &= (2 \cdot 50.3)(4.65 - 0.94)0.93 \\
 &= 3.474 \text{ cm}^2
 \end{aligned}$$

Area Available in Inward Nozzle [A3]:

$$\begin{aligned}
 &= 2 \cdot ti \cdot \min(h, TI, 2.5 \cdot ti) \cdot fr2 \\
 &= 2 \cdot 24 \cdot (50.3) \cdot 0.93 \\
 &= 22.525 \text{ cm}^2
 \end{aligned}$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$\begin{aligned}
 &= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2 \\
 &= 14^2 \cdot 0.93 + (9.76)^2 \cdot 0.93 \\
 &= 2.715 \text{ cm}^2
 \end{aligned}$$

Area Available in the Hub Section [A6]:

$$\begin{aligned}
 &= (2 \cdot \min(Tlnp, ho, Hubht)) \cdot (Hubtk - tn) \cdot fr2 \\
 &= (2 \cdot \min(50.3, 300, 20)) \cdot (30 - 7.65) \cdot 0.93 \\
 &= 12.922 \text{ cm}^2 \text{ Includes Bevel Area of } 4.585 \text{ cm}^2
 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.9435 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 65.7748 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 65.7748 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 6.4200 mm

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned}
 &= \min[tb3, \max(tb1, tb2)] \\
 &= \min[6.42, \max(65.8, 4.5)] \\
 &= 6.4200 \text{ mm}
 \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned}
 &= \max(ta, tb) \\
 &= \max(3.94, 6.42) \\
 &= 6.4200 \text{ mm}
 \end{aligned}$$

Available Nozzle Neck Thickness = 7.6454 mm -> OK

Weld Size Calculations, Description: K1 2"

Intermediate Calc. for nozzle/shell Welds	Tmin	19.0000 mm
Intermediate Calc. for Inward Weld	TminIns	19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	9.8980 = 0.7 * Wo mm
Inward Weld	6.0000 = Min per Code	6.8980 = 0.7 * Wi-Can mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned}
 &= \max(0, (A - A1 + 2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr))Sv) \\
 &= \max(0, (34.3 - 0.3 + 2 \cdot 27 \cdot 0.93 \cdot \\
 &\quad (1 \cdot 63 - 62.8))1509) \\
 &= 51531.12 \text{ kgf}
 \end{aligned}$$



For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
$$= (A2 + A4 - (W_i - Can / .707)^2 \cdot fr2) \cdot Sv$$
$$= (16.4 + 2.72 - 95.2 \cdot 0.93) \cdot 1509$$
$$= 27501.69 \text{ kgf}$$

Weld Load [W2]:
$$= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$$
$$= (16.4 + 22.5 + 2.72 + (31.7)) \cdot 1509$$
$$= 110700.02 \text{ kgf}$$

Weld Load [W3]:
$$= (A2 + A3 + A4 + A5 + (2 \cdot tn \cdot t \cdot fr1)) \cdot S$$
$$= (16.4 + 22.5 + 2.72 + 0 + (31.7)) \cdot 1509$$
$$= 110700.02 \text{ kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$
$$= (3.14/2.0) \cdot 105 \cdot 14 \cdot 0.49 \cdot 1407$$
$$= 15925. \text{ kgf}$$

Shear, Inward Nozzle Weld [Sinw]:
$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$
$$= (3.14/2.0) \cdot 105 \cdot 9.76 \cdot 0.49 \cdot 1407$$
$$= 11099. \text{ kgf}$$

Shear, Nozzle Wall [Snw]:
$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$
$$= (3.14 \cdot 39) \cdot (30 - 3) \cdot 0.7 \cdot 1407$$
$$= 32596. \text{ kgf}$$

Tension, Shell Groove Weld [Tngw]:
$$= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$$
$$= (3.14/2.0) \cdot 105 \cdot (60 - 0) \cdot 0.74 \cdot 1509$$
$$= 110548. \text{ kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
$$= (SONW + SNW) = (15925 + 32596) = 48520 \text{ kgf}$$

Failure Path [Path 2-2]:
$$= (Sonw + Tpgw + Tngw + Sinw)$$
$$= (15925 + 0 + 110548 + 11099) = 137571 \text{ kgf}$$

Failure Path [Path 3-3]:
$$= (Sonw + Tngw + Sinw)$$
$$= (15925 + 110548 + 11099) = 137571 \text{ kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 48520	kgf, must exceed W = 51531 kgf	or W1 = 27501 kgf
Path 2-2 = 137571	kgf, must exceed W = 51531 kgf	or W2 = 110700 kgf
Path 3-3 = 137571	kgf, must exceed W = 51531 kgf	or W3 = 110700 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 51.1 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 23.7280 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 6747.0483 mm

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Input, Nozzle Desc: M1 24"

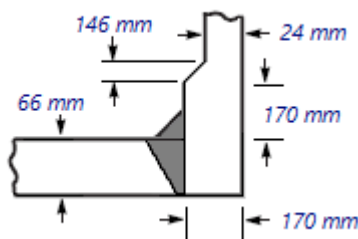
From: 60

Pressure for Reinforcement Calculations	P	50.897	kgf/cm ²
Temperature for Internal Pressure	Temp	275	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	1509.10	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1509.10	kgf/cm ²
Inside Diameter of Hemispherical Head	D	7414.00	mm
Head Finished (Minimum) Thickness	t	66.0000	mm
Head Internal Corrosion Allowance	cas	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	2000.0000	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1407.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.00	kgf/cm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	24.0000	mm
Flange Material		SA-182 F11	
Hub Height of Integral Nozzle	h	170.0000	mm
Height of Beveled Transition	L'	146.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		170.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		60.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.9	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle in the Meridional direction.

Reinforcement CALCULATION, Description: M1 24"



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 24.000 in.
Actual Thickness Used in Calculation 0.945 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
= $P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
= $50.9 \cdot 3710 / (2 \cdot 1509 \cdot 1 - 0.2 \cdot 50.9)$
= 62.7748 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $50.9 \cdot 305 / (1407 \cdot 1 + 0.4 \cdot 50.9)$
= 10.8686 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 146/0.5773
= 252.8795 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

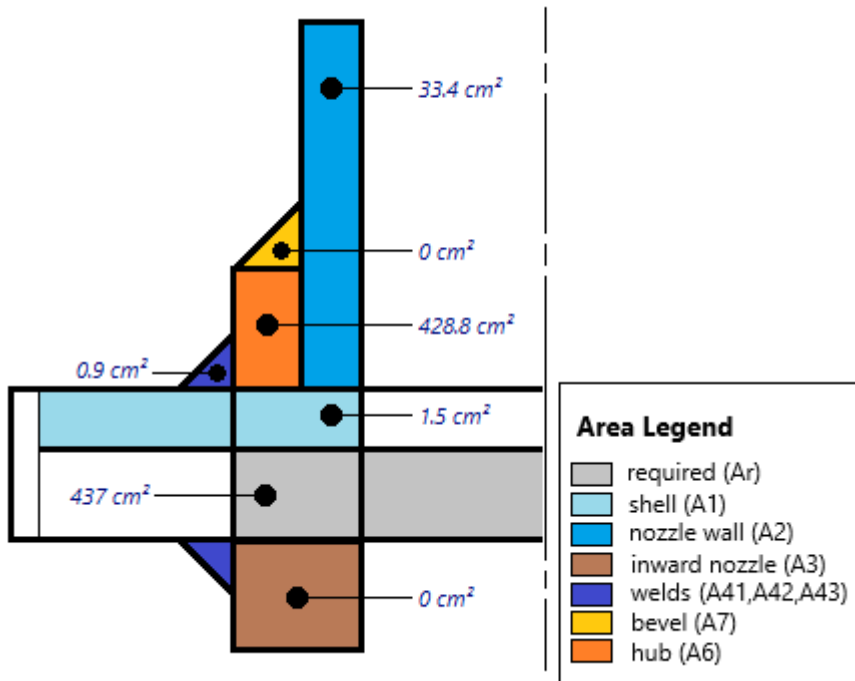
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1347.2386	mm
Parallel to Vessel Wall, opening length	d	673.6193	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	157.5000	mm

Weld Strength Reduction Factor [fr1]:
= min(1, S_n / S_v)
= min(1, 1407/1509)
= 0.932

Weld Strength Reduction Factor [fr2]:
= min(1, S_n / S_v)
= min(1, 1407/1509)
= 0.932

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(0.93, 1)
= 0.932



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	437.048	NA	NA
Area in Shell	A1	1.466	NA	NA
Area in Nozzle Wall	A2	33.411	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.932	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	428.785	NA	NA
TOTAL AREA AVAILABLE	Atot	464.595	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 57.42 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (674 \cdot 62.8 \cdot 1 + 2 \cdot 167 \cdot 62.8 \cdot 1 \cdot (1 - 0.93))$
 $= 437.048 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 674(1 \cdot 63 - 1 \cdot 62.8) - 2 \cdot 167$
 $(1 \cdot 63 - 1 \cdot 62.8) \cdot (1 - 0.93)$
 $= 1.466 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2/\sin(\alpha3)$
 $= (2 \cdot 158)(21 - 10.9)0.93/\sin(62.9)$
 $= 33.411 \text{ cm}^2$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10^2 \cdot 0.93 + (0)^2 \cdot 0.93$



= 0.932 cm²

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(158, 300, 170)) \cdot (170 - 24) \cdot 0.93$$

$$= 428.785 \text{ cm}^2$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, tg = 24, tr = 10.9, c = 3 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.52, Temp. Reduction = 30 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-2 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-32 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:B, Curve:A)

Govrn. thk, tg = 66, tr = 62.8, c = 3 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 1, Temp. Reduction = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	42 °C
Min Metal Temp. at Required thickness (UCS 66.1)	42 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 42 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 50.90/105.44 = 0.483

Weld Size Calculations, Description: M1 24"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A_1 + 2 \cdot t_n \cdot fr_1 \cdot (E_1 \cdot t - tr)) \cdot S_v)$$

$$= \max(0, (437 - 1.47 + 2 \cdot 167 \cdot 0.93 \cdot (1 \cdot 63 - 62.8)) \cdot 1509)$$

$$= 658394.69 \text{ kgf}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A_2 + A_4 - (W_i - Can / .707)^2 \cdot fr_2) \cdot S_v$$

$$= (462 + 0.93 - 0 \cdot 0.93) \cdot 1509$$

$$= 698906.19 \text{ kgf}$$

Weld Load [W2]:

$$= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S_v$$

$$= (462 + 0 + 0.93 + (196)) \cdot 1509$$

$$= 994966.75 \text{ kgf}$$

Weld Load [W3]:

$$= (A_2 + A_3 + A_4 + A_5 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S$$

$$= (462 + 0 + 0.93 + 0 + (196)) \cdot 1509$$

$$= 994966.75 \text{ kgf}$$



Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw \\ &= (3.14/2.0) \cdot 723 \cdot 10 \cdot 0.49 \cdot 1407 \\ &= 78348. \text{ kgf} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\ &= (3.14 \cdot 349) \cdot (170 - 3) \cdot 0.7 \cdot 1407 \\ &= 1804764. \text{ kgf} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng \\ &= (3.14/2.0) \cdot 723 \cdot (60 - 0) \cdot 0.74 \cdot 1509 \\ &= 761443. \text{ kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (78348 + 1804763) = 1883111 \text{ kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (78348 + 0 + 761443 + 0) = 839791 \text{ kgf} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (78348 + 761443 + 0) = 839791 \text{ kgf} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 1883111 kgf, must exceed W = 658394 kgf	or W1 = 698906 kgf
Path 2-2 = 839790 kgf, must exceed W = 658394 kgf	or W2 = 994966 kgf
Path 3-3 = 839790 kgf, must exceed W = 658394 kgf	or W3 = 994966 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 51.1 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: M1 24"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	24.000 in.
Actual Thickness Used in Calculation	0.945 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]

$$\begin{aligned} &= P \cdot R / (2 \cdot Sv \cdot E - 0.2 \cdot P) \text{ per UG-27 (d)} \\ &= 50.9 \cdot 3710 / (2 \cdot 1509 \cdot 1 - 0.2 \cdot 50.9) \\ &= 62.7748 \text{ mm} \end{aligned}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$\begin{aligned} &= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)} \\ &= 50.9 \cdot 305 / (1407 \cdot 1 + 0.4 \cdot 50.9) \\ &= 10.8686 \text{ mm} \end{aligned}$$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:

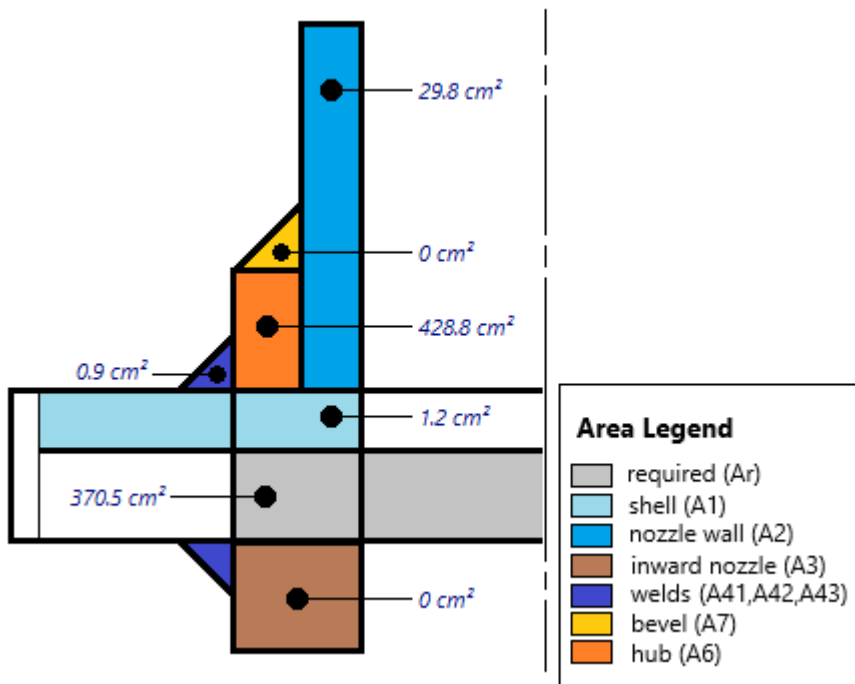
$$\begin{aligned} &= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30) \\ &= 146/0.5773 \end{aligned}$$

= 252.8795 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 1135.2001 mm
 Parallel to Vessel Wall, opening length d 567.6000 mm
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 157.5000 mm



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	370.495	NA	NA
Area in Shell	A1	1.228	NA	NA
Area in Nozzle Wall	A2	29.755	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.932	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	428.785	NA	NA
TOTAL AREA AVAILABLE	Atot	460.699	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (568 \cdot 62.8 \cdot 1 + 2 \cdot 167 \cdot 62.8 \cdot 1 \cdot (1 - 0.93))$
 $= 370.495 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 568 (1 \cdot 63 - 1 \cdot 62.8) - 2 \cdot 167$
 $(1 \cdot 63 - 1 \cdot 62.8) \cdot (1 - 0.93)$
 $= 1.228 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:



$$\begin{aligned}
 &= (2 \cdot t_{np}) (t_n - t_{rn}) fr_2 \\
 &= (2 \cdot 158) (21 - 10.9) 0.93 \\
 &= 29.755 \text{ cm}^2
 \end{aligned}$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$\begin{aligned}
 &= W_o^2 \cdot fr_2 + (W_i - can / 0.707)^2 \cdot fr_2 \\
 &= 10^2 \cdot 0.93 + (0)^2 \cdot 0.93 \\
 &= 0.932 \text{ cm}^2
 \end{aligned}$$

Area Available in the Hub Section [A6]:

$$\begin{aligned}
 &= (2 \cdot \min(t_{np}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot fr_2 \\
 &= (2 \cdot \min(158, 300, 170)) \cdot (170 - 24) \cdot 0.93 \\
 &= 428.785 \text{ cm}^2
 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 13.8686 \text{ mm}$
Wall Thickness per UG16(b),	$tr_{16b} = 4.5000 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$trb_1 = 65.7748 \text{ mm}$
Wall Thickness	$tb_1 = \max(trb_1, tr_{16b}) = 65.7748 \text{ mm}$
Wall Thickness	$tb_2 = \max(trb_2, tr_{16b}) = 4.5000 \text{ mm}$
Wall Thickness per table UG-45	$tb_3 = 11.3312 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned}
 &= \min[tb_3, \max(tb_1, tb_2)] \\
 &= \min[11.3, \max(65.8, 4.5)] \\
 &= 11.3312 \text{ mm}
 \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned}
 &= \max(t_a, t_b) \\
 &= \max(13.9, 11.3) \\
 &= 13.8686 \text{ mm}
 \end{aligned}$$

Available Nozzle Neck Thickness = 24.0000 mm -> OK

Weld Size Calculations, Description: M1 24"

Intermediate Calc. for nozzle/shell Welds	$T_{min} = 19.0000 \text{ mm}$
---	--------------------------------

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned}
 &= \max(0, (A - A_1 + 2 \cdot t_n \cdot fr_1 \cdot (E_1 \cdot t - tr)) Sv) \\
 &= \max(0, (370 - 1.23 + 2 \cdot 167 \cdot 0.93 \cdot (1 \cdot 63 - 62.8)) 1509) \\
 &= 558319.44 \text{ kgf}
 \end{aligned}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned}
 &= (A_2 + A_4 - (W_i - Can / 0.707)^2 \cdot fr_2) \cdot Sv \\
 &= (459 + 0.93 - 0 \cdot 0.93) \cdot 1509 \\
 &= 693388.19 \text{ kgf}
 \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned}
 &= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot Sv \\
 &= (459 + 0 + 0.93 + (196)) \cdot 1509 \\
 &= 989448.69 \text{ kgf}
 \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned}
 &= (A_2 + A_3 + A_4 + A_5 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S \\
 &= (459 + 0 + 0.93 + 0 + (196)) \cdot 1509 \\
 &= 989448.69 \text{ kgf}
 \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis



Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw \\ &= (3.14/2.0) \cdot 610 \cdot 10 \cdot 0.49 \cdot 1407 \\ &= 66017. \text{ kgf} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\ &= (3.14 \cdot 294) \cdot (170 - 3) \cdot 0.7 \cdot 1407 \\ &= 1520716. \text{ kgf} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng \\ &= (3.14/2.0) \cdot 610 \cdot (60 - 0) \cdot 0.74 \cdot 1509 \\ &= 641602. \text{ kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (66017 + 1520716) = 1586733 \text{ kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (66017 + 0 + 641602 + 0) = 707618 \text{ kgf} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (66017 + 641602 + 0) = 707618 \text{ kgf} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 1586733 kgf, must exceed W = 558319 kgf or W1 = 693388 kgf
Path 2-2 = 707618 kgf, must exceed W = 558319 kgf or W2 = 989448 kgf
Path 3-3 = 707618 kgf, must exceed W = 558319 kgf or W3 = 989448 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 51.1 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 217.7910 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 595.8986 mm

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Input, Nozzle Desc: K5 4"

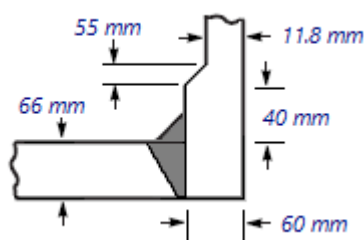
From: 60

Pressure for Reinforcement Calculations	P	50.897	kgf/cm ²
Temperature for Internal Pressure	Temp	275	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	1509.10	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1509.10	kgf/cm ²
Inside Diameter of Hemispherical Head	D	7414.00	mm
Head Finished (Minimum) Thickness	t	66.0000	mm
Head Internal Corrosion Allowance	cas	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	2000.0000	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1407.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.00	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		60.00	deg
Diameter		4.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-182 F11	
Hub Height of Integral Nozzle	h	40.0000	mm
Height of Beveled Transition	L'	55.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		60.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		60.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.9	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle in the Meridional direction.

Reinforcement CALCULATION, Description: K5 4"



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 3.571 in.
Actual Thickness Used in Calculation 0.465 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
= $P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
= $50.9 \cdot 3710 / (2 \cdot 1509 \cdot 1 - 0.2 \cdot 50.9)$
= 62.7748 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $50.9 \cdot 48.3 / (1407 \cdot 1 - 0.6 \cdot 50.9)$
= 1.7878 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 48.2/0.5773
= 83.4823 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

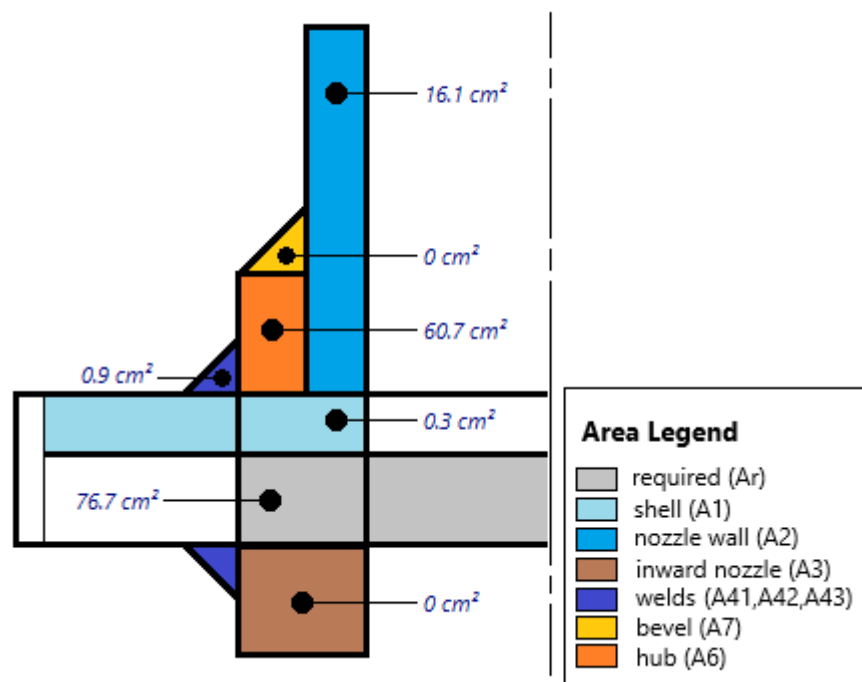
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	258.0297	mm
Parallel to Vessel Wall	Rn+tn+t	129.0148	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	105.4860	mm

Weld Strength Reduction Factor [fr1]:
= min(1, Sn/Sv)
= min(1, 1407/1509)
= 0.932

Weld Strength Reduction Factor [fr2]:
= min(1, Sn/Sv)
= min(1, 1407/1509)
= 0.932

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(0.93, 1)
= 0.932



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	76.673	NA	NA
Area in Shell	A1	0.306	NA	NA
Area in Nozzle Wall	A2	16.120	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.932	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	60.666	NA	NA
TOTAL AREA AVAILABLE	Atot	78.024	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 57.68 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (114 \cdot 62.8 \cdot 1 + 2 \cdot 57 \cdot 62.8 \cdot 1 \cdot (1 - 0.93)) \\
 &= 76.673 \text{ cm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned}
 &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 144(1 \cdot 63 - 1 \cdot 62.8) - 2 \cdot 57 \\
 &\quad (1 \cdot 63 - 1 \cdot 62.8) \cdot (1 - 0.93) \\
 &= 0.306 \text{ cm}^2
 \end{aligned}$$

Area Available in Nozzle Projecting Outward [A2]:

$$\begin{aligned}
 &= (2 \cdot tlnp)(tn - trn)fr2/\sin(\alpha3) \\
 &= (2 \cdot 105)(8.8 - 1.79)0.93/\sin(58.9) \\
 &= 16.120 \text{ cm}^2
 \end{aligned}$$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$\begin{aligned}
 &= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2 \\
 &= 10^2 \cdot 0.93 + (0)^2 \cdot 0.93
 \end{aligned}$$



= 0.932 cm²

Area Available in the Hub Section [A6]:

$$\begin{aligned} &= (2 \cdot \min(T_{lnp}, h_o, Hubht)) \cdot (Hubtk - t_n) \cdot fr2 \\ &= (2 \cdot \min(105, 300, 40)) \cdot (60 - 11.8) \cdot 0.93 \\ &= 60.666 \text{ cm}^2 \text{ Includes Bevel Area of } 24.716 \text{ cm}^2 \end{aligned}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, tg = 11.8, tr = 1.79, c = 3 mm, E* = 1

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.2$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-24 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:B, Curve:A)

Govrn. thk, tg = 60, tr = 1.79, c = 3 mm, E* = 1

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.031$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	40 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $50.90 / 105.44 = 0.483$

Weld Size Calculations, Description: K5 4"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned} &= \max(0, (A - A1 + 2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv) \\ &= \max(0, (76.7 - 0.31 + 2 \cdot 57 \cdot 0.93 \cdot (1 \cdot 63 - 62.8)) \cdot 1509) \\ &= 115606.25 \text{ kgf} \end{aligned}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned} &= (A2 + A4 - (Wi - Can / .707)^2 \cdot fr2) \cdot Sv \\ &= (76.8 + 0.93 - 0 \cdot 0.93) \cdot 1509 \\ &= 117284.18 \text{ kgf} \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned} &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv \\ &= (76.8 + 0 + 0.93 + (67)) \cdot 1509 \\ &= 218334.78 \text{ kgf} \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned} &= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S \\ &= (76.8 + 0 + 0.93 + 0 + (67)) \cdot 1509 \\ &= 218334.78 \text{ kgf} \end{aligned}$$



Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.14/2.0) \cdot 249 \cdot 10 \cdot 0.49 \cdot 1407 \\ &= 27001. \text{ kgf} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n \\ &= (3.14 \cdot 90.9) \cdot (60 - 3) \cdot 0.7 \cdot 1407 \\ &= 160386. \text{ kgf} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng} \\ &= (3.14/2.0) \cdot 249 \cdot (60 - 0) \cdot 0.74 \cdot 1509 \\ &= 262418. \text{ kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (27001 + 160386) = 187387 \text{ kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (27001 + 0 + 262418 + 0) = 289419 \text{ kgf} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (27001 + 262418 + 0) = 289419 \text{ kgf} \end{aligned}$$

Summary of Failure Path Calculations:

$$\begin{aligned} \text{Path 1-1} &= 187387 \text{ kgf, must exceed } W = 115606 \text{ kgf} & \text{or } W1 &= 117284 \text{ kgf} \\ \text{Path 2-2} &= 289418 \text{ kgf, must exceed } W = 115606 \text{ kgf} & \text{or } W2 &= 218334 \text{ kgf} \\ \text{Path 3-3} &= 289418 \text{ kgf, must exceed } W = 115606 \text{ kgf} & \text{or } W3 &= 218334 \text{ kgf} \end{aligned}$$

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 51.1 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: K5 4"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	3.571 in.
Actual Thickness Used in Calculation	0.465 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]

$$\begin{aligned} &= P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P) \text{ per UG-27 (d)} \\ &= 50.9 \cdot 3710 / (2 \cdot 1509 \cdot 1 - 0.2 \cdot 50.9) \\ &= 62.7748 \text{ mm} \end{aligned}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$\begin{aligned} &= P \cdot R / (S_n \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\ &= 50.9 \cdot 48.3 / (1407 \cdot 1 - 0.6 \cdot 50.9) \\ &= 1.7878 \text{ mm} \end{aligned}$$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:

$$\begin{aligned} &= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30) \\ &= 48.2 / 0.5773 \end{aligned}$$

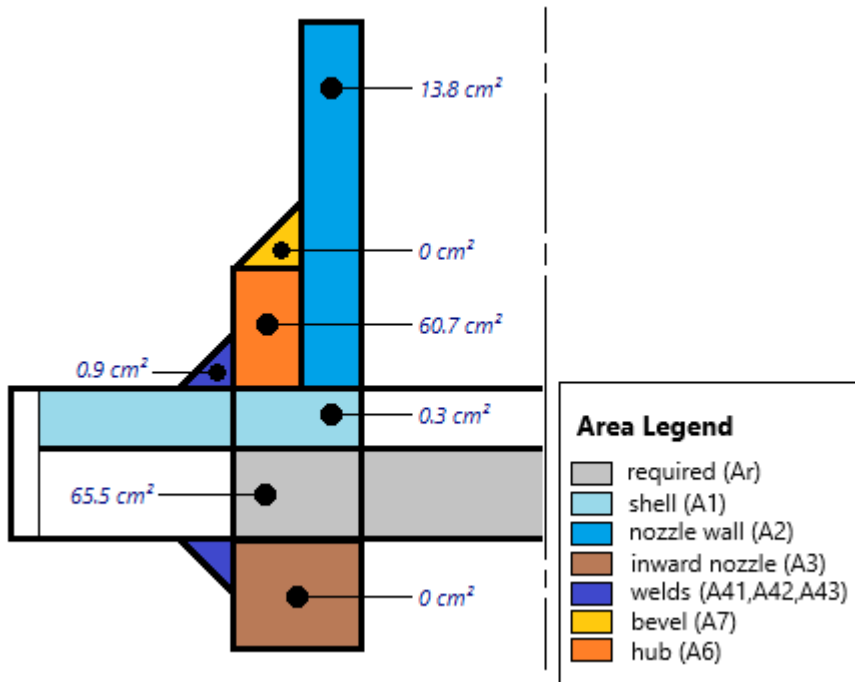


= 83.4823 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	240.3000	mm
Parallel to Vessel Wall	Rn+tn+t	120.1500	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	105.4860	mm



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	65.543	NA	NA
Area in Shell	A1	0.306	NA	NA
Area in Nozzle Wall	A2	13.796	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.932	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	60.666	NA	NA
TOTAL AREA AVAILABLE	Atot	75.700	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (96.7 \cdot 62.8 \cdot 1 + 2 \cdot 57 \cdot 62.8 \cdot 1 \cdot (1 - 0.93))$
 $= 65.543 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 144(1 \cdot 63 - 1 \cdot 62.8) - 2 \cdot 57$
 $(1 \cdot 63 - 1 \cdot 62.8) \cdot (1 - 0.93)$
 $= 0.306 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:



$$= (2 \cdot t_{np}) (t_n - t_{rn}) fr_2$$

$$= (2 \cdot 105) (8.8 - 1.79) 0.93$$

$$= 13.796 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can / 0.707)^2 \cdot fr_2$$

$$= 10^2 \cdot 0.93 + (0)^2 \cdot 0.93$$

$$= 0.932 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(t_{np}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(105, 300, 40)) \cdot (60 - 11.8) \cdot 0.93$$

$$= 60.666 \text{ cm}^2 \text{ Includes Bevel Area of } 24.716 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 4.7878 \text{ mm}$
Wall Thickness per UG16(b),	$tr_{16b} = 4.5000 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$trb_1 = 65.7748 \text{ mm}$
Wall Thickness	$tb_1 = \max(trb_1, tr_{16b}) = 65.7748 \text{ mm}$
Wall Thickness	$tb_2 = \max(trb_2, tr_{16b}) = 4.5000 \text{ mm}$
Wall Thickness per table UG-45	$tb_3 = 8.2578 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[8.26, \max(65.8, 4.5)]$$

$$= 8.2578 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(4.79, 8.26)$$

$$= 8.2578 \text{ mm}$$

Available Nozzle Neck Thickness = 11.8015 mm -> OK

Weld Size Calculations, Description: K5 4"

Intermediate Calc. for nozzle/shell Welds	T_{min}	19.0000 mm
---	-----------	----------------------

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A_1 + 2 \cdot t_n \cdot fr_1 \cdot (E_1 \cdot t - tr)) Sv)$$

$$= \max(0, (65.5 - 0.31 + 2 \cdot 57 \cdot 0.93 \cdot (1 \cdot 63 - 62.8)) 1509)$$

$$= 98810.34 \text{ kgf}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A_2 + A_4 - (W_i - Can / 0.707)^2 \cdot fr_2) \cdot Sv$$

$$= (74.5 + 0.93 - 0 \cdot 0.93) \cdot 1509$$

$$= 113776.85 \text{ kgf}$$

Weld Load [W2]:

$$= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot Sv$$

$$= (74.5 + 0 + 0.93 + (67)) \cdot 1509$$

$$= 214827.45 \text{ kgf}$$

Weld Load [W3]:

$$= (A_2 + A_3 + A_4 + A_5 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S$$

$$= (74.5 + 0 + 0.93 + 0 + (67)) \cdot 1509$$

$$= 214827.45 \text{ kgf}$$

Strength of Connection Elements for Failure Path Analysis



Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw \\ &= (3.14/2.0) \cdot 211 \cdot 10 \cdot 0.49 \cdot 1407 \\ &= 22817. \text{ kgf} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\ &= (3.14 \cdot 76.8) \cdot (60 - 3) \cdot 0.7 \cdot 1407 \\ &= 135535. \text{ kgf} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng \\ &= (3.14/2.0) \cdot 211 \cdot (60 - 0) \cdot 0.74 \cdot 1509 \\ &= 221758. \text{ kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (22817 + 135535) = 158353 \text{ kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (22817 + 0 + 221758 + 0) = 244575 \text{ kgf} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (22817 + 221758 + 0) = 244575 \text{ kgf} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 158352 kgf, must exceed W = 98810 kgf	or W1 = 113776 kgf
Path 2-2 = 244575 kgf, must exceed W = 98810 kgf	or W2 = 214827 kgf
Path 3-3 = 244575 kgf, must exceed W = 98810 kgf	or W3 = 214827 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 51.1 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 70.0696 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 448.1772 mm

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Input, Nozzle Desc: K2 2"

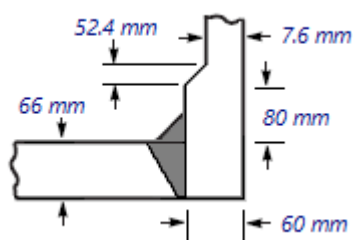
From: 60

Pressure for Reinforcement Calculations	P	50.897	kgf/cm ²
Temperature for Internal Pressure	Temp	275	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	1509.10	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1509.10	kgf/cm ²
Inside Diameter of Hemispherical Head	D	7414.00	mm
Head Finished (Minimum) Thickness	t	66.0000	mm
Head Internal Corrosion Allowance	cas	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	3499.9998	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1407.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.00	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-182 F11	
Hub Height of Integral Nozzle	h	80.0000	mm
Height of Beveled Transition	L'	52.3546	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		60.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	226.8480	mm
Weld leg size between Nozzle and Pad/Shell	Wo	12.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		60.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.9	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle in the Meridional direction.

Reinforcement CALCULATION, Description: K2 2"



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 1.773 in.
Actual Thickness Used in Calculation 0.301 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
= $P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
= $50.9 \cdot 3710 / (2 \cdot 1509 \cdot 1 - 0.2 \cdot 50.9)$
= 62.7748 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $50.9 \cdot 25.5 / (1407 \cdot 1 - 0.6 \cdot 50.9)$
= 0.9435 mm

Intermediate Hub Nozzle Calculations:

The Diameter Limit is less than the Hub OD !

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 52.4 / 0.5773
= 90.6809 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

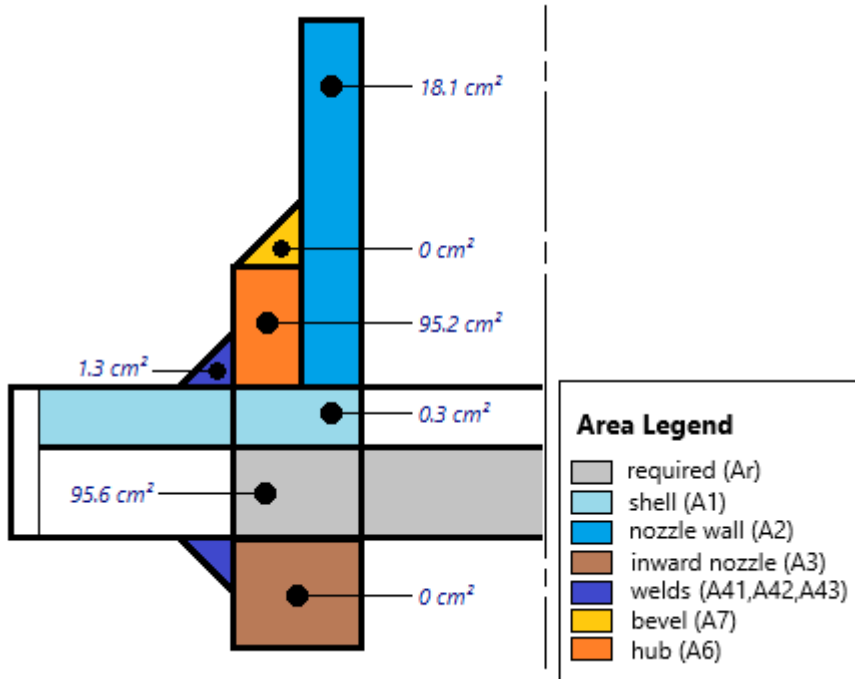
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	289.2350	mm
Parallel to Vessel Wall, opening length	d	144.6175	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	102.2944	mm

Weld Strength Reduction Factor [fr1]:
= min(1, S_n / S_v)
= min(1, 1407/1509)
= 0.932

Weld Strength Reduction Factor [fr2]:
= min(1, S_n / S_v)
= min(1, 1407/1509)
= 0.932

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(0.93, 1)
= 0.932



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	95.625	NA	NA
Area in Shell	A1	0.308	NA	NA
Area in Nozzle Wall	A2	18.113	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.343	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	95.231	NA	NA
TOTAL AREA AVAILABLE	Atot	114.994	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 20.66 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (145 \cdot 62.8 \cdot 1 + 2 \cdot 57 \cdot 62.8 \cdot 1 \cdot (1 - 0.93))$$

$$= 95.625 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 145(1 \cdot 63 - 1 \cdot 62.8) - 2 \cdot 57(1 \cdot 63 - 1 \cdot 62.8) \cdot (1 - 0.93)$$

$$= 0.308 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2/\sin(\alpha3)$$

$$= (2 \cdot 102)(4.65 - 0.94)0.93/\sin(22.9)$$

$$= 18.113 \text{ cm}^2$$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 12^2 \cdot 0.93 + (0)^2 \cdot 0.93$$



= 1.343 cm²

Area Available in the Hub Section [A6]:

$$\begin{aligned} &= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot f_{r2} \\ &= (2 \cdot \min(102, 227, 80)) \cdot (60 - 7.65) \cdot 0.93 \\ &= 95.231 \text{ cm}^2 \text{ Includes Bevel Area of } 17.131 \text{ cm}^2 \end{aligned}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, tg = 7.65, tr = 0.94, c = 3 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.2, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:B, Curve:A)

Govrn. thk, tg = 60, tr = 0.94, c = 3 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.017, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	40 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 50.90/105.44 = 0.483

Weld Size Calculations, Description: K2 2"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	8.4840 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned} &= \max(0, (A - A_1 + 2 \cdot t_n \cdot f_{r1} \cdot (E_1 \cdot t - t_r)) \cdot S_v) \\ &= \max(0, (95.6 - 0.31 + 2 \cdot 57 \cdot 0.93 \cdot (1 \cdot 63 - 62.8)) \cdot 1509) \\ &= 144203.48 \text{ kgf} \end{aligned}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned} &= (A_2 + A_4 - (W_i - C_{an} / .707)^2 \cdot f_{r2}) \cdot S_v \\ &= (113 + 1.34 - 0 \cdot 0.93) \cdot 1509 \\ &= 173072.67 \text{ kgf} \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned} &= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v \\ &= (113 + 0 + 1.34 + (67)) \cdot 1509 \\ &= 274123.28 \text{ kgf} \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned} &= (A_2 + A_3 + A_4 + A_5 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S \\ &= (113 + 0 + 1.34 + 0 + (67)) \cdot 1509 \\ &= 274123.28 \text{ kgf} \end{aligned}$$



Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.14/2.0) \cdot 468 \cdot 12 \cdot 0.49 \cdot 1407 \\ &= 60775. \text{ kgf} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n \\ &= (3.14 \cdot 153) \cdot (60 - 3) \cdot 0.7 \cdot 1407 \\ &= 269965. \text{ kgf} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng} \\ &= (3.14/2.0) \cdot 468 \cdot (60 - 0) \cdot 0.74 \cdot 1509 \\ &= 492214. \text{ kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (60775 + 269965) = 330740 \text{ kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (60775 + 0 + 492214 + 0) = 552989 \text{ kgf} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (60775 + 492214 + 0) = 552989 \text{ kgf} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 330739 kgf, must exceed W = 144203 kgf	or W1 = 173072 kgf
Path 2-2 = 552988 kgf, must exceed W = 144203 kgf	or W2 = 274123 kgf
Path 3-3 = 552988 kgf, must exceed W = 144203 kgf	or W3 = 274123 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 51.1 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: K2 2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.773 in.
Actual Thickness Used in Calculation	0.301 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]

$$\begin{aligned} &= P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P) \text{ per UG-27 (d)} \\ &= 50.9 \cdot 3710 / (2 \cdot 1509 \cdot 1 - 0.2 \cdot 50.9) \\ &= 62.7748 \text{ mm} \end{aligned}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$\begin{aligned} &= P \cdot R / (S_n \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\ &= 50.9 \cdot 25.5 / (1407 \cdot 1 - 0.6 \cdot 50.9) \\ &= 0.9435 \text{ mm} \end{aligned}$$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:

$$\begin{aligned} &= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30) \\ &= 52.4 / 0.5773 \end{aligned}$$

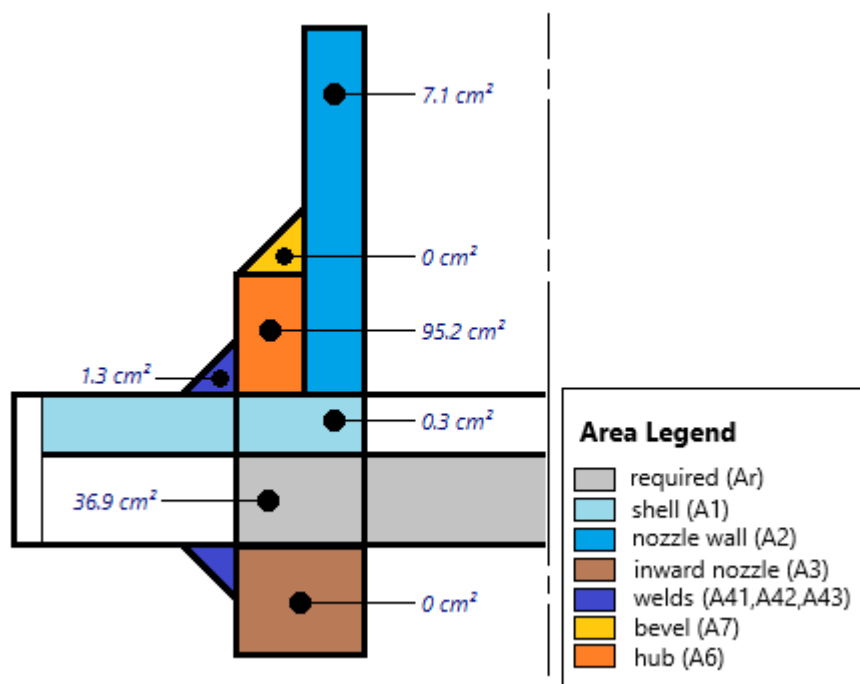


= 90.6809 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	186.3250	mm
Parallel to Vessel Wall	Rn+tn+t	93.1625	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	102.2944	mm



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	36.878	NA	NA
Area in Shell	A1	0.287	NA	NA
Area in Nozzle Wall	A2	7.061	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds A41+A42+A43		1.325	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	95.231	NA	NA
TOTAL AREA AVAILABLE	Atot	103.905	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (51 \cdot 62.8 \cdot 1 + 2 \cdot 57 \cdot 62.8 \cdot 1 \cdot (1 - 0.93))$
 $= 36.878 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 135 (1 \cdot 63 - 1 \cdot 62.8) - 2 \cdot 57$
 $(1 \cdot 63 - 1 \cdot 62.8) \cdot (1 - 0.93)$
 $= 0.287 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:



$$= (2 \cdot t_{np}) (t_n - t_{rn}) fr_2$$

$$= (2 \cdot 102) (4.65 - 0.94) 0.93$$

$$= 7.061 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= (W_o^2 - \text{Area Lost}) \cdot fr_2 + ((W_i - \text{can}/0.707)^2 - \text{Area Lost}) \cdot fr_2$$

$$= (12^2 - 0.018) \cdot 0.93 + (0^2 - 0) \cdot 0.93$$

$$= 1.325 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{np}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(102, 227, 80)) \cdot (60 - 7.65) \cdot 0.93$$

$$= 95.231 \text{ cm}^2 \text{ Includes Bevel Area of } 17.131 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 3.9435 \text{ mm}$
Wall Thickness per UG16(b),	$tr_{16b} = 4.5000 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$trb_1 = 65.7748 \text{ mm}$
Wall Thickness	$tb_1 = \max(trb_1, tr_{16b}) = 65.7748 \text{ mm}$
Wall Thickness	$tb_2 = \max(trb_2, tr_{16b}) = 4.5000 \text{ mm}$
Wall Thickness per table UG-45	$tb_3 = 6.4200 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[6.42, \max(65.8, 4.5)]$$

$$= 6.4200 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(3.94, 6.42)$$

$$= 6.4200 \text{ mm}$$

Available Nozzle Neck Thickness = 7.6454 mm -> OK

Weld Size Calculations, Description: K2 2"

Intermediate Calc. for nozzle/shell Welds $T_{min} = 19.0000 \text{ mm}$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$8.4840 = 0.7 \cdot W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A_1 + 2 \cdot t_n \cdot fr_1 \cdot (E_1 \cdot t - tr)) \cdot S_v)$$

$$= \max(0, (36.9 - 0.29 + 2 \cdot 57 \cdot 0.93 \cdot (1 \cdot 63 - 62.8)) \cdot 1509)$$

$$= 55580.66 \text{ kgf}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A_2 + A_4 - (W_i - \text{Can}/0.707)^2 \cdot fr_2) \cdot S_v$$

$$= (102 + 1.33 - 0 \cdot 0.93) \cdot 1509$$

$$= 156368.83 \text{ kgf}$$

Weld Load [W2]:

$$= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S_v$$

$$= (102 + 0 + 1.33 + (67)) \cdot 1509$$

$$= 257419.44 \text{ kgf}$$

Weld Load [W3]:

$$= (A_2 + A_3 + A_4 + A_5 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S$$

$$= (102 + 0 + 1.33 + 0 + (67)) \cdot 1509$$

$$= 257419.44 \text{ kgf}$$

Strength of Connection Elements for Failure Path Analysis



Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw \\ &= (3.14/2.0) \cdot 165 \cdot 12 \cdot 0.49 \cdot 1407 \\ &= 21447. \text{ kgf} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\ &= (3.14 \cdot 54) \cdot (60 - 3) \cdot 0.7 \cdot 1407 \\ &= 95268. \text{ kgf} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng \\ &= (3.14/2.0) \cdot 165 \cdot (60 - 0) \cdot 0.74 \cdot 1509 \\ &= 173698. \text{ kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (21447 + 95268) = 116715 \text{ kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (21447 + 0 + 173698 + 0) = 195145 \text{ kgf} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (21447 + 173698 + 0) = 195145 \text{ kgf} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 116715 kgf, must exceed W = 55580 kgf	or W1 = 156368 kgf
Path 2-2 = 195144 kgf, must exceed W = 55580 kgf	or W2 = 257419 kgf
Path 3-3 = 195144 kgf, must exceed W = 55580 kgf	or W3 = 257419 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 51.1 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 268.8260 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 683.3487 mm

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Input, Nozzle Desc: N2-104-D2A

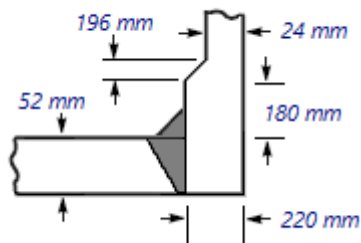
From: 120

Pressure for Reinforcement Calculations	P	50.897	kgf/cm ²
Temperature for Internal Pressure	Temp	275	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	1509.10	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1509.10	kgf/cm ²
Inside Diameter of Hemispherical Head	D	5648.00	mm
Head Finished (Minimum) Thickness	t	52.0000	mm
Head Internal Corrosion Allowance	cas	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1407.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.00	kgf/cm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		36.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	24.0000	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	180.0000	mm
Height of Beveled Transition	L'	196.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		220.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	150.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		52.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.9	
Flange Series		A	
Piping Exit Angle [North Clockwise reference]		0.0	deg
Bend Radius Multiplier		1.50	
Horizontal Run Length		3500.0	mm
Centerline Distance to Tangent		4500.0	mm

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N2-104-D2A

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	36.000 in.
Actual Thickness Used in Calculation	0.945 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
 $= P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
 $= 50.9 \cdot 2827 / (2 \cdot 1509 \cdot 1 - 0.2 \cdot 50.9)$
 $= 47.8340 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 50.9 \cdot 457 / (1407 \cdot 1 + 0.4 \cdot 50.9)$
 $= 16.3029 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 196 / 0.5773$
 $= 339.4821 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

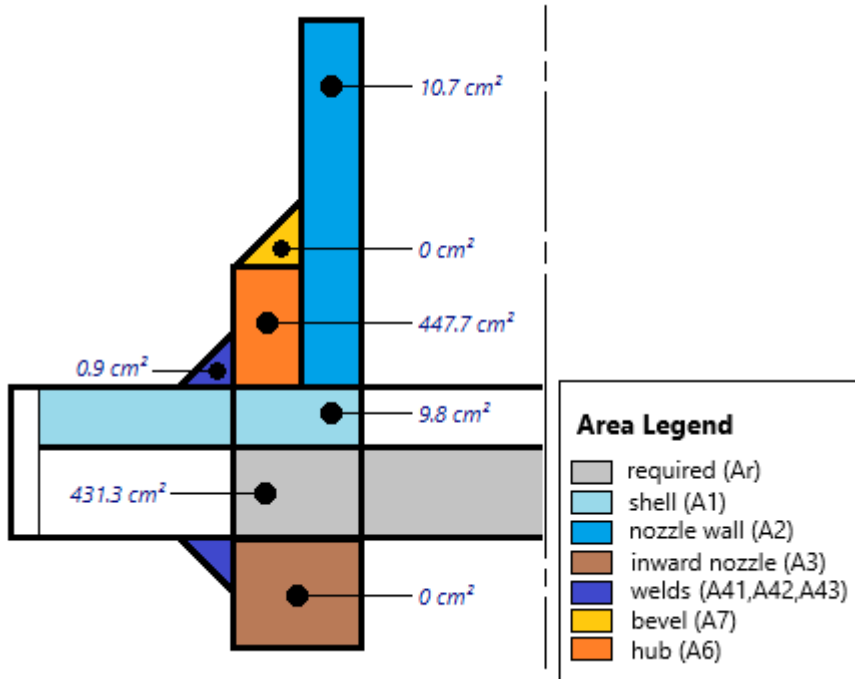
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	1744.8000 mm
Parallel to Vessel Wall, opening length	d	872.4000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	122.5000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1407 / 1509)$
 $= 0.932$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1407 / 1509)$
 $= 0.932$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.93, 1)$
 $= 0.932$



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	431.350	NA	NA
Area in Shell	A1	9.829	NA	NA
Area in Nozzle Wall	A2	10.729	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.932	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	447.711	NA	NA
TOTAL AREA AVAILABLE	Atot	469.202	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (872 \cdot 47.8 \cdot 1 + 2 \cdot 217 \cdot 47.8 \cdot (1 - 0.93))$
 $= 431.350 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 872(1 \cdot 49 - 1 \cdot 47.8) - 2 \cdot 217$
 $(1 \cdot 49 - 1 \cdot 47.8) \cdot (1 - 0.93)$
 $= 9.829 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 123)(21 - 16.3)0.93$
 $= 10.729 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10^2 \cdot 0.93 + (0)^2 \cdot 0.93$
 $= 0.932 \text{ cm}^2$



Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubt} - t_n) \cdot f_{r2}$$

$$= (2 \cdot \min(123, 150, 180)) \cdot (220 - 24) \cdot 0.93$$

$$= 447.711 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 19.3029 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 50.8340 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 50.8340 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.3, \max(50.8, 4.5)]$$

$$= 11.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(19.3, 11.3)$$

$$= 19.3029 \text{ mm}$$

Available Nozzle Neck Thickness = 24.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 24, tr = 16.3, c = 3 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.78, Temp. Reduction = 12 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	18 °C
Min Metal Temp. at Required thickness (UCS 66.1)	5 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:B, Curve:A)

Govrn. thk, tg = 52, tr = 47.8, c = 3 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.98, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	37 °C
Min Metal Temp. at Required thickness (UCS 66.1)	36 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 36 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 50.90/105.44 = 0.483

Weld Size Calculations, Description: N2-104-D2A

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - tr)) S_v)$$

$$= \max(0, (431 - 9.83 + 2 \cdot 217 \cdot 0.93 \cdot (1 \cdot 49 - 47.8)) 1509)$$

$$= 643235.12 \text{ kgf}$$

For hub type nozzles, A2 includes the area of the hub.



Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned} &= (A2+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv \\ &= (458 + 0.93 - 0 \cdot 0.93) \cdot 1509 \\ &= 693239.06 \text{ kgf} \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned} &= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv \\ &= (458 + 0 + 0.93 + (198)) \cdot 1509 \\ &= 992451.12 \text{ kgf} \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned} &= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S \\ &= (458 + 0 + 0.93 + 0 + (198)) \cdot 1509 \\ &= 992451.12 \text{ kgf} \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw \\ &= (3.14/2.0) \cdot 914 \cdot 10 \cdot 0.49 \cdot 1407 \\ &= 99025. \text{ kgf} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\ &= (3.14 \cdot 447) \cdot (220 - 3) \cdot 0.7 \cdot 1407 \\ &= 2999280. \text{ kgf} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng \\ &= (3.14/2.0) \cdot 914 \cdot (52 - 3) \cdot 0.74 \cdot 1509 \\ &= 785962. \text{ kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (99025 + 2999280) = 3098305 \text{ kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (99025 + 0 + 785962 + 0) = 884987 \text{ kgf} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (99025 + 785962 + 0) = 884987 \text{ kgf} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 3098305 kgf, must exceed W = 643235 kgf	or W1 = 693239 kgf
Path 2-2 = 884987 kgf, must exceed W = 643235 kgf	or W2 = 992451 kgf
Path 3-3 = 884987 kgf, must exceed W = 643235 kgf	or W3 = 992451 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 52.1 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 37.2557 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 239.2556 mm

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Input, Nozzle Desc: UC1-104-D2A

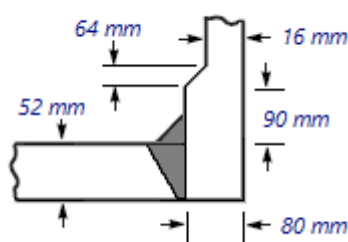
From: 120

Pressure for Reinforcement Calculations	P	50.897	kgf/cm ²
Temperature for Internal Pressure	Temp	275	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	1509.10	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1509.10	kgf/cm ²
Inside Diameter of Hemispherical Head	D	5648.00	mm
Head Finished (Minimum) Thickness	t	52.0000	mm
Head Internal Corrosion Allowance	cas	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	1800.0000	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1407.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.00	kgf/cm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		270.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.0000	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	90.0000	mm
Height of Beveled Transition	L'	64.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		80.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	1850.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	52.0000	mm
Inside Projection	h	600.0000	mm
Weld leg size, Inside Element to Shell	Wi	14.0000	mm
Flange Class		600	
Flange Grade		GR 1.9	
Flange Series		A	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: UC1-104-D2A



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 24.000 in.
Actual Thickness Used in Calculation 0.630 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
= $P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
= $50.9 \cdot 2827 / (2 \cdot 1509 \cdot 1 - 0.2 \cdot 50.9)$
= 47.8340 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $50.9 \cdot 305 / (1407 \cdot 1 + 0.4 \cdot 50.9)$
= 10.8686 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 64/0.5773
= 110.8513 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

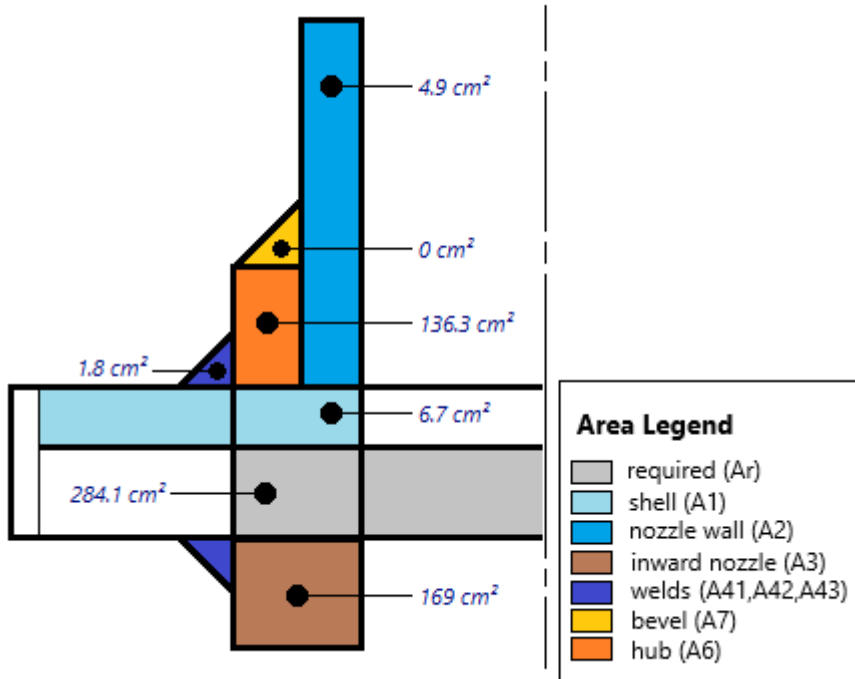
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	1167.2001	mm
Parallel to Vessel Wall, opening length	d	583.6000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	122.5000	mm
Normal to Vessel Wall, Inward		25.0000	mm

Weld Strength Reduction Factor [fr1]:
= min(1, S_n/S_v)
= min(1, 1407/1509)
= 0.932

Weld Strength Reduction Factor [fr2]:
= min(1, S_n/S_v)
= min(1, 1407/1509)
= 0.932

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(0.93, 1)
= 0.932



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	284.143	NA	NA
Area in Shell	A1	6.683	NA	NA
Area in Nozzle Wall	A2	4.869	NA	NA
Area in Inward Nozzle	A3	169.034	NA	NA
Area in Welds	A41+A42+A43	1.820	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	136.344	NA	NA
TOTAL AREA AVAILABLE	Atot	318.749	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (584 \cdot 47.8 \cdot 1 + 2 \cdot 77 \cdot 47.8 \cdot 1 \cdot (1 - 0.93))$$

$$= 284.143 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 584 (1 \cdot 49 - 1 \cdot 47.8) - 2 \cdot 77 (1 \cdot 49 - 1 \cdot 47.8) \cdot (1 - 0.93)$$

$$= 6.683 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp) (tn - trn) fr2$$

$$= (2 \cdot 123) (13 - 10.9) 0.93$$

$$= 4.869 \text{ cm}^2$$

Area Available in Inward Nozzle [A3]:

$$= 2 \cdot ti \cdot \min(h, TI, 2.5 \cdot ti) \cdot fr2$$

$$= 2 \cdot 74 \cdot (123) \cdot 0.93$$

$$= 169.034 \text{ cm}^2$$



Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$\begin{aligned} &= W_o^2 \cdot fr_2 + (W_i - can / 0.707)^2 \cdot fr_2 \\ &= 10^2 \cdot 0.93 + (9.76)^2 \cdot 0.93 \\ &= 1.820 \text{ cm}^2 \end{aligned}$$

Area Available in the Hub Section [A6]:

$$\begin{aligned} &= (2 \cdot \min(T_{hp}, h_o, Hubht)) \cdot (Hubtk - t_n) \cdot fr_2 \\ &= (2 \cdot \min(123, 1850, 90)) \cdot (80 - 16) \cdot 0.93 \\ &= 136.344 \text{ cm}^2 \text{ Includes Bevel Area of } 28.938 \text{ cm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 13.8686 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 50.8340 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 50.8340 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned} &= \min[tb3, \max(tb1, tb2)] \\ &= \min[11.3, \max(50.8, 4.5)] \\ &= 11.3312 \text{ mm} \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= \max(ta, tb) \\ &= \max(13.9, 11.3) \\ &= 13.8686 \text{ mm} \end{aligned}$$

Available Nozzle Neck Thickness = 16.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 16, tr = 10.9, c = 3 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.84, Temp. Reduction = 9 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	6 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-3 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:B, Curve:A)

Govrn. thk, tg = 52, tr = 47.8, c = 3 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.98, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	37 °C
Min Metal Temp. at Required thickness (UCS 66.1)	36 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 36 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 50.90/105.44 = 0.483

Weld Size Calculations, Description: UC1-104-D2A

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot t_n \cdot fr_1 \cdot (E1 \cdot t - tr)) S_v)$$



$$\begin{aligned} &= \max(0, (284 - 6.68 + 2 \cdot 77 \cdot 0.93 \cdot \\ &\quad (1 \cdot 49 - 47.8)) \cdot 1509) \\ &= 421241.22 \text{ kgf} \end{aligned}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

$$\begin{aligned} \text{Weld Load [W1]:} \\ &= (A2 + A4 - (W_i - \text{Can}) / .707)^2 \cdot f_r2 \cdot S_v \\ &= (141 + 1.82 - 95.2 \cdot 0.93) \cdot 1509 \\ &= 214510.11 \text{ kgf} \end{aligned}$$

$$\begin{aligned} \text{Weld Load [W2]:} \\ &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S_v \\ &= (141 + 169 + 1.82 + (70.4)) \cdot 1509 \\ &= 577110.50 \text{ kgf} \end{aligned}$$

$$\begin{aligned} \text{Weld Load [W3]:} \\ &= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S \\ &= (141 + 169 + 1.82 + 0 + (70.4)) \cdot 1509 \\ &= 577110.50 \text{ kgf} \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

$$\begin{aligned} \text{Shear, Outward Nozzle Weld [Sonw]:} \\ &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.14/2.0) \cdot 610 \cdot 10 \cdot 0.49 \cdot 1407 \\ &= 66017. \text{ kgf} \end{aligned}$$

$$\begin{aligned} \text{Shear, Inward Nozzle Weld [Sinw]:} \\ &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.14/2.0) \cdot 610 \cdot 9.76 \cdot 0.49 \cdot 1407 \\ &= 64415. \text{ kgf} \end{aligned}$$

$$\begin{aligned} \text{Shear, Nozzle Wall [Snw]:} \\ &= (\pi \cdot (D_{lr} + D_{lo}) / 4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n \\ &= (3.14 \cdot 298) \cdot (80 - 3) \cdot 0.7 \cdot 1407 \\ &= 710699. \text{ kgf} \end{aligned}$$

$$\begin{aligned} \text{Tension, Shell Groove Weld [Tngw]:} \\ &= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng} \\ &= (3.14/2.0) \cdot 610 \cdot (52 - 3) \cdot 0.74 \cdot 1509 \\ &= 523974. \text{ kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

$$\begin{aligned} \text{Failure Path [Path 1-1]:} \\ &= (\text{SONW} + \text{SNW}) = (66017 + 710699) = 776715 \text{ kgf} \end{aligned}$$

$$\begin{aligned} \text{Failure Path [Path 2-2]:} \\ &= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ &= (66017 + 0 + 523974 + 64415) = 654406 \text{ kgf} \end{aligned}$$

$$\begin{aligned} \text{Failure Path [Path 3-3]:} \\ &= (\text{Sonw} + \text{Tngw} + \text{Sinw}) \\ &= (66017 + 523974 + 64415) = 654406 \text{ kgf} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 776715 kgf, must exceed W = 421241 kgf or W1 = 214510 kgf
Path 2-2 = 654406 kgf, must exceed W = 421241 kgf or W2 = 577110 kgf
Path 3-3 = 654406 kgf, must exceed W = 421241 kgf or W3 = 577110 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 52.1 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

PV Elite 28 U2 Licensee: SL User
FileName : 104D2A-104-D2B-STACKED_R1
Nozzle Calcs.: UC1-104-D2A

Nozl: 47 2:45pm Aug 12, 2026

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HEXAGON

The Drop for this Nozzle is : 16.4971 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 2502.0000 mm

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Input, Nozzle Desc: H1-104-D2A

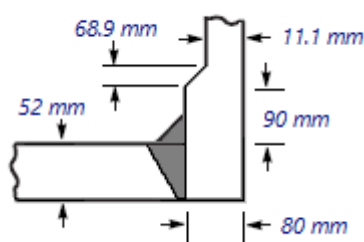
From: 120

Pressure for Reinforcement Calculations	P	50.897	kgf/cm ²
Temperature for Internal Pressure	Temp	275	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	1509.10	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1509.10	kgf/cm ²
Inside Diameter of Hemispherical Head	D	5648.00	mm
Head Finished (Minimum) Thickness	t	52.0000	mm
Head Internal Corrosion Allowance	cas	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	1000.0000	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-335 P11 +IT	
Material UNS Number		K11597	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1127.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1203.00	kgf/cm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		8.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		80	
Flange Material		SA-182 F11	
Hub Height of Integral Nozzle	h	90.0000	mm
Height of Beveled Transition	L'	68.8875	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		80.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	1850.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		52.0000	mm
Inside Projection	h	600.0000	mm
Weld leg size, Inside Element to Shell	Wi	14.0000	mm
This is a Manway, Coupling or Access Opening.			
Flange Class		600	
Flange Grade		GR 1.9	

The Pressure Design option was Design Pressure + static head.





ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 8.625 in.
Actual Thickness Used in Calculation 0.438 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
= $P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
= $50.9 \cdot 2827 / (2 \cdot 1509 \cdot 1 - 0.2 \cdot 50.9)$
= 47.8340 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $50.9 \cdot 110 / (1127 \cdot 1 + 0.4 \cdot 50.9)$
= 4.8591 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 68.9/0.5773
= 119.3167 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

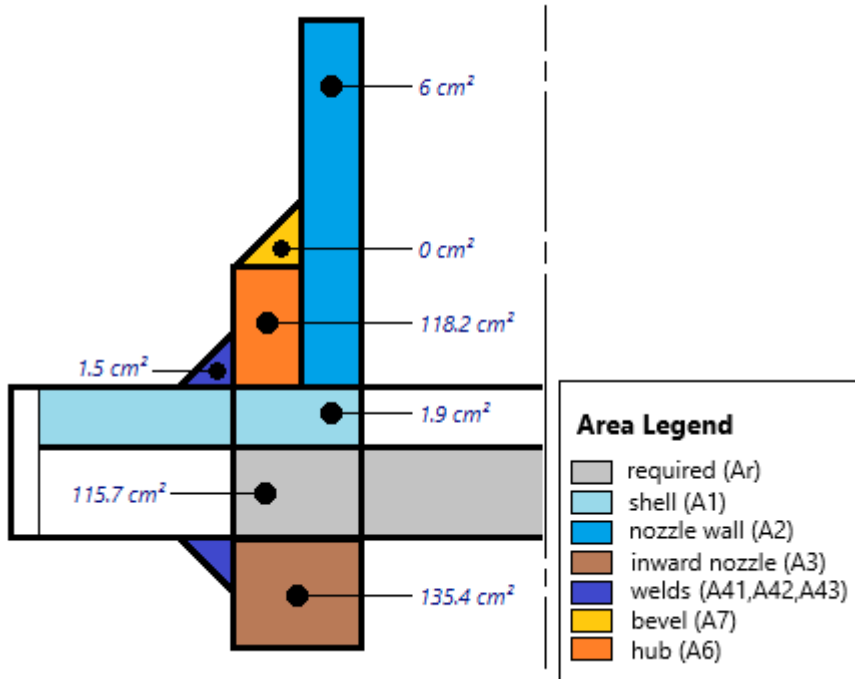
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	405.7000	mm
Parallel to Vessel Wall, opening length	d	202.8500	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	122.5000	mm
Normal to Vessel Wall, Inward		12.7812	mm

Weld Strength Reduction Factor [fr1]:
= min(1, S_n / S_v)
= min(1, 1127/1509)
= 0.747

Weld Strength Reduction Factor [fr2]:
= min(1, S_n / S_v)
= min(1, 1127/1509)
= 0.747

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(0.75, 1)
= 0.747



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	115.683	NA	NA
Area in Shell	A1	1.911	NA	NA
Area in Nozzle Wall	A2	5.953	NA	NA
Area in Inward Nozzle	A3	135.395	NA	NA
Area in Welds	A41+A42+A43	1.458	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	118.153	NA	NA
TOTAL AREA AVAILABLE	Atot	262.869	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (203 \cdot 47.8 \cdot 1 + 2 \cdot 77 \cdot 47.8 \cdot 1 \cdot (1 - 0.75))$
 $= 115.683 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 203 (1 \cdot 49 - 1 \cdot 47.8) - 2 \cdot 77$
 $(1 \cdot 49 - 1 \cdot 47.8) \cdot (1 - 0.75)$
 $= 1.911 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp) (tn - trn) fr2$
 $= (2 \cdot 123) (8.11 - 4.86) 0.75$
 $= 5.953 \text{ cm}^2$

Area Available in Inward Nozzle [A3]:
 $= 2 \cdot ti \cdot \min(h, TI, 2.5 \cdot ti) \cdot fr2$
 $= 2 \cdot 74 \cdot (123) \cdot 0.75$
 $= 135.395 \text{ cm}^2$



Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can / 0.707)^2 \cdot fr_2$$

$$= 10^2 \cdot 0.75 + (9.76)^2 \cdot 0.75$$

$$= 1.458 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{hp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(123, 1850, 90)) \cdot (80 - 11.1) \cdot 0.75$$

$$= 118.153 \text{ cm}^2 \text{ Includes Bevel Area of } 25.551 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 7.8591 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= t_a$$

$$= 7.8591 \text{ mm}$$

Available Nozzle Neck Thickness = 11.1125 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, $t_g = 11.1$, $t_r = 4.56$, $c = 3 \text{ mm}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.56$, Temp. Reduction = $26 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve B	-26 $^\circ\text{C}$
Min Metal Temp. at Required thickness (UCS 66.1)	-48 $^\circ\text{C}$

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: A

Govrn. thk, $t_g = 52$, $t_r = 47.8$, $c = 3 \text{ mm}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.98$, Temp. Reduction = $1 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve A	37 $^\circ\text{C}$
Min Metal Temp. at Required thickness (UCS 66.1)	36 $^\circ\text{C}$

Gov. MDMT of the nozzle to shell joint welded assembly : 36 $^\circ\text{C}$

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 $^\circ\text{C}$
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 $^\circ\text{C}$

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $50.90 / 105.44 = 0.483$

Weld Size Calculations, Description: H1-104-D2A

Intermediate Calc. for nozzle/shell Welds $T_{min} = 19.0000 \text{ mm}$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A_1 + 2 \cdot t_n \cdot fr_1 \cdot (E_1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (116 - 1.91 + 2 \cdot 77 \cdot 0.75 \cdot (1 \cdot 49 - 47.8)) \cdot 1509)$$

$$= 173717.47 \text{ kgf}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A_2 + A_4 - (W_i - Can / .707)^2 \cdot fr_2) \cdot S_v$$

$$= (124 + 1.46 - 95.2 \cdot 0.75) \cdot 1509$$



= 188414.69 kgf

Weld Load [W2]:

$$\begin{aligned} &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv \\ &= (124 + 135 + 1.46 + (56.4)) \cdot 1509 \\ &= 478855.78 \text{ kgf} \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned} &= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S \\ &= (124 + 135 + 1.46 + 0 + (56.4)) \cdot 1509 \\ &= 478855.78 \text{ kgf} \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw \\ &= (3.14/2.0) \cdot 219 \cdot 10 \cdot 0.49 \cdot 1127 \\ &= 19003. \text{ kgf} \end{aligned}$$

Shear, Inward Nozzle Weld [Sinw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw \\ &= (3.14/2.0) \cdot 219 \cdot 9.76 \cdot 0.49 \cdot 1127 \\ &= 18542. \text{ kgf} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\ &= (3.14 \cdot 105) \cdot (80 - 3) \cdot 0.7 \cdot 1127 \\ &= 201297. \text{ kgf} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng \\ &= (3.14/2.0) \cdot 219 \cdot (52 - 3) \cdot 0.74 \cdot 1509 \\ &= 188303. \text{ kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (19003 + 201297) = 220300 \text{ kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (19003 + 0 + 188303 + 18542) = 225849 \text{ kgf} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (19003 + 188303 + 18542) = 225849 \text{ kgf} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 220300 kgf, must exceed W = 173717 kgf	or W1 = 188414 kgf
Path 2-2 = 225849 kgf, must exceed W = 173717 kgf	or W2 = 478855 kgf
Path 3-3 = 225849 kgf, must exceed W = 173717 kgf	or W3 = 478855 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 52.1 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 2.1253 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 2502.0000 mm

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Input, Nozzle Desc: K4-104-D2A

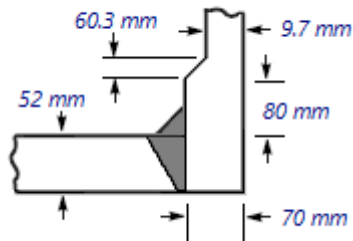
From: 120

Pressure for Reinforcement Calculations	P	50.897	kgf/cm ²
Temperature for Internal Pressure	Temp	275	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	1509.10	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1509.10	kgf/cm ²
Inside Diameter of Hemispherical Head	D	5648.00	mm
Head Finished (Minimum) Thickness	t	52.0000	mm
Head Internal Corrosion Allowance	cas	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	2500.0000	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-335 P11 +IT	
Material UNS Number		K11597	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1127.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1203.00	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-182 F11	
Flange Type		N/A	
Hub Height of Integral Nozzle	h	80.0000	mm
Height of Beveled Transition	L'	60.2654	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		70.0000	mm
Style of FVC Nozzle		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	800.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	52.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.9	
Piping Exit Angle [North Clockwise reference]		0.0	deg
Bend Radius Multiplier		1.50	
Horizontal Run Length		3000.0	mm
Centerline Distance to Tangent		3000.0	mm

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle in the Meridional direction.

Reinforcement CALCULATION, Description: K4-104-D2A

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.734 in.
Actual Thickness Used in Calculation	0.383 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
 $= P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
 $= 50.9 \cdot 2827 / (2 \cdot 1509 \cdot 1 - 0.2 \cdot 50.9)$
 $= 47.8340 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 50.9 \cdot 37.7 / (1127 \cdot 1 - 0.6 \cdot 50.9)$
 $= 1.7507 \text{ mm}$

Intermediate Hub Nozzle Calculations:

The Diameter Limit is less than the Hub OD !

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 60.3 / 0.5773$
 $= 104.3829 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

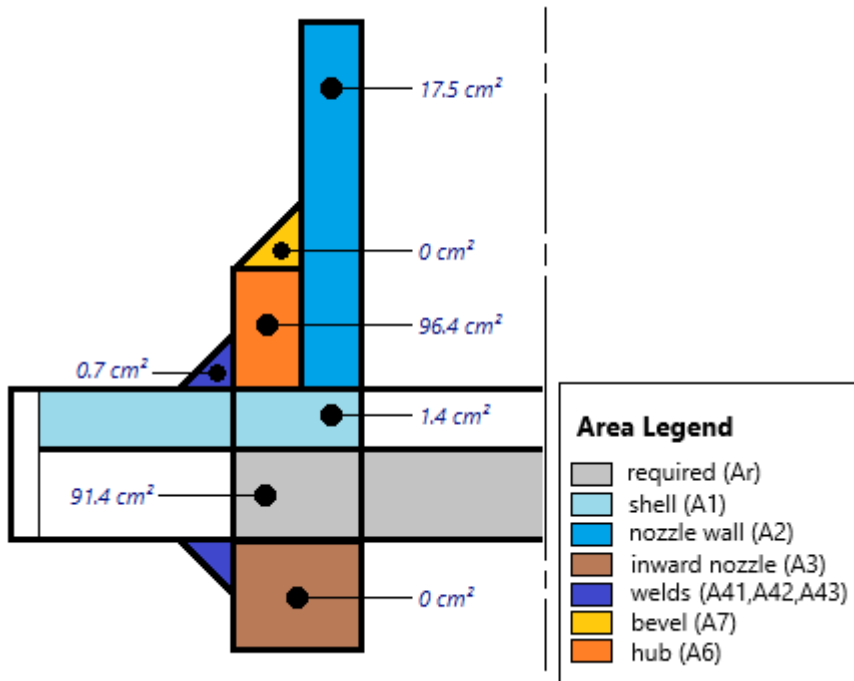
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	314.3065	mm
Parallel to Vessel Wall, opening length	d	157.1533	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	121.2193	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1127 / 1509)$
 $= 0.747$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1127 / 1509)$
 $= 0.747$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.75, 1)$
 $= 0.747$



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	91.402	NA	NA
Area in Shell	A1	1.437	NA	NA
Area in Nozzle Wall	A2	17.463	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.747	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	96.424	NA	NA
TOTAL AREA AVAILABLE	Atot	116.071	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 28.68 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (157 \cdot 47.8 \cdot 1 + 2 \cdot 67 \cdot 47.8 \cdot 1 \cdot (1 - 0.75))$$

$$= 91.402 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 157(1 \cdot 49 - 1 \cdot 47.8) - 2 \cdot 67(1 \cdot 49 - 1 \cdot 47.8) \cdot (1 - 0.75)$$

$$= 1.437 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2/\sin(\alpha3)$$

$$= (2 \cdot 121)(6.73 - 1.75)0.75/\sin(31.1)$$

$$= 17.463 \text{ cm}^2$$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10^2 \cdot 0.75 + (0)^2 \cdot 0.75$$



= 0.747 cm²

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot f_{r2}$$

$$= (2 \cdot \min(121, 800, 80)) \cdot (70 - 9.73) \cdot 0.75$$

$$= 96.424 \text{ cm}^2 \text{ Includes Bevel Area of } 24.414 \text{ cm}^2$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, tg = 9.73, tr = 1.64, c = 3 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.24, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: A

Govrn. thk, tg = 52, tr = 47.8, c = 3 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.98, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	37 °C
Min Metal Temp. at Required thickness (UCS 66.1)	36 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 36 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 50.90/105.44 = 0.483

Weld Size Calculations, Description: K4-104-D2A

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - tr)) \cdot S_v)$$

$$= \max(0, (91.4 - 1.44 + 2 \cdot 67 \cdot 0.75 \cdot (1 \cdot 49 - 47.8)) \cdot 1509)$$

$$= 137527.38 \text{ kgf}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A4 - (W_i - C_{an} / .707)^2 \cdot f_{r2}) \cdot S_v$$

$$= (114 + 0.75 - 0 \cdot 0.75) \cdot 1509$$

$$= 172993.64 \text{ kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$$

$$= (114 + 0 + 0.75 + (49)) \cdot 1509$$

$$= 246992.39 \text{ kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$

$$= (114 + 0 + 0.75 + 0 + (49)) \cdot 1509$$

$$= 246992.39 \text{ kgf}$$



Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.14/2.0) \cdot 436 \cdot 10 \cdot 0.49 \cdot 1127 \\ &= 37849. \text{ kgf} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n \\ &= (3.14 \cdot 148) \cdot (70 - 3) \cdot 0.7 \cdot 1127 \\ &= 246374. \text{ kgf} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng} \\ &= (3.14/2.0) \cdot 436 \cdot (52 - 3) \cdot 0.74 \cdot 1509 \\ &= 375042. \text{ kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (37849 + 246374) = 284223 \text{ kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (37849 + 0 + 375042 + 0) = 412891 \text{ kgf} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (37849 + 375042 + 0) = 412891 \text{ kgf} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 284222 kgf, must exceed W = 137527 kgf	or W1 = 172993 kgf
Path 2-2 = 412891 kgf, must exceed W = 137527 kgf	or W2 = 246992 kgf
Path 3-3 = 412891 kgf, must exceed W = 137527 kgf	or W3 = 246992 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 52.1 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: K4-104-D2A

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.734 in.
Actual Thickness Used in Calculation	0.383 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]

$$\begin{aligned} &= P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P) \text{ per UG-27 (d)} \\ &= 50.9 \cdot 2827 / (2 \cdot 1509 \cdot 1 - 0.2 \cdot 50.9) \\ &= 47.8340 \text{ mm} \end{aligned}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$\begin{aligned} &= P \cdot R / (S_n \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\ &= 50.9 \cdot 37.7 / (1127 \cdot 1 - 0.6 \cdot 50.9) \\ &= 1.7507 \text{ mm} \end{aligned}$$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:

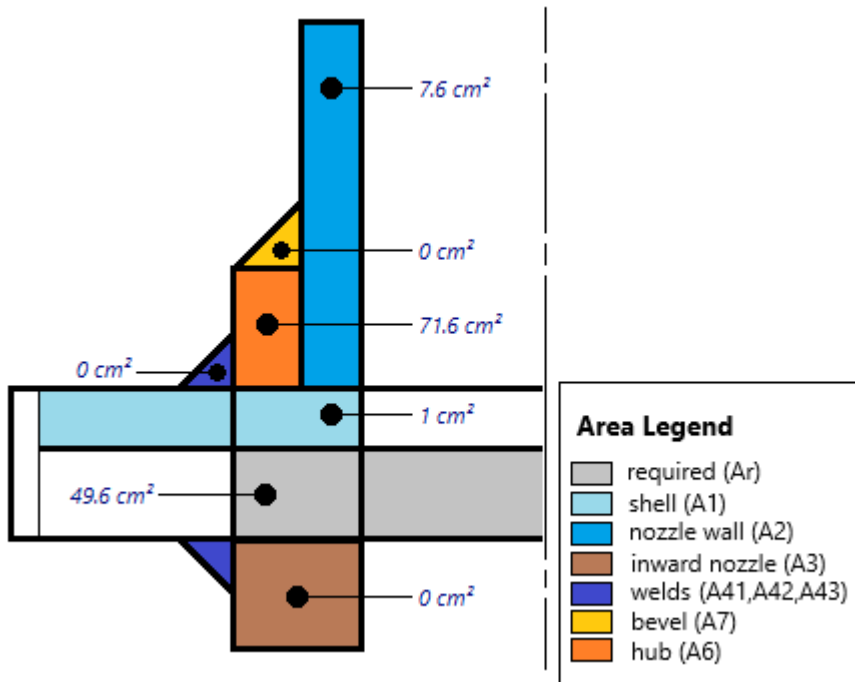
$$\begin{aligned} &= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30) \\ &= 49/0.5773 \end{aligned}$$

= 84.8705 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	186.9000	mm
Parallel to Vessel Wall	Rn+tn+t	93.4500	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	101.7069	mm



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	49.582	NA	NA
Area in Shell	A1	0.971	NA	NA
Area in Nozzle Wall	A2	7.571	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	71.575	NA	NA
TOTAL AREA AVAILABLE	Atot	80.116	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (75.4 \cdot 47.8 \cdot 1 + 2 \cdot 55.7 \cdot 47.8 \cdot 1 \cdot (1 - 0.75))$
 $= 49.582 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 111 (1 \cdot 49 - 1 \cdot 47.8) - 2 \cdot 55.7$
 $(1 \cdot 49 - 1 \cdot 47.8) \cdot (1 - 0.75)$
 $= 0.971 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:



$$= (2 \cdot t_{np}) (t_n - t_{rn}) fr_2$$

$$= (2 \cdot 102) (6.73 - 1.75) 0.75$$

$$= 7.571 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= (W_o^2 - \text{Area Lost}) \cdot fr_2 + ((W_i - \text{can}/0.707)^2 - \text{Area Lost}) \cdot fr_2$$

$$= (10^2 - 1) \cdot 0.75 + (0^2 - 0) \cdot 0.75$$

$$= 0.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{np}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(102, 800, 80)) \cdot (58.7 - 9.73) \cdot 0.75$$

$$= 71.575 \text{ cm}^2 \text{ Includes Bevel Area of } 13.025 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 4.7507 \text{ mm}$
 Wall Thickness per UG16(b), $tr_{16b} = 4.5000 \text{ mm}$
 Wall Thickness, shell/head, internal pressure $trb_1 = 50.8340 \text{ mm}$
 Wall Thickness $tb_1 = \max(trb_1, tr_{16b}) = 50.8340 \text{ mm}$
 Wall Thickness $tb_2 = \max(trb_2, tr_{16b}) = 4.5000 \text{ mm}$
 Wall Thickness per table UG-45 $tb_3 = 7.8000 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[7.8, \max(50.8, 4.5)]$$

$$= 7.8000 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(4.75, 7.8)$$

$$= 7.8000 \text{ mm}$$

Available Nozzle Neck Thickness = 9.7345 mm -> OK

Weld Size Calculations, Description: K4-104-D2A

Intermediate Calc. for nozzle/shell Welds $T_{min} = 19.0000 \text{ mm}$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot t_n \cdot fr_1 \cdot (E1 \cdot t - tr)) \cdot S_v)$$

$$= \max(0, (49.6 - 0.97 + 2 \cdot 55.7 \cdot 0.75 \cdot (1 \cdot 49 - 47.8)) \cdot 1509)$$

$$= 74824.39 \text{ kgf}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A4 - (W_i - \text{Can}/0.707)^2 \cdot fr_2) \cdot S_v$$

$$= (79.1 + 0 - 0 \cdot 0.75) \cdot 1509$$

$$= 119438.69 \text{ kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S_v$$

$$= (79.1 + 0 + 0 + (40.8)) \cdot 1509$$

$$= 180995.20 \text{ kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S$$

$$= (79.1 + 0 + 0 + 0 + (40.8)) \cdot 1509$$

$$= 180995.20 \text{ kgf}$$

Strength of Connection Elements for Failure Path Analysis



Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw \\ &= (3.14/2.0) \cdot 187 \cdot 10 \cdot 0.49 \cdot 1127 \\ &= 16212. \text{ kgf} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\ &= (3.14 \cdot 65.6) \cdot (58.7 - 3) \cdot 0.7 \cdot 1127 \\ &= 90591. \text{ kgf} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng \\ &= (3.14/2.0) \cdot 187 \cdot (52 - 3) \cdot 0.74 \cdot 1509 \\ &= 160648. \text{ kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (16212 + 90591) = 106803 \text{ kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (16212 + 0 + 160648 + 0) = 176860 \text{ kgf} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (16212 + 160648 + 0) = 176860 \text{ kgf} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 106803 kgf, must exceed W = 74824 kgf	or W1 = 119438 kgf
Path 2-2 = 176860 kgf, must exceed W = 74824 kgf	or W2 = 180995 kgf
Path 3-3 = 176860 kgf, must exceed W = 74824 kgf	or W3 = 180995 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 52.1 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 195.7997 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 1104.1672 mm

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Input, Nozzle Desc: K2

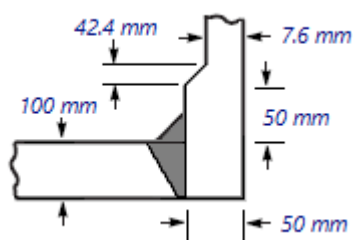
From: 130

Pressure for Reinforcement Calculations	P	50.897	kgf/cm ²
Temperature for Internal Pressure	Temp	275	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	1509.10	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1509.10	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	5600.00	mm
Shell Finished (Minimum) Thickness	t	100.0000	mm
Shell Internal Corrosion Allowance	cas	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1420.00	cm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1407.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.00	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	50.0000	mm
Height of Beveled Transition	L'	42.3546	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		52.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.9	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K2

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 1.773 in.
Actual Thickness Used in Calculation 0.301 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $50.9 \cdot 2803 / (1509 \cdot 1 - 0.6 \cdot 50.9)$
= 96.4885 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $50.9 \cdot 25.5 / (1407 \cdot 1 - 0.6 \cdot 50.9)$
= 0.9435 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 42.4/0.5773
= 73.3603 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

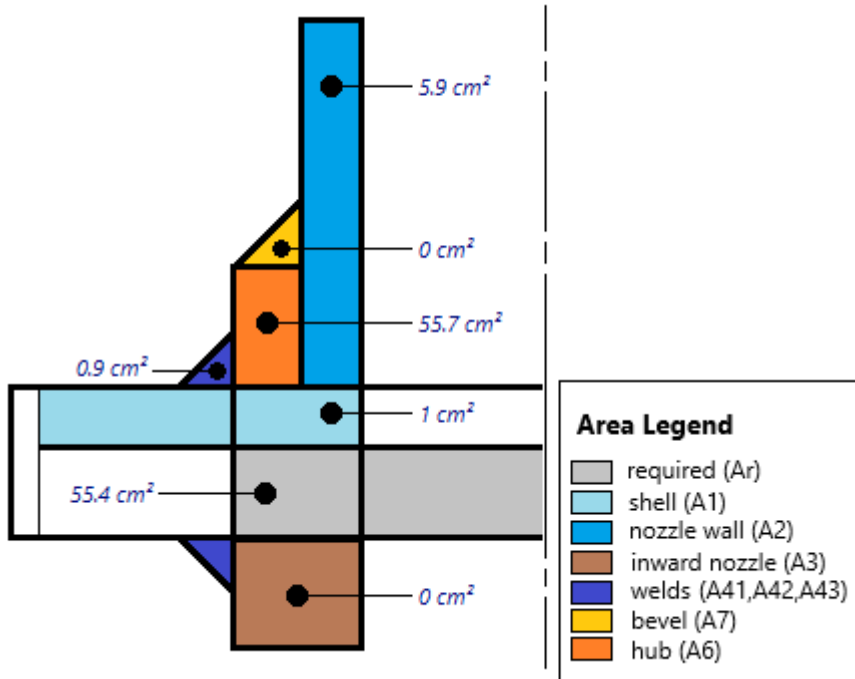
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	254.3250	mm
Parallel to Vessel Wall	Rn+tn+t	127.1625	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	84.9738	mm

Weld Strength Reduction Factor [fr1]:
= min(1, Sn/Sv)
= min(1, 1407/1509)
= 0.932

Weld Strength Reduction Factor [fr2]:
= min(1, Sn/Sv)
= min(1, 1407/1509)
= 0.932

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(0.93, 1)
= 0.932



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	55.379	NA	NA
Area in Shell	A1	1.007	NA	NA
Area in Nozzle Wall	A2	5.866	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.932	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	55.707	NA	NA
TOTAL AREA AVAILABLE	Atot	63.512	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (51 \cdot 96.5 \cdot 1 + 2 \cdot 47 \cdot 96.5 \cdot 1 \cdot (1 - 0.93))$
 $= 55.379 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 203 (1 \cdot 97 - 1 \cdot 96.5) - 2 \cdot 47$
 $(1 \cdot 97 - 1 \cdot 96.5) \cdot (1 - 0.93)$
 $= 1.007 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp) (tn - trn) fr2$
 $= (2 \cdot 85) (4.65 - 0.94) 0.93$
 $= 5.866 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can / 0.707)^2 \cdot fr2$
 $= 10^2 \cdot 0.93 + (0)^2 \cdot 0.93$
 $= 0.932 \text{ cm}^2$



Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubt} - t_n) \cdot f_{r2}$$

$$= (2 \cdot \min(85, 500, 50)) \cdot (50 - 7.65) \cdot 0.93$$

$$= 55.707 \text{ cm}^2 \text{ Includes Bevel Area of } 16.218 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.9435 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 99.4885 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 99.4885 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 6.4200 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[6.42, \max(99.5, 4.5)]$$

$$= 6.4200 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(3.94, 6.42)$$

$$= 6.4200 \text{ mm}$$

Available Nozzle Neck Thickness = 7.6454 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 7.65, tr = 0.94, c = 3 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.2, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:B, Curve:A)

Govrn. thk, tg = 50, tr = 0.94, c = 3 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.02, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	36 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 50.90/105.44 = 0.483

Weld Size Calculations, Description: K2

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - tr)) S_v)$$

$$= \max(0, (55.4 - 1.01 + 2 \cdot 47 \cdot 0.93 \cdot (1 \cdot 97 - 96.5)) 1509)$$

$$= 82728.09 \text{ kgf}$$

For hub type nozzles, A2 includes the area of the hub.



Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned} &= (A2+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv \\ &= (61.6 + 0.93 - 0 \cdot 0.93) \cdot 1509 \\ &= 94325.42 \text{ kgf} \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned} &= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv \\ &= (61.6 + 0 + 0.93 + (85)) \cdot 1509 \\ &= 222615.52 \text{ kgf} \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned} &= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S \\ &= (61.6 + 0 + 0.93 + 0 + (85)) \cdot 1509 \\ &= 222615.52 \text{ kgf} \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw \\ &= (3.14/2.0) \cdot 145 \cdot 10 \cdot 0.49 \cdot 1407 \\ &= 15707. \text{ kgf} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\ &= (3.14 \cdot 49) \cdot (50 - 3) \cdot 0.7 \cdot 1407 \\ &= 71283. \text{ kgf} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng \\ &= (3.14/2.0) \cdot 145 \cdot (52 - 0) \cdot 0.74 \cdot 1509 \\ &= 132295. \text{ kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (15707 + 71283) = 86990 \text{ kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (15707 + 0 + 132295 + 0) = 148001 \text{ kgf} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (15707 + 132295 + 0) = 148001 \text{ kgf} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 86989	kgf, must exceed W = 82728 kgf	or W1 = 94325 kgf
Path 2-2 = 148001	kgf, must exceed W = 82728 kgf	or W2 = 222615 kgf
Path 3-3 = 148001	kgf, must exceed W = 82728 kgf	or W3 = 222615 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 51.2 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 0.9393 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 600.9394 mm

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Input, Nozzle Desc: N1-104-D2A

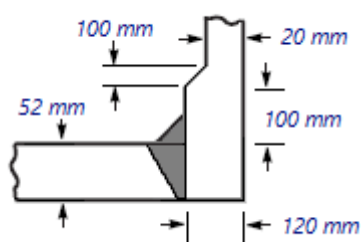
From: 140

Pressure for Reinforcement Calculations	P	50.897	kgf/cm ²
Temperature for Internal Pressure	Temp	275	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	1509.10	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1509.10	kgf/cm ²
Inside Diameter of Hemispherical Head	D	5648.00	mm
Head Finished (Minimum) Thickness	t	52.0000	mm
Head Internal Corrosion Allowance	cas	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1407.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.00	kgf/cm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		36.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	20.0000	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	100.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		120.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	600.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	52.0000	mm
Inside Projection	h	300.0000	mm
Weld leg size, Inside Element to Shell	Wi	14.0000	mm
Flange Class		600	
Flange Grade		GR 1.9	
Flange Series		B	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N1-104-D2A



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 36.000 in.
Actual Thickness Used in Calculation 0.787 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
= $P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
= $50.9 \cdot 2827 / (2 \cdot 1509 \cdot 1 - 0.2 \cdot 50.9)$
= 47.8340 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $50.9 \cdot 457 / (1407 \cdot 1 + 0.4 \cdot 50.9)$
= 16.3029 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / $\tan(30)$
= $100 / 0.5773$
= 173.2052 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

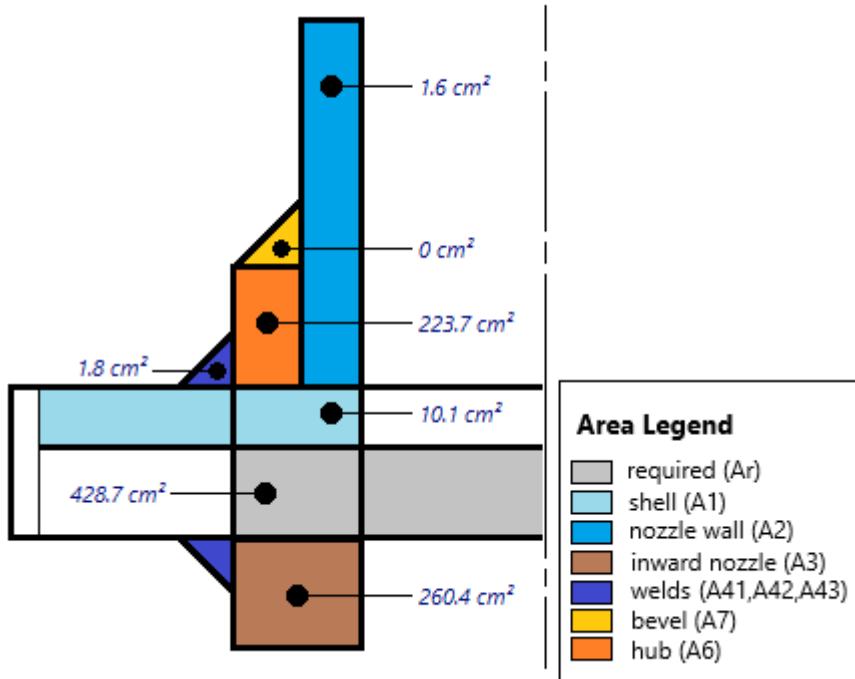
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	1760.8000	mm
Parallel to Vessel Wall, opening length	d	880.4000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	122.5000	mm
Normal to Vessel Wall, Inward		35.0000	mm

Weld Strength Reduction Factor [fr1]:
= $\min(1, S_n / S_v)$
= $\min(1, 1407 / 1509)$
= 0.932

Weld Strength Reduction Factor [fr2]:
= $\min(1, S_n / S_v)$
= $\min(1, 1407 / 1509)$
= 0.932

Weld Strength Reduction Factor [fr3]:
= $\min(fr2, fr4)$
= $\min(0.93, 1)$
= 0.932



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	428.704	NA	NA
Area in Shell	A1	10.081	NA	NA
Area in Nozzle Wall	A2	1.592	NA	NA
Area in Inward Nozzle	A3	260.404	NA	NA
Area in Welds	A41+A42+A43	1.820	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	223.704	NA	NA
TOTAL AREA AVAILABLE	Atot	497.601	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (880 \cdot 47.8 \cdot 1 + 2 \cdot 117 \cdot 47.8 \cdot 1 \cdot (1 - 0.93))$$

$$= 428.704 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 880(1 \cdot 49 - 1 \cdot 47.8) - 2 \cdot 117(1 \cdot 49 - 1 \cdot 47.8) \cdot (1 - 0.93)$$

$$= 10.081 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 123)(17 - 16.3)0.93$$

$$= 1.592 \text{ cm}^2$$

Area Available in Inward Nozzle [A3]:

$$= 2 \cdot ti \cdot \min(h, TI, 2.5 \cdot ti) \cdot fr2$$

$$= 2 \cdot 114 \cdot (123) \cdot 0.93$$

$$= 260.404 \text{ cm}^2$$



Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can/0.707)^2 \cdot fr_2$$

$$= 10^2 \cdot 0.93 + (9.76)^2 \cdot 0.93$$

$$= 1.820 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{hp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(123, 600, 100)) \cdot (120 - 20) \cdot 0.93$$

$$= 223.704 \text{ cm}^2 \text{ Includes Bevel Area of } 37.236 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 19.3029 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 50.8340 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 50.8340 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.3, \max(50.8, 4.5)]$$

$$= 11.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(19.3, 11.3)$$

$$= 19.3029 \text{ mm}$$

Available Nozzle Neck Thickness = 20.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 850.5,	Allowable	: 1407.0 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 2667.0 kgf/cm ²	Passed
Occasional	: 646.5,	Allowable	: 1871.3 kgf/cm ²	Passed
Shear	: 127.6,	Allowable	: 984.9 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 20, tr = 16.3, c = 3 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.96, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	12 °C
Min Metal Temp. at Required thickness (UCS 66.1)	10 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:A)

Govrn. thk, tg = 52, tr = 47.8, c = 3 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.98, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	37 °C
Min Metal Temp. at Required thickness (UCS 66.1)	36 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 36 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 50.90/105.44 = 0.483

Weld Size Calculations, Description: N1-104-D2A



Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))Sv)
= max(0, (429 - 10.1 + 2 • 117 • 0.93 •
(1 • 49 - 47.8))1509)
= 635582.56 kgf

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A4-(Wi-Can/.707)²•fr2)•Sv
= (225 + 1.82 - 95.2 • 0.93) • 1509
= 341401.69 kgf

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (225 + 260 + 1.82 + (107)) • 1509
= 897042.25 kgf

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (225 + 260 + 1.82 + 0 + (107)) • 1509
= 897042.25 kgf

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.14/2.0) • 914 • 10 • 0.49 • 1407
= 99025. kgf

Shear, Inward Nozzle Weld [Sinw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.14/2.0) • 914 • 9.76 • 0.49 • 1407
= 96622. kgf

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.14 • 449) • (120 - 3) • 0.7 • 1407
= 1624364. kgf

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • (T - cas) • 0.74 • Sng
= (3.14/2.0) • 914 • (52 - 3) • 0.74 • 1509
= 785962. kgf

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (99025 + 1624364) = 1723389 kgf

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (99025 + 0 + 785962 + 96622) = 981609 kgf

Failure Path [Path 3-3]:
= (Sonw + Tngw + Sinw)
= (99025 + 785962 + 96622) = 981609 kgf



Summary of Failure Path Calculations:

Path 1-1 = 1723389 kgf, must exceed W = 635582 kgf or W1 = 341401 kgf
 Path 2-2 = 981609 kgf, must exceed W = 635582 kgf or W2 = 897042 kgf
 Path 3-3 = 981609 kgf, must exceed W = 635582 kgf or W3 = 897042 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 52.1 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 37.2557 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 951.9999 mm

Input Echo, WRC107/537 Item 1, Description: N1-104-D2A :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	3.0000	mm
Vessel Diameter	Dv	5648.000	mm
Vessel Thickness	Tv	52.000	mm
Design Temperature	T1	275.0	C
Vessel Material		SA-387 11	
Vessel UNS Number		K11789	
Vessel Cold S.I. Allowable	Smc	1509.10	kgf/cm ²
Vessel Hot S.I. Allowable	Smh	1509.10	kgf/cm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	OD	
Corrosion Allowance for Nozzle	Can	3.0000	mm
Nozzle Diameter	Dn	1114.400	mm
Nozzle Thickness	Tn	120.000	mm
Nozzle Material		SA-182 F11	
Nozzle UNS Number		K11572	
Nozzle Cold S.I. Allowable	SNmc	1407.00	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh	1407.00	kgf/cm ²
Design Internal Pressure	Dp	50.897	kgf/cm ²
Include Pressure Thrust		No	

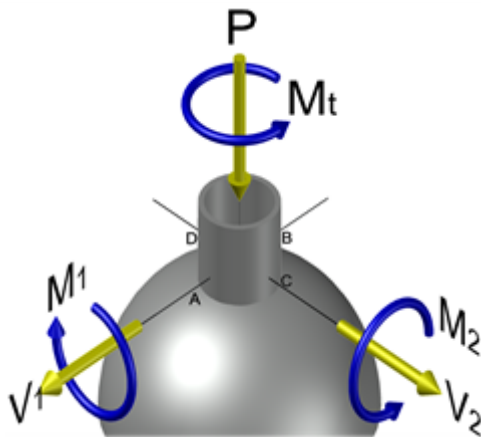
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-8256.9	kgf
Longitudinal Shear (SUS)	(V1) V1	5963.3	kgf
Circumferential Shear (SUS)	(Vc) V2	5963.3	kgf
Circumferential Moment (SUS)	(Mc) M1	16514.3	kgf-m.
Longitudinal Moment (SUS)	(Ml) M2	16514.3	kgf-m.
Torsional Moment (SUS)	Mt	18991.5	kgf-m.

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 1114.4 + 2 \cdot 1.65 \cdot \sqrt{(2851.5 (52.0 - 3.0))}$$

$$= 2347.927 \text{ mm}$$

WRC 107 Stress Calculation for SUSTAINED loads:

Radial Load	P	-8256.9	kgf
Circumferential Shear	(VC) V2	5963.3	kgf
Longitudinal Shear	(VL) V1	5963.3	kgf
Circumferential Moment	(MC) M1	16514.3	kgf-m.
Longitudinal Moment	(ML) M2	16514.3	kgf-m.
Torsional Moment	MT	18991.5	kgf-m.

Dimensionless Param: U = 1.49 TAU = 5.00 (4.26) RHO = 0.42

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 1	0.02638	(A,B,C,D)
$M(x) / P$	SP 1	0.02489	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.02602	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 1	0.03338	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.02602	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 1	0.03338	(A,B,C,D)
$N(y) * T / P$	SP 1	0.03827	(A,B,C,D)
$M(y) / P$	SP 1	0.00699	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.03473	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 1	0.01052	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.03473	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 1	0.01052	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (kgf/cm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb.	P	9.1	9.1	9.1	9.1	9.1	9.1	9.1	9.1
Rad. Bend.	P	51.3	-51.3	51.3	-51.3	51.3	-51.3	51.3	-51.3
Rad. Memb.	MC	0.0	0.0	0.0	0.0	-47.9	-47.9	47.9	47.9
Rad. Bend.	MC	0.0	0.0	0.0	0.0	-368.6	368.6	368.6	-368.6
Rad. Memb.	ML	-47.9	-47.9	47.9	47.9	0.0	0.0	0.0	0.0
Rad. Bend.	ML	-368.6	368.6	368.6	-368.6	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-356.0	278.4	476.9	-363.0	-356.0	278.4	476.9	-363.0

Tang. Memb. P	13.2	13.2	13.2	13.2	13.2	13.2	13.2	13.2
Tang. Bend. P	14.4	-14.4	14.4	-14.4	14.4	-14.4	14.4	-14.4
Tang. Memb. MC	0.0	0.0	0.0	0.0	-63.9	-63.9	63.9	63.9
Tang. Bend. MC	0.0	0.0	0.0	0.0	-116.2	116.2	116.2	-116.2
Tang. Memb. ML	-63.9	-63.9	63.9	63.9	0.0	0.0	0.0	0.0
Tang. Bend. ML	-116.2	116.2	116.2	-116.2	0.0	0.0	0.0	0.0
Tot. Tang. Str.	-152.5	51.0	207.7	-53.5	-152.5	51.0	207.7	-53.5
Shear VC	7.0	7.0	-7.0	-7.0	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-7.0	-7.0	7.0	7.0
Shear MT	19.9	19.9	19.9	19.9	19.9	19.9	19.9	19.9
Tot. Shear	26.8	26.8	12.9	12.9	12.9	12.9	26.8	26.8
Str. Int.	359.5	281.5	477.5	363.5	356.9	279.1	479.5	365.3

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (kgf/cm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	1468.5	1468.5	1468.5	1468.5	1468.5	1468.5	1468.5	1468.5
Rad.	Pl (SUS)	-38.8	-38.8	57.0	57.0	-38.8	-38.8	57.0	57.0
Rad.	Q (SUS)	-317.2	317.2	419.9	-419.9	-317.2	317.2	419.9	-419.9
Long.	Pm (SUS)	1468.5	1468.5	1468.5	1468.5	1468.5	1468.5	1468.5	1468.5
Long.	Pl (SUS)	-50.8	-50.8	77.1	77.1	-50.8	-50.8	77.1	77.1
Long.	Q (SUS)	-101.8	101.8	130.6	-130.6	-101.8	101.8	130.6	-130.6
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	7.0	7.0	-7.0	-7.0	-7.0	-7.0	7.0	7.0
Shear	Q (SUS)	19.9	19.9	19.9	19.9	19.9	19.9	19.9	19.9
Pm (SUS)		1468.5	1468.5	1468.5	1468.5	1468.5	1468.5	1468.5	1468.5
Pm+Pl (SUS)		1432.9	1432.9	1547.8	1547.8	1432.9	1432.9	1547.8	1547.8
Pm+Pl+Q (Total)		1319.5	1750.1	1946.0	1415.5	1316.8	1747.7	1948.0	1417.3

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1468.52	1509.10	Passed
Pm+Pl (SUS)	1547.76	2263.65	Passed
Pm+Pl+Q (TOTAL)	1948.04	4527.30	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: M1-104-D2A

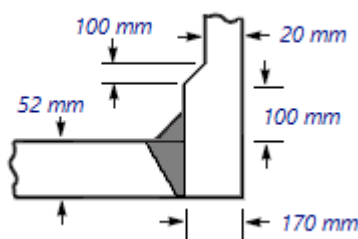
From: 140

Pressure for Reinforcement Calculations	P	50.897	kgf/cm ²
Temperature for Internal Pressure	Temp	275	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	1509.10	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1509.10	kgf/cm ²
Inside Diameter of Hemispherical Head	D	5648.00	mm
Head Finished (Minimum) Thickness	t	52.0000	mm
Head Internal Corrosion Allowance	cas	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	1450.0000	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1407.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.00	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	20.0000	mm
Flange Material		SA-182 F11	
Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	100.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		170.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	700.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		52.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
This is a Manway, Coupling or Access Opening.			
Flange Class		600	
Flange Grade		GR 1.9	
Flange Series		A	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle in the Meridional direction.



Reinforcement CALCULATION, Description: M1-104-D2A

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	24.000 in.
Actual Thickness Used in Calculation	0.787 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
= $P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
= $50.9 \cdot 2827 / (2 \cdot 1509 \cdot 1 - 0.2 \cdot 50.9)$
= 47.8340 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $50.9 \cdot 308 / (1407 \cdot 1 - 0.6 \cdot 50.9)$
= 11.3814 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= $150 / 0.5773$
= 259.8077 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

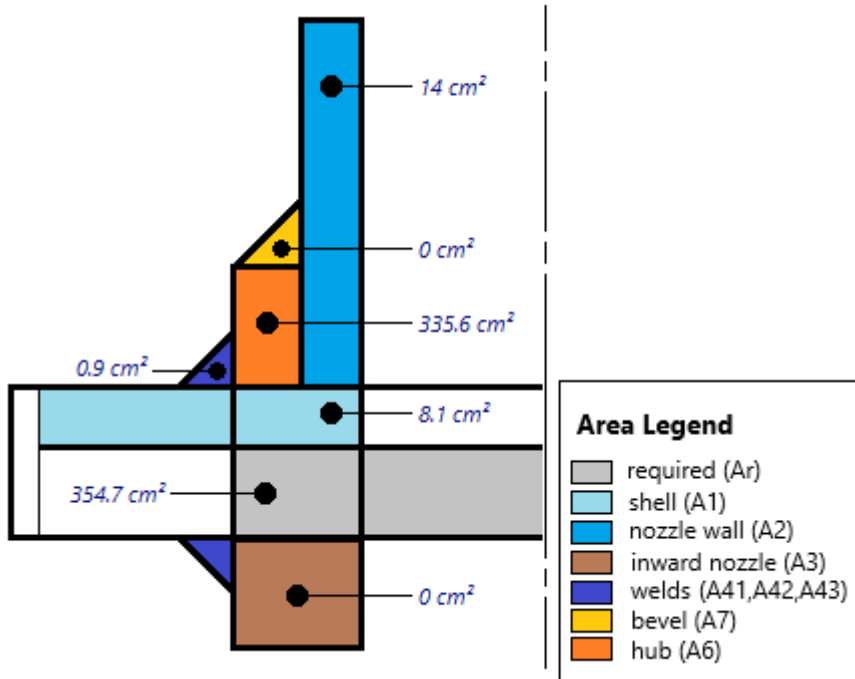
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1437.8009	mm
Parallel to Vessel Wall, opening length	d	718.9005	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	122.5000	mm

Weld Strength Reduction Factor [fr1]:
= min(1, S_n / S_v)
= min(1, $1407 / 1509$)
= 0.932

Weld Strength Reduction Factor [fr2]:
= min(1, S_n / S_v)
= min(1, $1407 / 1509$)
= 0.932

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(0.93, 1)
= 0.932



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	354.688	NA	NA
Area in Shell	A1	8.119	NA	NA
Area in Nozzle Wall	A2	13.984	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.932	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	335.556	NA	NA
TOTAL AREA AVAILABLE	Atot	358.592	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 58.90 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (719 \cdot 47.8 \cdot 1 + 2 \cdot 167 \cdot 47.8 \cdot 1 \cdot (1 - 0.93))$
 $= 354.688 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 719(1 \cdot 49 - 1 \cdot 47.8) - 2 \cdot 167$
 $(1 \cdot 49 - 1 \cdot 47.8) \cdot (1 - 0.93)$
 $= 8.119 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2/\sin(\alpha3)$
 $= (2 \cdot 123)(17 - 11.4)0.93/\sin(66.6)$
 $= 13.984 \text{ cm}^2$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10^2 \cdot 0.93 + (0)^2 \cdot 0.93$



= 0.932 cm²

Area Available in the Hub Section [A6]:

$$\begin{aligned} &= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2 \\ &= (2 \cdot \min(123,700, 100)) \cdot (170 - 20) \cdot 0.93 \\ &= 335.556 \text{ cm}^2 \text{ Includes Bevel Area of } 55.853 \text{ cm}^2 \end{aligned}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, tg = 20, tr = 11.4, c = 3 mm, E* = 1
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.67$, Temp. Reduction = 18 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-26 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:B, Curve:A)

Govrn. thk, tg = 52, tr = 47.8, c = 3 mm, E* = 1
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.98$, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	37 °C
Min Metal Temp. at Required thickness (UCS 66.1)	36 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 36 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $50.90 / 105.44 = 0.483$

Weld Size Calculations, Description: M1-104-D2A

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned} &= \max(0, (A - A_1 + 2 \cdot t_n \cdot fr_1 \cdot (E_1 \cdot t - tr)) \cdot S_v) \\ &= \max(0, (355 - 8.12 + 2 \cdot 167 \cdot 0.93 \cdot (1 \cdot 49 - 47.8)) \cdot 1509) \\ &= 528487.00 \text{ kgf} \end{aligned}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned} &= (A_2 + A_4 - (W_i - Can / .707)^2 \cdot fr_2) \cdot S_v \\ &= (350 + 0.93 - 0 \cdot 0.93) \cdot 1509 \\ &= 528898.31 \text{ kgf} \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned} &= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S_v \\ &= (350 + 0 + 0.93 + (153)) \cdot 1509 \\ &= 759167.62 \text{ kgf} \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned} &= (A_2 + A_3 + A_4 + A_5 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S \\ &= (350 + 0 + 0.93 + 0 + (153)) \cdot 1509 \\ &= 759167.62 \text{ kgf} \end{aligned}$$



Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.14/2.0) \cdot 1109 \cdot 10 \cdot 0.49 \cdot 1407 \\ &= 120094. \text{ kgf} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n \\ &= (3.14 \cdot 457) \cdot (170 - 3) \cdot 0.7 \cdot 1407 \\ &= 2361228. \text{ kgf} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng} \\ &= (3.14/2.0) \cdot 1109 \cdot (52 - 3) \cdot 0.74 \cdot 1509 \\ &= 953183. \text{ kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (120094 + 2361227) = 2481321 \text{ kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (120094 + 0 + 953183 + 0) = 1073276 \text{ kgf} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (120094 + 953183 + 0) = 1073276 \text{ kgf} \end{aligned}$$

Summary of Failure Path Calculations:

$$\begin{aligned} \text{Path 1-1} &= 2481321 \text{ kgf, must exceed } W = 528487 \text{ kgf} & \text{or } W1 &= 528898 \text{ kgf} \\ \text{Path 2-2} &= 1073276 \text{ kgf, must exceed } W = 528487 \text{ kgf} & \text{or } W2 &= 759167 \text{ kgf} \\ \text{Path 3-3} &= 1073276 \text{ kgf, must exceed } W = 528487 \text{ kgf} & \text{or } W3 &= 759167 \text{ kgf} \end{aligned}$$

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 51.2 kgf/cm²

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: M1-104-D2A

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	24.000 in.
Actual Thickness Used in Calculation	0.787 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]

$$\begin{aligned} &= P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P) \text{ per UG-27 (d)} \\ &= 50.9 \cdot 2827 / (2 \cdot 1509 \cdot 1 - 0.2 \cdot 50.9) \\ &= 47.8340 \text{ mm} \end{aligned}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$\begin{aligned} &= P \cdot R / (S_n \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\ &= 50.9 \cdot 308 / (1407 \cdot 1 - 0.6 \cdot 50.9) \\ &= 11.3814 \text{ mm} \end{aligned}$$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

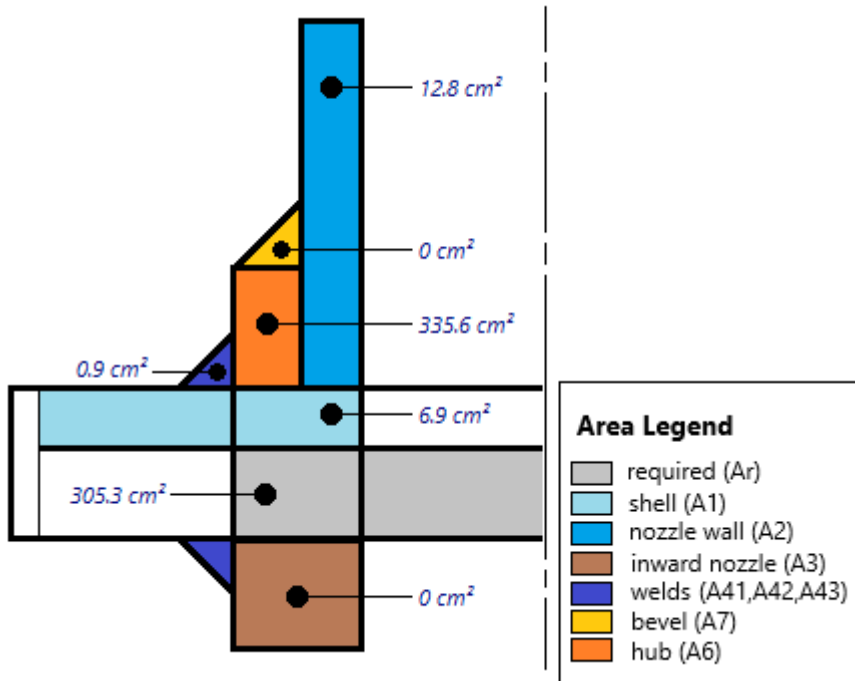
Height value from sketch (e-1) [te]:

$$\begin{aligned} &= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30) \\ &= 150 / 0.5773 \\ &= 259.8077 \text{ mm} \end{aligned}$$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 1231.2001 mm
 Parallel to Vessel Wall, opening length d 615.6000 mm
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 122.5000 mm



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	305.276	NA	NA
Area in Shell	A1	6.914	NA	NA
Area in Nozzle Wall	A2	12.834	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.932	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	335.556	NA	NA
TOTAL AREA AVAILABLE	Atot	356.237	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (616 \cdot 47.8 \cdot 1 + 2 \cdot 167 \cdot 47.8 \cdot 1 \cdot (1 - 0.93))$$

$$= 305.276 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 616(1 \cdot 49 - 1 \cdot 47.8) - 2 \cdot 167$$

$$(1 \cdot 49 - 1 \cdot 47.8) \cdot (1 - 0.93)$$

$$= 6.914 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 123)(17 - 11.4)0.93$$



$$= 12.834 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$\begin{aligned} &= W_o^2 \cdot fr_2 + (W_i - can/0.707)^2 \cdot fr_2 \\ &= 10^2 \cdot 0.93 + (0)^2 \cdot 0.93 \\ &= 0.932 \text{ cm}^2 \end{aligned}$$

Area Available in the Hub Section [A6]:

$$\begin{aligned} &= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2 \\ &= (2 \cdot \min(123, 700, 100)) \cdot (170 - 20) \cdot 0.93 \\ &= 335.556 \text{ cm}^2 \text{ Includes Bevel Area of } 55.853 \text{ cm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 14.3814 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= t_a \\ &= 14.3814 \text{ mm} \end{aligned}$$

Available Nozzle Neck Thickness = 20.0000 mm -> OK

Weld Size Calculations, Description: M1-104-D2A

Intermediate Calc. for nozzle/shell welds $T_{min} = 19.0000 \text{ mm}$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned} &= \max(0, (A - A_1 + 2 \cdot t_n \cdot fr_1 \cdot (E_1 \cdot t - tr)) \cdot S_v) \\ &= \max(0, (305 - 6.91 + 2 \cdot 167 \cdot 0.93 \cdot (1 \cdot 49 - 47.8)) \cdot 1509) \\ &= 455735.94 \text{ kgf} \end{aligned}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned} &= (A_2 + A_4 - (W_i - Can/0.707)^2 \cdot fr_2) \cdot S_v \\ &= (348 + 0.93 - 0 \cdot 0.93) \cdot 1509 \\ &= 527162.56 \text{ kgf} \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned} &= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S_v \\ &= (348 + 0 + 0.93 + (153)) \cdot 1509 \\ &= 757431.94 \text{ kgf} \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned} &= (A_2 + A_3 + A_4 + A_5 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S \\ &= (348 + 0 + 0.93 + 0 + (153)) \cdot 1509 \\ &= 757431.94 \text{ kgf} \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.14/2.0) \cdot 950 \cdot 10 \cdot 0.49 \cdot 1407 \\ &= 102837. \text{ kgf} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n \\ &= (3.14 \cdot 391) \cdot (170 - 3) \cdot 0.7 \cdot 1407 \\ &= 2021937. \text{ kgf} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$$



$$= (3.14/2.0) \cdot 950 \cdot (52 - 3) \cdot 0.74 \cdot 1509 \\ = 816218. \text{ kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (102837 + 2021937) = 2124774 \text{ kgf}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ = (102837 + 0 + 816218 + 0) = 919055 \text{ kgf}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw}) \\ = (102837 + 816218 + 0) = 919055 \text{ kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 2124774 kgf, must exceed W = 455735 kgf or W1 = 527162 kgf
Path 2-2 = 919054 kgf, must exceed W = 455735 kgf or W2 = 757431 kgf
Path 3-3 = 919054 kgf, must exceed W = 455735 kgf or W3 = 757431 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 52.1 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 356.8922 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 1117.2950 mm

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Input, Nozzle Desc: K1-104-D2A

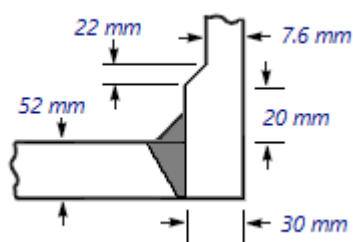
From: 140

Pressure for Reinforcement Calculations	P	50.897	kgf/cm ²
Temperature for Internal Pressure	Temp	275	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	1509.10	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1509.10	kgf/cm ²
Inside Diameter of Hemispherical Head	D	5648.00	mm
Head Finished (Minimum) Thickness	t	52.0000	mm
Head Internal Corrosion Allowance	cas	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	1050.0000	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1407.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.00	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-182 F11	
Hub Height of Integral Nozzle	h	20.0000	mm
Height of Beveled Transition	L'	22.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		30.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	900.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		52.0000	mm
Inside Projection	h	3800.0002	mm
Weld leg size, Inside Element to Shell	Wi	10.0000	mm
Flange Class		600	
Flange Grade		GR 1.9	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle in the Meridional direction.

Reinforcement CALCULATION, Description: K1-104-D2A



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 1.773 in.
Actual Thickness Used in Calculation 0.301 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
= $P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
= $50.9 \cdot 2827 / (2 \cdot 1509 \cdot 1 - 0.2 \cdot 50.9)$
= 47.8340 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $50.9 \cdot 25.5 / (1407 \cdot 1 - 0.6 \cdot 50.9)$
= 0.9435 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 22.4 / 0.5773
= 38.7193 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

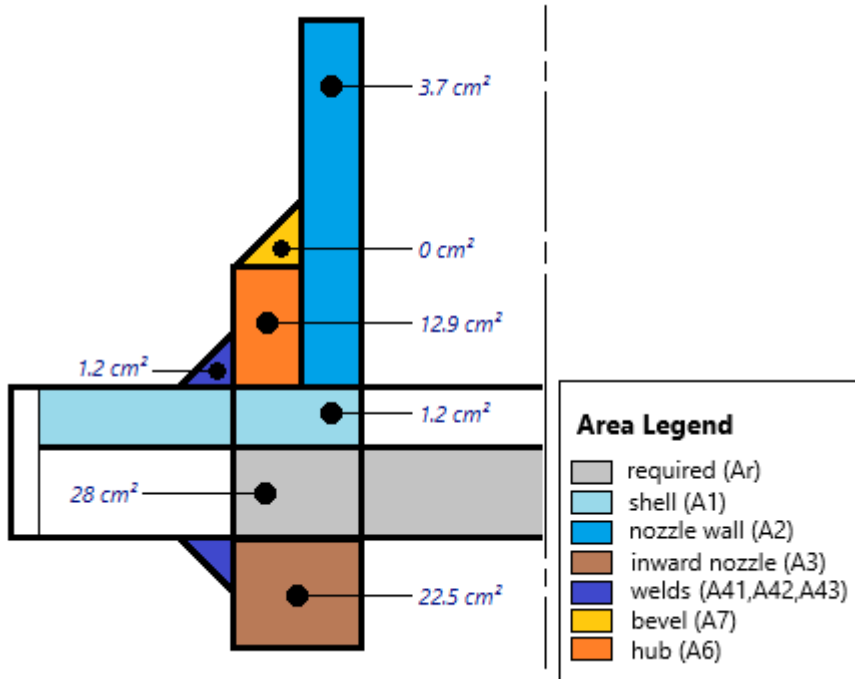
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	162.1850	mm
Parallel to Vessel Wall	Rn+tn+t	81.0925	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	50.3328	mm
Normal to Vessel Wall, Inward		4.1135	mm

Weld Strength Reduction Factor [fr1]:
= min(1, S_n / S_v)
= min(1, 1407/1509)
= 0.932

Weld Strength Reduction Factor [fr2]:
= min(1, S_n / S_v)
= min(1, 1407/1509)
= 0.932

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(0.93, 1)
= 0.932



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	28.006	NA	NA
Area in Shell	A1	1.208	NA	NA
Area in Nozzle Wall	A2	3.718	NA	NA
Area in Inward Nozzle	A3	22.525	NA	NA
Area in Welds	A41+A42+A43	1.241	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	12.922	NA	NA
TOTAL AREA AVAILABLE	Atot	41.615	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 68.39 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (54.9 \cdot 47.8 \cdot 1 + 2 \cdot 27 \cdot 47.8 \cdot 1 \cdot (1 - 0.93))$$

$$= 28.006 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 107 (1 \cdot 49 - 1 \cdot 47.8) - 2 \cdot 27 (1 \cdot 49 - 1 \cdot 47.8) \cdot (1 - 0.93)$$

$$= 1.208 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp) (tn - trn) fr2 / \sin(\alpha3)$$

$$= (2 \cdot 50.3) (4.65 - 0.94) 0.93 / \sin(69.1)$$

$$= 3.718 \text{ cm}^2$$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Nozzle [A3]:

$$= 2 \cdot ti \cdot \min(h, TI, 2.5 \cdot ti) \cdot fr2$$

$$= 2 \cdot 24 \cdot (50.3) \cdot 0.93$$



$$= 22.525 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr2 + (W_i - can/0.707)^2 \cdot fr2$$

$$= 10^2 \cdot 0.93 + (5.76)^2 \cdot 0.93$$

$$= 1.241 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr2$$

$$= (2 \cdot \min(50.3, 900, 20)) \cdot (30 - 7.65) \cdot 0.93$$

$$= 12.922 \text{ cm}^2 \text{ Includes Bevel Area of } 4.585 \text{ cm}^2$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, $t_g = 7.65$, $t_r = 0.94$, $c = 3 \text{ mm}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.2$, Temp. Reduction = 78°C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:B, Curve:A)

Govrn. thk, $t_g = 30$, $t_r = 0.94$, $c = 3 \text{ mm}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.035$, Temp. Reduction = 78°C

Min Metal Temp. w/o impact per UCS-66, Curve A	24 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104°C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $50.90/105.44 = 0.483$

Weld Size Calculations, Description: K1-104-D2A

Intermediate Calc. for nozzle/shell Welds $T_{min} \quad 19.0000 \text{ mm}$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$$

$$= \max(0, (28 - 1.21 + 2 \cdot 27 \cdot 0.93 \cdot (1 \cdot 49 - 47.8)) \cdot 1509)$$

$$= 41325.68 \text{ kgf}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A4 - (W_i - Can/0.707)^2 \cdot fr2) \cdot Sv$$

$$= (16.6 + 1.24 - 33.1 \cdot 0.93) \cdot 1509$$

$$= 26519.05 \text{ kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$$

$$= (16.6 + 22.5 + 1.24 + (24.7)) \cdot 1509$$

$$= 98207.31 \text{ kgf}$$



Weld Load [W3]:

$$\begin{aligned} &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f r 1)) \cdot S \\ &= (16.6 + 22.5 + 1.24 + 0 + (24.7)) \cdot 1509 \\ &= 98207.31 \text{ kgf} \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.14/2.0) \cdot 113 \cdot 10 \cdot 0.49 \cdot 1407 \\ &= 12235. \text{ kgf} \end{aligned}$$

Shear, Inward Nozzle Weld [Sinw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.14/2.0) \cdot 113 \cdot 5.76 \cdot 0.49 \cdot 1407 \\ &= 7044. \text{ kgf} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n \\ &= (3.14 \cdot 42) \cdot (30 - 3) \cdot 0.7 \cdot 1407 \\ &= 35061. \text{ kgf} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng} \\ &= (3.14/2.0) \cdot 113 \cdot (52 - 3) \cdot 0.74 \cdot 1509 \\ &= 97109. \text{ kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (12235 + 35061) = 47296 \text{ kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (12235 + 0 + 97109 + 7044) = 116388 \text{ kgf} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (12235 + 97109 + 7044) = 116388 \text{ kgf} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 47296	kgf, must exceed W = 41325 kgf	or W1 = 26519 kgf
Path 2-2 = 116388	kgf, must exceed W = 41325 kgf	or W2 = 98207 kgf
Path 3-3 = 116388	kgf, must exceed W = 41325 kgf	or W3 = 98207 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 52.1 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: K1-104-D2A

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.773 in.
Actual Thickness Used in Calculation	0.301 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]

$$\begin{aligned} &= P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P) \text{ per UG-27 (d)} \\ &= 50.9 \cdot 2827 / (2 \cdot 1509 \cdot 1 - 0.2 \cdot 50.9) \\ &= 47.8340 \text{ mm} \end{aligned}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$= P \cdot R / (S_n \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 50.9 \cdot 25.5 / (1407 \cdot 1 - 0.6 \cdot 50.9)$$

$$= 0.9435 \text{ mm}$$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:

$$= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$$

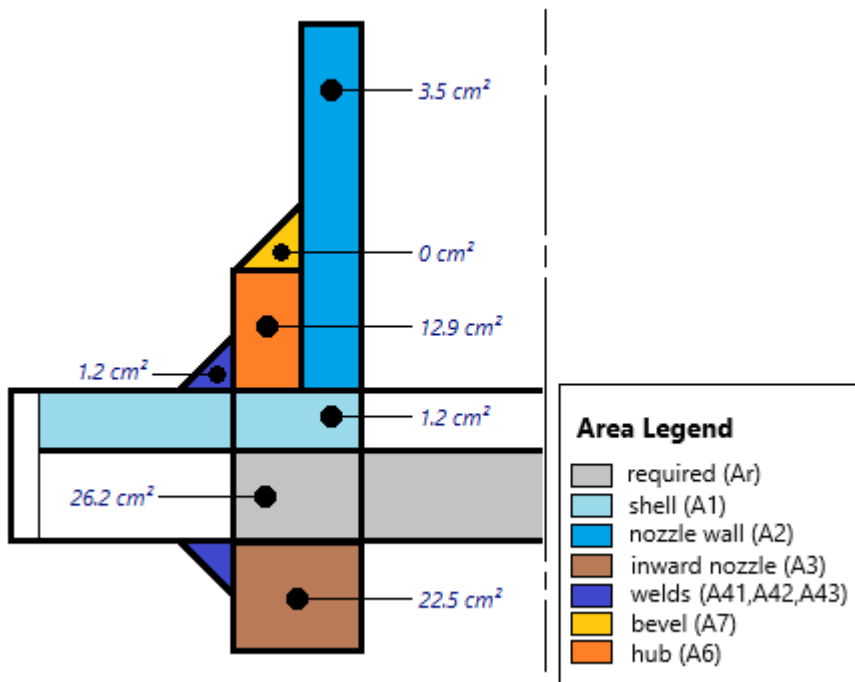
$$= 22.4 / 0.5773$$

$$= 38.7193 \text{ mm}$$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	158.3250	mm
Parallel to Vessel Wall	Rn+tn+t	79.1625	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	50.3328	mm
Normal to Vessel Wall, Inward		4.1135	mm



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	26.159	NA	NA
Area in Shell	A1	1.208	NA	NA
Area in Nozzle Wall	A2	3.474	NA	NA
Area in Inward Nozzle	A3	22.525	NA	NA
Area in Welds	A41+A42+A43	1.241	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	12.922	NA	NA
TOTAL AREA AVAILABLE	Atot	41.371	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (51 \cdot 47.8 \cdot 1 + 2 \cdot 27 \cdot 47.8 \cdot 1 \cdot (1 - 0.93))$$



= 26.159 cm²

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 107(1 \cdot 49 - 1 \cdot 47.8) - 2 \cdot 27$$

$$(1 \cdot 49 - 1 \cdot 47.8) \cdot (1 - 0.93)$$

$$= 1.208 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 50.3)(4.65 - 0.94)0.93$$

$$= 3.474 \text{ cm}^2$$

Area Available in Inward Nozzle [A3]:

$$= 2 \cdot ti \cdot \min(h, TI, 2.5 \cdot ti) \cdot fr2$$

$$= 2 \cdot 24 \cdot (50.3) \cdot 0.93$$

$$= 22.525 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10^2 \cdot 0.93 + (5.76)^2 \cdot 0.93$$

$$= 1.241 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(Tlnp, ho, Hubht)) \cdot (Hubtk - tn) \cdot fr2$$

$$= (2 \cdot \min(50.3, 900, 20)) \cdot (30 - 7.65) \cdot 0.93$$

$$= 12.922 \text{ cm}^2 \text{ Includes Bevel Area of } 4.585 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.9435 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 50.8340 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 50.8340 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 6.4200 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[6.42, \max(50.8, 4.5)]$$

$$= 6.4200 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(3.94, 6.42)$$

$$= 6.4200 \text{ mm}$$

Available Nozzle Neck Thickness = 7.6454 mm -> OK

Weld Size Calculations, Description: K1-104-D2A

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr))Sv)$$

$$= \max(0, (26.2 - 1.21 + 2 \cdot 27 \cdot 0.93 \cdot$$

$$(1 \cdot 49 - 47.8))1509)$$

$$= 38539.30 \text{ kgf}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.



Weld Load [W1]:

$$\begin{aligned} &= (A2+A4-(W_i\text{-Can}/.707)^2\cdot fr2)\cdot Sv \\ &= (16.4 + 1.24 - 33.1 \cdot 0.93) \cdot 1509 \\ &= 26150.97 \text{ kgf} \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned} &= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv \\ &= (16.4 + 22.5 + 1.24 + (24.7)) \cdot 1509 \\ &= 97839.23 \text{ kgf} \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned} &= (A2+A3+A4+A5+(2\cdot tn\cdot t\cdot fr1))\cdot S \\ &= (16.4 + 22.5 + 1.24 + 0 + (24.7)) \cdot 1509 \\ &= 97839.23 \text{ kgf} \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw \\ &= (3.14/2.0) \cdot 105 \cdot 10 \cdot 0.49 \cdot 1407 \\ &= 11375. \text{ kgf} \end{aligned}$$

Shear, Inward Nozzle Weld [Sinw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw \\ &= (3.14/2.0) \cdot 105 \cdot 5.76 \cdot 0.49 \cdot 1407 \\ &= 6549. \text{ kgf} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\ &= (3.14 \cdot 39) \cdot (30 - 3) \cdot 0.7 \cdot 1407 \\ &= 32596. \text{ kgf} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng \\ &= (3.14/2.0) \cdot 105 \cdot (52 - 3) \cdot 0.74 \cdot 1509 \\ &= 90281. \text{ kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (11375 + 32596) = 43970 \text{ kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (11375 + 0 + 90281 + 6549) = 108204 \text{ kgf} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (11375 + 90281 + 6549) = 108204 \text{ kgf} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 43970	kgf, must exceed W = 38539 kgf	or W1 = 26150 kgf
Path 2-2 = 108204	kgf, must exceed W = 38539 kgf	or W2 = 97839 kgf
Path 3-3 = 108204	kgf, must exceed W = 38539 kgf	or W3 = 97839 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 52.1 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 21.6500 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 4755.9351 mm

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Input, Nozzle Desc: K3-104-D2A

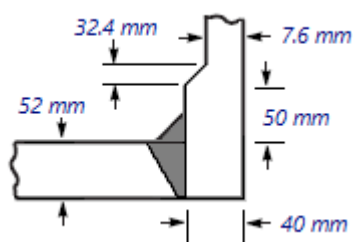
From: 140

Pressure for Reinforcement Calculations	P	50.897	kgf/cm ²
Temperature for Internal Pressure	Temp	275	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	1509.10	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1509.10	kgf/cm ²
Inside Diameter of Hemispherical Head	D	5648.00	mm
Head Finished (Minimum) Thickness	t	52.0000	mm
Head Internal Corrosion Allowance	cas	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	1050.0000	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1407.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.00	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-182 F11	
Hub Height of Integral Nozzle	h	50.0000	mm
Height of Beveled Transition	L'	32.3546	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		40.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	900.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		52.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.9	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle in the Meridional direction.

Reinforcement CALCULATION, Description: K3-104-D2A



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 1.773 in.
Actual Thickness Used in Calculation 0.301 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
= $P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
= $50.9 \cdot 2827 / (2 \cdot 1509 \cdot 1 - 0.2 \cdot 50.9)$
= 47.8340 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $50.9 \cdot 25.5 / (1407 \cdot 1 - 0.6 \cdot 50.9)$
= 0.9435 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 32.4 / 0.5773
= 56.0398 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

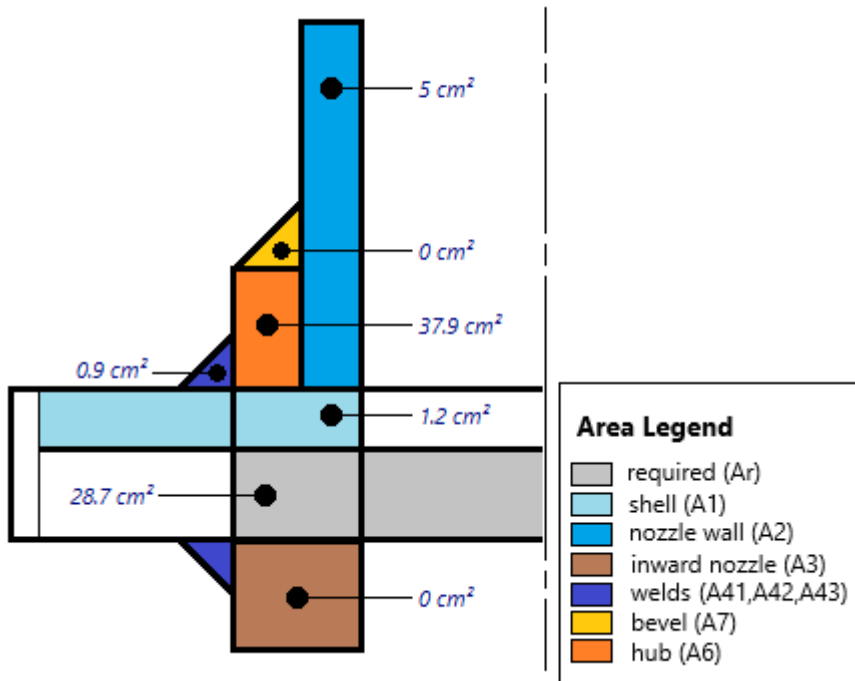
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	162.1850	mm
Parallel to Vessel Wall	Rn+tn+t	81.0925	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	67.6533	mm

Weld Strength Reduction Factor [fr1]:
= min(1, Sn/Sv)
= min(1, 1407/1509)
= 0.932

Weld Strength Reduction Factor [fr2]:
= min(1, Sn/Sv)
= min(1, 1407/1509)
= 0.932

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(0.93, 1)
= 0.932



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	28.653	NA	NA
Area in Shell	A1	1.193	NA	NA
Area in Nozzle Wall	A2	4.998	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.932	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	37.911	NA	NA
TOTAL AREA AVAILABLE	Atot	45.033	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 68.39 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (54.9 \cdot 47.8 \cdot 1 + 2 \cdot 37 \cdot 47.8 \cdot 1 \cdot (1 - 0.93))$
 $= 28.653 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 107(1 \cdot 49 - 1 \cdot 47.8) - 2 \cdot 37(1 \cdot 49 - 1 \cdot 47.8) \cdot (1 - 0.93)$
 $= 1.193 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2/\sin(\alpha3)$
 $= (2 \cdot 67.7)(4.65 - 0.94)0.93/\sin(69.1)$
 $= 4.998 \text{ cm}^2$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10^2 \cdot 0.93 + (0)^2 \cdot 0.93$



= 0.932 cm²

Area Available in the Hub Section [A6]:

$$\begin{aligned} &= (2 \cdot \min(T_{lnp}, h_o, Hubht)) \cdot (Hubtk - t_n) \cdot fr2 \\ &= (2 \cdot \min(67.7, 900, 50)) \cdot (40 - 7.65) \cdot 0.93 \\ &= 37.911 \text{ cm}^2 \text{ Includes Bevel Area of } 7.745 \text{ cm}^2 \end{aligned}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, tg = 7.65, tr = 0.94, c = 3 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.2, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:A)

Govrn. thk, tg = 40, tr = 0.94, c = 3 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.026, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	31 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 50.90/105.44 = 0.483

Weld Size Calculations, Description: K3-104-D2A

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned} &= \max(0, (A - A1 + 2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv) \\ &= \max(0, (28.7 - 1.19 + 2 \cdot 37 \cdot 0.93 \cdot (1 \cdot 49 - 47.8)) \cdot 1509) \\ &= 42654.36 \text{ kgf} \end{aligned}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned} &= (A2 + A4 - (Wi - Can / .707)^2 \cdot fr2) \cdot Sv \\ &= (42.9 + 0.93 - 0 \cdot 0.93) \cdot 1509 \\ &= 66159.91 \text{ kgf} \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned} &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv \\ &= (42.9 + 0 + 0.93 + (33.8)) \cdot 1509 \\ &= 117177.66 \text{ kgf} \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned} &= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S \\ &= (42.9 + 0 + 0.93 + 0 + (33.8)) \cdot 1509 \\ &= 117177.66 \text{ kgf} \end{aligned}$$



Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.14/2.0) \cdot 134 \cdot 10 \cdot 0.49 \cdot 1407 \\ &= 14565. \text{ kgf} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n \\ &= (3.14 \cdot 47.3) \cdot (40 - 3) \cdot 0.7 \cdot 1407 \\ &= 54204. \text{ kgf} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng} \\ &= (3.14/2.0) \cdot 134 \cdot (52 - 3) \cdot 0.74 \cdot 1509 \\ &= 115600. \text{ kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (14565 + 54204) = 68769 \text{ kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (14565 + 0 + 115600 + 0) = 130165 \text{ kgf} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (14565 + 115600 + 0) = 130165 \text{ kgf} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 68768	kgf, must exceed W = 42654 kgf	or W1 = 66159 kgf
Path 2-2 = 130165	kgf, must exceed W = 42654 kgf	or W2 = 117177 kgf
Path 3-3 = 130165	kgf, must exceed W = 42654 kgf	or W3 = 117177 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 52.1 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: K3-104-D2A

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.773 in.
Actual Thickness Used in Calculation	0.301 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]

$$\begin{aligned} &= P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P) \text{ per UG-27 (d)} \\ &= 50.9 \cdot 2827 / (2 \cdot 1509 \cdot 1 - 0.2 \cdot 50.9) \\ &= 47.8340 \text{ mm} \end{aligned}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$\begin{aligned} &= P \cdot R / (S_n \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\ &= 50.9 \cdot 25.5 / (1407 \cdot 1 - 0.6 \cdot 50.9) \\ &= 0.9435 \text{ mm} \end{aligned}$$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:

$$\begin{aligned} &= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30) \\ &= 32.4 / 0.5773 \end{aligned}$$

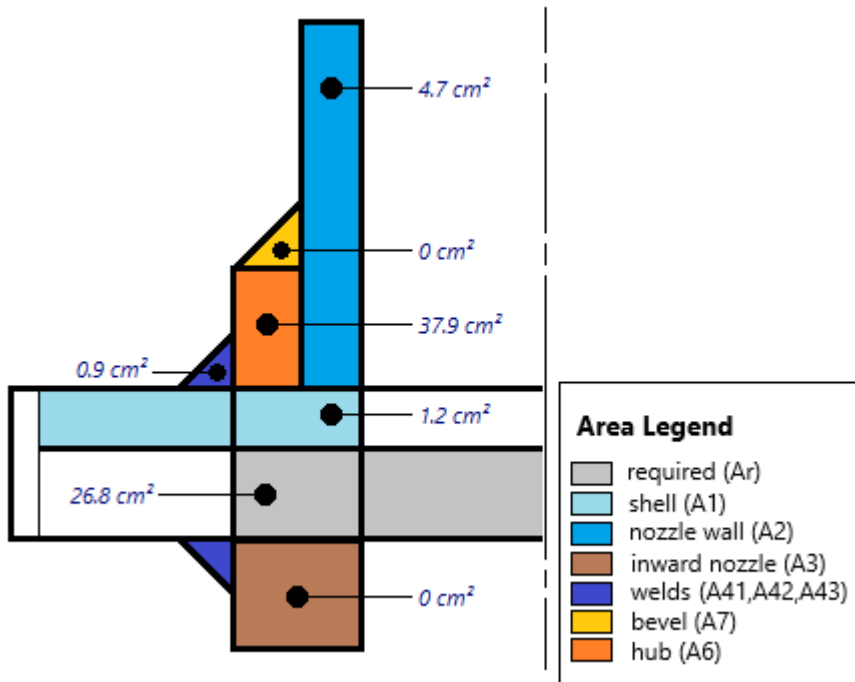


= 56.0398 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	158.3250	mm
Parallel to Vessel Wall	Rn+tn+t	79.1625	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	67.6533	mm



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	26.807	NA	NA
Area in Shell	A1	1.193	NA	NA
Area in Nozzle Wall	A2	4.670	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds A41+A42+A43		0.932	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	37.911	NA	NA
TOTAL AREA AVAILABLE	Atot	44.705	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (51 \cdot 47.8 \cdot 1 + 2 \cdot 37 \cdot 47.8 \cdot 1 \cdot (1 - 0.93))$
 $= 26.807 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 107(1 \cdot 49 - 1 \cdot 47.8) - 2 \cdot 37(1 \cdot 49 - 1 \cdot 47.8) \cdot (1 - 0.93)$
 $= 1.193 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:



$$= (2 \cdot t_{np}) (t_n - t_{rn}) \cdot fr2$$

$$= (2 \cdot 67.7) (4.65 - 0.94) \cdot 0.93$$

$$= 4.670 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr2 + (W_i - can/0.707)^2 \cdot fr2$$

$$= 10^2 \cdot 0.93 + (0)^2 \cdot 0.93$$

$$= 0.932 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(t_{np}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot fr2$$

$$= (2 \cdot \min(67.7, 900, 50)) \cdot (40 - 7.65) \cdot 0.93$$

$$= 37.911 \text{ cm}^2 \text{ Includes Bevel Area of } 7.745 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 3.9435 \text{ mm}$
Wall Thickness per UG16(b),	$tr16b = 4.5000 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$trb1 = 50.8340 \text{ mm}$
Wall Thickness	$tb1 = \max(trb1, tr16b) = 50.8340 \text{ mm}$
Wall Thickness	$tb2 = \max(trb2, tr16b) = 4.5000 \text{ mm}$
Wall Thickness per table UG-45	$tb3 = 6.4200 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[6.42, \max(50.8, 4.5)]$$

$$= 6.4200 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(3.94, 6.42)$$

$$= 6.4200 \text{ mm}$$

Available Nozzle Neck Thickness = 7.6454 mm -> OK

Weld Size Calculations, Description: K3-104-D2A

Intermediate Calc. for nozzle/shell Welds	$T_{min} = 19.0000 \text{ mm}$
---	--------------------------------

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$$

$$= \max(0, (26.8 - 1.19 + 2 \cdot 37 \cdot 0.93 \cdot (1 \cdot 49 - 47.8)) \cdot 1509)$$

$$= 39867.99 \text{ kgf}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A4 - (W_i - Can/0.707)^2 \cdot fr2) \cdot Sv$$

$$= (42.6 + 0.93 - 0 \cdot 0.93) \cdot 1509$$

$$= 65665.16 \text{ kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$$

$$= (42.6 + 0 + 0.93 + (33.8)) \cdot 1509$$

$$= 116682.92 \text{ kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$$

$$= (42.6 + 0 + 0.93 + 0 + (33.8)) \cdot 1509$$

$$= 116682.92 \text{ kgf}$$

Strength of Connection Elements for Failure Path Analysis



Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw \\ &= (3.14/2.0) \cdot 125 \cdot 10 \cdot 0.49 \cdot 1407 \\ &= 13541. \text{ kgf} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\ &= (3.14 \cdot 44) \cdot (40 - 3) \cdot 0.7 \cdot 1407 \\ &= 50392. \text{ kgf} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng \\ &= (3.14/2.0) \cdot 125 \cdot (52 - 3) \cdot 0.74 \cdot 1509 \\ &= 107472. \text{ kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (13541 + 50392) = 63933 \text{ kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (13541 + 0 + 107472 + 0) = 121012 \text{ kgf} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (13541 + 107472 + 0) = 121012 \text{ kgf} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 63932	kgf, must exceed W = 39867	kgf	or W1 = 65665	kgf
Path 2-2 = 121012	kgf, must exceed W = 39867	kgf	or W2 = 116682	kgf
Path 3-3 = 121012	kgf, must exceed W = 39867	kgf	or W3 = 116682	kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 52.1 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 25.9133 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 981.8483 mm

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Noted issues for the current model:

Listed below are the known problem areas for the current design. If one or more of the design flags are turned on, please re-run the analysis. Some of these issues may be resolved when using updated input values.

**** Warning: One or more nozzles do not meet proper Spacing Requirements!**

Please review all reports carefully!

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